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MODIFIED DOUBLE STRANDED OLIGONUCLEOTIDES

Abstract

One aspect of the present invention relates to double-stranded RNA (dsRNA) agent capable of inhibiting the expression of a target gene. Other aspects of the invention relate to pharmaceutical compositions comprising these dsRNA molecules suitable for therapeutic use, and methods of inhibiting the expression of a target gene by administering these dsRNA molecules, e.g., for the treatment of various disease conditions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a 35 U.S.C. § 371 National Phase Entry Application of International Patent Application No. PCT/US2021/057016 filed on Oct. 28, 2021, which designated the U.S., and which claims benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/140,714 filed, Jan. 22, 2021, U.S. Provisional Application No. 63/146,115 filed Feb. 5, 2021, U.S. Provisional Application No. 63/148,991 filed, Feb. 12, 2021, U.S. Provisional Application No. 63/153,983 filed Feb. 26, 2021, U.S. Provisional Application No. 63/156,476 filed Mar. 4, 2021, U.S. Provisional Application No. 63/161,313 filed Mar. 15, 2021, U.S. Provisional Application No. 63/164,467 filed Mar. 22, 2021, U.S. Provisional Application No. 63/179,607 filed Apr. 26, 2021, and U.S. Provisional Application No. 63/181,748 filed Apr. 29, 2021, the contents of each of which are incorporated herein by reference in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 10, 2021, is named O51058-O99300WOPT_SL.txt and is 837,545 bytes in size.

FIELD OF THE INVENTION

[0003] The invention relates to dsRNA molecules having particular motifs that are advantageous for inhibition of target gene expression, as well dsRNA agent compositions, suitable for therapeutic use. Additionally, the invention provides methods of inhibiting the expression of a target gene by administering these dsRNA agents, e.g., for the treatment of various diseases.

BACKGROUND

[0004] RNA interference or “RNAi” is a term initially coined by Fire and co-workers to describe the observation that double-stranded RNAi (dsRNA) can block gene expression (Fire et al. (1998)

Nature 391, 806-811; Elbashir et al. (2001) Genes Dev. 15, 188-200). Short dsRNA directs gene-specific, post-transcriptional silencing in many organisms, including vertebrates, and has provided a new tool for studying gene function. RNAi is mediated by RNA-induced silencing complex (RISC), a sequence-specific, multi-component nuclease that destroys messenger RNAs homologous to the silencing trigger. RISC is known to contain short RNAs (approximately 22 nucleotides) derived from the double-stranded RNA trigger, but the protein components of this activity remained unknown.

[0005] There remains a need in the art for effective nucleotide or chemical motifs for dsRNA molecules, which are advantageous for inhibition of target gene expression. This invention is directed to that effort.

SUMMARY

[0006] This invention provides effective nucleotide or chemical motifs for dsRNA molecules, which are advantageous for inhibition of target gene expression, as well as RNAi compositions suitable for therapeutic use.

[0007] Inventors have discovered inter alia that double stranded RNA (dsRNA) molecules having a 2'-fluoro nucleotide at least at position 10 of the sense strand unexpectedly and surprisingly have improved in vitro potency, i.e., increased RNA interference (RNAi) activity. Accordingly, in one aspect provided herein is a double stranded RNA (dsRNA) molecule comprising a sense strand and an antisense strand, each strand independently having a length of 15 to 35 nucleotides, and wherein the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand.

[0008] It is noted that the sense strand can further comprises one or more, e.g., 1, 2, 3, 4 or 5 additional 2'-fluoro nucleotides. Accordingly, in some embodiments, the sense strand comprises 1, 2, 3, 4, or 5 additional 2'-fluoro nucleotides. The additional 2'-fluoro nucleotides can be located anywhere in the sense strand. Thus, in some embodiments, the sense strand further comprises a 2'-fluoro nucleotide at position 8, 9, 11 or 12, counting from 5'-end of the sense strand. For example, the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand. In other words, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, counting from 5'-end of the sense strand. In another example, the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, counting from 5'-end of the sense strand.

[0009] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand. In some other embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 8, 9 and 10, counting from 5'-end of the sense strand. In yet some other embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10, 11 and 12, counting from 5'-end of the sense strand.

[0010] In some embodiments of any one of the aspects, the sense strand does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.

[0011] The antisense strand of the dsRNA molecules described herein can comprise one or more 2'-deoxy, e.g., 2'-H nucleotides. For example, the antisense strand comprises 1, 2, 3, 4, 5, 6 or more 2'-deoxy nucleotides. In some embodiments, the antisense strand comprises 2, 3, 4, 5 or 6 2'-deoxy nucleotides. The 2'-deoxy nucleotides can be located anywhere in the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3, 4, 5 or 6 of positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 2 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand.

[0012] In some embodiments, the antisense can comprise one or more, e.g., 1, 2, 3, 4, 5 or more of 2'-fluoro nucleotides. For example, the antisense strand can comprise a 2'-fluoro nucleotide at position 14, counting from 5'-end of the antisense strand.

[0013] In some embodiments, the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand

[0014] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and 2'-fluoro nucleotide at position 14, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from 5'-end of the antisense strand.

[0015] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at position 14, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 12 and 14, counting from the 5'-end of the antisense strand, and the sense strand comprises a 2'-fluoro nucleotide at position 10 and a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.

[0016] In various embodiments, the dsRNA molecule has a double stranded (duplex) region of between 19 to 25 base pairs. For example, the dsRNA molecule has a duplex region of 20, 21, 22, 23 or 24 basepairs. In some particular embodiments, the dsRNA molecule has a double duplex) region of 20, 21 or 22 base pairs.

[0017] In some embodiments, the dsRNA molecule comprises a ligand. For example, the sense strand of the dsRNA molecule comprises a ligand. Exemplary ligands include, but are not limited to, ASGPR ligands and ligands comprising a lipophilic group.

[0018] The dsRNA molecule can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8 or more phosphorothioate linkages. The phosphorothioate linkages can be present only in one of the strands or in both strands of the dsRNA. For example, the sense strand can comprise 1, 2, 3 or 4 phosphorothioate linkages. In another non-limiting example, the antisense strand can comprise 1, 2, 3, 4, 5 or 6 phosphorothioate linkages. In some embodiments, the sense strand comprises 1, 2, 3 or 4 phosphorothioate linkages and the antisense independently comprises 1, 2, 3, 4, 5, or 6 phosphorothioate linkages. For example, the sense strand comprises 1 or 2 phosphorothioate linkages and the antisense strand comprises 1, 2, 3 or 4 phosphorothioate linkages.

[0019] In some embodiments, the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand, the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand and the antisense further comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3'-end of the antisense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand, and the antisense strand comprises phosphorothioate linkages and between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand, and between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 3'-end of the antisense strand.

[0020] In some embodiments, the remaining nucleotides in the dsRNA are 2'-OMe nucleotides. For example, all of the remaining nucleotides in the sense strand are 2'-OMe nucleotides. In other words, the sense strand solely comprises 2'-fluoro and 2'-OMe nucleotides.

[0021] It is understood that the antisense strand has sufficient complementarity to a target sequence to mediate RNA interference. In other words, the dsRNA molecules of the invention are capable of inhibiting the expression of a target gene.

[0022] In another aspect, the invention further provides a method for delivering the dsRNA molecule of the invention to a specific target in a subject by subcutaneous or intravenous administration. The invention further provides the dsRNA molecules of the invention for use in a method for delivering said agents to a specific target in a subject by subcutaneous or intravenous administration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] This patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0024] FIGS. 1A and 1B are graphs showing dsRNAs according to exemplary embodiments of the invention have improved in vitro potency relative to the parent dsRNA molecules when dosed at 10 nM (FIG. 1A) or at 1 nM (FIG. 1).

[0025] FIGS. 2A-2D are graphs showing that dsRNA molecules according to embodiments of the invention have improved in vivo efficacy compared to the parent molecules. Parent duplexes are AD-1181401(Sequence 1, FIG. 2A); AD-1181410(Sequence 2, FIG. 2B); AD-1181426 (Sequence 3, FIG. 2C); and AD-1181451(Sequence 4, FIG. 2D)

[0026] FIGS. 3A-3D are graphs showing dsRNAs according to exemplary embodiments of the invention and targeting different targets have improved in vitro potency relative to the parent dsRNA molecules when dosed at 1 nM (FIGS. 3A and 3B) or at 0.1 nM (FIG. 3C).

[0027] FIGS. 4A-4C are graphs showing that presence of a 2'-fluoro nucleotide at position 10 of the sense strand, counting from the 5'-end of the sense strand, enhances the RNAi efficacy of the dsRNA molecule compared to the parent.

[0028] FIGS. 5A and 5B are schematic representation of some exemplary designs of dsRNA molecules according to embodiments of the invention.

[0029] FIGS. 6A and 6B are graphs showing improved in vitro target knockdown (FIG. 6A) and log 2 activity (FIG. 6B) of exemplary dsRNA according to some embodiments of the disclosure relative to exemplary parent dsRNA molecules.

[0030] FIGS. 7A and 7B are graphs showing improved in vitro target knockdown of AGT (FIG. 6A) and log 2 activity (FIG. 6B) of exemplary dsRNA according to some embodiments of the disclosure relative to exemplary parent dsRNA molecules targeting AGT.

[0031] FIGS. 8A-8H are showing similar or improved metabolic stability of sense strand (FIGS. 8A-8D) and antisense strand (FIGS. 8E-8H) of exemplary dsRNA molecules in mouse liver homogenate (FIGS. 8A and 8E), rat liver homogenate (FIGS. 8B and 8F), rat brain homogenate (FIGS. 8C and 8G), and cynomolgus liver homogenate (FIGS. 8D and 8H). Parent duplexes are AD-1181401(TTR Seq 1); AD-1181410(TTR Seq 2); and AD-74210(F12).

[0032] FIGS. 9A and 9B are showing similar or improved metabolic stability of exemplary dsRNA molecules in mouse. Parent duplexes are AD-1181401(TTR Seq 1); AD-1181410(TTR Seq 2); and AD-74210(F12).

[0033] FIGS. 10A-10D are graphs showing that dsRNA molecules according to embodiments of the invention have improved in vivo efficacy and/or duration in non-human primates, mice (FIGS. 10A and 10B) and in cynomolgus monkeys (FIGS. 10C and 10D) compared to the parent molecules AD-74210(FIGS. 10A and 10C) and AD-75885(FIGS. 10B and 10D).

DETAILED DESCRIPTION

[0034] In one aspect, the invention provides a double-stranded RNA (dsRNA) agent capable of inhibiting expression of a target gene. Without limitations, the dsRNA agents of the invention can be substituted for the dsRNA molecules and can be used in RNA interference based gene silencing techniques, including, but not limited to, in vitro or in vivo applications.

[0035] Generally, the dsRNA molecule comprises a sense strand (also referred to as passenger strand) and an antisense strand (also referred to as guide strand). Each strand of the dsRNA molecule can range from 15-35 nucleotides in length. For example, each strand can be between, 17-35 nucleotides in length, 17-30 nucleotides in length, 25-35 nucleotides in length, 27-30 nucleotides in length, 17-23 nucleotides in length, 17-21 nucleotides in length, 17-19 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length. Without limitations, the sense and antisense strands can be equal length or unequal length. For example, the sense strand and the antisense strand independently have a length of 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides.

[0036] In some embodiments, the antisense strand is of length 15-35 nucleotides. In some embodiments, the antisense strand is 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the antisense strand can be 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the antisense strand is 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the antisense strand is 21, 22, 23, 24 or 25 nucleotides in length. In some particular embodiments, the antisense strand is 22, 23 or 24 nucleotides in length. For example, the antisense strand is 23 nucleotides in length.

[0037] Similar to the antisense strand, the sense strand can be, in some embodiments, 15-35 nucleotides in length. In some embodiments, the sense strand is 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the sense strand can be 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the sense strand is 17, 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the sense strand is 19, 20, 21, 22 or 23 nucleotides in length. In some particular embodiments, the sense strand is 20, 21 or 22 nucleotides in length. For example, the sense strand is 21 nucleotides in length.

[0038] In some embodiments, the sense strand can be 15-35 nucleotides in length, and the antisense strand can be independent from the sense strand, 15-35 nucleotides in length. In some embodiments, the sense strand is 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length, and the antisense strand is independently 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the sense and the antisense strand can be independently 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the sense strand and the antisense strand are independently 17, 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the sense strand is 19, 20, 21, 22 or 23 nucleotides in length and the antisense strand is 21, 22, 23, 24 or 25 nucleotides in length. In some particular embodiments, the sense strand is 20, 21 or 22 nucleotides in length and the antisense strand is 22, 23 or 24 nucleotides in length. For example, the sense strand is 21 nucleotides in length and the antisense strand is 23 nucleotides in length.

[0039] The sense strand and antisense strand typically form a double-stranded or duplex region. Without limitations, the duplex region of a dsRNA agent described herein can be 12-35 nucleotide (or base) pairs in length. For example, the duplex region can be between 14-35 nucleotide pairs in length, 17-30 nucleotide pairs in length, 25-35 nucleotides in length, 27-35 nucleotide pairs in length, 17-23 nucleotide pairs in length, 17-21 nucleotide pairs in length, 17-19 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotide

pairs in length. In some embodiments, the duplex region is 18, 19, 20, 21, 22, 23, 24 or 25 nucleotide pairs in length. For example, the duplex region is 19, 20, 21, 22 or 23 nucleotide pairs in length. In some embodiments, the duplex region is 20, 21 or 22 nucleotide pairs in length. For example, the dsRNA molecule has a duplex region of 21 base pairs.

[0040] As described herein, the dsRNA molecule of the invention can further comprise at least one, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more 2'-deoxy, e.g., 2'-H nucleotides. For example, the dsRNA can comprise 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 2'-deoxy, e.g., 2'-H nucleotides. The 2'-deoxy nucleotide may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. In one non-limiting example, the sense strand does not comprise a 2'-deoxy nucleotide at position 11, counting from 5'-end of the sense strand.

[0041] In some embodiments, the antisense strand the antisense strand comprises 1, 2, 3, 4, 5 or 6 of 2'-deoxy nucleotides. For example, antisense strand can comprise 2, 3, 4, 5 or 6 of 2'-deoxy nucleotides. The 2'-deoxy nucleotides can be located anywhere in the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3, 4, 5 or 6 of positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand. In one non-limiting example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3 or 4 of positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand.

[0042] In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 5, 7 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14, counting, from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand

[0043] In some embodiments, the antisense comprises a 2'-deoxy nucleotide at position 2 or 12, counting from 5'-end of the antisense strand. For example, the antisense comprises a 2'-deoxy nucleotide at position 12, counting from 5'-end of the antisense strand.

[0044] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. In another example, the sense strand comprises a 2'-fluoro nucleotide at positions 8, 9 and 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.

[0045] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10, 11 and 12, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.

[0046] In some embodiments of any one of the aspects, the sense strand does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at at least one e.g., 1, 2 or 3 of positions 9, 10 and 11 but does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at position 10 and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand.

[0047] For example, the sense strand comprises a 2'-fluoro nucleotide position 10 and a 2'-OMe

nucleotide at position 7, counting from the 5'-end of the sense strand.

[0048] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0049] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0050] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0051] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the sense strand are 2'-OMe nucleotides.

2'-fluoro modifications (antisense strand)

[0052] The dsRNA molecules of the invention comprise one or more (e.g., one, two, three, four, five, six, seven, eight, nine, ten or more).sub.2'-fluoro nucleotides. Without limitations, the 2'-fluoro nucleotides all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least one, 2'-fluoro nucleotide. The 2'-fluoro modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the 2'-fluoro modification can occur on every nucleotide on the sense strand and/or antisense strand; each 2'-fluoro modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both 2'-fluoro modifications in an alternating pattern. The alternating pattern of the 2'-fluoro modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the 2'-fluoro modifications on the sense strand can have a shift relative to the alternating pattern of the 2'-fluoro modifications on the antisense strand.

[0053] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more).sub.2'-fluoro nucleotides. Without limitations, a 2'-fluoro modification in the antisense strand can be present at any positions.

[0054] In some embodiments, the antisense comprises one or more, e.g., 1, 2, 3, 4, 5 or more 2'-fluoro nucleotides. For example, the antisense strand comprises 1, 2, 3, 4, or 5 more 2'-fluoro nucleotides. In some embodiments, the antisense strand comprises 1, 2 or 3 2'-fluoro nucleotides. For example, the antisense strand comprises a single 2'-fluoro nucleotide. It is noted that a 2'-fluoro nucleotide can be located anywhere in the antisense strand. For example, a 2'-fluoro nucleotide can be at position 2 or 14, counting from 5'-end, of the antisense strand. In some embodiments, the antisense comprises a 2'-fluoro nucleotide at position 14, counting from 5'-end, of the antisense strand.

[0055] In some embodiments, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides at positions 5, 7 and 12, counting from 5'-end of the antisense strand. In a further example, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand.

[0056] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand further comprises a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and

a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand.

[0057] In some embodiments, the antisense strand comprises a nucleotide other than a 2'-deoxy nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, antisense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

[0058] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-fluoro at position 16, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

[0059] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12 and a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0060] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12 and a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0061] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0062] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0063] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 16, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0064] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 16, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0065] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0066] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0067] It is noted the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand and/or the antisense strand can be unmodified or modified nucleotides. Accordingly, in some

embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand are unmodified or modified nucleotides. For example, the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand can be modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, optionally provided no more than one modified nucleotide is an 2'-O—C10-30aliphatic group.

[0068] In some embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand are unmodified or modified nucleotides. For example, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand can be modified nucleotides. In some embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand can be selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

[0069] In some embodiments, the remaining nucleotides in the sense strand and/or the antisense strand are 2'-OMe nucleotides.

[0070] As described herein, the dsRNA agent can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides comprising a modified sugar. By a “modified sugar” is meant a sugar other than 2'-deoxy (i.e, 2'-H), 2'-OH, 2'-F or 2'-OMe ribose sugar. Some exemplary nucleotides comprising a modified sugar are locked nucleic acid (LNA), HNA, CeNA, 2'-methoxyethyl, 2'-O-allyl, 2'-C-allyl, 2'-O—N-methylacetamido (2'-O-NMA), a 2'-O-dimethylaminoethoxyethyl(2'-O-DMAEOE), 2'-O-aminopropyl(2'-O-AP), and 2'-ara-F. Accordingly, in some embodiments, the dsRNA agent can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides independently selected from the group consisting of acyclic nucleotides, locked nucleic acid (LNA), HNA, CeNA, 2'-methoxyethyl, 2'-O-allyl, 2'-C-allyl, 2'-O—N-methylacetamido (2'-O-NMA), a 2'-O-dimethylaminoethoxyethyl(2'-O-DMAEOE), 2'-O-aminopropyl(2'-O-AP), and 2'-ara-F. A nucleotide comprising modified sugar can be present anywherein the dsRNA molecule. For example, a nucleotide comprising a modified sugar can be present in the sense strand or a nucleotide comprising a modified sugar can be present in the antisense strand. When two or more nucleotides comprising a modified sugar are present in the dsRNA molecule, they can all be in the sense strand, antisense strand or both in the sense and antisense strands.

[0071] In some embodiments, an unmodified nucleotide is a 2'-OH nucleotide comprising an unmodified nucleobase, i.e., adenine, guanine, cytosine, or uracil.

[0072] In some embodiments, the dsRNA can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides comprising a non-natural nucleobase. By a “non-natural nucleobase” is meant a nucleobase other than adenine, guanine, cytosine, uracil, or thymine. Exemplary non-natural nucleobases include, but are not limited to, inosine, xanthine, hypoxanthine, nubularine, isoguanisine, tubercidine, and substituted or modified analogs of adenine, guanine, cytosine and uracil, such as 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 5-halouracil, 5-(2-aminopropyl) uracil, 5-amino allyl uracil, 8-halo, amino, thiol, thioalkyl, hydroxyl and other 8-substituted adenines and guanines, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine, 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine, dihydrouracil, 3-deaza-5-azacytosine, 2-aminopurine, 5-alkyluracil, 7-alkylguanine, 5-alkyl cytosine, 7-deazaadenine, N6, N6-dimethyladenine, 2,6-diaminopurine, 5-amino-allyl-uracil, N3-methyluracil, substituted 1,2,4-triazoles, 2-pyridinone, 5-nitroindole, 3-nitropyrrole, 5-methoxyuracil, uracil-5-oxyacetic acid, 5-methoxycarbonylmethyluracil, 5-methyl-2-thiouracil, 5-methoxycarbonylmethyl-2-thiouracil, 5-methylaminomethyl-2-thiouracil, 3-(3-amino-3carboxypropyl) uracil, 3-methylcytosine, 5-methylcytosine, N4-acetyl cytosine, 2-thiocytosine, N6-methyladenine, N6-isopentyladenine, 2-methylthio-N6-isopentenyladenine, N-methylguanines,

or O-alkylated bases. Further purines and pyrimidines include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in the Concise Encyclopedia of Polymer Science and Engineering, pages 858-859, Kroschwitz, J. I., ed. John Wiley & Sons, 1990, and those disclosed by Englisch et al., Angewandte Chemie, International Edition, 1991, 30, 613.

[0073] In some embodiments, the non-natural nucleobase can be selected from the group consisting of inosine, xanthine, hypoxanthine, nubularine, isoguanisine, tubercidine, 2-(halo) adenine, 2-(alkyl) adenine, 2-(propyl) adenine, 2-(amino) adenine, 2-(aminoalkyl) adenine, 2-(aminopropyl) adenine, 2-(methylthio)-N^{sup}.6-(isopentenyl) adenine, 6-(alkyl) adenine, 6-(methyl) adenine, 7-(deaza) adenine, 8-(alkenyl) adenine, 8-(alkyl) adenine, 8-(alkynyl) adenine, 8-(amino) adenine, 8-(halo) adenine, 8-(hydroxyl) adenine, 8-(thioalkyl) adenine, 8-(thiol) adenine, N^{sup}.6-(isopentyl) adenine, N^{sup}.6-(methyl) adenine, N^{sup}.6, N^{sup}.6-(dimethyl) adenine, 2-(alkyl) guanine, 2-(propyl) guanine, 6-(alkyl) guanine, 6-(methyl) guanine, 7-(alkyl) guanine, 7-(methyl) guanine, 7-(deaza) guanine, 8-(alkyl) guanine, 8-(alkenyl) guanine, 8-(alkynyl) guanine, 8-(amino) guanine, 8-(halo) guanine, 8-(hydroxyl) guanine, 8-(thioalkyl) guanine, 8-(thiol) guanine, N-(methyl) guanine, 2-(thio) cytosine, 3-(deaza)-5-(aza) cytosine, 3-(alkyl) cytosine, 3-(methyl) cytosine, 5-(alkyl) cytosine, 5-(alkynyl) cytosine, 5-(halo) cytosine, 5-(methyl) cytosine, 5-(propynyl) cytosine, 5-(propynyl) cytosine, 5-(trifluoromethyl) cytosine, 6-(azo) cytosine, N^{sup}.4-(acetyl) cytosine, 3-(3-amino-3-carboxypropyl) uracil, 2-(thio) uracil, 5-(methyl)-2-(thio) uracil, 5-(methylaminomethyl)-2-(thio) uracil, 4-(thio) uracil, 5-(methyl)-4-(thio) uracil, 5-(methylaminomethyl)-4-(thio) uracil, 5-(methyl)-2,4-(dithio) uracil, 5-(methylaminomethyl)-2,4-(dithio) uracil, 5-(2-aminopropyl) uracil, 5-(alkyl) uracil, 5-(alkynyl) uracil, 5-(allylamino) uracil, 5-(aminoallyl) uracil, 5-(aminoalkyl) uracil, 5-(guanidiniumalkyl) uracil, 5-(1,3-diazole-1-alkyl) uracil, 5-(cyanoalkyl) uracil, 5-(dialkylaminoalkyl) uracil, 5-(dimethylaminoalkyl) uracil, 5-(halo) uracil, 5-(methoxy) uracil, uracil-5-oxyacetic acid, 5-(methoxycarbonylmethyl)-2-(thio) uracil, 5-(methoxycarbonyl-methyl) uracil, 5-(propynyl) uracil, 5-(propynyl) uracil, 5-(trifluoromethyl) uracil, 6-(azo) uracil, dihydrouracil, N^{sup}.3-(methyl) uracil, 5-uracil (i.e., pseudouracil), 2-(thio) pseudouracil, 4-(thio) pseudouracil, 2,4-(dithio) pseudouracil, 5-(alkyl) pseudouracil, 5-(methyl) pseudouracil, 5-(alkyl)-2-(thio) pseudouracil, 5-(methyl)-2-(thio) pseudouracil, 5-(alkyl)-4-(thio) pseudouracil, 5-(methyl)-4-(thio) pseudouracil, 5-(alkyl)-2,4-(dithio) pseudouracil, 5-(methyl)-2,4-(dithio) pseudouracil, 1-substituted pseudouracil, 1-substituted 2-(thio)-pseudouracil, 1-substituted 4-(thio) pseudouracil, 1-substituted 2,4-(dithio) pseudouracil, 1-(aminocarbonylethylenyl)-pseudouracil, 1-(aminocarbonylethylenyl)-2-(thio)-pseudouracil, 1-(aminocarbonylethylenyl)-4-(thio) pseudouracil, 1-(aminocarbonylethylenyl)-2,4-(dithio) pseudouracil, 1-(aminoalkylaminocarbonylethylenyl)-pseudouracil, 1-(aminoalkylamino-carbonylethylenyl)-2-(thio)-pseudouracil, 1-(aminoalkylaminocarbonylethylenyl)-4-(thio) pseudouracil, 1-(aminoalkylaminocarbonylethylenyl)-2,4-(dithio) pseudouracil, 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 1,3-(diaz)-2-(oxo)-phenthiazin-1-yl, 1-(aza)-2-(thio)-3-(aza)-phenthiazin-1-yl, 7-substituted 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-substituted 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 7-substituted 1,3-(diaz)-2-(oxo)-phenthiazin-1-yl, 7-substituted 1-(aza)-2-(thio)-3-(aza)-phenthiazin-1-yl, 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenthiazin-1-yl, 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenthiazin-1-yl, 7-(guanidiniumalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 7-(guanidiniumalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenthiazin-1-yl, 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenthiazin-1-yl, 1,3,5-(triaz)-2,6-(diox)-naphthalene, inosine, xanthine, hypoxanthine, nubularine, tubercidine, isoguanisine, inosinyl, 2-aza-inosinyl, 7-deaza-inosinyl, nitroimidazolyl, nitropyrazolyl, nitrobenzimidazolyl, nitroindazolyl, aminoindolyl, pyrrolopyrimidinyl, 3-(methyl) isocarbostyryl, 5-(methyl) isocarbostyryl, 3-(methyl)-7-(propynyl) isocarbostyryl, 7-(aza) indolyl, 6-(methyl)-7-(aza) indolyl,

imidizopyridinyl, 9-(methyl)-imidizopyridinyl, pyrrolopyrizinyl, isocarbostyryl, 7-(propynyl) isocarbostyryl, propynyl-7-(aza) indolyl, 2,4,5-(trimethyl)phenyl, 4-(methyl) indolyl, 4,6-(dimethyl) indolyl, phenyl, naphthalenyl, anthracenyl, phenanthracenyl, pyrenyl, stilbenyl, tetracenyl, pentacenyl, difluorotolyl, 4-(fluoro)-6-(methyl)benzimidazole, 4-(methyl)benzimidazole, 6-(azo) thymine, 2-pyridinone, 5-nitroindole, 3-nitropyrrole, 6-(aza)pyrimidine, 2-(amino) purine, 2,6-(diamino) purine, 5-substituted pyrimidines, N.sup.2-substituted purines, N.sup.6-substituted purines, O.sup.6-substituted purines, substituted 1,2,4-triazoles, and any O-alkylated or N-alkylated derivatives thereof.

[0074] A nucleotide comprising a non-natural nucleobase can be present anywherein the dsRNA molecule. For example, a nucleotide comprising a non-natural nucleobase can be present in the sense strand or a nucleotide comprising a non-natural nucleobase can be present in the antisense strand. When two or more nucleotides comprising a non-natural nucleobase are present in the dsRNA molecule, they can all be in the sense strand, antisense strand or both in the sense and antisense strands.

[0075] The dsRNA molecule of the invention can further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand and/or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand.

[0076] In some embodiments, the dsRNA molecule comprises the phosphorothioate or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region comprises two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. Preferably, these terminal three nucleotides may be at the 3'-end of the antisense strand.

[0077] In some embodiments, the sense strand of the dsRNA molecule comprises 1-10 blocks of two to ten phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said sense strand is paired with an antisense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0078] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of two phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the

oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0079] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of three phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0080] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of four phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 or 14 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0081] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of five phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0082] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of six phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0083] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of seven phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7 or 8 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0084] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of eight phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5 or 6 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0085] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of nine phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3 or 4 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate,

methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0086] In some embodiments, the dsRNA molecule of the invention further comprises one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the termini position(s) of the sense and/or antisense strand. For example, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage at one end or both ends of the sense and/or antisense strand.

[0087] In some embodiments, the dsRNA molecule of the invention comprises one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the internal region of the duplex of each of the sense and/or antisense strand. For example, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides may be linked through phosphorothioate methylphosphonate internucleotide linkage at position 8-16 of the duplex region counting from the 5'-end of the sense strand; the dsRNA molecule can optionally further comprise one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the termini position(s).

[0088] In some embodiments, the dsRNA molecule of the invention further comprises one to five phosphorothioate or methylphosphonate internucleotide linkage modification(s) within position 1-5 and one to five phosphorothioate or methylphosphonate internucleotide linkage modification(s) within the last 3 positions of the sense strand (counting from the 5'-end), and one to five phosphorothioate or methylphosphonate internucleotide linkage modification at positions 1 and 2 and one to five phosphorothioate or methylphosphonate internucleotide linkage modification within the last six positions of the antisense strand (counting from the 5'-end).

[0089] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification within position 1-5 and one phosphorothioate or methylphosphonate internucleotide linkage modification within the last six positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate or methylphosphonate internucleotide linkage modifications within the last six the last six positions of the antisense strand (counting from the 5'-end).

[0090] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and one phosphorothioate internucleotide linkage modification within the last six positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within the last six positions of the antisense strand (counting from the 5'-end).

[0091] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and two phosphorothioate internucleotide linkage modifications within the last four positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within the last six positions of the antisense strand (counting from the 5'-end).

[0092] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and two phosphorothioate internucleotide linkage modifications within the last four positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and one phosphorothioate internucleotide linkage modification within the last six positions of the antisense strand (counting from the 5'-end).

[0093] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification within position 1-5 and one phosphorothioate internucleotide linkage modification within the last four positions of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications

antisense strand (counting from the 5'-end).

[0103] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification at position 1, and one phosphorothioate internucleotide linkage modification at position 21 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications at positions 21 and 22 the antisense strand (counting from the 5'-end).

[0104] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications at position 1 and 2, and two phosphorothioate internucleotide linkage modifications at position 22 and 23 of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and one phosphorothioate internucleotide linkage modification at position 21 of the antisense strand (counting from the 5'-end).

[0105] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification at position 1, and one phosphorothioate internucleotide linkage modification at position 21 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications at positions 22 and 23 the antisense strand (counting from the 5'-end).

[0106] In some embodiments, the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand.

[0107] In some embodiments, the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand. For example, the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand.

[0108] In some embodiments, the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand. For example, the antisense strand comprises phosphorothioate linkages between nucleotides n and $n-1$, and between nucleotides $n-1$ and $n-2$, where n is length of the antisense strand, i.e., number of nucleotides in the antisense strand. In other words, the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 3'-end of the antisense strand.

[0109] In some embodiments, the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3'-end of the antisense strand. For example, the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand and between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 3'-end of the antisense strand.

[0110] In some embodiments, the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand and the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand, and the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand.

[0111] In some embodiments, the sense strand comprises at least two phosphorothioate

internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand and the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3'-end of the antisense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand, and the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 3'-end of the antisense strand.

[0112] In some embodiments, compound of the invention comprises a pattern of backbone chiral centers. In some embodiments, a common pattern of backbone chiral centers comprises at least 5 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 6 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 7 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 8 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 9 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 10 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 11 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 12 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 13 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 14 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 15 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 16 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 17 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 18 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 19 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 8 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 7 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 6 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 5 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 4 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 3 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 2 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 1 internucleotidic linkages in the Rp configuration. In some embodiments, a common pattern of backbone chiral centers comprises no more than 8 internucleotidic linkages which are not chiral (as a non-limiting example, a phosphodiester). In some embodiments, a common pattern of backbone chiral centers comprises no more than 7 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 5 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 4

internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 3 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 2 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises no more than 1 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 10 internucleotidic linkages in the Sp configuration, and no more than 8 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 11 internucleotidic linkages in the Sp configuration, and no more than 7 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 12 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 13 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 14 internucleotidic linkages in the Sp configuration, and no more than 5 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 15 internucleotidic linkages in the Sp configuration, and no more than 4 internucleotidic linkages which are not chiral. In some embodiments, the internucleotidic linkages in the Sp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages in the Rp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages which are not chiral are optionally contiguous or not contiguous.

[0113] In some embodiments, compound of the invention comprises a block is a stereochemistry block. In some embodiments, a block is an Rp block in that each internucleotidic linkage of the block is Rp. In some embodiments, a 5'-block is an Rp block. In some embodiments, a 3'-block is an Rp block. In some embodiments, a block is an Sp block in that each internucleotidic linkage of the block is Sp. In some embodiments, a 5'-block is an Sp block. In some embodiments, a 3'-block is an Sp block. In some embodiments, provided oligonucleotides comprise both Rp and Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Rp but no Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Sp but no Rp blocks. In some embodiments, provided oligonucleotides comprise one or more PO blocks wherein each internucleotidic linkage in a natural phosphate linkage.

[0114] In some embodiments, compound of the invention comprises a 5'-block is an Sp block wherein each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block comprises 4 or more nucleoside units. In some embodiments, a 5'-block comprises 5 or more nucleoside units. In some embodiments, a 5'-block comprises 6 or more nucleoside units. In some embodiments, a 5'-block comprises 7 or more nucleoside units. In some embodiments, a 3'-block is an Sp block wherein each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block comprises 4 or more nucleoside units. In some embodiments, a 3'-block comprises 5 or more nucleoside units. In some embodiments, a 3'-block comprises 6 or more nucleoside units. In some embodiments, a 3'-block comprises 7 or more nucleoside units.

[0115] In some embodiments, compound of the invention comprises a type of nucleoside in a

region or an oligonucleotide is followed by a specific type of internucleotidic linkage, e.g., natural phosphate linkage, modified internucleotidic linkage, Rp chiral internucleotidic linkage, Sp chiral internucleotidic linkage, etc. In some embodiments, A is followed by Sp. In some embodiments, A is followed by Rp. In some embodiments, A is followed by natural phosphate linkage (PO). In some embodiments, U is followed by Sp. In some embodiments, U is followed by Rp. In some embodiments, U is followed by natural phosphate linkage (PO). In some embodiments, C is followed by Sp. In some embodiments, C is followed by Rp. In some embodiments, C is followed by natural phosphate linkage (PO). In some embodiments, G is followed by Sp. In some embodiments, G is followed by Rp. In some embodiments, G is followed by natural phosphate linkage (PO). In some embodiments, C and U are followed by Sp. In some embodiments, C and U are followed by Rp. In some embodiments, C and U are followed by natural phosphate linkage (PO). In some embodiments, A and G are followed by Sp. In some embodiments, A and G are followed by Rp.

[0116] Various publications describe multimeric siRNAs which can all be used with the dsRNA of the invention. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887 and WO2011/031520 which are hereby incorporated by their entirety.

Ligands

[0117] A wide variety of entities can be coupled to the dsRNA agents described herein. Preferred moieties are ligands, which are coupled, preferably covalently, either directly or indirectly via an intervening tether. Generally, a ligand alters the distribution, targeting or lifetime of the molecule, e.g., a dsRNA described herein, into which it is incorporated. In some embodiments a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, receptor e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. Ligands providing enhanced affinity for a selected target are also termed targeting ligands herein.

[0118] Some ligands can have endosomolytic properties. The endosomolytic ligands promote the lysis of the endosome and/or transport of the composition of the invention, or its components, from the endosome to the cytoplasm of the cell. The endosomolytic ligand may be a polyanionic peptide or peptidomimetic which shows pH-dependent membrane activity and fusogenicity. In some embodiments, the endosomolytic ligand assumes its active conformation at endosomal pH. The “active” conformation is that conformation in which the endosomolytic ligand promotes lysis of the endosome and/or transport of the composition of the invention, or its components, from the endosome to the cytoplasm of the cell. Exemplary endosomolytic ligands include the GALA peptide (Subbarao et al., *Biochemistry*, 1987, 26:2964-2972, which is incorporated by reference in its entirety), the EALA peptide (Vogel et al., *J. Am. Chem. Soc.*, 1996, 118:1581-1586, which is incorporated by reference in its entirety), and their derivatives (Turk et al., *Biochem. Biophys. Acta*, 2002, 1559:56-68, which is incorporated by reference in its entirety). In some embodiments, the endosomolytic component may contain a chemical group (e.g., an amino acid) which will undergo a change in charge or protonation in response to a change in pH. The endosomolytic component may be linear or branched.

[0119] Ligands can improve transport, hybridization, and specificity properties and can also improve nuclease resistance of the resultant natural or modified oligoribonucleotide, or a polymeric molecule comprising any combination of monomers described herein and/or natural or modified ribonucleotides.

[0120] Ligands in general can include therapeutic modifiers, e.g., for enhancing uptake; diagnostic compounds or reporter groups e.g., for monitoring distribution; cross-linking agents; and nuclease-resistance conferring moieties. General examples include lipids, steroids, vitamins, sugars, proteins, peptides, polyamines, and peptide mimics.

[0121] Ligands can include a naturally occurring substance, such as a protein (e.g., human serum

albumin (HSA), low-density lipoprotein (LDL), high-density lipoprotein (HDL), or globulin); a carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin or hyaluronic acid); or a lipid. The ligand may also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid, an oligonucleotide (e.g. an aptamer). Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic acid anhydride copolymer, poly(L-lactide-co-glycolide) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl) methacrylamide copolymer (HMPA), polyethylene glycol (PEG), polyvinyl alcohol (PVA), polyurethane, poly(2-ethylacrylic acid), N-isopropylacrylamide polymers, or polyphosphazine. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, protamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an alpha helical peptide.

[0122] Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, nanobody, or portion of an antibody of nanobody that binds to a specified cell type such as a kidney cell or a cell of the blood-brain barrier.

[0123] A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, glycosylated polyamino acids, multivalent galactose, transferrin-targeting group, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, biotin, an RGD peptide, an RGD peptide mimetic or an aptamer.

[0124] Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralen, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases or a chelating agent (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis —O (hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl) lithocholic acid, O3-(oleoyl) cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), MPEG, [MPEG].sub.2, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu³⁺ complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

[0125] Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a cancer cell, endothelial cell, or bone cell. Ligands may also include hormones and hormone receptors. They can also include non-peptide species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, or aptamers. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF- κ B.

[0126] The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, and/or intermediate filaments. The drug can be, for example, taxon, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

[0127] The ligand can increase the uptake of the dsRNA into the cell by activating an inflammatory response, for example. Exemplary ligands that would have such an effect include tumor necrosis factor alpha (TNF-alpha), interleukin-1 beta, or gamma interferon.

[0128] In some embodiments, the ligand is a lipid or lipid-based molecule. Such a lipid or lipid-

based molecule preferably binds a serum protein, e.g., human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the liver, including parenchymal cells of the liver. Other molecules that can bind HSA can also be used as ligands. For example, naproxen or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, and/or (c) can be used to adjust binding to a serum protein, e.g., HSA. A lipid based ligand can be used to modulate, e.g., control the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.

[0129] In a preferred embodiment, the lipid based ligand binds HSA. Preferably, it binds HSA with a sufficient affinity such that the conjugate will be preferably distributed to a non-kidney tissue. However, it is preferred that the affinity not be so strong that the HSA-ligand binding cannot be reversed.

[0130] In another preferred embodiment, the lipid based ligand binds HSA weakly or not at all, such that the conjugate will be preferably distributed to the kidney. Other moieties that target to kidney cells can also be used in place of or in addition to the lipid based ligand.

[0131] In some embodiments, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells.

Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include B vitamins, e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by cancer cells. Also included are HAS, low density lipoprotein (LDL) and high-density lipoprotein (HDL).

[0132] In another aspect, the ligand is a cell-permeation agent, preferably a helical cell-permeation agent. Preferably, the agent is amphipathic. An exemplary agent is a peptide such as tat or antennapedia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudo-peptide linkages, and use of D-amino acids. The helical agent is preferably an alpha-helical agent, which preferably has a lipophilic and a lipophobic phase.

[0133] The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long. A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or cross-linked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence

AAVALLPAVLLALLAP (SEQ ID NO: 1). An RFGF analogue (e.g., amino acid sequence AALLPVLLAAP (SEQ ID NO: 2)) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRRPPQ (SEQ ID NO: 3)) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO: 4)) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lam et al., Nature, 354:82-94, 1991, which is incorporated by reference in its entirety). Preferably the peptide or peptidomimetic tethered to an iRNA agent via an incorporated monomer unit is a cell targeting peptide such as an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to

about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized. An RGD peptide moiety can be used to target a tumor cell, such as an endothelial tumor cell or a breast cancer tumor cell (Zitzmann et al., Cancer Res., 62:5139-43, 2002, which is incorporated by reference in its entirety). An RGD peptide can facilitate targeting of an iRNA agent to tumors of a variety of other tissues, including the lung, kidney, spleen, or liver (Aoki et al., Cancer Gene Therapy 8:783-787, 2001, which is incorporated by reference in its entirety). Preferably, the RGD peptide will facilitate targeting of an iRNA agent to the kidney. The RGD peptide can be linear or cyclic, and can be modified, e.g., glycosylated or methylated to facilitate targeting to specific tissues. For example, a glycosylated RGD peptide can deliver an iRNA agent to a tumor cell expressing $\alpha\beta$.sub.3 (Haubner et al., Jour. Nucl. Med., 42:326-336, 2001, which is incorporated by reference in its entirety). Peptides that target markers enriched in proliferating cells can be used. For example, RGD containing peptides and peptidomimetics can target cancer cells, in particular cells that exhibit an integrin. Thus, one could use RGD peptides, cyclic peptides containing RGD, RGD peptides that include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand. Generally, such ligands can be used to control proliferating cells and angiogenesis. Preferred conjugates of this type ligands that targets PECAM-1, VEGF, or other cancer gene, e.g., a cancer gene described herein.

[0134] A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, an α -helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α -defensin, β -defensin or bactenecin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., Nucl. Acids Res. 31:2717-2724, 2003, which is incorporated by reference in its entirety).

[0135] In some embodiments, a targeting peptide can be an amphipathic α -helical peptide. Exemplary amphipathic α -helical peptides include, but are not limited to, cecropins, lycotoxins, paradaxins, buforin, CPF, bombinin-like peptide (BLP), cathelicidins, ceratotoxins, *S. clava* peptides, hagfish intestinal antimicrobial peptides (HFIAPs), magainines, brevinins-2, dermaseptins, melittins, pleurocidin, H.sub.2A peptides, *Xenopus* peptides, esculentinis-1, and caerins. A number of factors will preferably be considered to maintain the integrity of helix stability. For example, a maximum number of helix stabilization residues will be utilized (e.g., leu, ala, or lys), and a minimum number of helix destabilization residues will be utilized (e.g., proline, or cyclic monomeric units. The capping residue will be considered (for example Gly is an exemplary N-capping residue and/or C-terminal amidation can be used to provide an extra H-bond to stabilize the helix. Formation of salt bridges between residues with opposite charges, separated by $i\pm 3$, or $i\pm 4$ positions can provide stability. For example, cationic residues such as lysine, arginine, homo-arginine, ornithine or histidine can form salt bridges with the anionic residues glutamate or aspartate.

[0136] Peptide and peptidomimetic ligands include those having naturally occurring or modified peptides, e.g., D or L peptides; α , β , or γ peptides; N-methyl peptides; azapeptides; peptides having one or more amide, i.e., peptide, linkages replaced with one or more urea, thiourea, carbamate, or sulfonyl urea linkages; or cyclic peptides.

[0137] The targeting ligand can be any ligand that is capable of targeting a specific receptor. Examples are: folate, GalNAc, galactose, mannose, mannose-6P, clusters of sugars such as GalNAc cluster, mannose cluster, galactose cluster, or an aptamer. A cluster is a combination of two or more sugar units. The targeting ligands also include integrin receptor ligands, Chemokine receptor ligands, transferrin, biotin, serotonin receptor ligands, PSMA, endothelin, GCPII, somatostatin,

LDL and HDL ligands. The ligands can also be based on nucleic acid, e.g., an aptamer. The aptamer can be unmodified or have any combination of modifications disclosed herein.

[0138] Endosomal release agents include imidazoles, poly or oligoimidazoles, PEIs, peptides, fusogenic peptides, polycarboxylates, polycations, masked oligo or poly cations or anions, acetals, polyacetals, ketals/polyketals, orthoesters, polymers with masked or unmasked cationic or anionic charges, dendrimers with masked or unmasked cationic or anionic charges.

[0139] PK modulator stands for pharmacokinetic modulator. PK modulator include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, PEG, vitamins etc. Exemplary PK modulator include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin etc. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g. oligonucleotides of about 5 bases, 10 bases, 15 bases or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands).

[0140] In addition, aptamers that bind serum components (e.g. serum proteins) are also amenable to the present invention as PK modulating ligands.

[0141] Other ligand conjugates amenable to the invention are described in U.S. patent application Ser. No. 10/916,185, filed Aug. 10, 2004; U.S. Ser. No. 10/946,873, filed Sep. 21, 2004; USSN: U.S. Ser. No. 10/833,934, filed Aug. 3, 2007; USSN: U.S. Ser. No. 11/115,989 filed Apr. 27, 2005 and USSN: U.S. Ser. No. 11/944,227 filed Nov. 21, 2007, which are incorporated by reference in their entireties for all purposes.

[0142] In some embodiments, the dsRNA molecule can comprise two or more, e.g., 2, 3, 4 or 5 ligands. When two or more ligands are present, the ligands can all have same properties, all have different properties or some ligands have the same properties while others have different properties. For example, a ligand can have targeting properties, have endosomolytic activity or have PK modulating properties. In a preferred embodiment, all the ligands have different properties.

[0143] In some embodiments the dsRNA molecule comprises two ligands. For example, the sense strand of the dsRNA molecule comprises a first ligand attached at the 3'-end of the sense strand and a second ligand attached at the 5'-end of the sense strand. In some embodiments, the dsRNA molecule comprises two ligands linked to the sense strand, where the first ligand comprises an inverted abasic nucleotide (i.e., an abasic nucleotide linked via 3'->3' linkage) and the second ligand comprises an ASGPR ligand.

[0144] Ligands can be coupled to the dsRNA at various places, for example, 3'-end, 5'-end, and/or at an internal position of the sense and/or antisense strand. In preferred embodiments, the ligand is attached to the sense and/or antisense strand of the dsRNA via a linker or tether. The ligand or tethered ligand can be present on a monomer when said monomer is incorporated into the growing strand. In some embodiments, the ligand may be incorporated via coupling to a "precursor" monomer after said "precursor" monomer has been incorporated into the growing strand. For example, a monomer having, e.g., an amino-terminated tether (i.e., having no associated ligand), e.g., TAP-(CH₂)₂NH₂ may be incorporated into a growing oligonucleotide strand. In a subsequent operation, i.e., after incorporation of the precursor monomer into the strand, a ligand having an electrophilic group, e.g., a pentafluorophenyl ester or aldehyde group, can subsequently be attached to the precursor monomer by coupling the electrophilic group of the ligand with the terminal nucleophilic group of the precursor monomer's tether.

[0145] In another example, a monomer having a chemical group suitable for taking part in Click Chemistry reaction may be incorporated e.g., an azide or alkyne terminated tether/linker. In a subsequent operation, i.e., after incorporation of the precursor monomer into the strand, a ligand having complementary chemical group, e.g. an alkyne or azide can be attached to the precursor monomer by coupling the alkyne and the azide together.

[0146] The ligands can be attached to one or both strands. In some embodiments, a dsRNA described herein comprises a ligand conjugated to the sense strand. In some embodiments, a dsRNA described herein comprises a ligand conjugated to the antisense strand.

[0147] In some embodiments, the ligand is conjugated to the sense strand. As described herein, the ligand can be conjugated at the 3'-end, 5'-end or at an internal position of the sense strand. In some embodiments, the ligand is conjugated to the 3'-end of the sense strand. In some embodiments, the ligand is conjugated to the 5'-end of the sense strand. In some embodiments, the ligand is conjugated at an internal position of the sense strand. In other words, the ligand is conjugated to a non-terminal nucleotide of the sense strand. It is noted that the ligand can be conjugated to a nucleobase, sugar moiety or internucleotide linkage of the sense strand.

[0148] In some embodiments, the ligand is conjugated at the 2'-position of a nucleotide in the sense strand. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand.

[0149] In some embodiments, ligand can be conjugated to nucleobases, sugar moieties, or internucleosidic linkages of nucleic acid molecules. Conjugation to purine nucleobases or derivatives thereof can occur at any position including, endocyclic and exocyclic atoms. In some embodiments, the 2-, 6-, 7-, or 8-positions of a purine nucleobase are attached to a conjugate moiety. Conjugation to pyrimidine nucleobases or derivatives thereof can also occur at any position. In some embodiments, the 2-, 5-, and 6-positions of a pyrimidine nucleobase can be substituted with a conjugate moiety. Conjugation to sugar moieties of nucleosides can occur at any carbon atom. Example carbon atoms of a sugar moiety that can be attached to a conjugate moiety include the 2', 3', and 5' carbon atoms. The 1' position can also be attached to a conjugate moiety, such as in an abasic residue. Internucleosidic linkages can also bear conjugate moieties. For phosphorus-containing linkages (e.g., phosphodiester, phosphorothioate, phosphorodithioate, phosphoroamidate, and the like), the conjugate moiety can be attached directly to the phosphorus atom or to an O, N, or S atom bound to the phosphorus atom. For amine- or amide-containing internucleosidic linkages (e.g., PNA), the conjugate moiety can be attached to the nitrogen atom of the amine or amide or to an adjacent carbon atom.

[0150] In some embodiments, the ligand is conjugated to the sense strand. As described herein, the ligand can be conjugated at the 3'-end, 5'-end or at an internal position of the sense strand. In some embodiments, the ligand is conjugated to the 3'-end of the sense strand. Further, the ligand can be conjugated to a nucleobase, sugar moiety or internucleotide linkage of the sense strand.

[0151] Any suitable ligand in the field of RNA interference may be used, although the ligand is typically a carbohydrate e.g. monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, polysaccharide.

[0152] Linkers that conjugate the ligand to the nucleic acid include those discussed above. For example, the ligand can be one or more carbohydrates, e.g., GalNAc (N-acetylgalactosamine) derivatives attached through a monovalent, bivalent or trivalent branched linker.

[0153] In some embodiments, the dsRNA of the invention is conjugated to a bivalent and trivalent branched linkers include the structures shown in any of Formula (IV)-(VII):

##STR00001##

wherein: [0154] q.sup.2A, q.sup.2B, q.sup.3A, q.sup.3B, q.sup.4A, q.sup.4B, q.sup.5A, q.sup.5B and q.sup.5C represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different; [0155] P.sup.2A, P.sup.2B, P.sup.3A, P.sup.3B, P.sup.4A, P.sup.4B, P.sup.5A, P.sup.5B, P.sup.5C, T.sup.2A, T.sup.2B, T.sup.3A, T.sup.3B, T.sup.4A, T.sup.4B, T.sup.5A, T.sup.5B, T.sup.5C are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH.sub.2, CH.sub.2NH or CH.sub.2O; [0156] Q.sup.2A, Q.sup.2B, Q.sup.3A, Q.sup.3B, Q.sup.4A, Q.sup.4B, Q.sup.5A, Q.sup.5B, Q.sup.5C are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be interrupted or terminated by one or more of O, S, S(O), SO.sub.2, N(RN), C(R')=C(R''), C=C or

C(O); [0157] R.sup.2A, R.sup.2B, R.sup.3A, R.sup.3B, R.sup.4A, R.sup.4B, R.sup.5A, R.sup.5B, R.sup.5C are each independently for each occurrence absent, NH, O, S, CH.sub.2, C(O) O, C(O) NH, NHCH(R.sup.a) C(O), —C(O)—CH(R.sup.a)—NH—, CO, CH=N—O,
##STR00002##

or heterocyclyl; [0158] L.sup.2A, L.sup.2B, L.sup.3A, L.sup.3B, L.sup.4A, L.sup.4B, L.sup.5A, L.sup.5B and L.sup.5C represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and [0159] R.sup.a is H or amino acid side chain.

[0160] Trivalent conjugating GalNAc derivatives are particularly useful for use with dsRNA agents described herein for inhibiting the expression of a target gene, such as those of Formula (VII):

##STR00003##

wherein L.sup.5A, L.sup.5B and L.sup.5C represent a monosaccharide, such as GalNAc derivative.

[0161] Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the following compounds:

##STR00004## ##STR00005##

[0162] In some embodiments, a dsRNA described herein comprises Ligand 1, i.e., a ligand having the following structure:

##STR00006##

[0163] In some embodiments, a dsRNA described herein comprises a ligand described in U.S. Pat. No. 5,994,517 or U.S. Pat. No. 6,906,182, content of each of which is incorporated herein by reference in its entirety.

[0164] In some embodiments, the ligand can be a tri-antennary ligand described in FIG. 3 of U.S. Pat. No. 6,906,182. For example, a dsRNA described herein can comprise a ligand selected from the following tri-antennary ligands:

##STR00007##

[0165] In some embodiments of any one of the aspects, the antisense strand comprises a phosphoryl analog or phosphate mimic at the 5'-terminus. In some embodiments, the antisense strand comprises an alkenylphosphonates, i.e., a vinyl phosphonate at the 5'-terminus. For example, the antisense strand comprises a 5'-E-vinyl phosphonate.

[0166] In some embodiments, the antisense strand comprises a cyclopropylphosphonate at the

##STR00008##

5'-terminus. For example, the antisense comprises at the 5'-terminus, where * is a bond to C5 position of the nucleotide at the 5'-terminus.

[0167] In some embodiments of any one of the aspects, at least one of the strands, e.g., the sense and/or the antisense strand of the double-stranded RNA comprises a monomer or ligand selected from the following:

##STR00009##

where * is a bond to a 5', 3'-terminal hydroxyl group of the strand;

##STR00010##

where * is a bond to 2'-hydroxyl of a nucleotide in the strand;

##STR00011##

where * is a bond to a 5' or 3'-terminus of the strand;

##STR00012##

where * is a bond to a 5' or 3'-terminus of the strand;

##STR00013##

where —O—* is a connection to a 5' or 3'-terminus of the strand; and/or

##STR00014##

where —P—* is a connection to a 5' or 3'-hydroxyl group of the strand.

[0168] In some embodiments, the ligand comprises a lipophilic group. For example, the ligand can be a C.sub.6-30 aliphatic group or a C.sub.10-30 aliphatic group. "Aliphatic" as used herein means

a saturated or unsaturated and straight, branched, and/or cyclic hydrocarbon having the defined number of carbon atoms; examples include alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkylalkyl, cycloalkylalkenyl, and cycloalkylalkynyl, having the defined number of carbon atoms. In some embodiments, the ligand is a C₁₀₋₃₀alkyl group. For example, the ligand is a straight-chain or branched hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is a straight-chain hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is a straight-chain hexyl, octyl, decyl, dodecyl, hexadecyl, octadecyl, icosyl, or docosyl group. For example, the ligand is a straight-chain hexadecyl group. For example, the ligand is a straight-chain docosyl group.

[0169] In certain embodiments, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain or branched hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain hexadecyl, octadecyl, icosyl, or docosyl group.

[0170] The internal sense strand nucleotide position can be all positions except the three terminal positions from each end of the sense strand. In some embodiments, the internal positions exclude a cleavage site region of the sense strand. In some embodiments, the internal positions exclude positions 9-12 or positions 11-13, counting from the 5'-end of the sense strand. For example, the internal nucleotide position can be one or more of positions 4-8 and 13-18 on the sense strand, such as one or more of positions 5, 6, 7, 15, and 17 on the sense strand counting from the 5'-end of the sense strand. In one embodiment, the internal nucleotide position can be one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 6 or 7 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 6 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 7 of the sense strand, counting from the 5'-end. In certain embodiments, the internal nucleotide comprising the ligand has the formula,

##STR00015##

where B is a nucleotide base or a nucleotide base analog, optionally where B is adenine, guanine, cytosine, thymine or uracil.

[0171] In some embodiments, the ligand comprises an inverted nucleotide or an inverted abasic nucleotide. For example, the ligand comprises an abasic nucleotide linked via a 5'→5' or 3'→3' linkage to a strand of the dsRNA molecule. In some embodiments, the ligand comprises an abasic nucleotide linked via a 3'→3' linkage to the 3'-end of the sense strand.

[0172] In other embodiments, the ligand comprises a lipophilic group that comprises a steroidal fused ring system. For example, the ligand can comprise cholesterol or corticosterone. Examples of such ligands, include, for example,

##STR00016##

For example, such ligands may be attached to the 5' and/or 3' ends of a strand of the dsRNA molecule. In some embodiments, the ligand may be attached to the 5' or 3'-end of the sense strand. In some embodiments, the ligand may be attached to the 5'-end of the sense strand. In some embodiments, the ligand may be attached to the 3'-end of the sense strand. Attachment to the dsRNA molecule may be via a bond to the oxygen atom illustrated above having an open valence,

or a bond formed with the 4-hydroxyl group of the pyrrolidine ring.

[0173] The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one “backbone attachment point,” preferably two “backbone attachment points” and (ii) at least one “tethering attachment point.” A “backbone attachment point” as used herein refers to a functional group, e.g. a hydroxyl group, or generally, a bond available for, and that is suitable for incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A “tethering attachment point” (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g. monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide and polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

[0174] In one embodiment the dsRNA molecule of the invention is conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group; preferably, the cyclic group is selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl and decalin; preferably, the acyclic group is selected from serinol backbone or diethanolamine backbone.

[0175] The ligand can be attached to the sense strand, antisense strand or both strands, at the 3'-end, 5'-end or both ends. For instance, the ligand can be conjugated to the sense strand, in particular, the 3'-end of the sense strand.

[0176] The ligand can be conjugated to the sense strand or the antisense strand via a linker comprising a cleavable group. A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In a preferred embodiment of the dsRNA molecule according to the present invention, the cleavable linking group is cleaved at least 10 times or more, preferably at least 100 times faster in the target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

[0177] Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

[0178] A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a preferred pH, thereby releasing the cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

[0179] A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted.

[0180] For example, liver targeting ligands can be linked to the cationic lipids through a linker that

includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

[0181] Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synoviocytes.

[0182] In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It may be useful to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In preferred embodiments, useful candidate compounds are cleaved at least 2, 4, 10 or 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

[0183] One class of cleavable linking groups is redox cleavable linking groups, which may be used in the dsRNA molecule according to the present invention that are cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulfide linking group (—S—S—). To determine if a candidate cleavable linking group is a suitable “reductively cleavable linking group,” or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated by incubation with dithiothreitol (DTT), or other reducing agent using reagents known in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In a preferred embodiment, candidate compounds are cleaved by at most 10% in the blood. In preferred embodiments, useful candidate compounds are degraded at least 2, 4, 10 or 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

[0184] Phosphate-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)—O— , —O—P(S)(ORk)—O— , —O—P(S)(SRk)—O— , —S—P(O)(ORk)—O— , —O—P(O)(ORk)—S— , —S—P(O)(ORk)—S— , —O—P(S)(ORk)—S— , —S—P(S)(ORk)—O— , —O—P(O)(Rk)—O— , —O—P(S)(Rk)—O— , —S—P(O)(Rk)—O— , —S—P(S)(Rk)—O— , —S—P(O)(Rk)—S— , —O—P(S)(Rk)—S— , wherein Rk at each occurrence can be, independently, hydrogen, C1-C20 alkyl, C1-C20 haloalkyl, C6-C10 aryl, C7-C12 aralkyl. Preferred embodiments are —O—P(O)(OH)—O— , —O—P(S)(OH)—O— , —O—P(S)(SH)—O— , —S—P(O)(OH)—O— , —O—P(O)(OH)—S— , —S—P(O)(OH)—S— , —O—P(S)(OH)—S— , —S—P(S)(OH)—O— , —O—P(O)(H)—O— , —O—P(S)(H)—O— , —S—P(O)(H)—O— , —S—P(S)(H)—O— , —S—P(O)(H)—S— , —O—P(S)(H)—S— . A preferred embodiment is —O—P(O)(OH)—O— . These candidates can be evaluated using methods analogous to those described above.

[0185] Acid cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are linking groups that are cleaved under acidic conditions. In preferred

embodiments acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.5, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C=NN— , C(O)O— , or —OC(O)— . A preferred embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

[0186] Ester-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include but are not limited to esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O— , or —OC(O)— . These candidates can be evaluated using methods analogous to those described above.

[0187] Peptide-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynylene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula $\text{—NHCHR}_{\text{sup.A}}\text{C(O)NHCHR}_{\text{sup.B}}\text{C(O)—}$, where $\text{R}_{\text{sup.A}}$ and $\text{R}_{\text{sup.B}}$ are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above. As used herein, “carbohydrate” refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which may be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate moiety made up of one or more monosaccharide units each having at least six carbon atoms (which may be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri- and oligosaccharides containing from about 4-9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C.sub.5 and above (preferably C.sub.5-C.sub.8) sugars; di- and trisaccharides include sugars having two or three monosaccharide units (preferably C.sub.5-C.sub.8).

[0188] In some embodiments, the dsRNA molecule of the invention comprises one or more overhang regions and/or capping groups of dsRNA molecule at the 3'-end, or 5'-end or both ends of a strand. The overhang can be 1-10 nucleotides in length. For example, the overhang can be 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides in length. In some embodiments, the overhang is 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with the target sequence or it can be complementary to the gene sequences being targeted or it can be the other sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

[0189] In some embodiments, the nucleotides in the overhang region of the dsRNA molecule of the invention can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-Fluoro 2'-O-methyl, thymidine (T), 2'-O-methoxyethyl-5-

methylyridine, 2'-O-methoxyethyladenosine, 2'-O-methoxyethyl-5-methylcytidine, GNA, SNA, hGNA, hhGNA, mGNA, TNA, h'GNA, and any combinations thereof. For example, dTdT can be an overhang sequence for either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be other sequence.

[0190] The 5'- or 3'-overhangs at the sense strand, antisense strand or both strands of the dsRNA molecule of the invention may be phosphorylated. In some embodiments, the overhang region contains two nucleotides having a phosphorothioate between the two nucleotides, where the two nucleotides can be the same or different. In some embodiments, the overhang is present at the 3'-end of the sense strand, antisense strand or both strands. In some embodiments, this 3'-overhang is present in the antisense strand. In some embodiments, this 3'-overhang is present in the sense strand.

[0191] The dsRNA molecule of the invention may comprise only a single overhang, which can strengthen the interference activity of the dsRNA, without affecting its overall stability. For example, the single-stranded overhang is located at the 3'-terminal end of the sense strand or, alternatively, at the 3'-terminal end of the antisense strand. The dsRNA can also have a blunt end, located at the 5'-end of the antisense strand (or the 3'-end of the sense strand) or vice versa.

[0192] Generally, the antisense strand of the dsRNA has a nucleotide overhang at the 3'-end, and the 5'-end is blunt. While not bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC process. For example, the single overhang is at least one, two, three, four, five, six, seven, eight, nine, or ten nucleotides in length. In some embodiments, the dsRNA has a 2 nucleotide overhang on the 3'-end of the antisense strand and a blunt end at the 5'-end of the antisense strand.

[0193] The dsRNA of the invention can comprise one or more modified nucleotides. For example, every nucleotide in the sense strand and antisense strand of the dsRNA molecule can be modified. Each nucleotide can be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens and/or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar; replacement of the ribose sugar; wholesale replacement of the phosphate moiety with "dephospho" linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.

[0194] As nucleic acids are polymers of subunits, many of the modifications occur at a position which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases, the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3' or 5' terminal position, may only occur in a central region, may only occur at a non-terminal region, or may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of a RNA or may only occur in a single strand region of a RNA. For example, a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5' end or ends can be phosphorylated.

[0195] It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5' or 3' overhang, or in both. For example, it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3' or 5' overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of

deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

[0196] In some embodiments, the dsRNA molecule of the invention comprises modifications of an alternating pattern, particular in the B1, B2, B3, B1', B2', B3', B4' regions. The term "alternating motif" or "alternative pattern" as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be "ABABABABABAB . . ." "AABBAABBAABB . . ." "AABAABAABAAB . . ." "AAABAAABAAAB . . ." "AAABBBAAABBB . . .," or "ABCABCABCABC . . .," etc.

[0197] The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as "ABABAB . . .", "ACACAC . . ." "BDBDBD . . ." or "CDCDCD . . .," etc.

[0198] In some embodiments, the dsRNA molecule of the invention comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense strand in the dsRNA duplex, the alternating motif in the sense strand may start with "ABABAB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BABABA" from 3'-5' of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with "AABBAABB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BBAABBAA" from 3'-5' of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

5'-Modifications

[0199] In some embodiments dsRNA molecules of the invention are 5' phosphorylated or include a phosphoryl analog or phosphate mimic at the 5' terminus. 5'-phosphate modifications include those which are compatible with RISC mediated gene silencing. Suitable modifications include: 5'-monophosphate ((HO).sub.2(O)P—O—5'); 5'-diphosphate ((HO).sub.2(O)P—O—P(HO)(O)—O—5'); 5'-triphosphate ((HO).sub.2(O)P—O—(HO)(O)P—O—P(HO)(O)—O—5'); 5'-guanosine cap (7-methylated or non-methylated)(7m-G-O—5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O—5'); 5'-adenosine cap (Appp), and any modified or unmodified nucleotide cap structure (N—O—5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O—5'); 5'-monothiophosphate (phosphorothioate; (HO).sub.2(S)P—O—5'); 5'-monodithiophosphate (phosphorodithioate; (HO)(HS)(S)P—O—5'), 5'-phosphorothiolate ((HO).sub.2(O)P—S—5'); any additional combination of oxygen/sulfur replaced monophosphate, diphosphate and triphosphates (e.g. 5'-alpha-thiotriphosphate, 5'-gamma-thiotriphosphate, etc.), 5'-phosphoramidates ((HO).sub.2(O)P—NH—5', (HO)(NH2)(O)P—O—5'), 5'-alkylphosphonates (e.g. RP(OH)(O)—O—5', R=alkyl, such as methyl, ethyl, isopropyl, propyl, etc.), 5'-alkenylphosphonates (i.e. vinyl, substituted vinyl), (OH).sub.2(O)P—5'—CH.sub.2-), cyclopropylphosphonates, 5'-alkyletherphosphonates (R=alkylether-methoxymethyl(MeOCH2—), ethoxymethyl, etc., e.g. RP(OH)(O)—O—5'-). In one example, the modification can be placed in the antisense strand of a dsRNA molecule.

[0200] In some embodiments of any one of the aspects, the antisense strand comprises a phosphoryl analog or phosphate mimic at the 5'-terminus. In some embodiments, the antisense

strand comprises an alkenylphosphonates, i.e., a vinyl phosphonate at the 5'-terminus. For example, the antisense strand comprises a 5'-E-vinyl phosphonate. In exemplary embodiments, a 5' vinyl phosphonate modified nucleotide at the 5'-terminus may have the structure:

##STR00017##

wherein X is O or S; R is hydrogen, hydroxy, fluoro, or C_{sub.1-20}alkoxy (e.g., methoxy); R^{sup.5'} is =C(H)—P(O)(OH)_{sub.2} and the double bond between the C_{sub.5'} carbon and R^{sup.5'} is in the E or Z orientation (e.g., E orientation); and B is a nucleobase or a modified nucleobase, optionally where B is adenine, guanine, cytosine, thymine, or uracil.

[0201] In some embodiments, the antisense strand comprises a cyclopropylphosphonate at the 5'-terminus. For example, the antisense comprises

##STR00018##

at the 5'-terminus (which, for example, may replace the 4'-group in immediately preceding structure.

[0202] The dsRNA agents of the invention can comprise thermally destabilizing modifications in the seed region of the antisense strand (i.e., at positions 2-9 of the 5'-end of the antisense strand) to reduce or inhibit off-target gene silencing. Without wishing to be bound by a theory, dsRNAs with an antisense strand comprising at least one thermally destabilizing modification of the duplex within the first 9 nucleotide positions, counting from the 5' end, of the antisense strand have reduced off-target gene silencing activity. Accordingly, in some embodiments, the antisense strand comprises at least one (e.g., one, two, three, four, five or more) thermally destabilizing modification of the duplex within the first 9 nucleotide positions of the 5' region of the antisense strand. In some embodiments, thermally destabilizing modification of the duplex is located in positions 2-9, or preferably positions 4-8, from the 5'-end of the antisense strand. In some further embodiments, the thermally destabilizing modification of the duplex is located at position 5, 6, 7 or 8 from the 5'-end of the antisense strand.

[0203] In still some further embodiments, the thermally destabilizing modification of the duplex is located at position 7 from the 5'-end of the antisense strand.

[0204] The term "thermally destabilizing modification(s)" includes modification(s) that would result with a dsRNA with a lower overall melting temperature (T_m)(preferably a T_m with one, two, three or four degrees lower than the T_m of the dsRNA without having such modification(s). In some embodiments, the thermally destabilizing modification of the duplex is located at position 2, 3, 4, 5, 6, 7, 8 or 9 from the 5'-end of the antisense strand.

[0205] The thermally destabilizing modifications can include, but are not limited to, abasic modification; mismatch with the opposing nucleotide in the opposing strand; and sugar modification such as 2'-deoxy modification or acyclic nucleotide, e.g., unlocked nucleic acids (UNA) or glycol nucleic acid (GNA), or a 5'-2'-linked nucleotide (e.g., having 3'-OMe, 3'-F, 3'-H or 3'-OH, herein a "3'-RNA"). For example, the thermally destabilizing modifications can include, but are not limited to, mUNA and GNA building blocks as follows:

##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024##

[0206] In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide (e.g., having 3'-OMe, 3'-F, 3'-H or 3'-OH. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OMe, 3'-F, 3'-H or 3'-OH. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OMe. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-F. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-H. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OH, e.g., having the formula

##STR00025##

wherein B is a nucleotide base or a nucleotide base analog, optionally where B is adenine, guanine, cytosine, thymine or uracil.

[0207] In some embodiments, the destabilizing modification is selected from the group consisting

of GNA-isoC, GNA-isoG, 5'-mUNA, 4'-mUNA, 3'-mUNA, and 2'-mUNA.

[0208] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

##STR00026##

[0209] R=H, OH; OMe; Cl, F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino; [0210] R'=H, Me; [0211] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C5-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 2-thiouridine; 4-thiouridine; C5-modified pyrimidines; C2-modified purines; N8-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and [0212] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0213] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

##STR00027## [0214] R=H, OH; OMe; Cl, F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino; [0215] R'=H, Me; [0216] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and [0217] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0218] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

##STR00028## [0219] R=H, OMe; F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; O-nPr; O-alkyl; O-alkylamino; R'=H, Me; [0220] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 7-deazapurines; and Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0221] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

##STR00029## [0222] R=H, OH; OMe; Cl, F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino; [0223] R'=H, Me; [0224] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and [0225] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0226] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

##STR00030## [0227] R=H, OH; OMe; Cl, F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino; [0228] R'=H, Me; [0229] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic

heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modiifed purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and [0230] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0231] In some embodiments, the modification mUNA is selected from the group consisting of ##STR00031## [0232] R=H, OMe; F; OH; O—(CH.sub.2).sub.2OMe; SMe, NMe.sub.2; NH.sub.2; Me; O-nPr; O-alkyl; O-alkylamino; [0233] R'=H, Me; [0234] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C.sub.5-modified pyrimidines; C2-modified purines; N8-modiifed purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamninopurine; pseudocytosine; 2-aminopurine; xanthosine; N.sup.6-alkyl-A; O.sup.6-alkyl-G; 7-deazapurines; and Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0235] Exemplary abasic modifications include, but are not limited to the following:

##STR00032##

[0236] Wherein R=H, Me, Et or OMe; R'=H, Me, Et or OMe; R''=H, Me, Et or OMe

##STR00033##

wherein B is a modified or unmodified nucleobase and the asterisk on each structure represents either R, S or racemic.

[0237] Exemplified sugar modifications include, but are not limited to the following:

##STR00034##

wherein B is a modified or unmodified nucleobase and the asterisk on each structure represents either R, S or racemic.

[0238] In some embodiments the thermally destabilizing modification of the duplex is selected from the mUNA and GNA building blocks described in Examples 1-3 herein. In some embodiments, the destabilizing modification is selected from the group consisting of GNA-isoC, GNA-isoG, 5'-mUNA, 4'-mUNA, 3'-mUNA, and 2'-mUNA. In some further embodiments of this, the dsRNA molecule further comprises at least one thermally destabilizing modification selected from the group consisting of GNA, 2'-OMe, 3'-OMe, 5'-Me, Hy p-spacer, SNA, hGNA, hhGNA, mGNA, TNA and h'GNA (Mod A-Mod K).

[0239] The term “acyclic nucleotide” refers to any nucleotide having an acyclic ribose sugar, for example, where any of bonds between the ribose carbons (e.g., C1'-C2', C2'-C3', C3'-C4', C4'-O4', or C1'-O4') is absent and/or at least one of ribose carbons or oxygen (e.g., C1', C2', C3', C4' or O4') are independently or in combination absent from the nucleotide. In some embodiments, acyclic nucleotide is

##STR00035##

wherein B is a modified or unmodified nucleobase, R1 and R2 independently are H, halogen, OR3, or alkyl; and R3 is H, alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar). The term “UNA” refers to unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked “sugar” residue. In one example, UNA also encompasses monomers with bonds between C1'-C4' being removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2' and C3' carbons) of the sugar is removed (see Mikhailov et. al., Tetrahedron Letters, 26(17): 2059(1985); and Fluiter et al., Mol. Biosyst., 10:1039 (2009), which are hereby incorporated by reference in their entirety). The acyclic derivative provides greater backbone flexibility without affecting the Watson-Crick pairings. The acyclic nucleotide can be linked via 2'-5' or 3'-5' linkage.

[0240] The term ‘GNA’ refers to glycol nucleic acid which is a polymer similar to DNA or RNA but differing in the composition of its “backbone” in that is composed of repeating glycerol units linked by phosphodiester bonds:

##STR00036##

[0241] The thermally destabilizing modification of the duplex can be mismatches (i.e., noncomplementary base pairs) between the thermally destabilizing nucleotide and the opposing nucleotide in the opposite strand within the dsRNA duplex. Exemplary mismatch base pairs include G: G, G: A, G: U, G: T, A: A, A: C, C: C, C: U, C: T, U: U, T: T, U: T, or a combination thereof. Other mismatch base pairings known in the art are also amenable to the present invention. A mismatch can occur between nucleotides that are either naturally occurring nucleotides or modified nucleotides, i.e., the mismatch base pairing can occur between the nucleobases from respective nucleotides independent of the modifications on the ribose sugars of the nucleotides. In certain embodiments, the dsRNA molecule contains at least one nucleobase in the mismatch pairing that is a 2'-deoxy nucleobase; e.g., the 2'-deoxy nucleobase is in the sense strand.

[0242] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes nucleotides with impaired W-C H-bonding to complementary base on the target mRNA, such as:

##STR00037## ##STR00038##

[0243] More examples of abasic nucleotide, acyclic nucleotide modifications (including UNA and GNA), and mismatch modifications have been described in detail in WO 2011/133876, which is herein incorporated by reference in its entirety.

[0244] The thermally destabilizing modifications may also include universal base with reduced or abolished capability to form hydrogen bonds with the opposing bases, and phosphate modifications.

[0245] In some embodiments, the thermally destabilizing modification of the duplex includes nucleotides with non-canonical bases such as, but not limited to, nucleobase modifications with impaired or completely abolished capability to form hydrogen bonds with bases in the opposite strand. These nucleobase modifications have been evaluated for destabilization of the central region of the dsRNA duplex as described in WO 2010/0011895, which is herein incorporated by reference in its entirety. Exemplary nucleobase modifications are:

##STR00039##

[0246] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes one or more α -nucleotide complementary to the base on the target mRNA, such as:

##STR00040##

Wherein R is H, OH, OCH₃, F, NH₂, NHMe, NMe₂ or O-alkyl

[0247] Exemplary phosphate modifications known to decrease the thermal stability of dsRNA duplexes compared to natural phosphodiester linkages are:

##STR00041##

[0248] The alkyl for the R group can be a C₁-C₆alkyl. Specific alkyls for the R group include, but are not limited to methyl, ethyl, propyl, isopropyl, butyl, pentyl and hexyl.

[0249] It is noted a thermally destabilizing modification can replace a 2'-doeoxy nucleotide in the antisense strand. For example, a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and/or 16, counting from 5'-end, of the antisense strand can be replaced with a thermally destabilizing modification described herein. Thus, in some embodiments, the antisense strand comprises a thermally destabilizing modification at 1, 2, 3, 4, 5 and/or 6 of positions 2, 5, 7, 12, 14 and/or 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a thermally destabilizing modification at positions 5 and 7, counting from 5'-end of the antisense strand.

[0250] In addition to the antisense strand comprising a thermally destabilizing modification, the dsRNA can also comprise one or more stabilizing modifications. For example, the dsRNA can comprise at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, the stabilizing modifications all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least two stabilizing

modifications. The stabilizing modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the stabilizing modification can occur on every nucleotide on the sense strand and/or antisense strand; each stabilizing modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both stabilizing modification in an alternating pattern. The alternating pattern of the stabilizing modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the stabilizing modifications on the sense strand can have a shift relative to the alternating pattern of the stabilizing modifications on the antisense strand.

[0251] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the antisense strand can be present at any positions. In some embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 8, 9, 14 and 16 from the 5'-end. In some other embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 14 and 16 from the 5'-end. In still some other embodiments, the antisense comprises stabilizing modifications at positions 2, 14 and 16 from the 5'-end.

[0252] In some embodiments, the antisense strand comprises at least one stabilizing modification adjacent to the destabilizing modification. For example, the stabilizing modification can be the nucleotide at the 5'-end or the 3'-end of the destabilizing modification, i.e., at position -1 or +1 from the position of the destabilizing modification. In some embodiments, the antisense strand comprises a stabilizing modification at each of the 5'-end and the 3'-end of the destabilizing modification, i.e., positions -1 and +1 from the position of the destabilizing modification.

[0253] In some embodiments, the antisense strand comprises at least two stabilizing modifications at the 3'-end of the destabilizing modification, i.e., at positions +1 and +2 from the position of the destabilizing modification. In some embodiments, the sense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the sense strand can be present at any positions. In some embodiments, the sense strand comprises stabilizing modifications at positions 7, 10 and 11 from the 5'-end. In some other embodiments, the sense strand comprises stabilizing modifications at positions 7, 9, 10 and 11 from the 5'-end. In some embodiments, the sense strand comprises stabilizing modifications at positions opposite or complimentary to positions 11, 12 and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some other embodiments, the sense strand comprises stabilizing modifications at positions opposite or complimentary to positions 11, 12, 13 and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some embodiments, the sense strand comprises a block of two, three or four stabilizing modifications.

[0254] In some embodiments, the sense strand does not comprise a stabilizing modification in position opposite or complimentary to the thermally destabilizing modification of the duplex in the antisense strand.

[0255] Exemplary thermally stabilizing modifications include, but are not limited to 2'-fluoro modifications. Other thermally stabilizing modifications include, but are not limited to LNA.

[0256] It is noted a thermally stabilizing modification can replace a 2'-fluoro nucleotide in the sense and/or antisense strand. For example, a 2'-fluoro nucleotide at positions 8, 9, 10, 11 and/or 12, counting from 5'-end, of the sense strand, can be replaced with a thermally stabilizing modification. Similarly, a 2'-fluoro nucleotide at position 14, counting from 5'-end, of the antisense strand, can be replaced with a thermally stabilizing modification.

[0257] For the dsRNA molecules to be more effective in vivo, the antisense strand must have some metabolic stability. In other words, for the dsRNA molecules to be more effective in vivo, some amount of the antisense stand may need to be present in vivo after a period time after administration. Accordingly, in some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the

antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 5 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 6 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 7 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 8 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 9 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 10 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 11 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 12 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 13 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 14 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 15 after in vivo administration.

Uses of dsRNA

[0258] The present invention further relates to a use of a dsRNA molecule as defined herein for inhibiting expression of a target gene. In some embodiments, the present invention further relates to a use of a dsRNA molecule for inhibiting expression of a target gene in vitro.

[0259] The present invention further relates to a dsRNA molecule as defined herein for use in inhibiting expression of a target gene in a subject. The subject may be any animal, such as a mammal, e.g., a mouse, a rat, a sheep, a cattle, a dog, a cat, or a human

[0260] In some embodiments, the dsRNA molecule of the invention is administered in buffer. In some embodiments, siRNA compounds described herein can be formulated for administration to a subject. A formulated siRNA composition can assume a variety of states. In some examples, the composition is at least partially crystalline, uniformly crystalline, and/or anhydrous (e.g., less than 80, 50, 30, 20, or 10% water). In another example, the siRNA is in an aqueous phase, e.g., in a solution that includes water.

[0261] The aqueous phase or the crystalline compositions can, e.g., be incorporated into a delivery vehicle, e.g., a liposome (particularly for the aqueous phase) or a particle (e.g., a microparticle as can be appropriate for a crystalline composition). Generally, the siRNA composition is formulated in a manner that is compatible with the intended method of administration, as described herein. For example, in particular embodiments the composition is prepared by at least one of the following methods: spray drying, lyophilization, vacuum drying, evaporation, fluid bed drying, or a

combination of these techniques; or sonication with a lipid, freeze-drying, condensation and other self-assembly.

[0262] A dsRNA preparation can be formulated in combination with another agent, e.g., another therapeutic agent or an agent that stabilizes a dsRNA, e.g., a protein that complexes with dsRNA to form an iRNP. Still other agents include chelating agents, e.g., EDTA (e.g., to remove divalent cations such as Mg^{2+}), salts, RNase inhibitors (e.g., a broad specificity RNase inhibitor such as RNasin) and so forth.

[0263] In some embodiments, the dsRNA preparation includes another dsRNA compound, e.g., a second dsRNA that can mediate RNAi with respect to a second gene, or with respect to the same gene. Still other preparation can include at least 3, 5, ten, twenty, fifty, or a hundred or more different siRNA species. Such dsRNAs can mediate RNAi with respect to a similar number of different genes.

[0264] In some embodiments, the dsRNA preparation includes at least a second therapeutic agent (e.g., an agent other than a RNA or a DNA). For example, a dsRNA composition for the treatment of a viral disease, e.g., HIV, might include a known antiviral agent (e.g., a protease inhibitor or reverse transcriptase inhibitor). In another example, a dsRNA composition for the treatment of a cancer might further comprise a chemotherapeutic agent.

[0265] Exemplary formulations which can be used for administering the dsRNA molecule according to the present invention are discussed below.

[0266] Liposomes. A dsRNA preparation can be formulated for delivery in a membranous molecular assembly, e.g., a liposome or a micelle. As used herein, the term “liposome” refers to a vesicle composed of amphiphilic lipids arranged in at least one bilayer, e.g., one bilayer or a plurality of bilayers. Liposomes include unilamellar and multilamellar vesicles that have a membrane formed from a lipophilic material and an aqueous interior. The aqueous portion contains the siRNA composition. The lipophilic material isolates the aqueous interior from an aqueous exterior, which typically does not include the siRNA composition, although in some examples, it may. Liposomes are useful for the transfer and delivery of active ingredients to the site of action. Because the liposomal membrane is structurally similar to biological membranes, when liposomes are applied to a tissue, the liposomal bilayer fuses with bilayer of the cellular membranes. As the merging of the liposome and cell progresses, the internal aqueous contents that include the dsRNA are delivered into the cell where the dsRNA can specifically bind to a target RNA and can mediate RNAi. In some cases, the liposomes are also specifically targeted, e.g., to direct the dsRNA to particular cell types.

[0267] A liposome containing a dsRNA can be prepared by a variety of methods. In one example, the lipid component of a liposome is dissolved in a detergent so that micelles are formed with the lipid component. For example, the lipid component can be an amphipathic cationic lipid or lipid conjugate. The detergent can have a high critical micelle concentration and may be nonionic. Exemplary detergents include cholate, CHAPS, octylglucoside, deoxycholate, and lauroyl sarcosine. The dsRNA preparation is then added to the micelles that include the lipid component. The cationic groups on the lipid interact with the siRNA and condense around the dsRNA to form a liposome. After condensation, the detergent is removed, e.g., by dialysis, to yield a liposomal preparation of dsRNA.

[0268] If necessary a carrier compound that assists in condensation can be added during the condensation reaction, e.g., by controlled addition. For example, the carrier compound can be a polymer other than a nucleic acid (e.g., spermine or spermidine). pH can also be adjusted to favor condensation.

[0269] Further description of methods for producing stable polynucleotide delivery vehicles, which incorporate a polynucleotide/cationic lipid complex as structural components of the delivery vehicle, are described in, e.g., WO 96/37194. Liposome formation can also include one or more aspects of exemplary methods described in Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-

7417, 1987; U.S. Pat. Nos. 4,897,355; 5,171,678; Bangham, et al. *M. Mol. Biol.* 23:238, 1965; Olson, et al. *Biochim. Biophys. Acta* 557:9, 1979; Szoka, et al. *Proc. Natl. Acad. Sci.* 75:4194, 1978; Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984; Kim, et al. *Biochim. Biophys. Acta* 728:339, 1983; and Fukunaga, et al. *Endocrinol.* 115:757, 1984, which are incorporated by reference in their entirety. Commonly used techniques for preparing lipid aggregates of appropriate size for use as delivery vehicles include sonication and freeze-thaw plus extrusion (see, e.g., Mayer, et al. *Biochim. Biophys. Acta* 858:161, 1986, which is incorporated by reference in its entirety). Microfluidization can be used when consistently small (50 to 200 nm) and relatively uniform aggregates are desired (Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984, which is incorporated by reference in its entirety). These methods are readily adapted to packaging siRNA preparations into liposomes.

[0270] Liposomes that are pH-sensitive or negatively-charged entrap nucleic acid molecules rather than complex with them. Since both the nucleic acid molecules and the lipid are similarly charged, repulsion rather than complex formation occurs. Nevertheless, some nucleic acid molecules are entrapped within the aqueous interior of these liposomes. pH-sensitive liposomes have been used to deliver DNA encoding the thymidine kinase gene to cell monolayers in culture. Expression of the exogenous gene was detected in the target cells (Zhou et al., *Journal of Controlled Release*, 19, (1992) 269-274, which is incorporated by reference in its entirety).

[0271] One major type of liposomal composition includes phospholipids other than naturally-derived phosphatidylcholine. Neutral liposome compositions, for example, can be formed from dimyristoyl phosphatidylcholine (DMPC) or dipalmitoyl phosphatidylcholine (DPPC). Anionic liposome compositions generally are formed from dimyristoyl phosphatidylglycerol, while anionic fusogenic liposomes are formed primarily from dioleoyl phosphatidylethanolamine (DOPE). Another type of liposomal composition is formed from phosphatidylcholine (PC) such as, for example, soybean PC, and egg PC. Another type is formed from mixtures of phospholipid and/or phosphatidylcholine and/or cholesterol.

[0272] Examples of other methods to introduce liposomes into cells in vitro and include U.S. Pat. Nos. 5,283,185; 5,171,678; WO 94/00569; WO 93/24640; WO 91/16024; Felgner, *J. Biol. Chem.* 269:2550, 1994; Nabel, *Proc. Natl. Acad. Sci.* 90:11307, 1993; Nabel, *Human Gene Ther.* 3:649, 1992; Gershon, *Biochem.* 32:7143, 1993; and Strauss *EMBO J.* 11:417, 1992.

[0273] In some embodiments, cationic liposomes are used. Cationic liposomes possess the advantage of being able to fuse to the cell membrane. Non-cationic liposomes, although not able to fuse as efficiently with the plasma membrane, are taken up by macrophages in vivo and can be used to deliver siRNAs to macrophages.

[0274] Further advantages of liposomes include: liposomes obtained from natural phospholipids are biocompatible and biodegradable; liposomes can incorporate a wide range of water and lipid soluble drugs; liposomes can protect encapsulated siRNAs in their internal compartments from metabolism and degradation (Rosoff, in "Pharmaceutical Dosage Forms," Lieberman, Rieger and Banker (Eds.), 1988, volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size and the aqueous volume of the liposomes.

[0275] A positively charged synthetic cationic lipid, N—[1-(2,3-dioleoyloxy) propyl]-N,N,N-trimethylammonium chloride (DOTMA) can be used to form small liposomes that interact spontaneously with nucleic acid to form lipid-nucleic acid complexes which are capable of fusing with the negatively charged lipids of the cell membranes of tissue culture cells, resulting in delivery of siRNA (see, e.g., Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-7417, 1987 and U.S. Pat. No. 4,897,355 for a description of DOTMA and its use with DNA, which are incorporated by reference in their entirety).

[0276] A DOTMA analogue, 1,2-bis(oleoyloxy)-3-(trimethylammonia) propane (DOTAP) can be used in combination with a phospholipid to form DNA-complexing vesicles. Lipofectin™ (Bethesda Research Laboratories, Gaithersburg, Md.) is an effective agent for the delivery of highly

anionic nucleic acids into living tissue culture cells that comprise positively charged DOTMA liposomes which interact spontaneously with negatively charged polynucleotides to form complexes. When enough positively charged liposomes are used, the net charge on the resulting complexes is also positive. Positively charged complexes prepared in this way spontaneously attach to negatively charged cell surfaces, fuse with the plasma membrane, and efficiently deliver functional nucleic acids into, for example, tissue culture cells. Another commercially available cationic lipid, 1,2-bis(oleoyloxy)-3,3-(trimethylammonia) propane ("DOTAP")(Boehringer Mannheim, Indianapolis, Indiana) differs from DOTMA in that the oleoyl moieties are linked by ester, rather than ether linkages.

[0277] Other reported cationic lipid compounds include those that have been conjugated to a variety of moieties including, for example, carboxyspermine which has been conjugated to one of two types of lipids and includes compounds such as 5-carboxyspermylglycine dioctaoyleamide ("DOGS")(Transfectam™, Promega, Madison, Wisconsin) and dipalmitoylphosphatidylethanolamine 5-carboxyspermyl-amide ("DPPES")(see, e.g., U.S. Pat. No. 5,171,678).

[0278] Another cationic lipid conjugate includes derivatization of the lipid with cholesterol ("DC-Chol") which has been formulated into liposomes in combination with DOPE (See, Gao, X. and Huang, L., *Biochim. Biophys. Res. Commun.* 179:280, 1991). Lipopolylysine, made by conjugating polylysine to DOPE, has been reported to be effective for transfection in the presence of serum (Zhou, X. et al., *Biochim. Biophys. Acta* 1065:8, 1991, which is incorporated by reference in its entirety). For certain cell lines, these liposomes containing conjugated cationic lipids, are said to exhibit lower toxicity and provide more efficient transfection than the DOTMA-containing compositions. Other commercially available cationic lipid products include DMRIE and DMRIE-HP(Vical, La Jolla, California) and Lipofectamine (DOSPA)(Life Technology, Inc., Gaithersburg, Maryland). Other cationic lipids suitable for the delivery of oligonucleotides are described in WO 98/39359 and WO 96/37194.

[0279] Liposomal formulations are particularly suited for topical administration. Liposomes present several advantages over other formulations. Such advantages include reduced side effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer siRNA, into the skin. In some implementations, liposomes are used for delivering siRNA to epidermal cells and also to enhance the penetration of siRNA into dermal tissues, e.g., into skin. For example, the liposomes can be applied topically. Topical delivery of drugs formulated as liposomes to the skin has been documented (see, e.g., Weiner et al., *Journal of Drug Targeting*, 1992, vol. 2,405-410 and du Plessis et al., *Antiviral Research*, 18, 1992, 259-265; Mannino, R. J. and Fould-Fogerite, S., *Biotechniques* 6:682-690, 1988; Itani, T. et al. *Gene* 56:267-276, 1987; Nicolau, C. et al. *Meth. Enz.* 149:157-176, 1987; Straubinger, R. M. and Papahadjopoulos, D. *Meth. Enz.* 101:512-527, 1983; Wang, C. Y. and Huang, L., *Proc. Natl. Acad. Sci. USA* 84:7851-7855, 1987, which are incorporated by reference in their entirety).

[0280] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver a drug into the dermis of mouse skin. Such formulations with dsRNA described herein are useful for treating a dermatological disorder.

[0281] Liposomes that include dsRNA described herein can be made highly deformable. Such deformability can enable the liposomes to penetrate through pore that are smaller than the average radius of the liposome. For example, transfersomes are a type of deformable liposomes. Transfersomes can be made by adding surface edge activators, usually surfactants, to a standard

liposomal composition. Transfersomes that include dsRNA described herein can be delivered, for example, subcutaneously by infection in order to deliver dsRNA to keratinocytes in the skin. In order to cross intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. In addition, due to the lipid properties, these transfersomes can be self-optimizing (adaptive to the shape of pores, e.g., in the skin), self-repairing, and can frequently reach their targets without fragmenting, and often self-loading.

[0282] Other formulations amenable to the present invention are described in U.S. provisional application Ser. No. 61/018,616, filed Jan. 2, 2008; 61/018,611, filed Jan. 2, 2008; 61/039,748, filed Mar. 26, 2008; 61/047,087, filed Apr. 22, 2008 and 61/051,528, filed May 8, 2008. PCT application no PCT/US2007/080331, filed Oct. 3, 2007 also describes formulations that are amenable to the present invention.

[0283] Surfactants. The dsRNA compositions can include a surfactant. In some embodiments, the dsRNA is formulated as an emulsion that includes a surfactant. The most common way of classifying and ranking the properties of the many different types of surfactants, both natural and synthetic, is by the use of the hydrophile/lipophile balance (HLB). The nature of the hydrophilic group provides the most useful means for categorizing the different surfactants used in formulations (Rieger, in "Pharmaceutical Dosage Forms," Marcel Dekker, Inc., New York, NY, 1988, p. 285).

[0284] If the surfactant molecule is not ionized, it is classified as a nonionic surfactant. Nonionic surfactants find wide application in pharmaceutical products and are usable over a wide range of pH values. In general, their HLB values range from 2 to about 18 depending on their structure. Nonionic surfactants include nonionic esters such as ethylene glycol esters, propylene glycol esters, glyceryl esters, polyglyceryl esters, sorbitan esters, sucrose esters, and ethoxylated esters. Nonionic alkanolamides and ethers such as fatty alcohol ethoxylates, propoxylated alcohols, and ethoxylated/propoxylated block polymers are also included in this class. The polyoxyethylene surfactants are the most popular members of the nonionic surfactant class.

[0285] If the surfactant molecule carries a negative charge when it is dissolved or dispersed in water, the surfactant is classified as anionic. Anionic surfactants include carboxylates such as soaps, acyl lactylates, acyl amides of amino acids, esters of sulfuric acid such as alkyl sulfates and ethoxylated alkyl sulfates, sulfonates such as alkyl benzene sulfonates, acyl isethionates, acyl taurates and sulfosuccinates, and phosphates. The most important members of the anionic surfactant class are the alkyl sulfates and the soaps.

[0286] If the surfactant molecule carries a positive charge when it is dissolved or dispersed in water, the surfactant is classified as cationic. Cationic surfactants include quaternary ammonium salts and ethoxylated amines. The quaternary ammonium salts are the most used members of this class.

[0287] If the surfactant molecule has the ability to carry either a positive or negative charge, the surfactant is classified as amphoteric. Amphoteric surfactants include acrylic acid derivatives, substituted alkylamides, N-alkylbetaines and phosphatides.

[0288] The use of surfactants in drug products, formulations and in emulsions has been reviewed (Rieger, in "Pharmaceutical Dosage Forms," Marcel Dekker, Inc., New York, NY, 1988, p. 285).

[0289] Micelles and other Membranous Formulations. For ease of exposition the micelles and other formulations, compositions and methods in this section are discussed largely with regard to unmodified siRNA compounds. It may be understood, however, that these micelles and other formulations, compositions and methods can be practiced with other siRNA compounds, e.g., modified siRNA compounds, and such practice is within the invention. The siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, (e.g., a precursor, e.g., a larger siRNA compound which can be processed into a ssiRNA compound, or a DNA which encodes an siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, or precursor thereof))

composition can be provided as a micellar formulation. "Micelles" are defined herein as a particular type of molecular assembly in which amphipathic molecules are arranged in a spherical structure such that all the hydrophobic portions of the molecules are directed inward, leaving the hydrophilic portions in contact with the surrounding aqueous phase. The converse arrangement exists if the environment is hydrophobic.

[0290] A mixed micellar formulation suitable for delivery through transdermal membranes may be prepared by mixing an aqueous solution of the dsRNA composition, an alkali metal C.sub.8 to C.sub.22 alkyl sulphate, and a micelle forming compounds. Exemplary micelle forming compounds include lecithin, hyaluronic acid, pharmaceutically acceptable salts of hyaluronic acid, glycolic acid, lactic acid, chamomile extract, cucumber extract, oleic acid, linoleic acid, linolenic acid, monoolein, monooleates, monolaurates, borage oil, evening of primrose oil, menthol, trihydroxy oxo cholanyl glycine and pharmaceutically acceptable salts thereof, glycerin, polyglycerin, lysine, polylysine, triolein, polyoxyethylene ethers and analogues thereof, polidocanol alkyl ethers and analogues thereof, chenodeoxycholate, deoxycholate, and mixtures thereof. The micelle forming compounds may be added at the same time or after addition of the alkali metal alkyl sulphate. Mixed micelles will form with substantially any kind of mixing of the ingredients but vigorous mixing in order to provide smaller size micelles.

[0291] In one method, a first micellar composition is prepared which contains the dsRNA composition and at least the alkali metal alkyl sulphate. The first micellar composition is then mixed with at least three micelle forming compounds to form a mixed micellar composition. In another method, the micellar composition is prepared by mixing the dsRNA composition, the alkali metal alkyl sulphate and at least one of the micelle forming compounds, followed by addition of the remaining micelle forming compounds, with vigorous mixing.

[0292] Phenol and/or m-cresol may be added to the mixed micellar composition to stabilize the formulation and protect against bacterial growth. Alternatively, phenol and/or m-cresol may be added with the micelle forming ingredients. An isotonic agent such as glycerin may also be added after formation of the mixed micellar composition.

[0293] For delivery of the micellar formulation as a spray, the formulation can be put into an aerosol dispenser and the dispenser is charged with a propellant. The propellant, which is under pressure, is in liquid form in the dispenser. The ratios of the ingredients are adjusted so that the aqueous and propellant phases become one, i.e., there is one phase. If there are two phases, it is necessary to shake the dispenser prior to dispensing a portion of the contents, e.g., through a metered valve. The dispensed dose of pharmaceutical agent is propelled from the metered valve in a fine spray.

[0294] Propellants may include hydrogen-containing chlorofluorocarbons, hydrogen-containing fluorocarbons, dimethyl ether and diethyl ether. In certain embodiments, HFA 134a (1,1,1,2 tetrafluoroethane) may be used.

[0295] The specific concentrations of the essential ingredients can be determined by relatively straightforward experimentation. For absorption through the oral cavities, it is often desirable to increase, e.g., at least double or triple, the dosage for through injection or administration through the gastrointestinal tract.

[0296] Particles. In some embodiments, dsRNA preparations can be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

Pharmaceutical Compositions

[0297] The dsRNA agents of the invention can be formulated for pharmaceutical use. The present invention further relates to a pharmaceutical composition comprising the dsRNA molecule as defined herein. Pharmaceutically acceptable compositions comprise a therapeutically-effective amount of one or more of the dsRNA molecules in any of the preceding embodiments, taken alone

or formulated together with one or more pharmaceutically acceptable carriers (additives), excipient and/or diluents.

[0298] The pharmaceutical compositions may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, e.g., those targeted for buccal, sublingual, and systemic absorption, boluses, powders, granules, pastes for application to the tongue; (2) parenteral administration, for example, by subcutaneous, intramuscular, intravenous or epidural injection as, for example, a sterile solution or suspension, or sustained-release formulation; (3) topical application, for example, as a cream, ointment, or a controlled-release patch or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; (5) sublingually; (6) ocularly; (7) transdermally; or (8) nasally. Delivery using subcutaneous or intravenous methods can be particularly advantageous.

[0299] The phrase “therapeutically-effective amount” as used herein means that amount of a compound, material, or composition comprising a compound of the invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment.

[0300] The phrase “pharmaceutically acceptable” is employed herein to refer to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

[0301] The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, manufacturing aid (e.g., lubricant, talc magnesium, calcium or zinc stearate, or steric acid), or solvent encapsulating material, involved in carrying or transporting the subject compound from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the patient. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) lubricating agents, such as magnesium stearate, sodium lauryl sulfate and talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) pH buffered solutions; (21) polyesters, polycarbonates and/or polyanhydrides; (22) bulking agents, such as polypeptides and amino acids (23) serum component, such as serum albumin, HDL and LDL; and (22) other non-toxic compatible substances employed in pharmaceutical formulations.

[0302] The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 0.1 percent to about ninety-nine percent of active ingredient, preferably from about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

[0303] In certain embodiments, a formulation of the present invention comprises an excipient selected from the group consisting of cyclodextrins, celluloses, liposomes, micelle forming agents, e.g., bile acids, and polymeric carriers, e.g., polyesters and polyanhydrides; and a compound of the present invention. In certain embodiments, an aforementioned formulation renders orally bioavailable a compound of the present invention.

[0304] The dsRNA agent preparation can be formulated in combination with another agent, e.g., another therapeutic agent or an agent that stabilizes a dsRNA, e.g., a protein that complexes with the dsRNA to form an iRNP. Still other agents include chelating agents, e.g., EDTA (e.g., to remove divalent cations such as Mg^{2+}), salts, RNase inhibitors (e.g., a broad specificity RNase inhibitor such as RNasin) and so forth.

[0305] Methods of preparing these formulations or compositions include the step of bringing into association a compound of the present invention with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a compound of the present invention with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

[0306] In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accomplished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility.

[0307] The rate of absorption of the drug then depends upon its rate of dissolution which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

[0308] The compounds according to the invention may be formulated for administration in any convenient way for use in human or veterinary medicine, by analogy with other pharmaceuticals.

[0309] The term "treatment" is intended to encompass therapy and cure. The patient receiving this treatment is any animal in need, including primates, in particular humans, and other mammals such as equines, cattle, swine and sheep; and poultry and pets in general.

[0310] Double-stranded RNA agents are produced in a cell in vivo, e.g., from exogenous DNA templates that are delivered into the cell. For example, the DNA templates can be inserted into vectors and used as gene therapy vectors. Gene therapy vectors can be delivered to a subject by, for example, intravenous injection, local administration (U.S. Pat. No. 5,328,470, which is incorporated by reference in its entirety), or by stereotactic injection (see, e.g., Chen et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:3054-3057, which is incorporated by reference in its entirety). The pharmaceutical preparation of the gene therapy vector can include the gene therapy vector in an acceptable diluent, or can comprise a slow release matrix in which the gene delivery vehicle is imbedded. The DNA templates, for example, can include two transcription units, one that produces a transcript that includes the top strand of a dsRNA molecule and one that produces a transcript that includes the bottom strand of a dsRNA molecule. When the templates are transcribed, the dsRNA molecule is produced, and processed into siRNA agent fragments that mediate gene silencing.

[0311] The dsRNA molecule as defined herein or a pharmaceutical composition comprising a dsRNA molecule as defined herein can be administered to a subject using different routes of delivery. A composition that includes a dsRNA described herein can be delivered to a subject by a variety of routes. Exemplary routes include: intravenous, subcutaneous, topical, rectal, anal, vaginal, nasal, pulmonary, ocular.

[0312] The dsRNA molecule of the invention can be incorporated into pharmaceutical compositions suitable for administration. Such compositions typically include one or more species of dsRNAs and a pharmaceutically acceptable carrier. As used herein the language "pharmaceutically acceptable carrier" is intended to include any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration. The use of such media and agents for

pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the compositions is contemplated. Supplementary active compounds can also be incorporated into the compositions.

[0313] The compositions of the present invention may be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration may be topical (including ophthalmic, vaginal, rectal, intranasal, transdermal), oral or parenteral. Parenteral administration includes intravenous drip, subcutaneous, intraperitoneal or intramuscular injection, or intrathecal or intraventricular administration.

[0314] The route and site of administration may be chosen to enhance targeting. For example, to target muscle cells, intramuscular injection into the muscles of interest would be a logical choice. Lung cells might be targeted by administering the dsRNA in aerosol form. The vascular endothelial cells could be targeted by coating a balloon catheter with the dsRNA and mechanically introducing the dsRNA.

[0315] In one aspect, the invention features a method of administering a dsRNA molecule, e.g., a dsRNA agent described herein, to a subject (e.g., a human subject). In another aspect, the present invention relates to a dsRNA molecule as defined herein for use in inhibiting expression of a target gene in a subject. The method or the medical use includes administering a unit dose of the dsRNA molecule, e.g., a dsRNA agent described herein. In some embodiments, the unit dose is less than 10 mg per kg of bodyweight, or less than 10, 5, 2, 1, 0.5, 0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00005 or 0.00001 mg per kg of bodyweight, and less than 200 nmole of RNA agent (e.g., about 4.4×10^{16} copies) per kg of bodyweight, or less than 1500, 750, 300, 150, 75, 15, 7.5, 1.5, 0.75, 0.15, 0.075, 0.015, 0.0075, 0.0015, 0.00075, 0.00015 nmole of RNA agent per kg of bodyweight.

[0316] The defined amount can be an amount effective to treat or prevent a disease or disorder, e.g., a disease or disorder associated with the target gene. The unit dose, for example, can be administered by injection (e.g., intravenous, subcutaneous or intramuscular), an inhaled dose, or a topical application. In some embodiments dosages may be less than 10, 5, 2, 1, or 0.1 mg/kg of body weight.

[0317] In some embodiments, the unit dose is administered less frequently than once a day, e.g., less than every 2, 4, 8 or 30 days. In another embodiment, the unit dose is not administered with a frequency (e.g., not a regular frequency). For example, the unit dose may be administered a single time.

[0318] In some embodiments, the effective dose is administered with other traditional therapeutic modalities. In some embodiments, the subject has a viral infection and the modality is an antiviral agent other than a dsRNA molecule, e.g., other than a siRNA agent. In another embodiment, the subject has atherosclerosis and the effective dose of a dsRNA molecule, e.g., a siRNA agent, is administered in combination with, e.g., after surgical intervention, e.g., angioplasty.

[0319] In some embodiments, a subject is administered an initial dose and one or more maintenance doses of a dsRNA molecule, e.g., a siRNA agent, (e.g., a precursor, e.g., a larger dsRNA molecule which can be processed into a siRNA agent, or a DNA which encodes a dsRNA molecule, e.g., a siRNA agent, or precursor thereof). The maintenance dose or doses can be the same or lower than the initial dose, e.g., one-half less of the initial dose. A maintenance regimen can include treating the subject with a dose or doses ranging from 0.01 μ g to 15 mg/kg of body weight per day, e.g., 10, 1, 0.1, 0.01, 0.001, or 0.00001 mg per kg of bodyweight per day. The maintenance doses are, for example, administered no more than once every 2, 5, 10, or 30 days. Further, the treatment regimen may last for a period of time which will vary depending upon the nature of the particular disease, its severity and the overall condition of the patient. In certain embodiments the dosage may be delivered no more than once per day, e.g., no more than once per 24, 36, 48, or more hours, e.g., no more than once for every 5 or 8 days. Following treatment, the patient can be monitored for changes in his condition and for alleviation of the symptoms of the

disease state. The dosage of the compound may either be increased in the event the patient does not respond significantly to current dosage levels, or the dose may be decreased if an alleviation of the symptoms of the disease state is observed, if the disease state has been ablated, or if undesired side-effects are observed.

[0320] The effective dose can be administered in a single dose or in two or more doses, as desired or considered appropriate under the specific circumstances. If desired to facilitate repeated or frequent infusions, implantation of a delivery device, e.g., a pump, semi-permanent stent (e.g., intravenous, intraperitoneal, intracisternal or intracapsular), or reservoir may be advisable.

[0321] In some embodiments, the composition includes a plurality of dsRNA molecule species. In another embodiment, the dsRNA molecule species has sequences that are non-overlapping and non-adjacent to another species with respect to a naturally occurring target sequence. In another embodiment, the plurality of dsRNA molecule species is specific for different naturally occurring target genes. In another embodiment, the dsRNA molecule is allele specific.

[0322] The dsRNA molecules of the invention described herein can be administered to mammals, particularly large mammals such as nonhuman primates or humans in a number of ways.

[0323] In some embodiments, the administration of the dsRNA molecule, e.g., a siRNA agent, composition is parenteral, e.g., intravenous (e.g., as a bolus or as a diffusible infusion), intradermal, intraperitoneal, intramuscular, intrathecal, intraventricular, intracranial, subcutaneous, transmucosal, buccal, sublingual, endoscopic, rectal, oral, vaginal, topical, pulmonary, intranasal, urethral or ocular. Administration can be provided by the subject or by another person, e.g., a health care provider. The medication can be provided in measured doses or in a dispenser which delivers a metered dose. Selected modes of delivery are discussed in more detail below.

[0324] The invention provides methods, compositions, and kits, for rectal administration or delivery of dsRNA molecules described herein

[0325] In particular embodiments, the present invention relates to the dsRNA molecules of the present invention for use in the methods described above.

[0326] Methods of inhibiting expression of the target gene

[0327] Embodiments of the invention also relate to methods for inhibiting the expression of a target gene. The method comprises the step of administering the dsRNA molecules in any of the preceding embodiments, in an amount sufficient to inhibit expression of the target gene. The present invention further relates to a use of a dsRNA molecule as defined herein for inhibiting expression of a target gene in a target cell. In a preferred embodiment, the present invention further relates to a use of a dsRNA molecule for inhibiting expression of a target gene in a target cell in vitro.

[0328] Another aspect the invention relates to a method of modulating the expression of a target gene in a cell, comprising providing to said cell a dsRNA molecule of this invention. In some embodiments, the target gene is selected from the group consisting of Factor VII, Eg5, PCSK9, TPX2, apoB, SAA, TTR, RSV, PDGF beta gene, Erb-B gene, Src gene, CRK gene, GRB2 gene, RAS gene, MEKK gene, JNK gene, RAF gene, Erk1/2 gene, PCNA (p21) gene, MYB gene, JUN gene, FOS gene, BCL-2 gene, hepcidin, Activated Protein C, Cyclin D gene, VEGF gene, EGFR gene, Cyclin A gene, Cyclin E gene, WNT-1 gene, beta-catenin gene, c-MET gene, PKC gene, NFkB gene, STAT3 gene, survivin gene, Her2/Neu gene, topoisomerase I gene, topoisomerase II alpha gene, mutations in the p73 gene, mutations in the p21 (WAF1/CIP1) gene, mutations in the p27 (KIP1) gene, mutations in the PPMID gene, mutations in the RAS gene, mutations in the caveolin I gene, mutations in the MIB I gene, mutations in the MTAI gene, mutations in the M68 gene, mutations in tumor suppressor genes, and mutations in the p53 tumor suppressor gene.

[0329] In particular embodiments, the present invention relates to the dsRNA molecules of the present invention for use in the methods described above.

[0330] Exemplary embodiments of the various aspects can be described by the following lettered

embodiments: [0331] Embodiment A: A dsRNA agent comprising a sense strand and antisense,

each strand independently having a length of 15-35 nucleotides, wherein the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and wherein the antisense strand comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. [0332] Embodiment B: The dsRNA agent of Embodiment A, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand. [0333] Embodiment C: The dsRNA agent of Embodiment A or B, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand. [0334] Embodiment D: The dsRNA of any one of Embodiments A to C, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5-end of the sense strand. [0335] Embodiment E: The dsRNA agent of any one of Embodiments A to D, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9, counting from 5-end of the sense strand. [0336] Embodiment F: The dsRNA agent of any one of Embodiments A to E, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 11 and 12, counting from 5-end of the sense strand. [0337] Embodiment G: The dsRNA agent of any one of Embodiments A to F, wherein the sense strand comprises at least one 2'-OMe nucleotide. [0338] Embodiment H: The dsRNA agent of any one of Embodiments A to G, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, counting from 5'-end of the antisense strand. [0339] Embodiment I: The dsRNA agent of any one of Embodiments A to H, wherein the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5 and 7, counting from 5'-end of the antisense strand [0340] Embodiment J: The dsRNA agent of any one of Embodiments A to I, wherein the antisense strand comprises at least one 2'-fluoro nucleotide. [0341] Embodiment K: The dsRNA agent of any one of Embodiments A to J, wherein the antisense strand comprises a 2'-fluoro nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand. [0342] Embodiment L: The dsRNA agent of any one of Embodiments A to K, wherein the dsRNA agent comprises a ligand. [0343] Embodiment M: The dsRNA agent of any one of Embodiments A to L, wherein the sense strand comprises a ligand. [0344] Embodiment N: The dsRNA agent of Embodiment L or M, wherein the ligand is an ASGPR ligand. [0345] Embodiment O: The dsRNA agent of any one of Embodiments A to N, wherein the dsRNA agent comprises at least two phosphorothioate internucleotide linkages. [0346] Embodiment P: The dsRNA agent of any one of Embodiments A to O, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand. [0347] Embodiment Q: The dsRNA agent of any one of Embodiments A to P, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand. [0348] Embodiment R: The dsRNA agent of any one of Embodiments A to Q, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs. [0349] Embodiment S: The dsRNA agent of any one of Embodiments A to R, wherein the sense strand is 18-23 nucleotides in length. [0350] Embodiment T: The dsRNA agent of any one of Embodiments A to S, wherein the antisense strand is 18-25 nucleotides in length. [0351] Embodiment U: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and wherein the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. [0352] Embodiment V: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and wherein the antisense is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at position 5 and 7, counting from 5'-end of the antisense strand. [0353] Additional exemplary embodiments can be described by one or more of the following

numbered embodiments: [0354] Embodiment 1: dsRNA agent comprising a sense strand and antisense, each strand independently having a length of 15-35 nucleotides wherein each nucleotide is independently modified or unmodified, wherein: the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand, and wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand. [0355] Embodiment 2: The dsRNA agent of Embodiment 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand. [0356] Embodiment 3: The dsRNA agent of Embodiment 1 or 2, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand. [0357] Embodiment 4: The dsRNA of any one of Embodiments 1-3, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5-end of the sense strand. [0358] Embodiment 5: The dsRNA agent of any one of Embodiments 1-4, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9, counting from 5-end of the sense strand. [0359] Embodiment 6: The dsRNA agent of any one of Embodiments 1-5, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 11 and 12, counting from 5-end of the sense strand. [0360] Embodiment 7: The dsRNA agent of any one of Embodiments 1-6, wherein the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand. [0361] Embodiment 8: The dsRNA of any one of Embodiments 1-7, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand [0362] Embodiment 9: The dsRNA agent of any one of Embodiments 1-8, wherein the sense strand comprises at least one 2'-OMe nucleotide. [0363] Embodiment 10: The dsRNA agent of any one of Embodiments 1-9, wherein the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand. [0364] Embodiment 11: The dsRNA agent of any one of Embodiments 1-10, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand. [0365] Embodiment 12: The dsRNA agent of any one of Embodiments 1-11, wherein the antisense strand comprises a 2'-fluoro nucleotide at position 14 of the antisense strand, and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from 5'-end of the antisense strand. [0366] Embodiment 13: The dsRNA agent of any one of Embodiments 1-12, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. [0367] Embodiment 14: The dsRNA agent of any one of Embodiments 1-13, wherein the antisense strand comprises a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand. [0368] Embodiment 15: The dsRNA agent of any one of Embodiments 1-14, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand. [0369] Embodiment 16: The dsRNA agent of any one of Embodiments 1-11, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. [0370] Embodiment 17: The dsRNA agent of any one Embodiments 1-11 or 16, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand. [0371] Embodiment 18: The dsRNA agent of any one Embodiments 1-11 or 16-17, wherein the

antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12 and 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. [0372]

Embodiment 19: The dsRNA agent of any one of Embodiments 1-18, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. [0373]

Embodiment 20: The dsRNA agent of any one of Embodiments 1-19, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand. [0374]

Embodiment 21: The dsRNA agent of any one of Embodiments 1-20, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12 and 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. [0375]

Embodiment 22: The dsRNA agent of any one of Embodiments 1-11 or 16-21, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, 14 and 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. [0376]

Embodiment 23: The dsRNA agent of any one of Embodiments 1-11 or 16-22, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, 14 and 16, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. [0377]

Embodiment 24: The dsRNA agent of any one of Embodiments 1-15 or 19-20, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, and 16 and a 2'-fluoro at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. [0378]

Embodiment 25: The dsRNA agent of any one of Embodiments 1-15, 19-20 or 24, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, and 16 and a 2'-fluoro at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand. [0379]

Embodiment 26: The dsRNA agent of any one of Embodiments 1-25, wherein the dsRNA agent comprises a ligand. [0380]

Embodiment 27: The dsRNA agent of any one of Embodiments 1-26, wherein the sense strand comprises a ligand. [0381]

Embodiment 28: The dsRNA agent of Embodiment 27, wherein the ligand is at 3'-end of the sense strand. [0382]

Embodiment 29: The dsRNA agent of Embodiment 27, wherein the ligand is at 5'-end of the sense strand. [0383]

Embodiment 30: The dsRNA agent of any one of Embodiments 26-29, wherein the ligand comprises an ASGPR ligand. [0384]

Embodiment 31: The dsRNA agent of any one of Embodiments 26-29, wherein the ligand is lipophilic group. [0385]

Embodiment 32: The dsRNA agent of Embodiment 31, wherein the ligand is a C10-30aliphatic group. [0386]

Embodiment 33: The dsRNA agent of Embodiment 32, wherein the C10-30aliphatic [0387] group is a C.sub.10-30alkyl group. [0388]

Embodiment 34: The dsRNA agent of Embodiment 33, wherein the C.sub.10-30alkyl group is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. [0389]

Embodiment 35: The dsRNA agent of any one of Embodiments 27, wherein the ligand is conjugated to a non-terminal nucleotide of the sense strand. [0390]

Embodiment 36: The dsRNA agent of Embodiment 35, wherein the ligand is conjugated to the 2'-position of a non-terminal nucleotide of the sense strand, optionally conjugated to one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5' end). [0391]

Embodiment 37: The dsRNA agent of Embodiment 26-36, wherein the ligand comprises an abasic nucleotide, optionally the abasic nucleotide is an inverted nucleotide and linked via a 5'->5' or a 3'->3' linkage to a strand of the dsRNA agent. [0392]

Embodiment 38: The dsRNA agent of any one of Embodiments 26-37, wherein the ligand is attached at the 3'-end of the sense strand. [0393]

Embodiment 39: The dsRNA agent of Embodiment 38, wherein the ligand is attached at the 3'-end of the sense strand via a 3'->3' linkage. [0394] Embodiment 40: The dsRNA agent of any one of Embodiments 1-39, wherein the dsRNA comprises two ligands. [0395] Embodiment 41: The dsRNA of Embodiment 40, wherein the sense strand comprises a first ligand attached at the 3'-end of the sense strand and a second ligand attached at the 5'-end of the sense strand. [0396] Embodiment 42: The dsRNA of Embodiment 41, wherein the first ligand comprises an abasic nucleotide and the second ligand comprises an ASGPR ligand, optionally the abasic nucleotide is an inverted nucleotide and linked via a 3'->3' linkage to the sense strand. [0397] Embodiment 43: The dsRNA agent of any one of Embodiments 1-42, wherein the dsRNA agent comprises at least two phosphorothioate internucleotide linkages. [0398] Embodiment 44: The dsRNA agent of any one of Embodiments 1-43, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand. [0399] Embodiment 45: The dsRNA agent of any one of Embodiments 1-44, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand. [0400] Embodiment 46: The dsRNA agent of any one of Embodiments 1-45, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs. [0401] Embodiment 47: The dsRNA agent of any one of Embodiments 1-46, wherein the sense strand is 18-23 nucleotides in length. [0402] Embodiment 48: The dsRNA agent of any one of Embodiments 1-47, wherein the antisense strand is 18-25 nucleotides in length. [0403] Embodiment 49: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand, wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand. [0404] Embodiment 50: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the antisense is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, counting from 5'-end of the antisense strand, wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand. [0405] Embodiment 51: The dsRNA agent of any one of Embodiments 1-50, comprising a phosphate mimic at the 5'-end of the antisense strand. [0406] Embodiment 52: The dsRNA agent of Embodiment 51, wherein the phosphate mimic is a 5'-E-vinyl phosphonate. [0407] Embodiment 53: The dsRNA agent of Embodiment 52, wherein the phosphate mimic is a 5'-cyclopropylphosphonate having the structure:

##STR00042##

where * is a bond to C.sub.5 position of the nucleotide at the 5'-terminus. [0408] Embodiment 54: The dsRNA agent of any one of Embodiments 1-53, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are unmodified nucleotides or modified nucleotides, optionally selected from the groups consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C10-

30aliphatic group. [0409] Embodiment 55: The dsRNA agent of any one of Embodiments 1-54, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C10-30aliphatic group. [0410] Embodiment 56: The dsRNA agent of any one of Embodiments 1-55, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the antisense strand are unmodified nucleotides or modified nucleotides, optionally selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA. [0411] Embodiment 57: The dsRNA agent of any one of Embodiments 1-56, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the antisense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA, and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

Some Selected Definitions

[0412] For convenience, certain terms employed herein, in the specification, examples and appended claims are collected herein. Unless stated otherwise, or implicit from context, the following terms and phrases include the meanings provided below. Unless explicitly stated otherwise, or apparent from context, the terms and phrases below do not exclude the meaning that the term or phrase has acquired in the art to which it pertains. The definitions are provided to aid in describing particular embodiments, and are not intended to limit the claimed invention, because the scope of the invention is limited only by the claims. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.

[0413] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as those commonly understood to one of ordinary skill in the art to which this invention pertains. Although any known methods, devices, and materials may be used in the practice or testing of the invention, the methods, devices, and materials in this regard are described herein.

[0414] Further, the practice of the present invention can employ, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry, and immunology, which are within the skill of the art. Such techniques are explained fully in the literature, such as, “*Molecular Cloning: A Laboratory Manual*”, second edition (Sambrook et al., 1989); “*Oligonucleotide Synthesis*” (M. J. Gait, ed., 1984); “*Animal Cell Culture*” (R. I. Freshney, ed., 1987); “*Methods in Enzymology*” (Academic Press, Inc.); “*Current Protocols in Molecular Biology*” (F. M. Ausubel et al., eds., 1987, and periodic updates); “*PCR: The Polymerase Chain Reaction*”, (Mullis et al., ed., 1994); “*A Practical Guide to Molecular Cloning*” (Perbal Bernard V., 1988); “*Phage Display: A Laboratory Manual*” (Barbas et al., 2001).

[0415] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0416] Certain ranges are presented herein with numerical values being preceded by the term “about.” The term “about” is used herein to provide literal support for the exact number that it precedes, as well as a number that is near to or approximately the number that the term precedes. In determining whether a number is near to or approximately a specifically recited number, the near or approximating unrecited number may be a number which, in the context in which it is presented, provides the substantial equivalent of the specifically recited number.

[0417] As used herein the term “comprising” or “comprises” is used in reference to compositions,

methods, and respective component(s) thereof, that are essential to the invention, yet open to the inclusion of unspecified elements, whether essential or not.

[0418] The singular terms “a,” “an,” and “the” include plural referents unless context clearly indicates otherwise. Similarly, the word “or” is intended to include “and” unless the context clearly indicates otherwise. It is further noted that the claims can be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

[0419] As used herein, the terms “dsRNA”, “siRNA”, and “iRNA agent” are used interchangeably to refer to agents that can mediate silencing of a target RNA, e.g., mRNA, e.g., a transcript of a gene that encodes a protein. For convenience, such mRNA is also referred to herein as mRNA to be silenced. Such a gene is also referred to as a target gene. In general, the RNA to be silenced is an endogenous gene, exogenous gene or a pathogen gene. In addition, RNAs other than mRNA, e.g., tRNAs, and viral RNAs, can also be targeted.

[0420] As used herein, the phrase “mediates RNAi” refers to the ability to silence, in a sequence specific manner, a target gene, e.g., mRNA. While not wishing to be bound by theory, it is believed that silencing uses the RNAi machinery or process and a guide RNA, e.g., antisense strand of a dsRNA, where the antisense strand is 21 to 23 nucleotides in length.

[0421] As used herein, “specifically hybridizable” and “complementary” are terms which are used to indicate a sufficient degree of complementarity such that stable and specific binding occurs between a compound of the invention and a target RNA molecule. Specific binding requires a sufficient degree of complementarity to avoid non-specific binding of the oligomeric compound to non-target sequences under conditions in which specific binding is desired, i.e., under physiological conditions in the case of assays or therapeutic treatment, or in the case of in vitro assays, under conditions in which the assays are performed. The non-target sequences typically differ by at least 5 nucleotides.

[0422] In some embodiments, a dsRNA molecule of the invention is “sufficiently complementary” to a target RNA, e.g., a target mRNA, such that the dsRNA molecule silences production of protein encoded by the target mRNA. In another embodiment, the dsRNA molecule of the invention is “exactly complementary” to a target RNA, e.g., the target RNA and the dsRNA duplex agent anneal, for example to form a hybrid made exclusively of Watson-Crick base pairs in the region of exact complementarity. A “sufficiently complementary” target RNA can include an internal region (e.g., of at least 10 nucleotides) that is exactly complementary to a target RNA. Moreover, in some embodiments, the dsRNA molecule of the invention specifically discriminates a single-nucleotide difference. In this case, the dsRNA molecule only mediates RNAi if exact complementarity is found in the region (e.g., within 7 nucleotides of) the single-nucleotide difference.

[0423] The term ‘BNA’ refers to bridged nucleic acid, and is often referred as constrained or inaccessible RNA. BNA can contain a 5-, 6-membered, or even a 7-membered bridged structure with a “fixed” C3'-endo sugar pucker. The bridge is typically incorporated at the 2'-, 4'-position of the ribose to afford a 2', 4'-BNA nucleotide (e.g., LNA, or ENA). Examples of BNA nucleotides include the following nucleosides:

##STR00043##

[0424] The term ‘LNA’ refers to locked nucleic acid, and is often referred as constrained or inaccessible RNA. LNA is a modified RNA nucleotide. The ribose moiety of an LNA nucleotide is modified with an extra bridge (e.g., a methylene bridge or an ethylene bridge) connecting the 2' hydroxyl to the 4' carbon of the same ribose sugar. For instance, the bridge can “lock” the ribose in the 3'-endo North) conformation:

##STR00044##

[0425] The term ‘ENA’ refers to ethylene-bridged nucleic acid, and is often referred as constrained or inaccessible RNA.

[0426] The “cleavage site” herein means the backbone linkage in the target gene or the sense strand that is cleaved by the RISC mechanism by utilizing the iRNA agent. And the target cleavage site region comprises at least one or at least two nucleotides on both side of the cleavage site. For the sense strand, the cleavage site is the backbone linkage in the sense strand that would get cleaved if the sense strand itself was the target to be cleaved by the RNAi mechanism. The cleavage site can be determined using methods known in the art, for example the 5'-RACE assay as detailed in Soutschek et al., *Nature* (2004) 432, 173-178, which is incorporated by reference in its entirety. As is well understood in the art, the cleavage site region for a conical double stranded RNAi agent comprising two 21-nucleotides long strands (wherein the strands form a double stranded region of 19 consecutive base pairs having 2-nucleotide single stranded overhangs at the 3'-ends), the cleavage site region corresponds to positions 9-12 from the 5'-end of the sense strand.

[0427] The terms “decrease”, “reduced”, “reduction”, or “inhibit” are all used herein to mean a decrease by a statistically significant amount. In some embodiments, “reduce,” “reduction” or “decrease” or “inhibit” typically means a decrease by at least 10% as compared to a reference level (e.g. the absence of a given treatment) and can include, for example, a decrease by at least about 10%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 98%, at least about 99%, or more. As used herein, “reduction” or “inhibition” does not encompass a complete inhibition or reduction as compared to a reference level. “Complete inhibition” is a 100% inhibition as compared to a reference level. A decrease can be preferably down to a level accepted as within the range of normal for an individual without a given disorder.

[0428] As used herein, a “central region” of a strand refers to positions 5-17, e.g., positions 6-16, positions 6-15, positions 6-14, positions 6-13, positions 6-12, positions 7-15, positions 7-14, positions 7-13, positions 7-12, positions 8-16, positions 8-15, positions 8-14, positions 8-13, positions 8-12, positions 9-16, positions 9-15, positions 9-14, positions 9-13, positions 9-12, positions 10-16, positions 10-15, positions 10-14, positions 10-13 or positions 10-12, counting from the 5'-end of the strand. For example, the central region of a strand means positions 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 or 17 of the strand. A preferred central region for the sense strand is positions 6, 7, 8, 9, 10, 11, 12, 13, and 14, counting from the 5'-end of the sense strand. A more preferred central region for the sense strand is positions 7, 8, 9, 10, 11, 12 and 13, counting from the 5'-end of the sense strand. A preferred central region for the antisense strand is positions 9, 10, 11, 12, 13, 14, 15 16 and 17, counting from 5'-end of the antisense strand. A more preferred central region for the antisense strand is positions 10, 11, 12, 13, 14, 15 and 16, counting from 5'-end of the antisense strand.

[0429] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual aspects described and illustrated herein has discrete components and features which can be readily separated from or combined with the features of any of the other several aspects without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

[0430] The invention is further illustrated by the following examples, which should not be construed as further limiting. The contents of all references, pending patent applications and published patents, cited throughout this application are hereby expressly incorporated by reference.

Examples

Oligonucleotide Synthesis and Purification

[0431] All oligonucleotides were prepared on a MerMade 192 synthesizer on a 1 μ mole scale using universal or custom supports. All phosphoramidites were used at a concentration 100 mM in 100% Acetonitrile or 9:1 Acetonitrile: DMF with a standard protocol for 2-cyanoethyl phosphoramidites, except that the coupling time was extended to 400 seconds. Oxidation of the newly formed linkages

was achieved using a solution of 50 mM 12 in 9:1 Acetonitrile: Water to create phosphate linkages and 100 mM DDTT in 9:1 Pyridine: Acetonitrile to create phosphorothioate linkages. After the trityl-off synthesis, columns were incubated with 150 μ L of 40% aqueous Methylamine for 45 minutes and the solution drained via vacuum into a 96-well plate. After repeating the incubation and draining with a fresh portion of aqueous Methylamine, the plate containing crude oligonucleotide solution was sealed and shaken at room temperature for an additional 60 minutes to completely remove all protecting groups. Precipitation of the crude oligonucleotides was accomplished via the addition of 1.2 mL of 9:1 Acetonitrile: EtOH to each well followed by incubation at -20° C. overnight. The plate was then centrifuged at 3000 RPM for 45 minutes, the supernatant removed from each well, and the pellets resuspended in 950 μ L of 20 mM aqueous NaOAc. Each crude solution was finally desalted over a GE Hi-Trap Desalting Column (Sephadex G25 Superfine) using water to elute the final oligonucleotide products. All identities and purities were confirmed using ESI-MS and IEX HPLC, respectively.

Cell Culture and Transfections

[0432] Primary Mouse or Cyno Hepatocytes (Thermo Fisher Scientific/Gibco) were transfected by adding 4.9 μ L of Opti-MEM plus 0.1 μ L of Lipofectamine RNAiMax per well (Invitrogen, cat #13778-150) to 5 μ L of siRNA duplexes per well into a 384-well plate and incubated at room temperature for 15 minutes. 40 μ L of Dulbecco's Modified Eagle Medium (PCH) or William's Medium (PMH) containing $\sim 5 \times 10^5$ cells were then added to the siRNA mixture. Cells were incubated for 24 hours at 37° C. and then processed for RNA purification. Experiments were performed at 10 nM, 1 nM, and 0.1 nM doses of siRNA.

[0433] Sequences of parent dsRNA molecules are shown in Tables 1-3 and abbreviations used in the sequences are summarized in Table 4.

TABLE-US-00001	TABLE	1	Sequences of parent dsRNA molecules	SEQ ID	SEQ ID	Duplex
ID	ID	target ID	Sense Sequence (5'.fwdarw.3')	NO:	Antisense Sequence (5'.fwdarw.3')	NO:
AGT AD-67327	uscsuccAfcCfUfUfuucuucuaauL96	5	asUfsuagAfagaaaagGfuGfggagascsu	37		
AGT AD-85435	csascaauGfaGfAfGfuaccugugaaL96	6	usUfscacagguacucUfcAfuugugsgsa	38		
AGT AD-85438	cscsucaaCfuGfGfAfugaagaaacuL96	7	asGfsuuucuucuccAfgUfugaggsgsa	39		
AGT AD-85446	gscsugagAfaGfAfUfugacagguuaL96	8	usAfsaccugucaaucUfuCfucagcsasg	40		
F9 AD-674282	gsasagcaCfcUfGfAfacaccgucauL96	9	asUfsgacGfgUfGfuucaGfgUfgcuucsasu	41		
F9 AD-674283	asasugugCfaGfUfUfuuccuugauuL96	10	asAfsucaAfgGfAfaaacUfgCfacauuscsg	42		
F9 AD-674287	uscscuugUfuUfAfAfaugguggaauL96	11	asUfsuccAfcCfAfuuuuAfaCfaaggasusu	43		
F9 AD-674289	asasaugGfcAfGfGfugcaagcaguL96	12	asCfsugcUfuGfCfaccuGfcCfauuuususa	44		
F9 AD-674293	uscsauucAfaGfAfUfgacaucacuuL96	13	asAfsgugAfuGfUfcaucUfuGfaaugasasu	45		
F9 AD-674299	gsasacugGfaUfAfAfaccuuuaauaL96	14	usAfsuuaAfaGfGfuuuuUfcCfaguucscsa	46		
F9 AD-674303	usgsaaggAfgGfCfAfaagauucguuL96	15	asAfscgaAfuCfUfuugcCfuCfcuucascsg	47		
F9 AD-674305	asasuauaUfaCfUfAfagguuucccuL96	16	asGfsggaAfaCfCfuuaUfaUfauuuuscsc	48		
F9 AD-674307	ascsagacUfaGfUfCfuuuuacuuL96	17	asAfsgguAfgAfAfagacUfaGfucugusasa	49		
F9 AD-674310	csasuaagGfcGfUfGfaugguucuuuL96	18	asAfsagaAfcCfAfucacGfcCfuuaugsgsg	50		
F9 AD-674312	ususuaugGfcCfAfGfaagaauacaaL96	19	usUfsguaUfuCfUfucugGfcCfaaaaasug	51		
F9 AD-674314	csuscuccGfuCfAfUfucucaccuuuL96	20	asAfsaggUfgAfGfaaugAfcGfgagagsgsu	52		
F9 AD-674323	uscsucauCfuGfAfGfacuuggugaaL96	21	usUfscacCfaAfGfucucAfgAfugagasasa	53		
F9 AD-674325	gsascucuGfcUfAfAfcguuccacuaL96	22	usAfsgugGfaAfCfuuuaGfcAfgagucsasg	54		
F9 AD-68435	usgsugcaAfuGfAfAfggcaaaauauL96	23	asUfsauuUfgCfCfuuuuAfuUfgcacascsu	55		
mTTR AD-	csasuguUfcUfUfGfcucuauaaaaL96	24	usUfsuuaUfagagcaaGfaAfcacugsusu	56		
1181392 mTTR AD-	uscsuugcUfcUfAfUfaaaccguguuL96	25	asAfsacGfguuuuaGfaGfcaagasasc	57		
1181401 (Sequence 1) mTTR AD-	asgsuguuUfcUfGfCfucuauaaacaL96	26				
usGfsuuuAfuagagcaAfgAfacacusgsu	58	1181410 (Sequence 2) mTTR AD-				
asgsaacuGfgAfCfAfccaaucguaL96	27	usAfsagaUfuugguguCfcAfguucuscasc	59	1181417 mTTR		
AD- ascsagugUfuCfUfUfgcucuauaaaL96	28	usUfsuauAfgagcaagAfaCfacugususu	60	1181426		

(Sequence 3) mTTR AD- ususcuugCfuCfUfAfuaaacguguL96 29
asCfsacgGfuuuauagAfgCfaagaascsa 61 1181443 mTTR AD- gsusguucUfuGfCfUfcuaaaaaccaL96
30 usGfsguuUfauagagcAfaGfaacacsusg 62 1181451 (Sequence 4) mTTR AD-
asascuggAfcAfCfCfaaaucguacuL96 31 asGfsuacGfauuugguGfuCfcaguuscsu 63 1181460 mTTR
AD csasggagGfaCfCfAfggaucuugcaL96 32 usGfscAAGfauccuggUfcCfuccugsgsg 64 1181469 TTR
AD-157448 csasauaaAfaCfAfUfuccugugaaaL96 33 usUfsucacaggaaugUfuUfuauugsusc 65 TTR
AD-157464 csusaaagCfaGfUfGfuuuucaccuaL96 34 usAfsggugaaaaacacUfgCfuuuagsusa 66 TTR
AD-157468 csasgagaCfaAfUfAfaaacauuccuL96 35 asGfsgaauguuuuuauUfgUfcucugscsc 67 none
AD-64972 uscscucuGfaUfGfGfucaaaguccuL96 36 asGfsgacUfuugaccaUfcAfgaggascsa 68
TABLE-US-00002 TABLE 2 Additional exemplary sequences of parent dsRNA
molecules. SEQ SEQ Duplex ID ID ID Target Sense sequence (5'.fwdarw.3') NO: Antisense
sequence (5'.fwdarw.3') NO: AD- Marc1 asasaaauGfuUfCfUfcaaaaugaaL96 69
usUfscauUfuUfUfgagaAfcAfuuuuusasa 237 1531684 AD- Marc1
asasaucAfcCfAfCfucuuugggcaL96 70 usGfscAAcAfaAfGfagugGfuGfauuuuscsc 238 1531719 AD-
Marc1 asasagugGfgAfGfAfcccuguguaaL96 71 usUfsacaCfaGfGfgucuCfcCfacuuusgsa 239
1531682 AD-75247 PNPLA3 asasaugaAfaGfAfCfaaagguggauL96 72
asUfscCAcfcUfUfugucUfuUfcuuuuscsc 240 AD- C3 asasaugaGfgGfUfUfucacagucuaaL96 73
usUfsgacUfgUfGfaaacCfcUfcuuuuscsc 241 1531655 AD-67589 PNPLA3
asascuugCfuAfCfCfcuuaggauaL96 74 usAfsuccUfaAfUfggguAfgCfaaguusgsc 242 AD- Marc1
asasucacCfaCfUfCfuuuugggcaguL96 75 asCfsugcCfcAfAfagagUfgGfugauususu 243 1531703 AD-
C3 ascsauggGfcCfAfGfuggaagaucaL96 76 usGfsaucUfuCfCfacugGfcCfcaugusug 244 1531665
AD- C3 ascscaggAfaCfUfGfaaccuugauaL96 77 usAfsucaAfgGfUfucagUfuCfcuggusgsg 245
1531660 AD- C3 ascscaggAfuGfCfCfacuaugucuaL96 78 usAfsgacAfuAfGfuggcAfuCfcugguscsu
246 1531672 AD- PNPLA3 ascscuaaCfuAfAfAfaaauguuuuL96 79
usAfsaacAfuUfAfuuuuAfgUfuaggusgsa 247 1010735 AD- Marc1
ascscucgCfcUfGfGfuccugauuuuL96 80 usAfsaaUcfaGfGfaccagfgCfaggususc 248 1531721 AD-
67605 PNPLA3 ascscuguUfgAfAfUfuuuuguauuuuL96 81 asUfsaaUcfaAfAfaauUcfaAfcaggusasa
249 AD- C3 ascsuacaUfgAfAfCfcuacagagauL96 82 asUfscucUfgUfAfgguUcfaUfuguagusug 250
571552 AD- Marc1 ascsgauUfaUfGfGfaauaguucuuL96 83
asAfsgaaCfuAfUfuccaUfaAfucagususa 251 1531716 AD- C3 asgsaaaUfcUfAfCfuacaucauuL96
84 asAfsuagAfuGfUfaguaGfaAfuuuuscsc 252 569269 AD- Marc1
asgsacagGfaUfUfCfugaaaacucaL96 85 usGfsaguUfuUfCfagaaUfcCfugucusug 253 1531718 AD-
C3 asgsagcgGfgUfAfCfcucuucaucaL96 86 usGfsaugAfaGfAfgguaCfcCfcgucuscsc 254 1531663
AD- SCN9A asgscaaaGfgUfCfAfcauuuccuaL96 87 usAfsggaAfaUfUfgugaCfcUfuugcuscsa 255
1531765 AD- SCN9A asgscauaAfaUfGfUfuuuucgaaauaL96 88
usAfsuuUcfaAfAfaaUfuUfaugcususc 256 1531735 AD- PNPLA3
asgscaugAfgGfUfUfcuuagaaguL96 89 asCfsauUcfaAfAfgaacCfuCfaugcusgsg 257 519933 AD-
PNPLA3 asgsaagCfaAfCfCfuuuccgcuL96 90 asCfsaggCfgAfAfagguUfgCfuuccusasg 258
519780 AD- Marc1 asgsgaccAfgAfUfUfgcuuacucaL96 91
usUfsgagUfaAfGfcauUcfaGfguccusug 259 1531679 AD- Marc1
asgsgagaAfgAfAfAfagugauucaL96 92 usUfsgaaUfcAfCfuuuUcfaUfcuccuscsc 260 1531692 AD-
SCN9A asgsucuuCfaAfGfUfuggcaaaaauaL96 93 usAfsuuUfgCfCfaacuUfgAfagacusgsc 261
1531741 AD- C3 asgsuggaCfuAfUfGfuguacaagauL96 94 asUfscuuGfuAfCfacuAfgUfccacuscsc
262 572575 AD- SCN9A asgsuuccUfaUfCfUfcuuucagaaL96 95
usUfscugAfaAfGfgagaUfaGfgaacusasc 263 1531744 AD- PNPLA3
asusaaugUfcUfUfAfuguaaugcuL96 96 asAfsgcaUfuAfCfauaaGfaCfauuuscsc 264 519354 AD-
67551 PNPLA3 asusacauGfaGfCfAfagauuugcaL96 97 usUfsgcaAfaUfCfuugcUfcAfuguauscsc
265 AD- Marc1 asuscaacCfaGfGfAfgggaaacauaL96 98 usAfsuguUfuCfCfcuccUfgGfuugauscsa
266 1531687 AD- SCN9A asuscaucUfuUfGfGfgucauucuuuL96 99
asAfsagaAfuGfAfcccaAfaGfaugausasa 267 1531736 AD- Marc1

asuscugaUfgAfAfGfauuuuuuuL96 100 asAfsaaaAfuAfUfacuuUfcagauscsu 268 1531688
AD- SCN9A asuscuucUfuUfGfUfcguagugauuL96 101 asAfsucaCfuAfCfagacaAfaGfaagauscsa 269
1531747 AD- SCN9A asusgcugAfgAfAfAfuugucgaaaaL96 102
usUfsuucGfaCfAfauuuCfuCfagcauscsu 270 1531756 AD- Marc1
asusggcuUfgUfUfCfcagaugcauuL96 103 asAfsugcAfuCfUfggaaCfaAfgccauscsa 271 1531696 AD-
C3 asusguacCfaUfGfCfuaaggccaaaL96 104 usUfsuggCfcUfUfagcaUfgGfuacaususg 272 1531658
AD- SCN9A asusgucgAfgUfAfCfacuuuuacuuL96 105 asAfsguaAfaAfGfuguaCfuCfagacaususu 273
1531757 AD-67565 PNPLA3 asusguuaGfuAfGfAfauaagccuuL96 106
usAfsaggCfuUfAfuucuAfcUfaacauscsu 274 AD- PNPLA3 asusuaggAfuAfAfUfgucuuanguaL96
107 usAfscauAfaGfAfcauuAfuCfcuaausgsg 275 1010714 AD- SCN9A
asusucucUfuCfGfUfucacagaugaL96 108 usCfsaucUfgUfGfaacgAfaGfagaauscsc 276 1531728 AD-
Marc1 csasacacUfuGfAfAfgcaugguguuL96 109 asAfscaCfaUfGfcuucAfaGfuguugsusc 277
1531686 AD-67583 PNPLA3 csasagauUfuGfCfAfacuugcuacaL96 110
usGfsuagCfaAfGfuugcAfaAfucuuugscsu 278 AD- C3 csasagguCfuAfCfGfccuauuacauL96 111
asUfsguaAfuAfGfgcguAfgAfccuugsasc 279 572388 AD- C3 csascccuCfaUfCfAfucuaaccuggaL96
112 usCfscagGfuAfGfaugaUfgAfgggugsusu 280 1531662 AD- C3
csasccguAfuCfCfAfcugggaucuuL96 113 asGfsauuCfcCfAfguggAfuAfcggugsgsg 281 571901 AD-
PNPLA3 csasccuuUfuUfCfAfccuaacuaaaaL96 114 usUfsuagUfuAfGfgugaAfaAfaggugsusu 282
1010732 AD- PNPLA3 csasccuuUfuUfCfAfccuaacuaauL96 115
asUfsuagUfuAfGfgugaAfaAfaggugsusu 283 520061 AD- SCN9A
csascuccUfuCfCfUfgauuguguuuL96 116 asAfsacaCfaAfUfcaggAfaGfgagugsgsa 284 1531770 AD-
Marc1 csasgaacGfaAfAfGfuuaauaggaaL96 117 usUfsccaUfaUfAfacuuUfcGfuucugsasa 285
1531705 AD- C3 csasgagaAfaUfUfCfuacuacaucuuL96 118 asGfsaugUfaGfUfagaaUfuUfcucugsusa
286 569266 AD- SCN9A csasugaaCfGfAfCfUfucuuaccacuuL96 119
asAfsugugGfaAfGfaaguCfGfUfucaugsusg 287 1531740 AD-75265 PNPLA3
csasugagCfaAfGfAfuuugcaacuuL96 120 asAfsguuGfcAfAfaucUfgCfucaugsus 288 AD- Marc1
csasugguGfuUfUfCfagaacugagaL96 121 usCfsucaGfuUfCfugaaAfcAfccaugscsu 289 1531709 AD-
C3 csasuugaGfaAfCfCfcggaaggcauL96 122 asUfsgccUfuCfCfggguUfcUfcaugsusu 290 569048
AD- C3 cscsacagCfcAfAfAfgauaagaacuuL96 123 asGfsuucUfuAfUfcuuuGfgCfuguggsusc 291
571633 AD- C3 cscsagauCfcAfCfUfucaccaagauL96 124 asUfscuuGfgUfGfaaguGfgAfucuggsusa
292 569516 AD- PNPLA3 cscsauuaGfgAfUfAfaugucuuauuL96 125
asAfsuaaGfaCfAfuuauCfcUfaauggsgsu 293 519346 AD- C3 cscscgucGfuGfCfGfuuggcucaauL96
126 asUfsugaGfcCfAfacgcAfcGfacgggsasg 294 571748 AD- Marc1
cscsgaccCfaAfGfGfaccagauuguL96 127 asCfsauuCfuGfGfuccuUfgGfgucggsasa 295 1531708 AD-
C3 cscsgagcCfGfUfUfCfucuaaauuuL96 128 asAfsauuGfuAfGfagaaCfGfucggsasu 296 570712
AD- Marc1 cscsguauGfuCfCfUfggaauauuuL96 129 asUfsauuAfuUfCfcaggAfcAfuacggsusu 297
1531706 AD-67584 PNPLA3 cscsuaacUfaAfAfAfuaaanguuuuuL96 130
usUfsaaaCfaUfUfauuuUfaGfuuaaggsusg 298 AD- C3 cscsuuguCfuUfCfUfcagaaccagaL96 131
usCfsuggUfuCfUfgagaAfgAfcaaggsasg 299 1531668 AD-67577 PNPLA3
csgsacauCfuGfCfCfcuaaagucaL96 132 usUfsgacUfuUfAfgggcAfgAfugucgsusa 300 AD- C3
csgsugcgUfuGfGfCfucaaugaacuL96 133 asGfsuucAfuUfGfagccAfaCfgcacgsasc 301 571753 AD-
C3 csusacccUfaCfUfCfuguuguucgaL96 134 usCfsgaaCfaAfCfagagUfaGfgguagscsc 302 1531657
AD- C3 csusacugCfaGfCfUfaaaagacuuuL96 135 asAfsaguCfuUfUfuagcUfgCfaguagsgsg 303
571715 AD- C3 csusagugCfuGfUfCfcagugagaaaL96 136 usUfsucuCfaCfUfggacAfgCfacuagsusu
304 1531659 AD-67560 PNPLA3 csusauuaAfuGfGfUfcagacuguuaL96 137
usAfsacaGfuCfUfgaccAfuUfaauagsgsg 305 AD- PNPLA3 csusccauGfgCfGfGfggguaacaaaL96
138 usUfsuguUfaCfCfccccGfCfAfuggagsasc 306 1531674 AD- PNPLA3
csusgaguUfgGfUfUfuaugaaaaauL96 139 asUfsuuuCfaUfAfaaacCfaAfcucagscsu 307 519757 AD-
SCN9A csusgcccAfaAfAfUfacugauaauL96 140 usAfsuaaUfcAfGfuauuUfuGfggcagscsa 308
1531732 AD- C3 csusgggaGfgAfCfCfcugguaagcaL96 141 usGfscuuAfcCfAfggggUfcUfcccagscsg

309 1531656 AD- Marc1 csusguggAfgGfAfGfaagaaaagualL96 142
usAfsccuuUfuCfUfucucCfuCfcacagsasa 310 1531678 AD- Marc1
csusucuuAfuUfGfGfugacguggaaL96 143 usUfscCaCfGfUfCfaccAfuAfagaagscsu 311 1531701 AD-
Marc1 csusuguuCfcAfGfAfugcauuuuuaaL96 144 usUfsaaaAfuGfCfaucuGfgAfacaagscsc 312
1531680 AD- SCN9A csusuuguCfGfUfAfGfugauuuuccuL96 145
asGfsgaaAfaUfCfaccuAfgAfaaagsasa 313 1531734 AD- C3 gsascagaCfaAfGfAfccaucuacauL96
146 asUfsguaGfaUfGfgucuUfgUfcugucsusg 314 568977 AD- C3
gsasgaaaUfuCfUfAfcuacaucauL96 147 asUfsagaUfgUfAfguagAfaUfuucucsusg 315 569268 AD-
C3 gsasgaacCfaGfAfAfacaugccauL96 148 asUfsggcAfuUfGfuucUfgGfuucucsusu 316 572818
AD- SCN9A gsasggucAfaGfAfCfaucuuuaugaL96 149 usCfsauaAfaGfAfugucUfuGfaccucscsa 317
1531729 AD- Marc1 gsasgugcUfcCfUfUfcuccaguucuL96 150
asGfsaacUfgGfAfgaagGfaGfcacucscsg 318 1531689 AD-75275 PNPLA3
gsasuuugCfaAfCfUfugcuacccauL96 151 asUfsgggUfaGfCfaaguUfgCfaaauacsusu 319 AD-67568
PNPLA3 gscsacagGfgAfAfCfcucuaccuuaL96 152 usAfsaggUfaGfAfgguuCfcCfugugcsasg 320
AD- C3 gscsaggcAfgAfGfCfGcagcgggauL96 153 asUfscCgGfcUfGfGfcuUfuGfccugcsasc 321
569494 AD- Marc1 gscscuuCfcCfCfUfcagcuuaugaL96 154
usCfsauuAfgCfUfgaggGfgAfauggcsasa 322 1531722 AD- C3 gscscucuUfcUfUfAfacaaauuucuL96
155 asGfsaaaUfuUfGfuuaaGfaAfgaggcscsc 323 571610 AD- SCN9A
gscsguugUfaGfUfUfccuauccuL96 156 asGfsgagAfuAfGfgaacUfaCfaacgcscsu 324 1531753 AD-
SCN9A gscsucauUfaGfUfGfgacuauuucuL96 157 asGfsauuAfgUfGfcacaUfgAfugagcsasu 325
1531759 AD- C3 gscsugagGfaGfAfAfuugcuucauuL96 158 asAfsugaAfgCfAfauucUfcCfucagcsasc
326 572495 AD- PNPLA3 gscsugagUfuGfGfUfuuaugaaaaL96 159
usUfsuucAfuAfAfaaccAfaCfucagcsusc 327 1010719 AD- Marc1
gscsuucuCfaGfAfCfagcauuggauL96 160 asUfscCaAfuGfCfugucUfgAfgaagcsasg 328 1531683 AD-
Marc1 gsgsaccaGfaUfUfGfcuuacucagaL96 161 usCfsugaGfuAfAfgcaaUfcUfgguccsusu 329
1531711 AD- C3 gsgsagguUfgUfGfCfugagccggaaL96 162 usUfscCgGfcUfCfagcaCfaAfccuccscsc
330 1531664 AD- SCN9A gsgscacaUfgAfAfCfGacuucuuaL96 163
usGfsaagAfaGfUfcguuCfaUfgugccsasc 331 1531762 AD-67526 PNPLA3
gsgscuuAfuCfCfCfuccuuccuuaL96 164 usAfsaggAfaGfGfagggAfuAfaggccsasc 332 AD- SCN9A
gsgscuggAfuUfUfCfcuaauuguuuL96 165 asAfsacaAfuUfAfggaaAfuCfcagccsasa 333 1531725
AD- SCN9A gsgsgaaaAfcAfAfUfcuuccguuuL96 166 usAfsaacGfgAfAfgauuGfuUfuucccsusu 334
1531737 AD- PNPLA3 gsgsgguaAfcAfAfGfaugauaaucuL96 167
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gsgsugucUfcAfAfUfgcuucaauguL96 168 asCfsauuGfaAfGfcauuGfaGfacaccsasc 336 1531717 AD-
C3 gsusaaugCfaGfGfAfcuucuucuuL96 169 asAfsugaAfgAfAfguccUfgCfauuacsusg 337 570644
AD- C3 gsusaccuUfuUfCfAfuccagacaguL96 170 asCfsuguCfuGfGfaugaAfgAfgguacscsc 338
568962 AD- Marc1 gsusauaaCfuCfUfAfagaucugauuL96 171
asAfsucaGfaUfCfuuaGfgUfuuaacsasa 339 1531690 AD- SCN9A
gsusccucUfaAfGfAfagaauaaucauL96 172 usAfsgauAfuUfCfuucuUfaGfaggacsusg 340 1531750
AD- Marc1 gsusgaccCfuUfCfAfgaacgaaaguL96 173 asCfsuuuCfGfUfUfcugaAfgGfgucacsasc 341
1531710 AD-67578 PNPLA3 gsusgaguGfaCfAfAfcguaccuuaL96 174
usAfsaggGfuAfCfGuugUfcAfcucacsusc 342 AD-67582 PNPLA3
gsusgcuaAfaGfUfUfucccaucuuaL96 175 asAfsagaUfgGfGfaaacUfuUfagcacsusu 343 AD- C3
gsusgggaGfaAfGfUfucggccuagaL96 176 usCfsuagGfcCfGfaacuUfcUfcccacsusg 344 1531666 AD-
Marc1 usasacucUfaAfGfAfucugaugaauL96 177 asUfsucaUfcAfGfaucuUfaGfaguuasusa 345
1531695 AD- SCN9A usascaugAfuCfUfUfcuuugucguaL96 178
usAfsccaCfaAfAfgaagAfuCfauguasgsg 346 1531723 AD-75270 PNPLA3
usasccugUfuGfAfAfuuuuuguauuaL96 179 usAfsauaCfaAfAfauucAfaCfagguasasc 347 AD- C3
usascgugCfuGfCfCfcaguucgaaL96 180 usUfscgaAfaCfUfgggcAfgCfacguascsu 348 1531671 AD-
PNPLA3 usasggauAfaUfGfUfcuuauguaauL96 181 asUfsuacAfuAfAfgacaUfuAfuccuasasu 349

519350 AD- SCN9A usasugccAfaAfUfccuuuuuuauL96 182
usAfsuaaAfaAfGfgauuUfuGfgcauasgsa 350 1531764 AD- SCN9A
usasugugAfaAfCfAfaaccuuacgaL96 183 usCfsguaAfgGfUfuuguUfuCfacauasasu 351 1531739
AD- Marc1 usasuuguAfaUfUfUfcaggaugcgaL96 184 usCfsgcaUfcCfUfgaaaUfuAfcauasusu 352
1531681 AD- Marc1 uscsaaugCfuUfCfAfaugucccaguL96 185
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uscsagaaCfGfAfAfAfguuauauggaL96 187 usCfscauAfuAfAfcuuuUfcUfucugasasg 355 1531676
AD- Marc1 uscsaggaUfgCfGfAfugucuauL96 188 usGfscauAfgAfCfaucgCfaUfccugasasa 356
1531700 AD- Marc1 uscscauaGfaUfCfUfggaucuggcaL96 189
usGfscgaGfaUfCfcagaUfcUfauggasasa 357 1531702 AD- SCN9A
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SCN9A uscsacuuAfaGfAfAfgaaauaucuaL96 191 asUfsagaUfaUfUfcuucUfuAfgaggascsu 359
1531738 AD- SCN9A uscsaggaAfaGfUfCfaaguuccaaaL96 192
usUfsuggAfaCfUfugacUfuGfcaggasasa 360 1531733 AD-67554 PNPLA3
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usgsacccUfuCfAfGfaacgaaaguL96 194 asAfsuuUfcGfUfucugAfaGfggucascsa 362 1531685 AD-
75269 PNPLA3 usgsagugAfaGfAfAfaugaaagacaL96 195 usGfsucuUfuCfAfuuuUfuCfacucasgsu
363 AD- SCN9A usgsauagUfuAfCfCfuaguugcaL96 196 usUfsgcaAfaCfUfagguAfaCfuaucasasa
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usAfsaguGfaAfGfaucuUfgUfcugcasasa 365 1531754 AD- PNPLA3
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572022 AD- SCN9A usgscucuCfcAfUfAfuuggauaaaaL96 201
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usgscucuCfcUfUfUfgugguuuauL96 202 asUfsgaaAfcCfAfcaaaGfgAfgagcasusc 370 1531766 AD-
C3 usgsgacuAfuGfUfGfuacaagaccuL96 203 asGfsgucUfuGfUfacacAfuAfguccascsu 371 572577
AD- Marc1 usgsgaggAfgAfAfGfaaaagugauuL96 204 asAfsucaCfuUfUfucuuCfuCfuccascsa 372
1531697 AD-67561 PNPLA3 usgsgauaCfaUfGfAfgcaagauuuL96 205
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SCN9A usgsggucAfuUfCfUfucacuuugaaL96 208 usUfscaaAfgUfGfaagaAfuGfaccasasa 376
1531726 AD- SCN9A usgsgucuUfuAfCfUfggaucuuuuL96 209
asAfsaagAfuUfCfcaguAfaAfgaccasasa 377 1531742 AD-67573 PNPLA3
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Marc1 usgsguguCfuCfAfAfgucuucauaL96 212 usAfsuugAfaGfCfauugAfgAfcaccasgsa 380
1531712 AD- C3 usgsuacaAfgAfCfCfcgacuggucaL96 213 usGfsaccAfgUfCfggguCfuUfguacascsa
381 1531669 AD- SCN9A usgsuaggAfgAfAfUfucacuuuL96 214
usGfsaaaAfgUfGfaauuUfuCfcuacascsa 382 1531745 AD- SCN9A
usgsuaggAfgAfAfUfucacuuuL96 215 asGfsaaaAfgUfGfaauuUfuCfcuacascsa 383 1531724 AD-
SCN9A usgsucgaGfuAfCfAfcuuuuacugaL96 216 usCfsaguAfaAfAfguguAfcUfcgacacusu 384
1531731 AD- C3 usgsuucgUfgCfUfGfaauaagaaguL96 217 asCfsuucUfuAfUfucagCfaCfgaacascg
385 570132 AD- SCN9A usgsuucuGfuCfUfGfaguguguuuL96 218
usAfsaacAfcAfCfucagAfcAfgaacascsa 386 1531769 AD- Marc1
usgsuuuaAfaAfCfCfaauaucuaL96 219 asUfsagaUfaUfUfgggUfuUfaaacasasc 387 1531694 AD-
Marc1 ususaaaCfuGfUfGfaauaaauggaL96 220 usCfscauUfuAfUfucacAfgUfuuaasasa 388
1531677 AD- SCN9A ususaccuAfuCfUfCfugcuucaaguL96 221

asCfsuugAfaGfCfagagAfuAfgguaascsa 389 1531727 AD-67575 PNPLA3
 ussaccuGfuUfGfAfauuuuguauuL96 222 asAfsuacAfaAfAfuucaAfcAfgguaascsa 390 AD-75274
 PNPLA3 ussauguAfaUfGfCfugcccuguaaL96 223 usUfsacaGfgGfCfagcaUfuAfcuaaasgsa 391
 AD- SCN9A ususccucAfaGfGfAfaaaagauaaaL96 224 usUfsuauCfuUfUfuuccUfuGfaggaasasu 392
 1531768 AD- C3 ususcgugCfuGfAfAfuaagaagaauL96 225 asUfsucuUfcUfUfauucAfgCfacgaascsa
 393 570134 AD- Marc1 ususgccaUfuUfUfGfuccuuuguauuL96 226
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 PNPLA3 ususgguuUfuAfUfGfaaaagcuagaL96 228 usCfsuagCfuUfUfucauAfaAfaccaascsu 396
 AD- Marc1 ususguuaUfuUfCfAfggaugcgauaL96 229 usAfsucgCfaUfCfcugaAfaUfuacaasusa 397
 1531714 AD- Marc1 ususguucCfaGfAfUfgcauuuuauuL96 230
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 ususuaguAfcAfCfUfccuuauuauL96 231 asUfsgaaUfaAfGfgaguGfuAfcuaaasasu 399 1531746
 AD- SCN9A ususuaucAfuCfUfUfugggucauuL96 232 usAfsaugAfcCfCfaagAfuGfauaaasgsa 400
 1531755 AD- Marc1 ususuccaUfaGfAfUfcuggaucugaL96 233
 usCfsagaUfcCfAfgaucUfaUfggaaasasu 401 1531675 AD- SCN9A
 ususuguaGfaUfCfUfugcauuuacaL96 234 usGfsuaaUfuGfCfaagaUfcUfacaasasg 402 1531767 AD-
 PNPLA3 ususuagAfaCfAfCfcuuuuucacuL96 235 asGfsugaAfaAfAfggugUfuCfuaaaasusu 403
 520053 AD- PNPLA3 ususuucAfcCfUfAfacuaaaaauaaL96 236
 usUfsauuUfuAfGfuuaagGfuGfaaaaasgsg 404 1010734
 TABLE-US-00003 TABLE 3 More additional exemplary sequences of parent
 dsRNA molecules. SEQ SEQ Duplex ID ID ID Target Sense sequence (5'.fwdarw.3') NO:
 Antisense sequence (5'.fwdarw.3') NO: AD- AGT asgscugAfgGfGfCfcaccauccuuL96 405
 asAfsग्gaUfgGfUfggccCfuCfaggcuscsa 752 1632799 AD- AGT
 cscsugagGfgCfCfAfccauccucuuL96 406 asAfsग्gaGfaUfGfguggCfcCfucaggscsu 753 1632801 AD-
 AGT asgsgggCfcCfAfUfccucugccuuL96 407 asAfsग्gaAfgAfGfgaugGfuGfgccuscsa 754
 1632805 AD- AGT gsusgaccGfgGfUfGfuacauacacuL96 408
 asGfsuguAfuGfUfacacCfcGfgucacscsu 755 1632838 AD- AGT
 cscsgggGfuAfCfAfuacaccccuuL96 409 asAfsग्gaGfuGfUfauguAfcAfcccgsusc 756 1684490
 AD- AGT csgsggugUfaCfAfUfacaccccuuuL96 410 asAfsग्gaGfgUfGfuaugUfaCfaccgsgsu 757
 1684492 AD- AGT gsgsguguAfcAfUfAfcaccccuucuL96 411
 asGfsaagGfgGfUfguauGfuAfcacccsgsg 758 1684494 AD- AGT
 gsgsuguaCfaUfAfCfaccccuuccuL96 412 asGfsग्gaGfgGfGfuguaUfgUfacaccscsg 759 1684496 AD-
 AGT gsusguacAfuAfCfAfcacccuuccauL96 413 asUfsggaAfgGfGfguguAfuGfuacacscsc 760
 1684498 AD- AGT usgsuacaUfaCfAfCfcccuccacuL96 414
 asGfsuggAfaGfGfggugUfaUfguacacscsc 761 1684500 AD- AGT
 gsusacauAfcAfCfCfcuuccaccuL96 415 asGfsग्gaGfaAfGfgggGfuAfuguacsasc 762 1684502 AD-
 AGT usascauaCfaCfCfCfcuuccaccuL96 416 asAfsग्gaGfgAfAfggggUfgUfauguascsa 763
 1684504 AD- AGT ascsauacAfcCfCfCfuuccaccuL96 417
 asGfsग्gaUfgGfAfagggGfuGfuaugusasc 764 1684506 AD- AGT
 csasuacaCfcCfCfUfuccaccucguL96 418 asCfsgagGfuGfGfaaggGfgUfguaugsusa 765 1684508 AD-
 AGT asusacacCfcCfUfUfccaccucguuL96 419 asAfsग्gaGfgUfGfgaagGfgGfuguausgsu 766
 1684510 AD- AGT cscscuucCfaCfCfUfcgucauccauL96 420
 asUfsggaUfgAfCfagggUfgGfaagggsgsu 767 1684512 AD- AGT
 cscsuuccAfcCfUfCfugauccacuL96 421 asGfsग्gaAfuGfAfcgagGfuGfgaagggsgsg 768 1684514
 AD- AGT csusuccaCfcUfCfGfucauccacauL96 422 asUfsgugGfaUfGfacgaGfgUfggaagsgsg 769
 1632840 AD- AGT ususccacCfuCfGfUfcauccacaauL96 423
 asUfsuguGfgAfUfgacgAfgGfuggaasgsg 770 1632841 AD- AGT
 uscscaccUfcGfUfCfauccacaauuL96 424 asAfsuugUfgGfAfugacGfaGfguggasasg 771 1632842 AD-
 AGT cscsaccUfcGfUfCfauccacaauuL96 425 asCfsauuGfuGfGfaugaCfgAfggugsasa 772

1632843 AD- AGT csascucGfuCfAfUfccacaugauL96 426
asUfscauUfgUfGfgaugAfcGfaggugsgsa 773 1632844 AD- AGT
cscsucguCfaUfCfCfacaugagauL96 427 asUfscucAfuUfGfuggaUfgAfcgaggsusg 774 1632846 AD-
AGT csuscgucAfuCfCfAfaaugagaguL96 428 asCfsucuCfaUfUfguggAfuGfacgagsgsu 775
1632847 AD- AGT uscsgucaUfcCfAfCfaaugagaguL96 429
asAfsucUfcAfUfugugGfaUfgacgasgsg 776 1632848 AD- AGT
csgsucauCfcAfCfAfaugagaguauL96 430 asUfsacuCfuCfAfuuguGfgAfugacgsasg 777 1632849
AD- AGT gsuscaucCfaCfAfAfugagaguacuL96 431 asGfsuacUfcUfCfauugUfgGfaugacsgsa 778
1632836 AD- AGT uscsauccAfcAfAfUfgagaguaccuL96 432
asGfsguaCfuCfUfcuuGfuGfgaugascsg 779 1632850 AD- AGT
csasuccaCfaAfUfGfagaguaccuuL96 433 asAfsnguAfcUfCfucaUfgUfggaugsasc 780 1632851 AD-
AGT asusccacAfaUfGfAfgaguaccuguL96 434 asCfsaggUfaCfUfcucaUfuGfuggausgsa 781
1632852 AD- AGT uscsacaAfuGfAfGfaguaccuguL96 435
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cscsacaaUfgAfGfAfguaccuguguL96 436 asCfsacaGfgUfAfcucuCfaUfuguggsas 783 1632854 AD-
AGT csascaauGfaGfAfGfuaccugugauL96 437 asUfscacAfgGfUfacucUfcAfuugugsgsa 784
1632855 AD- AGT ascsaaugAfgAfGfUfaccugugaguL96 438
asCfsucaCfaGfGfuacuCfuCfauugusgsg 785 1632856 AD- AGT
csasaugaGfaGfUfAfccugugagcuL96 439 asGfscucAfcAfGfguacUfcUfcuuugsusg 786 1632857
AD- AGT asasugagAfgUfAfCfcugugagcauL96 440 asUfsgcuCfaCfAfgguaCfuCfucauusgsu 787
1632858 AD- AGT asusgagaGfuAfCfCfugugagcaguL96 441
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usgsagagUfaCfCfUfgugagcagcuL96 442 asGfscugCfuCfAfcaggUfaCfucucasusu 789 1632860 AD-
AGT gsasgaguAfcCfUfGfugagcagcuL96 443 asAfsnguGfcUfCfacagGfuAfcucucsasu 790
1632861 AD- AGT asgsaguaCfcUfGfUfgagcagcuguL96 444
asCfsagcUfgCfUfcacaGfgUfacucusc 791 1632862 AD- AGT
gsasguacCfuGfUfGfagcagcugguL96 445 asCfsagCfuGfCfucacAfgGfuacucsusc 792 1632863 AD-
AGT asgsuaccUfgUfGfAfgcagcugguL96 446 asGfscgaGfcUfGfcucaCfaGfguacusc 793
1632864 AD- AGT gsusaccuGfuGfAfGfagcagcugcauL96 447
asUfsgccAfgCfUfgcucAfcAfgguacsusc 794 1632865 AD- AGT
usasccugUfgAfGfCfagcuggcacuL96 448 asUfsugcCfaGfCfugcuCfaCfagguasc 795 1632866 AD-
AGT asasugguCfgGfGfAfgcuggcacuL96 449 asUfsggcCfaGfCfauccCfgAfccauusgsc 796
1632991 AD- AGT asusggucGfgGfAfUfgcuggcacuL96 450
asUfsuggCfcAfGfcaucCfcGfaccusug 797 1632992 AD- AGT
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AGT gsgsucggGfaUfGfCfuggccaacuL96 452 asAfsnguGfgCfCfagcaUfcCfagaccasusu 799
1632994 AD- AGT gsuscgggAfuGfCfUfggccaacuL96 453
asAfsaguUfgGfCfcagcAfuCfccagacsc 800 1632995 AD- AGT
uscsgggaUfgCfUfGfgccaacuL96 454 asGfsaagUfuGfGfccagCfaUfcccagasc 801 1632996 AD-
AGT csgsggauGfcUfGfGfccaacuL96 455 asAfsгааGfuUfGfgccaGfcAfuccgsasc 802
1632997 AD- AGT gsgsgaugCfuGfGfCfcaacuL96 456
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csusggccAfaCfUfUfcuugggcuL96 460 asAfsagcCfcAfAfgaagUfuGfgccagscsa 807 1633004 AD-
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1633010 AD- AGT ascsuucuUfgGfGfCfuuccguauauL96 465
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AD- AGT ususcuugGfgCfUfUfccguauauauL96 467 asUfsauaUfaCfGfgaagCfcCfaagaasgsu 814
1633013 AD- AGT uscsuuggGfcUfUfCfcguauauauuL96 468
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1633016 AD- AGT gsgsgcuuCfcGfUfAfuaauuggcauL96 471
asUfsgccAfuAfUfauacGfgAfagcccsasa 818 1633018 AD- AGT
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AGT usasuauGfgCfAfUfGfcacagugaguL96 476 asCfsucaCfuGfUfgcauGfcCfauauasusa 823
1633029 AD- AGT asusauggCfaUfGfCfacagugagcuL96 477
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827 1633032 AD- AGT gsgscaugCfaCfAfGfugagcuauL96 481
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1633048 AD- AGT gsgscaccCfuGfGfCfcucucucuuL96 487
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AD- AGT ascsaggcUfaCfAfGfgcauuccuguL96 489 asCfsaggAfuUfGfccugUfaGfccuguscsa 836
1633095 AD- AGT ususccuuGfgAfAfGfgacaagaacuL96 490
asGfsuucUfuGfUfccuuCfcAfaggaascsa 837 1633119 AD- AGT
cscsuuggAfaGfGfAfaagaacuguL96 491 asCfsaguUfcUfUfguccUfuCfcaaggsasa 838 1633121 AD-
AGT csusuggaAfgGfAfCfaagaacugcuL96 492 asGfscagUfuCfUfugucCfuUfccaagsgsa 839
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asAfsacaAfgCfUfggucGfgUfuggaasusu 1003 AD- AGT uscscaacCfgAfCfCfagcuuguuuuL96 655
asAfsaacAfaGfCfugguCfGfuuggasasu 1004 1634119 AD- AGT
cscsaaccGfaCfCfAfgcuuguuuuguL96 656 asCfsaaaCfaAfGfcuggUfcGfguuggsasa 1005 1634120
AD- AGT csasaccgAfcCfAfGfcuuguuuuguuL96 657 asAfscaaAfcAfAfgcugGfuCfggguugsgsa 1006
1634121 AD- AGT asasccgaCfcAfGfCfuuguuuuguguL96 658
asCfsacaAfaCfAfagcuGfgUfcgguusgsg 1007 1634122 AD- AGT
ascscgacCfaGfCfUfuguuugugauL96 659 asUfscacAfaAfCfaagcUfgGfucggususg 1008 1634123
AD- AGT cscsgaccAfgCfUfUfguuugugaauL96 660 asUfsucaCfaAfAfcaagCfuGfgucgggsusu 1009
1634124 AD- AGT csgsaccaGfcUfUfGfuuugugaaaL96 661
asUfsuucAfcAfAfacaaGfcUfggucgsgsu 1010 1634125 AD- AGT
gsasccagCfuUfGfUfuugugaaaacuL96 662 asGfsuuuCfaCfAfaacaAfgCfuggucsgsg 1011 1634126
AD- AGT ascscagcUfuGfUfUfugugaaacauL96 663 asUfsguuUfcAfCfaaacAfaGfcugguscsg 1012
1634127 AD- AGT cscsagcuUfgUfUfUfgugaaacaauL96 664
asUfsuguUfuCfAfcaaaCfaAfgcuggsusc 1013 1634128 AD- AGT
csasgcuuGfuUfUfGfugaaacaaaL96 665 asUfsuugUfuUfCfcaaaAfcAfagcugsgsu 1014 1634129
AD- AGT asgscuugUfuUfGfUfgaaacaaaaL96 666 asUfsuuuGfuUfUfcacaAfaCfaagcusgsg 1015
1634130 AD- AGT gscsuuguUfuGfUfGfaaacaaaaL96 667
asUfsuuuUfgUfUfucacAfaAfcaagcsusg 1016 1684562 AD- AGT
csusuguuUfgUfGfAfaacaaaaaaL96 668 asUfsuuuUfuGfUfuucaCfaAfacaagscsu 1017 1684563
AD- AGT ususguuuGfuGfAfAfacaaaaaaguL96 669 asCfsuuuUfuUfGfuuuuCfaAfaacaasgsc 1018
1684564 AD- AGT usgsuuccCfuUfUfUfcaaguugaguL96 670
asCfsucaAfcUfUfgaaaAfgGfgaacascsu 1019 1634135 AD- AGT
gsusucccUfuUfUfCfaaguugagauL96 671 asUfscucAfaCfUfugaaAfaGfggaacsasc 1020 1634136
AD- AGT ususcccuUfuUfCfAfaguugagaaL96 672 asUfsucuCfaAfCfuugaAfaAfgggaascsa 1021
1634137 AD- AGT csasaguGfaGfAfAfcaaaaauuguL96 673
asCfsaaUfuUfUfguucUfcAfacuugsasa 1022 1634146 AD- AGT
asasguugAfgAfAfCfaaaaauugguL96 674 asCfscaaUfuUfUfuguuCfuCfaacuusgsa 1023 1634147
AD-68585 AGT asgsuugaGfaAfCfAfaaaauuggguL96 675 asCfsccaAfuUfUfuuguUfcUfcaacusug
1024 AD- AGT gsusugagAfaCfAfAfaaaauuggguL96 676 asAfscccAfaUfUfuuuUfuCfucaacsusu
1025 1634148 AD- AGT ususgagaAfcAfAfAfaaauggguuuL96 677
asAfsaccCfaAfUfuuuuGfuUfcucaascsu 1026 1634149 AD- AGT
usgsagaaCfaAfAfAfauuggguuuuL96 678 asAfsaacCfcAfAfuuuuUfgUfucucasasc 1027 1634150
AD- AGT gsasgaacAfaAfAfAfauuggguuuuL96 679 asAfsaaaCfcCfAfauuuUfuGfuucucsasa 1028
1634151 AD- AGT asgsaacaAfaAfAfUfuggguuuuauL96 680
asUfsaaaAfcCfCfaauUfuUfguucuscasa 1029 1634152 AD- AGT
gsasacaaAfaAfUfUfggguuuuauL96 681 asUfsuaaAfaCfCfaauUfuUfuguucsusc 1030 1634153
AD- AGT asgsuauaCfaUfUfUfuugcauugcuL96 682 asGfscaaUfgCfAfaaaaUfgUfauacususu 1031
1634162 AD- AGT gsusauacAfuUfUfUfugcauugccuL96 683
asGfsgcaAfuGfCfaaaaAfuGfuauacsusu 1032 1634163 AD- AGT
usasuaaUfuUfUfUfgcauugccuL96 684 asAfsggcAfaUfGfcaaaAfaUfguauascsu 1033 1634164
AD- AGT asusacauUfuUfUfGfcuugccuuL96 685 asAfsaggCfaAfUfgcaaAfaAfuguausc 1034
1634165 AD- AGT asusuuuuGfcAfUfUfgccuucgguuL96 686
asAfsccgAfaGfGfcaauGfcAfaaaausgsu 1035 1634169 AD- AGT
ususuuugCfaUfUfGfccuucgguuuL96 687 asAfsaccGfaAfGfgcaaUfgCfaaaaausug 1036 1634170
AD- AGT ususuugcAfuUfGfCfcuucgguuuuL96 688 asAfsaacCfGfAfAfggcaAfuGfcaaaasasu 1037

1634171 AD- AGT ususugcaUfuGfCfCfuucgguuuL96 689
asCfsaaaCfcGfAfagggcAfaUfgcaaasasa 1038 1634172 AD- AGT
ususgcauUfgCfCfUfucgguuuL96 690 asAfscaaAfcCfGfaaggCfaAfugcaasasa 1039 1634173
AD- AGT usgscauuGfcCfUfUfcgguuuL96 691 asUfsacaAfaCfCfagaagGfcAfaugcasasa 1040
1634174 AD- AGT gscsauugCfcUfUfCfpgguuuL96 692
asAfsuacAfaAfCfcgaaGfgCfaaugcsasa 1041 1634175 AD- AGT
csasuugcCfuUfCfGfguuuuL96 693 asAfsauaCfaAfAfccgaAfgGfcaaugscsa 1042 1634176
AD- AGT asusugccUfuCfGfGfuuuuuL96 694 asAfsaauAfcAfAfaccgAfaGfgcaausgsc 1043
1634177 AD- AGT ususgccuUfcGfGfUfuuguuuuL96 695
asUfsaaaUfaCfAfaaccGfaAfggcaasusg 1044 1634178 AD- AGT
usgsccuuCfgGfUfUfuguuuuaguL96 696 asCfsuaaAfuAfCfaaacCfgAfagggcasasu 1045 1634179
AD- AGT gscscuucGfgUfUfUfguuuuaguL96 697 asAfscaaAfaUfAfcaaaCfcGfaaggcsasa 1046
1634180 AD- AGT cscsuucgGfuUfUfGfuuuuaguL96 698
asCfsacuAfaAfUfacaaAfcCfagaaggscsa 1047 1634181 AD- AGT
csusucggUfuUfGfUfauuuaguguuL96 699 asAfscauUfaAfAfuacaAfaCfcgaagsgsc 1048 1634182
AD- AGT ususcggUfuUfGfUfauuuagugucuL96 700 asGfsacaCfuAfAfaucAfaAfccgaasgsg 1049
1634183 AD- AGT uscsgguUfgUfAfUfuagugucuL96 701
asAfsagacAfcUfAfaaaCfaAfaccgasasg 1050 1634184 AD- AGT
csgsguuUfuAfUfUfuagugucuL96 702 asAfsagaCfaCfUfaaaAfcAfaaccgsasa 1051 1634185
AD- AGT gsgsuuugUfaUfUfUfagugucuL96 703 asCfsaagAfcAfCfuaaaUfaCfaaacsgsa 1052
1634186 AD- AGT gsusuuguAfuUfUfAfgugucuL96 704
asUfscaaGfaCfAfcuaaAfuAfcaaacscsg 1053 1634187 AD- AGT
ususuguaUfuUfAfGfugucuL96 705 asUfsucaAfgAfCfacuaAfaUfacaaascsc 1054 1634188
AD- AGT ususguauUfuAfGfUfgucuL96 706 asAfsuucAfaGfAfcacuAfaAfuacaasasc 1055
1634189 AD- AGT usgsuauUfaGfUfGfucuL96 707
asCfsauUfaAfGfacacUfaAfaucasasa 1056 1634190 AD- AGT
gsusauuuAfgUfGfUfcuugaaguL96 708 asAfscauUfcAfAfgacaCfuAfaauacsasa 1057 1634191
AD- AGT usasuuaGfuGfUfCfuugaaguL96 709 asUfsacaUfuCfAfagacAfcUfaaauscsa 1058
1634192 AD- AGT asusuagUfgUfCfUfugaaguL96 710
asUfsuacAfuUfCfaagaCfaCfuaaaasasc 1059 1634193 AD- AGT
ususuaguGfuCfUfUfgaaguaguL96 711 asCfsuaaCfaUfUfcaagAfcAfcuaaasasa 1060 1634194
AD- AGT usasagugUfcUfUfGfaauguaguL96 712 asUfscuuAfcAfUfucaaGfaCfacuaasasa 1061
1634195 AD- AGT usasguguCfuUfGfAfauguaaguL96 713
asUfsucuUfaCfAfuucaAfgAfcacuasasa 1062 1634196 AD- AGT
asgsugucUfuGfAfAfuguaagaacuL96 714 asGfsuucUfuAfCfaucAfaGfacacusasa 1063 1634197
AD- AGT usgsucuGfaAfUfGfuaagaacuL96 715 asAfsuguUfcUfUfacauUfcAfagacascsu 1064
1634199 AD- AGT gsuscuugAfaUfGfUfaagaaguL96 716
asCfsaugUfuCfUfuacaUfuCfaagacsasc 1065 1634200 AD- AGT
ususgaauGfuAfAfGfaacugaccuL96 717 asGfsgucAfuGfUfucuuAfcAfuucaasgsa 1066 1634203
AD- AGT gsusaagaAfcAfUfGfaccuccguguL96 718 asCfsacgGfaGfGfucauGfuUfcuuacsasu 1067
1634209 AD- AGT usasagaaCfaUfGfAfccuccguguL96 719
asAfscauGfgAfGfgucaUfgUfucuuascsa 1068 1634210 AD- AGT
asasgaacAfuGfAfCfcuccguguL96 720 asUfsacaCfgGfAfggucAfuGfuucuasasc 1069 1634211
AD- AGT asgsaacaUfgAfCfCfuccguguaguL96 721 asCfsuacAfcGfGfagguCfaUfguucusa 1070
1634212 AD- AGT gsasacauGfaCfCfUfccguguaguL96 722
asAfscauCfaCfGfgaggUfcAfuguucsusu 1071 1634213 AD- AGT
asascaugAfcCfUfCfcguguaguL96 723 asCfsacuAfcAfCfaggGfuCfauguuscsu 1072 1634214
AD- AGT ascsaugaCfcUfCfCfuguguaguL96 724 asAfscauUfaCfAfcggaGfgUfcaugususc 1073
1634215 AD- AGT csasugacCfuCfCfGfuguaguL96 725
asGfsacaCfuAfCfacggAfgGfucaugsusu 1074 1634216 AD- AGT

asugaccUfGfUfguagucuuL96 726 asAfsagAfcUfAfcagGfaGfgucausgsu 1075 1634217
 AD- AGT csusuaguUfuUfUfUfccacagauguL96 727 asCfsaucUfgUfGfgaaaAfaAfcuaagsgsu 1076
 1684565 AD- AGT ususaguuUfuUfUfCfcacagaugcuL96 728
 asGfscauCfuGfUfggaaAfaAfacuaasgsg 1077 1684566 AD- AGT
 usasguuuUfuUfCfCfacagaugcuuL96 729 asAfsagcaUfcUfGfuggaAfaAfaacuasasg 1078 1684567
 AD- AGT gsusuuuuUfcCfAfCfagaugcuuguL96 730 asCfsaagCfaUfCfugugGfaAfaaaacsusa 1079
 1684568 AD- AGT ususuuuCfCfAfGfagaugcuuguguL96 731
 asCfsacaAfgCfAfucugUfgGfaaaaasasc 1080 1684569 AD- AGT
 ususuuccAfcAfGfAfugcuugugauL96 732 asUfscacAfaGfCfaucuGfuGfgaaaasasa 1081 1684570
 AD- AGT ususuccaCfaGfAfUfgcuugugauuL96 733 asAfsucaCfaAfGfcaucUfgUfggaaasasa 1082
 1634234 AD- AGT ususccacAfgAfUfGfcuugugauuuL96 734
 asAfsaucAfcAfAfgcauCfuGfuggaasasa 1083 1634235 AD- AGT
 uscscacaGfaUfGfCfuugugauuuuL96 735 asAfsaauCfaCfAfagcaUfcUfguggasasa 1084 1634236
 AD- AGT cscsacagAfuGfCfUfugugauuuuL96 736 asAfsaaaUfcAfCfaagcAfuCfuguggsasa 1085
 1634237 AD- AGT csascagaUfgCfUfUfgugauuuuuL96 737
 asAfsaaaAfuCfAfcaagCfaUfcugugsgsa 1086 1634238 AD- AGT
 ascscugaAfuUfUfCfuguuugaauuL96 738 asAfsuucAfaAfCfagaaAfuUfcaggusgsc 1087 1634282
 AD- AGT cscsugaaUfuUfCfUfguuugaauuL96 739 asCfsauuCfaAfAfcagaAfaUfucaggsusg 1088
 1634283 AD- AGT gsgsaaccAfuAfGfCfugguuauuuuL96 740
 asAfsaauAfaCfCfagcuAfuGfguuccsgsc 1089 1634304 AD- AGT
 gsasaccaUfaGfCfUfgguuauuuuL96 741 asGfsaaaUfaAfCfcagcUfaUfgguuccsgsc 1090 1634305
 AD- AGT asasccauAfgCfUfGfguuauuuuL96 742 asAfsagaaAfuAfAfccagCfuAfugguuscsc 1091
 1634306 AD- AGT ascscauaGfcUfGfGfuauuuucucuL96 743
 asGfsagaAfaUfAfaccaGfcUfauggususc 1092 1634307 AD- AGT
 cscsauagCfuGfGfUfuauuuucuccuL96 744 asGfsgagAfaAfUfaaccAfgCfuaugggsusu 1093 1634308
 AD- AGT cscsuuguGfuUfAfGfuauuaaacguL96 745 asCfsguuUfaUfUfacuaAfcAfcaagsgsgsa 1094
 1634327 AD- AGT csusugugUfuAfGfUfaauuaaacguuL96 746
 asAfsccguUfuAfUfuacuAfaCfacaagsgsg 1095 1634328 AD- AGT
 ususguguUfaGfUfAfauuaaacgucuL96 747 asGfsacgUfuUfAfuuacUfaAfcacaasgsg 1096 1634329
 AD- AGT usgsuguuAfgUfAfAfuaaacgucuuL96 748 asAfsagcGfuUfUfaauaCfuAfacacasasg 1097
 1634330 AD- AGT gsusguuaGfuAfAfUfaaacgucuuuL96 749
 asAfsagaCfGfUfUfauuAfcUfaacacsasa 1098 1634331 AD- AGT
 usgsuuagUfaAfUfAfaacgucuuuL96 750 asCfsaagAfcGfUfuuuUfaCfuaacascsa 1099 1634332
 AD- AGT gsusuaguAfaUfAfAfacgucuuuL96 751 asGfscaaGfaCfGfuuuUfuAfcuaacsasc 1100
 1634333

TABLE-US-00004 TABLE 4 Abbreviations of nucleotide monomers used in nucleic acid sequence representation* Abbreviation Nucleotide(s) A Adenosine-3'-phosphate Ab beta-L-adenosine-3'-phosphate Abs beta-L-adenosine-3'-phosphorothioate Af 2'-fluoroadenosine-3'-phosphate Afs 2'-fluoroadenosine-3'-phosphorothioate As adenosine-3'-phosphorothioate C cytidine-3'-phosphate Cb beta-L-cytidine-3'-phosphate Cbs beta-L-cytidine-3'-phosphorothioate Cf 2'-fluorocytidine-3'-phosphate Cfs 2'-fluorocytidine-3'-phosphorothioate Cs cytidine-3'-phosphorothioate G guanosine-3'-phosphate Gb beta-L-guanosine-3'-phosphate Gbs beta-L-guanosine-3'-phosphorothioate Gf 2'-fluoroguanosine-3'-phosphate Gfs 2'-fluoroguanosine-3'-phosphorothioate Gs guanosine-3'-phosphorothioate T 5'-methyluridine-3'-phosphate Tf 2'-fluoro-5-methyluridine-3'-phosphate Tfs 2'-fluoro-5-methyluridine-3'-phosphorothioate Ts 5-methyluridine-3'-phosphorothioate U Uridine-3'-phosphate Uf 2'-fluorouridine-3'-phosphate Ufs 2'-fluorouridine-3'-phosphorothioate Us uridine-3'-phosphorothioate N any nucleotide, modified or unmodified a 2'-O-methyladenosine-3'-phosphate as 2'-O-methyladenosine-3'-phosphorothioate c 2'-O-methylcytidine-3'-phosphate CS 2'-O-methylcytidine-3'-phosphorothioate g 2'-O-methylguanosine-3'-phosphate gs 2'-O-methylguanosine-3'-phosphorothioate t 2'-O-methyl-5-methyluridine-3'-phosphate ts 2'-O-methyl-

5-methyluridine-3'-phosphorothioate u 2'-O-methyluridine-3'-phosphate us 2'-O-methyluridine-3'-phosphorothioate s phosphorothioate linkage L96 N-[tris(GalNAc-alkyl)-amido-dodecanoyl]-4-hydroxyprolinol [Hyp-(GalNAc-alkyl).sub.3] [00045]  embedded image (Agn) Adenosine-glycol nucleic acid (GNA) (Cgn) Cytidine-glycol nucleic acid (GNA) (Ggn) Guanosine-glycol nucleic acid (GNA) (Tgn) Thymidine-glycol nucleic acid (GNA) S-Isomer P Phosphate VP Vinyl-phosphonate (e.g., 5'-E-vinylphosphonate) dA 2'-deoxyadenosine-3'-phosphate dAs 2'-deoxyadenosine-3-phosphorothioate dC 2'-deoxycytidine-3-phosphate dCs 2'-deoxycytidine-3-phosphorothioate dG 2'-deoxyguanosine-3'-phosphate dGs 2'-deoxyguanosine-3'-phosphorothioate dT 2'-deoxythymidine-3'-phosphate dTs 2'-deoxythymidine-3'-phosphorothioate dU 2'-deoxyuridine dUs 2'-deoxyuridine-3-phosphorothioate (Ahd) 2'-O-hexadecyl-adenosine-3'-phosphate (Ahds) 2'-O-hexadecyl-adenosine-3'-phosphorothioate (Chd) 2'-O-hexadecyl-cytidine-3'-phosphate (Chds) 2'-O-hexadecyl-cytidine-3'-phosphorothioate (Ghd) 2'-O-hexadecyl-guanosine-3'-phosphate (Ghds) 2'-O-hexadecyl-guanosine-3'-phosphorothioate (Uhd) 2'-O-hexadecyl-uridine-3'-phosphate (Uhds) 2'-O-hexadecyl-uridine-3'-phosphorothioate *It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5'-3'-phosphodiester bonds; and it is understood that when the nucleotide contains a 2'-fluoro modification, then the fluoro replaces the hydroxy at that position of the parent nucleotide (i.e., it is a 2'-deoxy-2'-fluoronucleotide.

[0434] Some exemplary dsRNA molecule designs according to embodiments of the disclosure are shown schematically in FIGS. 5A and 5B. Exemplary dsRNA molecules according to some embodiments of the disclosure are listed in Table 5 and 6.

TABLE-US-00005 TABLE 5 Exemplary dsRNA molecule according to some embodiments of the disclosure

SEQ ID	SEQ ID	Duplex ID	Target	Sense sequence (5'.fwdarw.3')	Antisense sequence (5'.fwdarw.3')	NO
AD- Marc1	asasaaaugu	UfCfUfcaaaaaugaaL96	1101	usdTscadTudTuugadGaAfcauuuuusasa	1269	1636769
AD- Marc1	asasaucacCfAfCfucuuugggcaL96	1102	usdGscddCadAagagdTgGfugauuuuscsc	1270	1636805	
AD- Marc1	asasagugggAfGfAfccuguguaaL96	1103	usdTscadAcddAgggudCuCfccacuuusgsa	1271	1636782	
AD- PNPLA3	asasaugaaaGfAfCfaaagguggauL96	1104	asdTscddAcddCuugdTcUfuucauuuscsc	1272	1636754	
AD- C3	asasaugaggGfUfUfucacagucuaL96	1105	usdTsgadCudGugaadAcCfcucauuuscsc	1273	1636695	
AD- PNPLA3	asascuugcuAfCfCfcauuaggauaL96	1106	usdAsucdCudAauggdGuAfgcaaguusgsc	1274	1636733	
AD- Marc1	asasucaccaCfUfCfuugggagcaL96	1107	asdCsugdCcdCaaagdAgUfggugauususu	1275	1636798	
AD- C3	ascsaugggcCfAfGfuggaagaucalL96	1108	usdGsaudCudTccacdTgGfcccaugusug	1276	1636720	
AD- C3	ascscaggaaCfUfGfaaccuugauaL96	1109	usdAsucdAadGguucdAgUfuccuggusgsg	1277	1636701	
AD- C3	ascscaggauGfCfCfacuaugucuaL96	1110	usdAsgadCadTagugdGcAfuccugguscsu	1278	1636716	
AD- PNPLA3	ascscuaacuAfAfAfauaauguuuuL96	1111	usdAsaadCadTuauudTuAfguuaggusgsa	1279	1636729	
AD- Marc1	ascscucgccUfGfGfuccugauuuuL96	1112	usdAsaadTcdAggacdCaGfgcgaggususc	1280	1636806	
AD- PNPLA3	ascscuguugAfAfUfuuuuguauuuL96	1113	asdTsaadTadCaaaadTuCfaacaggusasa	1281	1636735	
AD- C3	ascsuacaugAfAfCfcuacagagauL96	1114	asdTscudCudGuaggdTcCfauguagusug	1282	1636714	
AD- Marc1	ascsugauuaUfGfGfaauaguucuuL96	1115	asdAsgadAcddTauucdCaUfaaucagususa	1283	1636778	
AD- C3	asgsaaaauucUfAfCfuacaucauuL96	1116	asdAsuadGadTguagdTaGfaauuucscsu	1284	1479350	
AD- Marc1	asgsacaggaUfUfCfugaaaacucaL96	1117	usdGsagdTudTucagdAaUfccugucusug	1285	1636785	
AD- C3	asgsagcgggUfAfCfcucucaucaL96	1118	usdGsaudGadAgaggdTcCfccgcucuscsc	1286	1636763	
AD- SCN9A	asgscaaaggUfCfAfcauuuuccuaL96	1119	usdAsggdAadAuugudGaCfcuuugcuscsc	1287	1636834	
AD- SCN9A	asgscauaaaaUfGfUfuuucgaaauaL96	1120	usdAsuudTcdGaaaadCaUfuuaugcuscsc	1288	1636811	
AD- PNPLA3	asgscaugagGfUfUfcuuagaauL96	1121	asdCsaudTcdTaagadAcCfucaugcuscsg	1289	1636757	
AD- PNPLA3						

asgsgaagcaAfCfCfuucgcugul96 1122 asdCsagdGcdGaaagdGuUfgcuuccusag 1290 1636758
AD- Marc1 asgsgaccagAfUfUfgcuuacucaaL96 1123 usdTsgadGudAagcadAuCfugguccusug 1291
1636791 AD- Marc1 asgsgagaagAfAfAfagugauucaL96 1124
usdTsgadAudCacuudTuCfuuccuccuscsc 1292 1636777 AD- SCN9A
asgsucuucaAfGfUfuggcaaaaual96 1125 usdAsuudTudGccaadCuUfgaagacuscs 1293 1636838
AD- C3 asgsuggacuAfUfGfuguacaagauL96 1126 asdTscudTgdTacacdAuAfguccacuscs 1294
1636703 AD- SCN9A asgsuuccuaUfCfUfccuuucagaaL96 1127
usdTscudGadAaggadGaUfaggaacusasc 1295 1636831 AD- PNPLA3
asusaaugucUfUfAfuguaaugcuuL96 1128 asdAsgcdAudTacaadAaGfacauuauscsc 1296 1636732
AD- PNPLA3 asusacaugaGfCfAfagauuugcaaL96 1129 usdTsgcdAadAucuudGcUfcaugauuscsc
1297 1636737 AD- Marc1 asuscaaccaGfGfAfgggaaacauaL96 1130
usdAsugdTudTcccudCcUfgguugauscsa 1298 1636779 AD- SCN9A
asuscaucuuUfGfGfgucauucuuuL96 1131 asdAsagdAadTgaccdCaAfagaugausasa 1299 1636843
AD- Marc1 asuscugaugAfAfGfuauuuuuuuL96 1132 asdAsaadAadTauacdTuCfaucagauscsc 1300
1636772 AD- SCN9A asuscuucuuUfGfUfcguagugauuL96 1133
asdAsucdAcdTacgadCaAfagaugauscsa 1301 1636836 AD- SCN9A
asusgcugagAfAfAfugucgaaaaL96 1134 usdTsuudCgdAcaudTuCfucagcauscsc 1302 1636813
AD- Marc1 asusggcuugUfUfCfcagaugcauuL96 1135 asdAsugdCadTcuggdAaCfaagccauscsa 1303
1636783 AD- C3 asusguaccaUfGfCfuaaggccaaaL96 1136 usdTsgudGcdCuuagdCaUfgguacausug
1304 1636692 AD- SCN9A asusgucgagUfAfCfacuuuuacuuL96 1137
asdAsgudAadAagugdTaCfucgacaususu 1305 1636844 AD- PNPLA3
asusguuaguAfGfAfauaagccuuL96 1138 usdAsagdGcdTuauudCuAfcuaacauscsu 1306 1636726
AD- PNPLA3 asusuaggauAfAfUfgucuuauaL96 1139 usdAscadTadAgacadTuAfuccuaausgsg
1307 1636760 AD- SCN9A asusucucuuCfGfUfucacagaugaL96 1140
usdCsaudCudGugaadCgAfagagaauscsc 1308 1636841 AD- Marc1
csasacacuuGfAfAfgcagguguuL96 1141 asdAscadCcdAugcudTcAfaguguugsusc 1309 1636789
AD- PNPLA3 csasagauuuGfCfAfacuugcuacaL96 1142 usdGsuadGcdAaguudGcAfaaucuugsusu
1310 1636762 AD- C3 csasaggucuAfCfGfccuauuacauL96 1143
asdTsgudAadTaggcdGuAfgaccuugsasc 1311 1636766 AD- C3 csascccucaUfCfAfucuaaccuggaL96
1144 usdCscadGgdTagaudGaUfgagggugsusu 1312 1636718 AD- C3
csasccguauCfCfAfcugggaauuL96 1145 asdGsaudTcdCcagudGgAfuacggugsgsg 1313 1636707
AD- PNPLA3 csasccuuuuUfCfAfccuaacuaaaaL96 1146 usdTsuadGudTaggudGaAfaaaggugsusu
1314 1636727 AD- PNPLA3 csasccuuuuUfCfAfccuaacuaauL96 1147
asdTsuadGudTaggudGaAfaaaggugsusu 1315 1636741 AD- SCN9A
csascuccuuCfCfUfgauuguguuuL96 1148 asdAsacdAcdAaucadGgAfaggagugsgsa 1316 1636830
AD- Marc1 csasgaacgaAfAfGfuauauaggaaL96 1149 usdTscddAudAuaacdTuUfcguucugsasa 1317
1636780 AD- C3 csasgagaaaUfUfCfuacuacauuL96 1150 asdGsaudGudAguagdAaUfuucucugsusa
1318 1636689 AD- SCN9A csasugaacgAfCfUfucuuuccacuuL96 1151
asdAsgudGgdAagaadGuCfguucaugsug 1319 1636839 AD- PNPLA3
csasugagcaAfGfAfuuugcaacuuL96 1152 asdAsgudTgdCaaudCuUfgcucaugsusa 1320 1636753
AD- Marc1 csasugguguUfUfCfagaacugagaL96 1153 usdCsucdAgdTucugdAaAfcaccaugscsu 1321
1636792 AD- C3 csasuugagaAfCfCfcggaaggcauL96 1154 asdTsgcdCudTccggdGuUfcucaaugsusu
1322 1636709 AD- C3 cscsacagccAfAfAfgauaagaacuL96 1155
asdGsaudCudTaucudTuGfgcuguggsusc 1323 1636721 AD- C3 cscsagauccAfCfUfucaccaagauL96
1156 asdTscudTgdGugaadGuGfgaucuggsusa 1324 1636719 AD- PNPLA3
cscsauuaggAfUfAfaugucuuuuL96 1157 asdAsuadAgdAcaudAuCfcuaauggsgsu 1325 1636765
AD- C3 cscscgucguGfCfGfuuggcucaauL96 1158 asdTsgudAgdCcaacdGcAfcgacgggsasg 1326
1636693 AD Marc1 cscsgaccaAfGfGfaccagauuguL96 1159
asdCsaadTcdTggucdCuUfgggucggsasa 1327 1636807 AD- C3 cscsgagccgUfUfCfucuaaauuuL96
1160 asdAsaudTgdTagagdAaCfggcucggsasu 1328 1479283 AD- Marc1

cscsguauguCfCfUfggaauuuL96 1161 asdTsaadTadTuccadGgAfcauacggsusu 1329 1636774 AD-
PNPLA3 cscsuaacuaAfAfAfuaauguuuuL96 1162 usdTsaadAcdAuuauTuUfaguuaaggsusg 1330
1636725 AD- C3 cscsuugucuUfCfUfcagaaccagaL96 1163 usdCsugdGudTcugadGaAfgacaaggsasg
1331 1636715 AD- PNPLA3 csgsacaucuGfCfCfcuaaagucaL96 1164
usdTsgadCudTuaggdGcAfgaugucgsusa 1332 1636750 AD- C3 csgsugcguuGfGfCfucaaugaacuL96
1165 asdGsuudCadTugagdCcAfacgcacgsasc 1333 1636702 AD- C3
csusaccuaCfUfCfuguuguucgaL96 1166 usdCsgadAcdAacagdAgUfaggguaagscsc 1334 1636705
AD- C3 csusacugcaGfCfUfaaaagacuuuL96 1167 asdAsagdTcdTuuuadGcUfgcaguagsgsg 1335
1636690 AD- C3 csusagugcuGfUfCfcagugagaaaL96 1168 usdTsuudTcdAcuggdAcAfgcacuagsusu
1336 1636694 AD- PNPLA3 csusauuaauGfGfUfcagacuguaL96 1169
usdAsacdAgdTcugadCcAfuaauagsgsg 1337 1636743 AD PNPLA3
csusccauggCfGfGfgguuaacaaL96 1170 usdTsgdTudAcccdCgCfcauggagsasc 1338 1636756 AD-
PNPLA3 csusgaguugGfUfUfuaugaaaL96 1171 asdTsuudTcdAuaaadAcCfaacucagscsu 1339
1636734 AD- SCN9A csusgccccAfAfUfacugauaauL96 1172
usdAsuudAudCaguadTuUfugggcagscsa 1340 1636840 AD- C3 csusgggaggAfCfCfcugguaagcaL96
1173 usdGscudTadCaggdGuCfcucccagscsg 1341 1636712 AD- Marc1
csusguggagGfAfGfaagaaaagual96 1174 usdAscudTudTcuudTcCfuccacagsasa 1342 1636800 AD-
Marc1 csusucuauUfGfGfugacguggaaL96 1175 usdTscdAcdGucacdCaAfuaagaagscsu 1343
1636790 AD- Marc1 csusuguuccAfGfAfugcauuuuL96 1176
usdTsaadAadTgcaudCuGfgaacaagscsc 1344 1636770 AD- SCN9A
csusuugucgUfAfGfugaauuuuccuL96 1177 asdGsgadAadAucacdTaCfagaaaagsasa 1345 1636837
AD- C3 gsascagacaAfGfAfccaucuacauL96 1178 asdTsgudAgdAuggdCuUfgucugucsusg 1346
1636699 AD- C3 gsasgaaauUfUfAfcaucaucuauL96 1179 asdTsgadAudGuagudAgAfauuucucsusg
1347 1479341 AD- C3 gsasgaaccaGfAfAfacaauGCCauL96 1180
asdTsggdCadTuguudTcUfgguucucsusu 1348 1636691 AD- SCN9A
gsasggucaaGfAfCfauuuuaugaL96 1181 usdCsaudAadAgaugdTcUfugaccucscsa 1349 1636815
AD- Marc1 gsasgugcucCfUfUfcuccaguucL96 1182 asdGsaadCudGgagadAgGfagcacucscsg 1350
1636795 AD- PNPLA3 gsasuugcaAfCfUfugcuaccuL96 1183
asdTsggdGudAgcaadGuUfgcaauucsusu 1351 1636731 AD- PNPLA3
gscsacagggAfAfCfcucuaccuL96 1184 usdAsagdGudAgaggdTuCfccugugcsasg 1352 1636744
AD- C3 gscsaggcagAfGfCfcagcggggaL96 1185 asdTscdCgdCugcgdCuCfugccugcsasc 1353
1636717 AD- Marc1 gscscauuccCfCfUfcagcuauaL96 1186
usdCsaudTadGcugadGgGfgaauGCCasa 1354 1636796 AD- C3 gscscucuucUfUfAfacaaaauucL96
1187 asdGsaadAudTuguudAaGfaagaggcscsc 1355 1636710 AD- SCN9A
gscsguuguaGfUfUfcuauuccuL96 1188 asdGsgadGadTaggadAcUfacaacgcscsu 1356 1636822
AD- SCN9A gscsucaucaUfGfUfgcuaauucL96 1189 asdGsaadTadGugcadCaUfgaugagcsasu 1357
1636828 AD- C3 gscsugaggaGfAfAfuuGcuucauuL96 1190 asdAsugdAadGcaudTcUfccucagcsasc
1358 1636767 AD- PNPLA3 gscsugaguGfGfUfuuuaugaaaaL96 1191
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gscsuucuaGfAfCfagcauuggauL96 1192 asdTscdAadTgcugdTcUfgagaagcsasg 1360 1636803 AD-
Marc1 gsgsaccagaUfUfGfcuucacagaL96 1193 usdCsugdAgdTaaagdAaUfcugguucsusu 1361
1636801 AD- C3 gsgsagguugUfGfCfugagccggaaL96 1194 usdTscdGgdCucagdCaCfaaccuccscsc
1362 1636713 AD- SCN9A gsgscacauGfAfCfagcuucuuL96 1195
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AD- SCN9A gsgscuggauUfUfCfcuaauuguuuL96 1197 asdAsacdAadTuaggdAaAfuccagccsasa
1365 1636823 AD- SCN9A gsgsgaaaacAfAfUfcuuccguuuL96 1198
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gsgsgguaacAfAfGfaugauaaucuL96 1199 asdGsaudTadTcaudTuGfuuaaccscscsg 1367 1636749 AD-
Marc1 gsgsugucucAfAfUfgcuucaauguL96 1200 asdCsaudTgdAagcadTuGfagacaccsasg 1368

136799 AD- C3 gsusaugcaGfGfAfcuucuuL96 1201 asdAsugdAadGaagudCcUfgcauuacsusg
1369 1636700 AD- C3 gsusaccucuUfCfAfuccagacaguL96 1202
asdCsugdTcdTggaudGaAfgagguacscsc 1370 1636706 AD- Marc1
gsusauaacuCfUfAfagaucugauuL96 1203 asdAsucdAgdAucuudAgAfguuauacsasa 1371 1636771
AD- SCN9A gsusccucuaAfGfAfagaauaucuaL96 1204 usdAsgadTadTucuudCuUfagaggacsusg 1372
1636826 AD- Marc1 gsusgaccuUfCfAfgaacgaaaguL96 1205
asdCsuudTcdGuucudGaAfgggucacsasc 1373 1636848 AD- PNPLA3
gsusgagugaCfAfAfcguaccuuL96 1206 usdAsagdGgdTacgudTgUfcacucacsusc 1374 1636746
AD- PNPLA3 gsusgcuaaaGfUfUfucccaucuuL96 1207 asdAsagdAudGggaadAcUfuuagcacsu
1375 1636747 AD- C3 gsusgggagaAfGfUfucggccuagaL96 1208
usdCsudGgdCcgaadCuUfcuccacsusg 1376 1636723 AD- Marc1
usasacucuaAfGfAfucugaugauL96 1209 asdTsucdAudCagaudCuUfagaguuasusa 1377 1636847
AD- SCN9A usascaugauCfUfUfcuuugucguaL96 1210 usdAscgdAcdAagadAgAfucauguasgsg
1378 1636824 AD- PNPLA3 usasccuguuGfAfAfuuuuguauuL96 1211
usdAsaudAcdAaaudTcAfacagguasasc 1379 1636748 AD- C3 usascgugcuGfCfCfcaguuucgaaL96
1212 usdTscgdAadAcuggdGcAfgcacguascsu 1380 1636704 AD- PNPLA3
usasggauaaUfGfUfcuuanguaaL96 1213 asdTsudCadTaagadCaUfuauccuasasu 1381 1636724 AD-
SCN9A usasugccaaAfAfUfccuuuuuauL96 1214 usdAsudAadAaggadTuUfuggcauasgsa 1382
1636832 AD- SCN9A usasugugaaAfCfAfaaccuuacgaL96 1215
usdCsgudAadGguudGuUfucacauasasu 1383 1636835 AD- Marc1
usasuuuaaUfUfUfcaggaugcgaL96 1216 usdCsgcdAudCcugadAaUfuacaauasusu 1384 1636788
AD- Marc1 uscsaaugcuUfCfAfauguccaguL96 1217 asdCsugdGgdAcaudGaAfgcauugasgsa 1385
1636787 AD- PNPLA3 uscsacuugaGfGfAfggcgagucuaL96 1218
usdAsgadCudCgccudCcUfcaagugascsu 1386 1636761 AD- Marc1
uscsagaacgAfAfAfguuauauggaL96 1219 usdCscadTadTaacadTuCfguucugasasg 1387 1636794
AD- Marc1 uscsaggaugCfGfAfugucuaugcaL96 1220 usdGscadTadGacaudCgCfauccugasasa 1388
1636804 AD- Marc1 uscscauagaUfCfUfggauucggcaL96 1221
usdGscddAgdAuccadGaUfcuauggasasa 1389 1636802 AD- SCN9A
uscscauuguCfUfUfgacaucuuauL96 1222 asdTsaadGadTgucadAgAfcaauggasusc 1390 1636814
AD- SCN9A uscscucuaaGfAfAfgaauaucuaL96 1223 asdTsgdAudAuucudTcUfuagaggascsu 1391
1636812 AD- SCN9A uscscugcaaGfUfCfaaguuccaaaL96 1224
usdTsgdGadAcuugdAcUfugcaggasasa 1392 1636850 AD- PNPLA3
uscsugagcuGfAfGfuugguuuuauL96 1225 asdTsaadAadCcaacdTcAfgcucagasgsg 1393 1636738
AD- Marc1 usgsacccuUfCfAfGfaacgaaaguL96 1226 asdAscudTudCguudTgAfagggucascsa 1394
1636797 AD- PNPLA3 usgsagugaaGfAfAfaugaaagacaL96 1227
usdGsudTudTcaudTcUfucacucasgsu 1395 1636728 AD- SCN9A
usgsauaguUfCfCfuaguuuugcaaL96 1228 usdTsgcdAadAcuagdGuAfacuauacasasa 1396 1636842
AD- SCN9A usgscagacaAfGfAfucuuacuuL96 1229 usdAsagdTgdAgaudCuUfgucugasusa
1397 1636852 AD- PNPLA3 usgscaaaaCfAfAfcaucaccguL96 1230
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1231 asdCsuudGgdTcucudTcUfgaucgcasgsg 1399 1636698 AD- C3
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AD- SCN9A usgscucuccAfUfAfuuggauaaaaL96 1233 usdTsuudAudCcaudAuGfgagagcasasu
1401 1636816 AD- SCN9A usgscucuccUfUfUfgugguuucauL96 1234
asdTsgadAadCcacadAaGfgagagcasusc 1402 1636846 AD- C3 usgsgacuauGfUfGfuacaagaccuL96
1235 asdGsgudCudTguacdAcAfuaguccascsu 1403 1636711 AD- Marc1
usgsgaggagAfAfGfaaaagugauuL96 1236 asdAsucdAcdTuucudTuCfuccuccascsa 1404 1636793
AD- PNPLA3 usgsgauacaUfGfAfgcaagauuuL96 1237 usdAsaadTcdTugcudCaUfguauccascsc
1405 1636742 AD- PNPLA3 usgsggagagAfUfAfugccuucgaaL96 1238
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1239 asdGsuadAudTguugdGaGfuugcccasg 1407 1636696 AD- SCN9A
 usgsggucauUfCfUfucacuuugaaL96 1240 usdTscadAadGugaadGaAfugacccasasa 1408 1636821
 AD- SCN9A usgsgucuuuAfCfUfggaauuuuL96 1241 asdAsaadGadTuccadGuAfaagaccasasa 1409
 1636851 AD- PNPLA3 usgsgugacaUfGfGfcuuccagauaL96 1242
 usdAsucdTgdGaagcdCaUfgucaccasgsu 1410 1636751 AD- SCN9A
 usgsgugucaUfCfAfugauaaauuL96 1243 asdAsaudTadTcuadGaUfgacaccasasu 1411 1636818
 AD- Marc1 usgsgugucuCfAfAfugcuucaauaL96 1244 usdAsuudGadAgcaudTgAfgacaccasgsa 1412
 1636781 AD- C3 usgsuacaagAfCfCfcgacuggucaL96 1245 usdGsacdCadGucggdGuCfuuguacascsa
 1413 1636722 AD- SCN9A usgsuaggagAfAfUfucacuuuucalL96 1246
 usdGsaadAadGugaadTuCfuccuacascsa 1414 1636820 AD- SCN9A
 usgsuaggagAfAfUfucacuuuucalL96 1247 asdGsaadAadGugaadTuCfuccuacascsa 1415 1636817
 AD- SCN9A usgsucgaguAfCfAfcuuuuacugaL96 1248 usdCsagdTadAaagudGuAfcucgacasusu
 1416 1636825 AD- C3 usgsuucgugCfUfGfaauaagaaguL96 1249
 asdCsuudCudTauucdAgCfacgaacascsg 1417 1636708 AD- SCN9A
 usgsuucuguCfUfGfaguguguuuuL96 1250 usdAsaadCadCacucdAgAfcagaacascsa 1418 1636819
 AD- Marc1 usgsuuuaaaAfCfCfcauauucuaL96 1251 asdTsagdAudAuuggdGuUfuuaaacasasc 1419
 1636775 AD- Marc1 ususaaaacuGfUfGfaauaaauggaL96 1252
 usdCscadTudTauucdAcAfguuuuasasa 1420 1636784 AD- SCN9A
 ususaccuauCfUfCfugcuucaaguL96 1253 asdCsuudGadAgcagdAgAfuagguaascsc 1421 1636829
 AD- PNPLA3 ususaccuguUfGfAfauuuuguauuL96 1254 asdAsuadCadAauudCaAfcagguaascsa
 1422 1636768 AD- PNPLA3 ususauguuaUfGfCfugcccuguaaL96 1255
 usdTsacdAgdGgcagdCaUfuacauaasgsa 1423 1636745 AD- SCN9A
 ususccucaaGfGfAfaaaagauaaaL96 1256 usdTsuadTcdTuuuudCcUfugaggaasasu 1424 1636809 AD-
 C3 ususcgugcuGfAfAfuaagaagaauL96 1257 asdTsuadTudCuuaudTcAfgcacgaascsa 1425 1636688
 AD- Marc1 ususgccauuUfUfGfuccuuugauuL96 1258 asdAsucdAadAggacdAaAfauggcaasusa 1426
 1636773 AD- PNPLA3 ususgcuaccCfAfUfuaggauaaualL96 1259
 usdAsuudAudCcuaadTgGfuguagcaasgsu 1427 1636752 AD- PNPLA3
 ususgguuuuAfUfGfaaaagcuagaL96 1260 usdCsuaadGcdTuuucdAuAfaaaccaascsu 1428 1636759
 AD- Marc1 ususguauuuUfCfAfggauugcgaualL96 1261 usdAsucdGcdAuccudGaAfauuacaasusa 1429
 1636849 AD- Marc1 ususguuccaGfAfUfgcauuuuauL96 1262
 asdTsuadAadAugcadTcUfggaacaasgsc 1430 1636776 AD- SCN9A
 ususuaguacAfCfUfccuuauucauL96 1263 asdTsgadAudAaggadGuGfuacuaaasasu 1431 1636845
 AD- SCN9A ususuaucauUfUfUfugggucauuuL96 1264 usdAsaudGadCccaadAgAfugauaaaasgsa
 1432 1636810 AD- Marc1 ususuccauaGfAfUfcuggaucugaL96 1265
 usdCsagdAudCcagadTcUfauggaaasasu 1433 1636786 AD- SCN9A
 ususuguagaUfCfUfugcauuuacalL96 1266 usdGsuadAudTgcaadGaUfcuacaaasasg 1434 1636808
 AD- PNPLA3 ususuugaaCfAfCfcuuuuucacuL96 1267 asdGsugdAadAaaggdTgUfucuaaaaasusu
 1435 1636736 AD- PNPLA3 ususuucacCfUfAfacuaaaaauaL96 1268
 usdTsaudTudTaguudAgGfugaaaaasgsg 1436 1636730

TABLE-US-00006 TABLE 6 More Exemplary dsRNA molecule according to some embodiments of the disclosure

SEQ SEQ Duplex ID ID ID Target Sense sequence
 (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD- AGT
 asgsgcugagGfGfCfcaccauccuuL96 1437 asdAsggdAudGguggdCcCfucaggcuscsa 1786 1657992
 AD- AGT cscsugagggCfCfAfccauccucuuL96 1438 asdAsgadGgdAuggdGgCfccucaggscsu 1787
 1657994 AD- AGT asgsggccacCfAfUfccucugccuuL96 1439
 asdAsggdCadGaggadTgGfuggcccuscsa 1788 1657998 AD- AGT
 gsgsaccggGfUfGfuacauacacuL96 1440 asdGsugdTadTguacdAcCfcggucacscsu 1789 1658030
 AD- AGT cscsggguguAfCfAfucacccccuuL96 1441 asdAsggdGgdTguadGuAfcacccggsusc 1790
 1684491 AD- AGT csgsgguguaCfAfUfacacccccuuL96 1442
 asdAsagdGgdGuguadTgUfacacccgsgsu 1791 1684493 AD- AGT

gsgsguguacAfUfAfcacccuL96 1443 asdGsaadGgugudAuGfuacacccsgsg 1792 1684495
AD- AGT gsgsuguacaUfAfCfaccuccuL96 1444 asdGsgadAgdGggugdTaUfguacaccscsg 1793
1684497 AD- AGT gsgsuguacauAfCfAfcuccuL96 1445
asdTsggdAadGgggudGuAfuguacaccsc 1794 1684499 AD- AGT
usgsuacauaCfAfCfccccuccacuL96 1446 asdGsugdGadAggggdTgUfauguacaccsc 1795 1684501
AD- AGT gsgsacauacAfCfCfccccuccacuL96 1447 asdGsgudGgdAagggdGuGfuanguacsasc 1796
1684503 AD- AGT usascauacaCfCfCfccccuccacuL96 1448
asdAsggdTgdGaaggdGgUfguanguascsa 1797 1684505 AD- AGT
ascsauacacCfCfCfccccuccacuL96 1449 asdGsagdGudGgaagdGgGfuguanguasc 1798 1684507
AD- AGT csasuacaccCfCfUfuccaccucguL96 1450 asdCsgadGgdTggaadGgGfguguaugsusa 1799
1684509 AD- AGT asusacaccCfUfUfccaccucguuL96 1451
asdAscgdAgdGuggadAgGfgguguausgsu 1800 1684511 AD- AGT
cscscuuccaCfCfUfugucauccauL96 1452 asdTsggdAudGacgadGgUfggaaggsgsgsu 1801 1684513
AD- AGT cscsuuccacCfUfCfugucauccacuL96 1453 asdGsugdGadTgacgdAgGfuggaaggsgsg 1802
1684515 AD- AGT csusuccaccUfCfGfucauccacauL96 1454
asdTsgudGgdAugacdGaGfguggaaggsgsg 1803 1658032 AD- AGT
ususccaccuCfGfUfcauccacauL96 1455 asdTsgdTgdGaugadCgAfgguggaasgsu 1804 1658033
AD- AGT uscscaccucGfUfCfauccacauuL96 1456 asdAsuudGudGgaugdAcGfagguggasasg 1805
1658034 AD- AGT cscsaccucGfUfCfAfucauccauL96 1457
asdCsaudTgdTggaudGaCfagguggasasa 1806 1658035 AD- AGT
csasccucguCfAfUfccacauL96 1458 asdTscadTudGuggadTgAfcgaggugsgsa 1807 1658036
AD- AGT cscsucgucaUfCfCfcauagagauL96 1459 asdTscudCadTugugdGaUfgacgaggsusg 1808
1658038 AD- AGT csuscgucaUfCfCfcauagagauL96 1460
asdCsucdTcdAuugudGgAfugacgagsgsu 1809 1658039 AD- AGT
uscsgucaucCfAfCfaugagaguL96 1461 asdAscudCudCauugdTgGfaugacgasgsu 1810 1658040
AD- AGT csgsucauccAfCfAfaugagaguL96 1462 asdTscdTcdTcaudGuGfauagacgsasg 1811
1658041 AD- AGT gsgsucauccAfAfAfugagaguL96 1463
asdGsuadCudCucaudTgUfggaugacgsa 1812 1658042 AD- AGT
uscсаuccacAfAfUfgagaguL96 1464 asdGsgudAcdTcucadTuGfuggaugascsg 1813 1658043
AD- AGT csasuccacaAfUfGfagaguL96 1465 asdAsggdTadCucucdAuUfguggaugasc 1814
1658044 AD- AGT asusccacaUfGfAfgaguL96 1466
asdCsagdGudAcucudCaUfuguggausgsa 1815 1658045 AD- AGT
uscscacaauGfAfGfaguaccuguuL96 1467 asdAscadGgdTacudTcAfuuggugasusg 1816 1658046
AD- AGT cscsacaugAfGfAfguaccuguguL96 1468 asdCsacdAgdGuacudCuCfauugggsasu 1817
1658047 AD- AGT csascaaugaGfAfGfuaccugugauL96 1469
asdTscadCadGguacdTcUfcauugugsgsa 1818 1658048 AD- AGT
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cscsguuucuCfCfUfuggucuaaguL96 1641 asdCsudAgdAccaadGgAfgaaacggscsu 1990 1659208
AD- AGT csgsuuucucCfUfUfggucuaaguuL96 1642 asdAscudTadGaccadAgGfagaaacgsgsc 1991
1659209 AD- AGT gsusuucccUfUfGfgucuaaguguL96 1643
asdCsacdTudAgaccdAaGfgagaaacsgsg 1992 1659210 AD- AGT
gsusuugcugGfGfUfuauuuuaguL96 1644 asdCsudAadAuaaadCcCfagcaaacsusg 1993 1659282
AD- AGT ususugcuggGfUfUfuauuuuagauL96 1645 asdTscudAadAuaaadAcCfcagcaaascsu 1994
1659283 AD- AGT ususgcugggUfUfUfauuuuagaguL96 1646
asdCsudTadAaaadAaCfccagcaasasc 1995 1659284 AD- AGT
usgscuggguUfUfAfuuuuagagauL96 1647 asdTscudCudAaaadAaAfcccagcasasa 1996 1659285
AD- AGT gscsuggguUfAfUfuuuagagaauL96 1648 asdTscudTcdTaaaadTaAfaccagcsasa 1997
1659286 AD- AGT csusggguuuAfUfUfuuagagaauL96 1649
asdAsuudCudCuaaadAuAfaaccagcsca 1998 1659287 AD- AGT
usgsgguuuUfUfUfuagagaanguL96 1650 asdCsaudTcdTcuaadAaUfaaaccasgsc 1999 1659288
AD- AGT gsgsguuuauUfUfUfagagaanguL96 1651 asdCscadTudCucuadAaAfuaaaccsasg 2000
1659289 AD- AGT gsgsuuuuuUfUfAfgagaanguL96 1652
asdCscadAudTcucudAaAfaaaaaccscsa 2001 1684537 AD- AGT
gsusuauuuUfAfGfagaanguL96 1653 asdCscadCadTucudTaAfaauaaacscsc 2002 1684539
AD- AGT ususuauuuAfGfAfgaanguL96 1654 asdCscadCcdAuucudCuAfaaaauaaacscsc 2003
1684541 AD- AGT asgsaanguGfGfUfggggaggcauL96 1655
asdTsgcdCudCcccadCcCfccauucscsu 2004 1684543 AD- AGT
gsasauggggGfUfGfgggaggcaauL96 1656 asdTsgdCcdTcccdAcCfcccauucsc 2005 1684545
AD- AGT asasuggggUfGfGfggaggcaaguL96 1657 asdCsudGcdCucccdCaCfcccauucscsu 2006
1684547 AD- AGT asusggggguGfGfGfgaggcaagauL96 1658
asdTscudTgdCucccdCcAfccccaususc 2007 1684549 AD- AGT
usgsggggugGfGfGfaggcaagaauL96 1659 asdTscudTudGccudCcCfcccccasusu 2008 1684551
AD- AGT gsgsgggugGfGfAfggcaagaacuL96 1660 asdGsuudCudTgccudCcCfcccccsasu 2009
1684553 AD- AGT gsusggggagGfCfAfagaaccaguL96 1661
asdAscudGgdTucuudGcCfuccccacscsc 2010 1684555 AD- AGT
usgsgggaggCfAfAfgaaccaguguL96 1662 asdCsacdTgdGuucudTgCfcuccccacscsc 2011 1684557
AD- AGT gsgsggaggCfAfGfaaccaguguL96 1663 asdAscudCudGguudTuGfccuccccasasc 2012
1684559 AD- AGT gsgsgaggcaAfGfAfaccaguguuuL96 1664
asdAsacdAcdTgguudCuUfgccuccscsa 2013 1684560 AD- AGT
gsgsaggcaaGfAfAfccaguguuuuL96 1665 asdAsaadCadCuggudTcUfugccuccscsc 2014 1684561
AD- AGT gsasggcaagAfAfCfcaguguuuauL96 1666 asdTsaadAcAcuggdTcUfuugccuccscsc 2015
1659290 AD- AGT asgsgcaagaAfCfCfaguguuuaguL96 1667
asdCsudAadCacugdGuUfcuugccscsc 2016 1659291 AD- AGT
gsgscaagaaCfCfAfguguuuagcuL96 1668 asdGscudAadAcacudGgUfucuugccscsc 2017 1659292

AD- AGT gscsaagaaccCfAfGfuguuagcgcuL96 1669 asdCsgcdTadAaacadTgGfuucuugcscsu 2018
1659293 AD- AGT csasagaaccAfGfUfguuuagcgcuL96 1670
asdGscgdCudAaacadCuGfguuuugscsc 2019 1659294 AD- AGT
asasgaaccaGfUfGfuuuagcgcgcuL96 1671 asdCsgcdGcdTaaacdAcUfgguucuusgsc 2020 1659295
AD- AGT asgsaaccagUfGfUfuuagcgcgguL96 1672 asdCscgdCgdCuaaadCaCfugguucususg 2021
1659296 AD- AGT gsasaccaguGfUfUfuagcgcgguL96 1673
asdCscddGcdGcuaadAcAfcugguucsusu 2022 1659297 AD- AGT
asasccagugUfUfUfagcgcgggauL96 1674 asdTscddCgdCgcuadAaCfacugguuscsu 2023 1659298
AD- AGT csusguuccaAfAfAfagaauuccauL96 1675 asdTsggdAadTucuudTuUfgggaacagsusa 2024
1659321 AD- AGT gsusuccaaaAfAfGfaauuccaacuL96 1676
asdGsuudGgdAauucdTuUfuuggaacsasg 2025 1659323 AD- AGT
uscscaaaaaGfAfAfuuccaaccguL96 1677 asdCsggdTudGgaudTcUfuuuuggasasc 2026 1659325
AD- AGT cscsaaaaagAfAfUfuccaaccgauL96 1678 asdTscgdGudTggaadTuCfuuuuuggsasa 2027
1659326 AD- AGT csasaaaagaAfUfUfccaaccgacuL96 1679
asdGsudGgdTuggadAuUfcuuuuugsgsa 2028 1659327 AD- AGT
asasaaagaaUfUfCfcaaccgaccuL96 1680 asdGsgudCgdGuuggdAaUfucuuuuusgsg 2029 1659328
AD- AGT asasaagaauUfCfCfaaccgaccuL96 1681 asdTsggdTcdGguugdGaAfuucuuuusgsg 2030
1659329 AD- AGT asasagaauUfCfAfaccgaccaguL96 1682
asdCsugdGudCgguudGgAfaucuuuususu 2031 1659330 AD- AGT
asasgaauucCfAfAfccgaccagcuL96 1683 asdGscudGgdTcgudTgGfaauucuuuususu 2032 1659331
AD- AGT asgsaauuccAfAfCfcgaccagcuL96 1684 asdAsgcdTgdGucggdTuGfgaaucuuuususu 2033
1659332 AD- AGT gsaauuccaAfCfCfaccagcuuuL96 1685
asdAsagdCudGgucgdGuUfggaauucsusu 2034 1659333 AD- AGT
asasuuccaaCfCfGfaccagcuuguL96 1686 asdCsaadGcdTggudGgUfuggaauuscsu 2035 1659334
AD- AGT asusuccaaccCfGfAfccagcuuguuL96 1687 asdAscadAgdCuggudCgGfuuggaauususc 2036
1659335 AD- AGT ususccaaccGfAfCfcagcuuguuuL96 1688
asdAsacdAadGcuggdTcGfguuggaasusu 2037 1659336 AD- AGT
uscscaccgAfCfCfagcuuguuuuL96 1689 asdAsaadCadAgcugdGuCfpgguuggasasu 2038 1659337
AD- AGT cscsaaccgaCfCfAfgcuuguuuuL96 1690 asdCsaadAcdaagcudGgUfcgguuggsasa 2039
1659338 AD- AGT csasaccgacCfAfGfcuuguuuuL96 1691
asdAscdaadCaagcdTgGfucgguugsgsa 2040 1659339 AD- AGT
asasccgaccAfGfCfuuguuuguguL96 1692 asdCsacdAadAcaagdCuGfgucgguusgsg 2041 1659340
AD- AGT ascscgaccaGfCfUfuguuugugauL96 1693 asdTscadCadAacaadGcUfpggucggusug 2042
1659341 AD- AGT cscsgaccagCfUfUfguuugugaauL96 1694
asdTsudAcdaaacadAgCfuggucggsusu 2043 1659342 AD- AGT
csgsaccagCfUfGfuugugugaaaL96 1695 asdTsuudCadCaaacdAaGfcuggucgsgsu 2044 1659343
AD- AGT gsasccagcuUfGfUfuugugaaacuL96 1696 asdGsuudTcdAcaaadCaAfcuggucgsgsg 2045
1659344 AD- AGT ascscagcuUfUfUfugugaaacuL96 1697
asdTsgdTudCacaadAcAfagcugguscsg 2046 1659345 AD- AGT
cscsagcuugUfUfUfgugaaacaauL96 1698 asdTsgdTudTcacadAaCfaagcuggsusc 2047 1659346 AD-
AGT csasgcuuguUfUfGfugaaacaaaL96 1699 asdTsuudGudTucacdAaAfcaagcugsgsu 2048
1659347 AD- AGT asgscuuguUfGfUfgaaacaaaauL96 1700
asdTsuudTgdTuucadCaAfacaagcugsgsg 2049 1659348 AD- AGT
gscsuuguuuGfUfGfaaacaaaaauL96 1701 asdTsuudTudGuuucdAcAfaacaagcsusg 2050 1659349
AD- AGT csusuguuuUfGfAfaacaaaaauL96 1702 asdTsuudTudTguuudCaCfaaacagaascu 2051
1659350 AD- AGT ususguuguGfAfAfacaacaaaaguL96 1703
asdCsudTudTuguudTcAfcaaacagsgsc 2052 1659351 AD- AGT
usgsuucccuUfUfUfcaaguugaguL96 1704 asdCsudAadCuugadAaAfgggaacascu 2053 1659371
AD- AGT gsusucccuUfUfCfaaguugagauL96 1705 asdTscudCadAcuugdAaAfaggaacasc 2054
1659372 AD- AGT ususcccuUfCfAfaguugagaauL96 1706

asdTsuadTcdGaAaagggaascsa 2055 1659373 AD- AGT csasaguGaGfAfAfcaaaaauuguL96
1707 asdCsaadTudTuugudTcUfcaacuugsasa 2056 1659382 AD- AGT
asasguugagAfAfCfaaaaauugguL96 1708 asdCscadAudTuuugdTuCfucaacuusgsa 2057 1659383
AD- AGT asgsuugagaAfCfAfaaaaauuggguL96 1709 asdCscddAadTuuuudGuUfcucaacusug 2058
1659384 AD- AGT gsusugagaaCfAfAfaaaauggguuL96 1710
asdAsccdCadAuuuudTgUfucucaacsusu 2059 1659385 AD- AGT
ususgagaacAfAfAfaauuggguuuL96 1711 asdAsacdCcdAuuuudTuGfuucucaascsu 2060 1659386
AD- AGT usgsagaacaAfAfAfaauuggguuuuL96 1712 asdAsaadCcdCaauudTuUfguucucasasc 2061
1659387 AD- AGT gsaagaacaaAfAfAfuuggguuuuuL96 1713
asdAsaadAcdCcaudTuUfuguucucsasa 2062 1659388 AD- AGT
asgsaacaaaAfAfUfuggguuuuuauL96 1714 asdTsaadAadCccaadTuUfuuguucuscsc 2063 1659389
AD- AGT gsaacaaaaAfUfUfgggguuuuuauL96 1715 asdTsuadAadAcccadAuUfuuguucuscsc 2064
1659390 AD- AGT asgsuauacaUfUfUfuugcauugcuL96 1716
asdGscadAudGcaaadAaUfguauacususu 2065 1659399 AD- AGT
gsusauacauUfUfUfugcauugccuL96 1717 asdGsgcdAadTgcaadAaAfuguauacsusu 2066 1659400
AD- AGT usasuacauUfUfUfgcauugccuuL96 1718 asdAsggdCadAugcadAaAfauguauascsu 2067
1659401 AD- AGT asusacauUfUfGfcauugccuuuL96 1719
asdAsagdGcdAaugcdAaAfaauguasasc 2068 1659402 AD- AGT
asusuuuugcAfUfUfgccuucgguuL96 1720 asdAsccdGadAggcadAuGfcaaaaausgsu 2069 1659406
AD- AGT ususuugcaUfUfGfccuucgguuuL96 1721 asdAsacdCgdAaggcdAaUfgcaaaaausug 2070
1659407 AD- AGT ususuugcaUfUfGfccuucgguuuL96 1722
asdAsaadCcdGaaggdCaAfugcaaaaasasu 2071 1659408 AD- AGT
ususugcauUfUfGfccuucgguuuL96 1723 asdCsaadAcdCgaagdGcAfaugcaasasa 2072 1659409
AD- AGT ususgcauUfUfUfugcauugccuuL96 1724 asdAscadAadCcgaadGgCfaaugcaasasa 2073
1659410 AD- AGT usgscuugcCfUfUfugcauugccuuL96 1725
asdTsacdAadAccgadAgGfcaaugcasasa 2074 1659411 AD- AGT
gscsauugccUfUfCfggguuuuguauL96 1726 asdAsuadCadAaccgdAaGfgcaaugcsasa 2075 1659412
AD- AGT csasuugccUfUfGfgguuuuguauL96 1727 asdAsaudAcdAaacddGaAfggcaaugscsa 2076
1659413 AD- AGT asusugccUfUfGfgguuuuguauL96 1728
asdAsaadTadCaaacdCgAfaggcaausgsc 2077 1659414 AD- AGT
ususgcuucGfGfUfuuguauuuauL96 1729 asdTsaadAudAcaaadCcGfaaggcaasug 2078 1659415
AD- AGT usgscuucGfGfUfuuguauuuaguL96 1730 asdCsudAadTacaadAcCfgaaggcasasu 2079
1659416 AD- AGT gscscuucGfGfUfuuguauuuaguL96 1731
asdAscudAadAuacadAaCfcgaaggcsasa 2080 1659417 AD- AGT
cscsuucgguUfUfGfuauuuaguguL96 1732 asdCsacdTadAauacdAaAfccgaaggscsa 2081 1659418
AD- AGT csusucgguUfUfGfuauuuaguguL96 1733 asdAscudAadAauadCaAfaccgaaggsc 2082
1659419 AD- AGT ususcgguUfUfGfuauuuaguguL96 1734
asdGsacdAcdTaaudAcAfaaccgaasgsc 2083 1659420 AD- AGT
uscsgguuUfUfUfuugugucuL96 1735 asdAsgadCadCuuaadTaCfaaaccgasag 2084 1659421
AD- AGT csgsguuUfUfUfuugugucuL96 1736 asdAsagdAcdAcuaadAuAfcaaaccgsasa 2085
1659422 AD- AGT gsgsuuUfUfUfugugucuL96 1737
asdCsaadGadCacuadAaUfacaaccsgsa 2086 1659423 AD- AGT
gsusuuguUfUfAfgugucuugauL96 1738 asdTscadAgdAcacudAaAfuacaaacscsg 2087 1659424
AD- AGT ususuguauUfUfGfugucuugaauL96 1739 asdTsuadAadGacacdTaAfaucacaaascsc 2088
1659425 AD- AGT ususguauUfUfGfugucuugaauL96 1740
asdAsuudCadAgacadCuAfaauacaasasc 2089 1659426 AD- AGT
usgsuauuUfUfGfucugauguL96 1741 asdCsaudTcdAagacdAcUfaaaucacasa 2090 1659427
AD- AGT gsusauuUfUfGfucugauguL96 1742 asdAscudTudCaagadCaCfuauuacacasa 2091
1659428 AD- AGT usasuuguUfUfGfucugauguL96 1743
asdTsacdAudTcaagdAcAfcuaauascsa 2092 1659429 AD- AGT

asusuagugUfCfUfugaaguaauL96 1744 asdTsuadCadTucaadGaCfacuaaasasc 2093 1659430 AD-
AGT ususuaguguCfUfUfgaaguaaguL96 1745 asdCsuudAcdAuucadAgAfcacuaaasusa 2094
1659431 AD- AGT ususagugucUfUfGfaaguaagauL96 1746
asdTscudTadCauucdAaGfacacuaasasu 2095 1659432 AD- AGT
usasgugucuUfGfAfauguaagaauL96 1747 asdTsucdTudAcauudCaAfgacacuasasa 2096 1659433
AD- AGT asgsugucuuGfAfAfuguaagaacuL96 1748 asdGsuudCudTacaudTcAfagacacusasa 2097
1659434 AD- AGT usgsucuugaAfUfGfuaagaacauuL96 1749
asdAsugdTudCuuacdAuUfcaagacascsu 2098 1659436 AD- AGT
gsuscuugaaUfGfUfaagaacauguL96 1750 asdCsaudGudTcuuadCaUfucaagacsasc 2099 1659437
AD- AGT ususgaauguAfAfGfaacaugaccuL96 1751 asdGsgudCadTguucdTuaAfcauucasgsa 2100
1659440 AD- AGT gsusaagaacAfUfGfaccuccguguL96 1752
asdCsacdGgdAggucdAuGfuucuuacsasu 2101 1659446 AD- AGT
usasagaacaUfGfAfccuccguguuL96 1753 asdAscadCgdGaggudCaUfguucuuascsa 2102 1659447
AD- AGT asasgaacauGfAfCfcuccguguauL96 1754 asdTscadAcidGgaggdTcAfuguucuasasc 2103
1659448 AD- AGT asgsaacaugAfCfCfuccguguaguL96 1755
asdCsuadCadCggagdGuCfauguucususa 2104 1659449 AD- AGT
gsasacaugaCfCfUfccguguaguL96 1756 asdAscudAcidAcggadGgUfcauguucsusu 2105 1659450
AD- AGT asascaugacCfUfCfcguguaguguL96 1757 asdCsacdTadCacggdAgGfucauguuscusu 2106
1659451 AD- AGT ascsaugaccUfCfCfuguaguaguL96 1758
asdAscadCudAcacgdGaGfgucaugususc 2107 1659452 AD- AGT
csasugaccuCfCfGfuguagugucuL96 1759 asdGsacdAcidTacacdGgAfggucaugsusu 2108 1659453
AD- AGT asusgaccucCfGfUfuguagugucuL96 1760 asdAsgadCadCuacadCgGfaggucausgsu 2109
1659454 AD- AGT csusuaguuuUfUfUfccacagauguL96 1761
asdCsaudCudGuggadAaAfaacuaagsgsu 2110 1659481 AD- AGT
ususaguuuUfUfCfcacagaugcuL96 1762 asdGscadTcdTguggdAaAfaaacuaasgsg 2111 1659482
AD- AGT usasguuuuuUfCfCfacagaugcuL96 1763 asdAsgcdAudCugugdGaAfaaaacuasasg 2112
1659483 AD- AGT gsusuuuuucCfAfCfagaugcuuguL96 1764
asdCsaadGcdAucugdTgGfaaaaaacsusa 2113 1659485 AD- AGT
ususuuuccaCfAfGfaugcuuguguL96 1765 asdCsacdAadGcaudTgUfggaaaaasasc 2114 1659487
AD- AGT ususuuccacAfGfAfugcuugugauL96 1766 asdTscadCadAgcaudCuGfuggaaaasasa 2115
1659488 AD- AGT ususuccacaGfAfUfgcuugugauuL96 1767
asdAsucdAcidAagcadTcUfguggaaaasasa 2116 1659489 AD- AGT
ususccacagAfUfGfucugugauuuL96 1768 asdAsaudCadCaagcdAuCfuguggaasasa 2117 1659490
AD- AGT uscsacagaUfGfCfuugugauuuuL96 1769 asdAsaadTcdAcaagdCaUfcuguggasasa 2118
1659491 AD- AGT cscsacagauGfCfUfugugauuuuuL96 1770
asdAsaadAudCacaadGcAfucuguggsasa 2119 1659492 AD- AGT
csascagaugCfUfUfgugauuuuuuL96 1771 asdAsaadAadTcacadAgCfaucugugsgsa 2120 1659493
AD- AGT ascscugaauUfUfCfuguuugaauuL96 1772 asdAsuudCadAacagdAaAfuucaggusgsc 2121
1659537 AD- AGT cscsugaauUfCfUfguuugauguL96 1773
asdCsaudTcdAaacadGaAfaucaggsusg 2122 1659538 AD- AGT
gsgsaaccuAfGfCfugguauuuuuL96 1774 asdAsaadTadAccagdCuAfugguuccsgsc 2123 1659559
AD- AGT gsaaccauaGfCfUfgguauuuucuL96 1775 asdGsaadAudAaccadGcUfaugguucscsg 2124
1659560 AD- AGT asasccauagCfUfGfuguauuuucuL96 1776
asdAsgadAadTaaccdAgCfuaugguuscsc 2125 1659561 AD- AGT
ascscuagcUfGfGfuauuuucucuL96 1777 asdGsagdAadAuaacdCaGfcuauggususc 2126 1659562
AD- AGT cscsauagcuGfGfUfuauuuuccuL96 1778 asdGsgadGadAuaadCcAfgcuaggsusu 2127
1659563 AD- AGT cscsuuguguUfAfGfuaauaaacguL96 1779
asdCsugdTudAuuacdTaAfcacaaggsusa 2128 1659582 AD- AGT
csusuguguUfGfUfaauaaacguuL96 1780 asdAscgdTudTauuadCuAfacacaagsgsg 2129 1659583
AD- AGT ususguguuaGfUfAfauaaacgucuL96 1781 asdGsacdGudTuauudAcUfaacacaasgsg 2130

1659584 AD- AGT usgsuguuagUfAfAfuaacgucuuL96 1782
 asdAsgadCgdTuuaudTaCfuaacacasasg 2131 1659585 AD- AGT
 gsusguuaguAfAfUfaaacgucuuuL96 1783 asdAsagdAcdGuuuadTuAfcuaacacsasa 2132 1659586
 AD- AGT usgsuuaguaAfUfAfaacgucuuL96 1784 asdCsaadGadCguuudAuUfacuaacascsa 2133
 1659587 AD- AGT gsusuaguuaUfAfAfacgucuuL96 1785
 asdGscadAgdAcguudTaUfuacuaacsasc 2134 1659588

[0435] In some embodiments, the dsRNA molecule is not a dsRNA molecule listed in Table 7.
 TABLE-US-00007 TABLE 7 Additional dsRNA molecules SEQ SEQ Duplex ID ID ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-
 cscsgcu(Chd)agGfUfUfcugcuuuuaaL96 2135 VPusdTsaadAadGcagadAcCfudGagcggscsc 2136
 1289862.1 AD- uscsugagCfuGfAfGfuugguuuuauL96 193
 asdTsaadAadCcaacdTcdAgdCucagascsg 2137 1193318.1 AD-
 uscsugagCfuGfAfGfuugguuuuauL96 193 asdTsaadAadCcaacdTcdAgdCucagascsc 2138
 1193320.1 AD- ususuagAfaCfAfCfcuuuuucacuL96 235
 asdGsugdAadAaaggdTgUfuCfuaaaasusu 2139 1193410.1 AD-
 ususuagAfaCfAfCfcuuuuucacuL96 235 asdGsugdAadAaaggdTgUfuCfuaaaascsc 2140
 1193413.1

Total RNA isolation using DYNABEADS mRNA Isolation Kit (Invitrogen™, part #: 610-12)
 [0436] Cells were lysed in 75 µL of Lysis/Binding Buffer containing 3 µL of beads per well and
 mixed for 10 minutes on an electrostatic shaker. The washing steps were automated on a Biotek
 EL406, using a magnetic plate support. Beads were washed (in 90 µL) once in Buffer A, once in
 Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration,
 complete 10 µL RT mixture was added to each well, as described below.

Cdna Synthesis Using ABI High Capacity cDNA Reverse Transcription Kit (Applied Biosystems,
 Foster City, CA, Cat #4368813)

[0437] A master mix of 1 µL 10× Buffer, 0.4 µL 25× dNTPs, 1 µL Random primers, 0.5 L Reverse
 Transcriptase, 0.5 µL RNase inhibitor and 6.6 µL of H.sub.2O per reaction were added per well.
 Plates were sealed, agitated for 10 minutes on an electrostatic shaker, and then incubated at 37° C.
 for 2 hours. Following this, the plates were agitated at 80° C. for 8 minutes

Real Time PCR

[0438] Two µL of cDNA were added to a master mix containing 0.5 µL of human GAPDH TaqMan
 Probe (4326317E), 0.5 µL human AGT (Hs00174854 ml), 2 µL nuclease-free water and 5 µL
 Lightcycler 480 probe master mix (Roche Cat #O4887301001) per well in a 384 well plates (Roche
 cat #O4887301001). Real time PCR was done in a LightCycler480 Real Time PCR system
 (Roche).

[0439] To calculate relative fold change, data were analyzed using the $\Delta\Delta C_t$ method and
 normalized to assays performed with cells transfected with 10 nM AD-1955, or mock transfected
 cells.

[0440] Results are shown in FIGS. 1A-1B and 3A-4C and summarized in Tables 8-14. The results
 show a consistently improved in vitro activity relative to the parent design across several
 exemplary designs and across a large set of exemplary sequences, targets, and cell lines.

TABLE-US-00008 TABLE 8 In vitro activity for various designs in primary mouse hepatocytes
 Parent Duplex ID Dose Parent StDev AS5-7 StDev AS14 StDev 2015 StDev D1 StDev AD-
 1181392 10 nM 4.2 0.5 4.9 0.3 5.6 0.5 4.7 0.6 4.4 0.3 AD-1181401 7.4 1.0 5.9 1.2 10.9 3.2 6.1 0.5
 4.6 0.2 AD-1181410 5.8 0.3 4.4 0.5 5.8 0.9 6.1 1.1 6.0 0.7 AD-1181417 5.8 0.8 6.1 1.8 8.1 1.9 10.1
 2.3 6.3 1.4 AD-1181426 6.2 1.5 6.4 0.2 5.3 0.7 6.4 0.6 4.2 0.7 AD-64972 5.6 0.9 5.2 0.6 5.7 1.7 7.7
 0.7 4.9 1.1 AD-1181443 5.3 0.4 4.7 1.0 6.6 0.4 3.8 0.7 4.6 0.6 AD-1181451 6.3 1.0 4.2 0.2 5.4 0.4
 6.7 1.0 5.8 0.7 AD-1181460 5.5 0.7 5.4 0.5 7.5 0.4 7.5 0.5 5.6 0.8 AD-1181469 11.2 1.8 8.2 1.8
 10.3 2.8 8.0 1.3 7.1 0.8 AD-674282 5.3 4.9 10.9 0.9 27.0 10.3 AD-674283 1.3 0.4 2.2 0.3 1.4 0.4
 AD-674287 4.1 1.2 4.5 0.7 12.4 2.0 AD-674289 9.2 6.3 11.8 2.0 14.0 2.8 AD-674293 3.1 3.4 7.7

TABLE-US-00009 TABLE 9 In vitro activity for various designs in primary mouse hepatocytes

Parent Duplex ID	Dose	Parent StDev	D2 StDev	D3 StDev	D4 StDev	v2 StDev	AD-1181392	10 nM																																																																																																																										
4.2 0.5 5.0 0.1 9.1 1.2 9.6 0.3 7.0 0.4	AD-1181401	7.4 1.0 6.2 1.8 8.2 0.8 7.8 1.2 7.4 0.5	AD-1181410	5.8 0.3 4.7 0.2 5.8 0.8 7.5 0.2 5.7 1.5	AD-1181417	5.8 0.8 5.0 1.4 11.8 2.1 12.1 1.5 8.6 0.6	AD-1181426	6.2 1.5 5.9 0.8 5.9 0.8 9.0 2.1 7.2 1.8	AD-64972	5.6 0.9 5.9 1.1 7.8 1.7 10.5 0.8 13.0 0.5	AD-1181443	5.3 0.4 6.0 1.4 6.8 0.5 4.0 0.5 5.2 0.3	AD-1181451	6.3 1.0 4.9 0.5 8.8 4.9 7.6 0.6 7.2 0.3	AD-1181460	5.5 0.7 6.0 0.7 8.9 1.1 12.4 3.3 10.7 2.0	AD-1181469	11.2 1.8 6.0 1.2 7.6 1.2 9.9 2.3 18.3 5.2	AD-674282	5.3 4.9	AD-674283	1.3 0.4	AD-674287	4.1 1.2	AD-674289	9.2 6.3	AD-674293	3.1 3.4	AD-674299	1.5 1.0	AD-674303	29.5 6.4	AD-674305	2.1 1.7	AD-674307	2.9 3.0	AD-674310	4.8 0.9	AD-674312	6.3 3.7	AD-674314	16.8 2.9	AD-674323	21.4 5.9	AD-674325	14.4 2.6	AD-68435	4.2 2.0	AD-1181392	1 nM	2.8 0.2 2.9 0.1 5.7 0.7 9.7 0.6 4.1 0.4	AD-1181401	8.9 1.6 4.6 0.8 6.6 0.5 7.8 0.4 5.0 0.2	AD-1181410	3.8 0.4 3.0 0.2 3.6 0.7 3.4 0.9 4.2 0.4	AD-1181417	11.5 2.0 4.3 0.5 8.3 1.2 43.7 5.6 17.7 4.3	AD-1181426	3.6 0.4 2.8 0.5 5.3 0.6 14.5 3.2 5.3 0.4	AD-64972	5.3 0.5 5.4 2.3 8.3 0.8 17.7 1.1 19.3 1.5	AD-1181443	4.5 0.5 4.5 0.5 5.4 0.3 4.0 0.5 4.9 0.8	AD-1181451	10.8 1.9 18.0 5.2 11.1 0.2 11.0 1.0 34.9 3.8	AD-1181460	18.7 2.0 5.9 0.7 17.6 2.0 38.2 6.6 22.1 2.6	AD-1181469	33.5 3.1 21.2 0.9 33.8 0.8 56.1 2.2 74.3 5.8	AD-674282	11.5 11.6	AD-674283	11.8 6.2	AD-674287	14.8 2.2	AD-674289	67.0 11.6	AD-674293	30.4 9.4	AD-674299	16.0 4.4	AD-674303	87.2 4.5	AD-674305	28.8 8.2	AD-674307	15.9 2.0	AD-674310	18.7 4.5	AD-674312	29.1 5.0	AD-674314	36.5 9.1	AD-674323	63.6 7.5	AD-674325	36.9 6.4	AD-68435	14.9 1.7	AD-674282	0.1 nM	3.5 2.0	AD-674283	5.9 2.9	AD-674287	13.4 7.8	AD-674289	53.5 6.6	AD-674293	22.9 1.1	AD-674299	13.5 4.3	AD-674303	57.4 5.9	AD-674305	25.7 4.9	AD-674307	14.8 1.4	AD-674310	23.3 11.8	AD-674312	35.9 12.0	AD-674314	34.9 2.5	AD-674323	70.3 6.8	AD-674325	28.8 2.4	AD-68435	15.8 2.4

11.8 1.8 AD-1181443 4.0 0.5 1.5 0.6 5.1 1.0 10.8 2.0 5.5 0.3 AD-1181451 13.1 0.7 6.3 1.4 13.9 1.8
 17.2 2.0 14.8 1.9 AD-1181460 22.6 4.7 3.0 0.5 4.9 0.2 16.4 1.8 7.7 1.1 AD-1181392 1 nM 5.5 0.5
 3.7 0.2 7.0 2.1 22.7 4.9 10.9 2.5 AD-1181401 47.8 13.8 15.0 2.9 18.6 4.6 39.1 2.5 17.9 1.3 AD-
 1181410 16.7 2.9 9.2 1.3 8.9 0.9 12.0 3.2 14.0 2.3 AD-1181417 49.0 14.9 27.9 3.2 42.6 9.8 88.7
 11.6 34.8 7.4 AD-1181426 15.8 3.8 8.1 0.8 8.6 3.5 20.5 2.6 12.0 2.9 AD-64972 39.9 6.4 17.9 2.7
 32.0 8.5 36.2 8.5 31.2 6.0 AD-1181443 16.9 2.3 9.5 4.1 21.4 4.0 28.3 4.3 20.8 4.2 AD-1181451
 44.9 3.3 31.9 9.0 49.0 6.0 53.4 6.9 43.3 4.4 AD-1181460 74.8 15.4 23.9 0.7 23.7 0.8 46.7 6.7 28.4
 6.8 AD-1181392 0.1 nM 23.5 1.4 18.7 1.4 26.1 7.9 56.7 7.6 27.7 3.2 AD-1181401 61.8 6.4 33.9 3.7
 37.7 4.1 75.4 15.2 45.9 8.7 AD-1181410 48.7 4.0 28.3 3.9 20.9 1.8 32.0 2.3 35.5 7.8 AD-1181417
 62.5 7.4 56.2 14.9 88.6 13.4 132.8 33.8 84.7 22.3 AD-1181426 71.0 16.7 53.9 4.2 46.1 17.4 136.9
 20.8 69.9 16.6 AD-64972 153.9 12.2 94.2 14.1 108.8 22.1 56.3 2.6 54.8 13.3 AD-1181443 80.2 5.8
 20.2 2.4 42.0 2.4 45.0 9.6 34.3 5.8 AD-1181451 69.6 6.4 58.7 7.8 71.2 9.0 80.8 25.3 65.2 10.0 AD-
 1181460 72.3 4.0 44.3 5.7 48.3 2.9 53.6 7.2 51.1 1.9

TABLE-US-00011 TABLE 11 In vitro activity for various designs in primary mouse hepatocytes
 Parent Duplex ID Dose Parent StDev D8 StDev D9 StDev D10 StDev D11 StDev D12 StDev AD-
 1181392 10 nM 1.2 0.1 11.0 2.4 9.3 1.7 4.7 0.7 10.5 3.0 14.7 2.5 AD-1181401 24.2 2.3 21.5 3.2
 13.6 2.6 11.4 2.0 24.1 3.0 23.4 3.6 AD-1181410 5.3 1.9 3.0 0.3 4.3 0.6 3.6 0.7 10.5 3.6 5.7 1.7 AD-
 1181417 15.4 3.6 48.9 10.6 48.8 13.3 23.4 11.8 54.9 16.4 53.8 6.7 AD-1181426 2.6 0.6 10.7 3.6 6.1
 1.3 6.9 2.2 21.0 5.4 18.3 4.6 AD-64972 12.1 2.9 13.7 3.1 13.4 4.4 15.1 5.6 27.0 5.7 30.5 3.6 AD-
 1181443 4.0 0.5 11.4 3.1 9.2 1.8 6.9 1.0 16.8 3.9 14.9 3.3 AD-1181451 13.1 0.7 19.6 3.2 20.4 4.1
 15.2 3.3 30.9 3.4 39.0 6.9 AD-1181460 22.6 4.7 24.3 5.2 23.0 2.8 15.0 1.8 35.8 3.9 47.1 2.6 AD-
 1181392 1 nM 5.5 0.5 29.8 1.0 27.1 3.3 12.6 0.9 28.1 8.1 32.2 3.7 AD-1181401 47.8 13.8 44.7 5.3
 30.3 4.6 29.8 6.7 43.3 0.4 48.2 11.0 AD-1181410 16.7 2.9 13.2 1.4 15.2 4.2 12.4 3.0 24.2 6.3 16.2
 2.3 AD-1181417 49.0 14.9 68.4 6.6 74.2 16.5 40.2 5.1 76.1 7.5 109.0 13.3 AD-1181426 15.8 3.8
 30.7 9.4 20.1 2.9 19.6 6.0 41.7 3.8 30.4 7.2 AD-64972 39.9 6.4 33.8 5.9 41.3 5.1 34.7 8.0 62.8 3.6
 62.5 7.1 AD-1181443 16.9 2.3 34.9 4.2 28.8 5.8 23.2 6.5 45.6 11.2 39.0 3.7 AD-1181451 44.9 3.3
 66.8 4.3 62.9 15.2 55.0 13.0 76.2 10.5 82.0 19.4 AD-1181460 74.8 15.4 70.3 13.8 49.4 6.2 50.2 7.3
 77.5 9.4 79.8 13.4 AD-1181392 0.1 nM 23.5 1.4 54.1 10.4 47.7 2.3 28.9 1.1 49.2 3.2 54.4 2.2 AD-
 1181401 61.8 6.4 60.0 5.6 47.9 5.6 51.8 7.7 66.5 6.4 68.6 2.0 AD-1181410 48.7 4.0 36.9 4.3 34.1
 7.6 33.4 1.8 62.0 9.3 43.6 6.5 AD-1181417 62.5 7.4 160.5 30.4 98.3 27.8 118.4 35.9 76.4 7.7 110.2
 13.7 AD-1181426 71.0 16.7 46.9 6.8 48.4 12.4 77.7 25.4 107.5 28.7 88.3 36.8 AD-64972 153.9
 12.2 55.6 11.8 68.1 24.6 69.6 22.2 130.8 26.1 123.2 18.6 AD-1181443 80.2 5.8 60.7 9.7 63.6 8.4
 46.4 5.7 56.6 7.5 58.8 1.6 AD-1181451 69.6 6.4 53.8 2.3 60.8 1.8 51.2 1.0 68.4 8.3 77.0 4.5 AD-
 1181460 72.3 4.0 64.1 9.0 71.2 3.6 64.4 3.5 68.4 3.2 68.6 3.3

TABLE-US-00012 TABLE 12 In vitro activity for various designs in primary cynomolgus
 hepatocytes Parent Duplex ID Dose Parent StDev D1 StDev D2 StDev AD-157464.8 10 nM 1.0
 0.0 1.0 0.3 1.2 0.3 AD-157448.7 1.5 0.3 2.4 0.3 3.0 1.5 AD-157468.11 2.2 1.2 1.6 0.2 1.8 0.1 AD-
 67327.28 8.3 11.8 1.7 0.5 1.8 0.1 AD-85446.5 1.9 0.3 1.7 0.7 1.8 0.5 AD-85435.13 2.9 0.1 2.1 0.2
 1.7 0.2 AD-85438.5 1.7 0.3 1.2 0.1 1.8 0.1 AD-157464.8 1 nM 3.0 0.6 1.4 0.4 1.6 0.4 AD-157448.7
 3.7 0.7 3.5 0.6 2.8 0.8 AD-157468.11 5.7 0.6 4.4 0.2 3.6 0.4 AD-67327.28 1.6 0.0 1.9 0.1 2.2 0.2
 AD-85446.5 4.5 0.6 2.6 0.7 2.7 0.7 AD-85435.13 6.7 0.5 2.8 0.3 2.8 0.2 AD-85438.5 3.5 0.1 3.2
 0.5 5.9 1.0 AD-157464.8 0.1 nM 10.2 0.1 1.9 0.5 2.9 0.9 AD-157448.7 9.0 1.0 8.9 1.7 6.9 1.0 AD-
 157468.11 16.9 3.9 13.1 1.5 10.0 1.3 AD-67327.28 4.6 0.7 3.9 0.4 4.8 0.4 AD-85446.5 15.9 1.5 5.9
 1.1 6.8 0.6 AD-85435.13 18.4 2.2 5.4 1.3 7.2 0.1 AD-85438.5 10.8 1.7 10.2 2.1 14.2 2.5

TABLE-US-00013 TABLE 13 In vitro activity for various designs in Hep3B Parent D1 Target
 Duplex ID 10 nM 1 nM 0.1 nM Duplex ID 10 nM 1 nM 0.1 nM Marc1 AD-1531684.2 8 ± 1.2
 18.4 ± 3.7 12 ± 1.8 AD-1636769.1 6.6 ± 0.8 11.4 ± 1.4 16.4 ± 1.3 Marc1 AD-1531719.2 26.8 ±
 2.8 61.9 ± 6.6 69 ± 8.4 AD-1636805.1 10.1 ± 1.7 32.9 ± 5.2 53.2 ± 3.4 Marc1 AD-1531682.2
 16.4 ± 2.8 42.4 ± 7.2 27.3 ± 2.4 AD-1636782.1 11.9 ± 1.1 25.9 ± 1.8 38.2 ± 5.1 PNPLA3 AD-
 75247.4 53.6 ± 9 88.8 ± 9.5 90.3 ± 10.4 AD-1636754.1 52.2 ± 3.3 71.2 ± 17.2 69.6 ± 13.7 C3

AD-1531655.2 14.3 ± 2.4 46 ± 2.2 54.8 ± 11.1 AD-1636695.1 14.6 ± 2.4 26.6 ± 4.6 30.5 ± 3.1
PNPLA3 AD-67589.6 40.1 ± 10.3 68.2 ± 16.9 57.6 ± 7.6 AD-1636733.1 29.9 ± 4.7 31 ± 5 37.9 ± 1.7 Marc1 AD-1531703.2 15.4 ± 1.5 47.9 ± 2.3 54.5 ± 6.6 AD-1636798.1 9 ± 1.8 16.6 ± 3.2
34.3 ± 3.5 C3 AD-1531665.2 46.4 ± 5.3 83.4 ± 12.2 94.3 ± 7 AD-1636720.1 35.7 ± 5 52.4 ± 3.4 71.4 ± 3.5 C3 AD-1531660.2 14.8 ± 4.1 39.2 ± 8.6 54.6 ± 8.7 AD-1636701.1 13.9 ± 4.4 30.3 ± 6.8 41.1 ± 4.1 C3 AD-1531672.2 25.6 ± 4.4 62.5 ± 3.8 82 ± 9.7 AD-1636716.1 19.2 ± 2.9 38.5 ± 5.4 55.1 ± 2.6 PNPLA3 AD-1010735.4 35.2 ± 8.8 55.1 ± 21.7 52.6 ± 7.4 AD-1636729.1 48.4 ± 3.4 45 ± 4.2 56.2 ± 11 Marc1 AD-1531721.2 38.2 ± 4.3 61.6 ± 5.7 69.9 ± 10.8 AD-1636806.1 16.7 ± 1.1 30.4 ± 2 53.6 ± 9.8 PNPLA3 AD-67605.10 30.7 ± 3.2 44.9 ± 3.4 42.2 ± 2.6 AD-1636735.1 41.9 ± 2.3 51.9 ± 11.3 64.6 ± 11.6 C3 AD-571552.3 33.6 ± 7.3 73.4 ± 4.7 87.3 ± 7.4 AD-1636714.1 25.1 ± 2.2 42.2 ± 7.6 50.4 ± 1.3 Marc1 AD-1531716.2 8 ± 1.4 20.3 ± 4.7 15.6 ± 2.3 AD-1636778.1 7.5 ± 0.8 10.5 ± 1.5 13.2 ± 1.1 C3 AD-569269.6 7.3 ± 1 12.3 ± 1.4 12.3 ± 3.2 AD-1479350.2 8.7 ± 1.2 10.5 ± 1.2 14.4 ± 1.7 Marc1 AD-1531718.2 9.1 ± 1.4 29.1 ± 5.1 23.1 ± 2.7 AD-1636785.1 7.6 ± 1.3 12.6 ± 1.9 24.1 ± 2.7 C3 AD-1531663.3 14.9 ± 1.6 26.9 ± 4.6 56.5 ± 2.7 AD-1636763.1 12.6 ± 2 14.5 ± 4.5 27.4 ± 3.6 SCN9A AD-1531765.2 40.2 ± 2.8 55.3 ± 7.9 68.2 ± 1.9 AD-1636834.1 34.7 ± 3.3 38.6 ± 2.2 47.7 ± 10.4 SCN9A AD-1531735.2 33.2 ± 4 46.1 ± 5.2 34.9 ± 4 AD-1636811.1 28.7 ± 3 42.5 ± 5.4 52.8 ± 4.3 PNPLA3 AD-519933.3 54.6 ± 8 99.2 ± 17.3 91.4 ± 18.6 AD-1636757.1 52 ± 6.2 56.2 ± 4.3 60.4 ± 2.7 PNPLA3 AD-519780.2 50.1 ± 3.9 80.1 ± 12.2 79.5 ± 10.2 AD-1636758.1 44.8 ± 7.9 55.5 ± 8 74.1 ± 6.4 Marc1 AD-1531679.2 28.2 ± 2.9 60.8 ± 10.5 61.4 ± 4.4 AD-1636791.1 16.4 ± 1.4 46.6 ± 8.1 59 ± 7.7 Marc1 AD-1531692.2 13.5 ± 0.7 23.7 ± 1.7 17.9 ± 2.9 AD-1636777.1 9.8 ± 2.7 20 ± 1.4 28.7 ± 3.8 SCN9A AD-1531741.2 72.7 ± 1.3 83.5 ± 8.9 69.8 ± 2.9 AD-1636838.1 47.3 ± 10.9 76.9 ± 4 106.3 ± 11.5 C3 AD-572575.3 20.8 ± 4.2 57.3 ± 9.8 68.9 ± 11.1 AD-1636703.1 26.4 ± 6 38 ± 10.6 58.9 ± 14.5 SCN9A AD-1531744.2 49.2 ± 3.9 65.5 ± 8.5 63.3 ± 7.5 AD-1636831.1 34.5 ± 5.7 66.5 ± 5.5 89.5 ± 7.2 PNPLA3 AD-519354.4 62.3 ± 16.1 77.2 ± 22.9 84.6 ± 15.6 AD-1636732.1 37.5 ± 7.3 44.5 ± 6.7 64.5 ± 13.5 PNPLA3 AD-67551.7 22.5 ± 1.9 38.4 ± 12 41 ± 4.2 AD-1636737.1 30.5 ± 4.4 39.2 ± 8.3 60 ± 7.1 Marc1 AD-1531687.2 14.7 ± 2.6 34.6 ± 3.9 21.1 ± 1.9 AD-1636779.1 14.3 ± 2.8 23.6 ± 1.9 32.5 ± 7.6 SCN9A AD-1531736.2 73.3 ± 10.3 78.8 ± 7.7 68.3 ± 6.1 AD-1636843.1 82.1 ± 9.5 95.3 ± 12.7 107.5 ± 11.9 Marc1 AD-1531688.2 5 ± 1.1 13 ± 2.3 9.2 ± 1.3 AD-1636772.1 6.5 ± 1 10.1 ± 0.5 11.2 ± 1.7 SCN9A AD-1531747.2 69.4 ± 2 77.8 ± 3.6 80 ± 5.3 AD-1636836.1 45.7 ± 2.9 60.4 ± 5.5 93.2 ± 3.3 SCN9A AD-1531756.2 38.3 ± 4.9 46.5 ± 2.8 46.4 ± 5.2 AD-1636813.1 29.8 ± 1.1 46.7 ± 4.7 53.2 ± 2.7 Marc1 AD-1531696.2 9.4 ± 0.8 22.3 ± 4.3 18.5 ± 4.2 AD-1636783.1 7.5 ± 0.7 12.7 ± 1.3 25 ± 2.2 C3 AD-1531658.2 13.3 ± 1.8 30.7 ± 0.8 39.7 ± 3.7 AD-1636692.1 12.1 ± 1.2 24.2 ± 5.8 40.9 ± 10.6 SCN9A AD-1531757.2 64.6 ± 4 75.8 ± 7.1 70.3 ± 3.8 AD-1636844.1 30.1 ± 2.4 46.6 ± 5.4 59.5 ± 3.3 PNPLA3 AD-67565.3 62.4 ± 17.9 69.5 ± 15 68.4 ± 8.1 AD-1636726.1 71.3 ± 5.6 50 ± 2.7 76.6 ± 18.7 PNPLA3 AD-1010714.3 52.6 ± 19 99.7 ± 26.3 85.9 ± 13 AD-1636760.1 28.4 ± 5.4 40 ± 6 54.8 ± 4.3 SCN9A AD-1531728.2 60.8 ± 4.2 69.8 ± 9.1 76.8 ± 4.7 AD-1636841.1 33.6 ± 5.1 53.3 ± 8.2 65.3 ± 8.5 Marc1 AD-1531686.2 16.4 ± 2.6 52.5 ± 3.5 50.2 ± 8.3 AD-1636789.1 12.3 ± 2.3 25.2 ± 4.6 45.5 ± 4.1 PNPLA3 AD-67583.7 45.2 ± 4.6 81.4 ± 24.3 64.5 ± 13.5 AD-1636762.1 46.4 ± 12.4 48.8 ± 8.4 48.8 ± 10.2 C3 AD-572388.7 20.2 ± 1.8 49.7 ± 7.8 68.7 ± 5.5 AD-1636766.1 71 ± 9.6 81.7 ± 9.4 90.1 ± 12.4 C3 AD-1531662.2 37.3 ± 5.1 83.1 ± 8.3 91.9 ± 5.3 AD-1636718.1 31.5 ± 1.6 68.9 ± 9.3 77.7 ± 5.9 C3 AD-571901.2 26.2 ± 7.2 62.6 ± 5.1 78.2 ± 7.4 AD-1636707.1 32.1 ± 7.9 60 ± 7.6 61 ± 6.7 PNPLA3 AD-1010732.4 30.3 ± 6.7 63.6 ± 15 52.1 ± 4.7 AD-1636727.1 39.3 ± 3.8 55 ± 13.3 59.5 ± 11 PNPLA3 AD-520061.6 32.7 ± 8.1 49.5 ± 8.4 51.3 ± 5.3 AD-1636741.1 42.2 ± 5.7 44.4 ± 2.4 57.3 ± 4.9 SCN9A AD-1531770.2 50.6 ± 8.6 74.1 ± 0.9 76.6 ± 3.5 AD-1636830.1 38.3 ± 1.7 59.3 ± 7.3 72.6 ± 7.7 Marc1 AD-1531705.2 5.8 ± 0.7 21.9 ± 2.7 11.9 ± 2.1 AD-1636780.1 6 ± 0.7 14.2 ± 5.2 27.9 ± 2.5 C3 AD-569266.3 7.5 ± 1.3 19 ± 4.3 24.5 ± 5.1 AD-1636689.1 12.2 ± 2.1 18.5 ± 3.2 22.2 ± 4.3 SCN9A AD-1531740.2 68.1 ± 10.5 84.9 ± 5.5 66.9 ± 2.4 AD-1636839.1 41.3 ± 6 56.5 ± 9.4 72 ± 4.9

PNPLA3 AD-75265.5 48.9 ± 5.5 83.9 ± 20.7 82.7 ± 8.8 AD-1636753.1 36.1 ± 10.1 55.8 ± 8.4
 67.9 ± 6.8 Marc1 AD-1531709.2 13.9 ± 3.1 37.2 ± 7.5 46.9 ± 13.3 AD-1636792.1 10.5 ± 0.7 20.9 ± 2.8
 30.7 ± 6.5 C3 AD-569048.2 20.9 ± 4 56.4 ± 11.3 64.2 ± 11.2 AD-1636709.1 34.3 ± 0.8
 50.6 ± 10 75.3 ± 7.9 C3 AD-571633.3 34.2 ± 5.7 72.1 ± 7.6 81.3 ± 7 AD-1636721.1 29.7 ± 2.1
 51.3 ± 3 78.2 ± 9.8 C3 AD-569516.3 28.8 ± 2.2 76.2 ± 9.6 91.9 ± 10.4 AD-1636719.1 32.5 ± 5.2
 51.9 ± 12.9 72.2 ± 11.7 PNPLA3 AD-519346.5 26.2 ± 4.5 53 ± 6.2 51 ± 4.9 AD-1636765.1
 27 ± 5.3 28.3 ± 3.6 41.8 ± 5.1 C3 AD-571748.3 8.2 ± 2.8 26.5 ± 2.2 38.3 ± 5.8 AD-1636693.1
 13.5 ± 2.4 32.2 ± 7.1 37.7 ± 6.8 Marc1 AD-1531708.2 27.5 ± 1.5 61.3 ± 6.1 70.9 ± 12.7 AD-
 1636807.1 18.9 ± 2.9 26.2 ± 4.7 49.5 ± 5.9 C3 AD-570712.7 7.3 ± 0.5 17.3 ± 1.6 28.1 ± 3.6 AD-
 1479283.2 10.2 ± 0.7 16.5 ± 1.3 27.9 ± 4.1 Marc1 AD-1531706.2 5.4 ± 0.4 14.1 ± 1.1 8.5 ± 1.1
 AD-1636774.1 5.7 ± 0.9 10.8 ± 2.4 13.8 ± 1.1 PNPLA3 AD-67584.8 49.1 ± 8.5 57.2 ± 5.6 49.8 ± 13.5
 AD-1636725.1 57.4 ± 13 57.4 ± 16.8 67.4 ± 9.5 C3 AD-1531668.2 17.2 ± 2.9 49.8 ± 8.7
 61.8 ± 9.7 AD-1636715.1 13.8 ± 1.8 28.2 ± 4.4 45.9 ± 4.9 PNPLA3 AD-67577.7 27.2 ± 3.3 41.7 ± 6.5
 44.1 ± 7.5 AD-1636750.1 30.7 ± 7.3 38.2 ± 4.1 58.8 ± 7.1 C3 AD-571753.4 11.7 ± 2.3 30 ± 6.6
 47.8 ± 10 AD-1636702.1 12.3 ± 0.9 17.9 ± 3.4 31.6 ± 3.1 C3 AD-1531657.2 20.6 ± 5.8 58.1 ± 9.5
 73 ± 7.9 AD-1636705.1 16 ± 2 30.2 ± 5.1 42 ± 4.2 C3 AD-571715.5 5.7 ± 0.6 11.7 ± 0.9
 24.5 ± 4.2 AD-1636690.1 6.4 ± 1.5 13.3 ± 2.2 19.1 ± 2.2 C3 AD-1531659.2 12.1 ± 1.9 29.4 ± 3.5
 38.2 ± 7.2 AD-1636694.1 12.8 ± 1.8 22.3 ± 6.6 25.6 ± 2.1 PNPLA3 AD-67560.7 52.7 ± 14.7
 63.9 ± 17.1 68.5 ± 10 AD-1636743.1 43.9 ± 6.2 46.6 ± 7.3 51.7 ± 9.1 PNPLA3 AD-1531674.2
 51.8 ± 5.6 55.2 ± 5.4 47.2 ± 8.1 AD-1636756.1 47.3 ± 8.3 61.1 ± 10.9 60.3 ± 6.7 PNPLA3 AD-
 519757.6 48.2 ± 8.9 43.1 ± 14.7 59.6 ± 5.1 AD-1636734.1 36.9 ± 8.1 65.6 ± 10.7 51.3 ± 10.8
 SCN9A AD-1531732.2 65.4 ± 1.9 43.7 ± 8.3 57.1 ± 4.6 AD-1636840.1 54.3 ± 6.5 63.4 ± 5.7 56.7 ± 12.5
 C3 AD-1531656.2 57.5 ± 10.1 104.8 ± 19.6 90.9 ± 15.1 AD-1636712.1 44.6 ± 2.5 59.1 ± 11.9
 94.7 ± 8.9 Marc1 AD-1531678.2 36.1 ± 3.9 66 ± 4.3 58.6 ± 11.2 AD-1636800.1 15.4 ± 3.1
 22.7 ± 1.6 38.5 ± 5.7 Marc1 AD-1531701.2 15.3 ± 1.7 32.8 ± 2.4 31.4 ± 3.2 AD-1636790.1 16 ± 2.9
 31.8 ± 3.5 49.4 ± 7.7 Marc1 AD-1531680.2 5.1 ± 0.6 16.4 ± 1.8 9.7 ± 1.1 AD-1636770.1
 6.5 ± 0.8 14.5 ± 3.6 19.7 ± 5 SCN9A AD-1531734.2 56.4 ± 4.3 67.5 ± 10.7 82 ± 4 AD-
 1636837.1 48.4 ± 6.4 78.6 ± 3.6 107.1 ± 13.3 C3 AD-568977.3 14.4 ± 1.7 38 ± 4.8 41.4 ± 11.1
 AD-1636699.1 17.7 ± 1.1 24.9 ± 3.3 39.3 ± 2.8 C3 AD-569268.10 6.8 ± 0.4 10.2 ± 1.9 14.2 ± 1.9
 AD-1479341.2 8.7 ± 1.6 12 ± 0.7 16.6 ± 2.5 C3 AD-572818.2 10.5 ± 1.5 27.2 ± 4.4 42 ± 7.4
 AD-1636691.1 13.1 ± 1.8 23.3 ± 2.2 36.9 ± 6.1 SCN9A AD-1531729.2 45 ± 3.7 60.9 ± 4.8 49.6 ± 7.2
 AD-1636815.1 35.8 ± 5.6 49.7 ± 6 51.2 ± 1.5 Marc1 AD-1531689.2 36.6 ± 7.1 61.4 ± 7.9
 69.7 ± 4.9 AD-1636795.1 22.5 ± 3.4 33.4 ± 1.5 51.1 ± 5.9 PNPLA3 AD-75275.3 56 ± 8.3 60.8 ± 11.8
 44.6 ± 2.6 AD-1636731.1 48.6 ± 4.2 61 ± 8.4 70.7 ± 11.9 PNPLA3 AD-67568.3 33.8 ± 2.8
 35 ± 4.9 50.8 ± 8.1 AD-1636744.1 36.2 ± 6.1 43.9 ± 5.5 48.5 ± 2.2 C3 AD-569494.2 98.1 ± 8
 113.9 ± 3.9 105.2 ± 8.5 AD-1636717.1 105.6 ± 13.2 92.8 ± 7.2 90.9 ± 14.1 Marc1 AD-
 1531722.2 26.5 ± 0.9 54.4 ± 5 54.4 ± 6.6 AD-1636796.1 12.3 ± 2.2 21.8 ± 0.4 27.7 ± 3.4 C3 AD-
 571610.3 21.1 ± 7.8 49 ± 4.8 69.1 ± 9.5 AD-1636710.1 25.4 ± 1.7 45.3 ± 5.6 61.5 ± 8.6 SCN9A
 AD-1531753.2 42.9 ± 1.7 59.3 ± 4.7 54.2 ± 5.1 AD-1636822.1 33 ± 2.8 53.9 ± 6.7 62 ± 7.7
 SCN9A AD-1531759.2 46.3 ± 3.4 50.6 ± 5.4 70.6 ± 2.3 AD-1636828.1 30.8 ± 2.4 41.1 ± 4.7 52.9 ± 7.6
 C3 AD-572495.3 8.7 ± 0.9 10.7 ± 1.7 17 ± 2.5 AD-1636767.1 11.5 ± 1.2 15 ± 1.5 21.2 ± 5.2
 PNPLA3 AD-1010719.3 30 ± 4.2 79.4 ± 24.3 28.9 ± 4 AD-1636739.1 20.9 ± 2.1 24.4 ± 5.8
 33.7 ± 8.2 Marc1 AD-1531683.2 28.6 ± 5.8 64.7 ± 6.8 54.8 ± 3 AD-1636803.1 32.4 ± 4.6
 58.5 ± 9.1 85.2 ± 5 Marc1 AD-1531711.2 22.6 ± 6.1 46.2 ± 9.7 41.4 ± 4.7 AD-1636801.1 11.7 ± 1
 20.4 ± 5.1 40.3 ± 4.2 C3 AD-1531664.2 19.1 ± 5.3 50.6 ± 8.8 59.4 ± 2.3 AD-1636713.1 32.7 ± 5.9
 47.5 ± 4.9 53.4 ± 2.6 SCN9A AD-1531762.2 57.9 ± 4 75.4 ± 6.4 73.8 ± 5.9 AD-1636833.1
 35.9 ± 3.5 49.3 ± 5.4 78.1 ± 13.9 PNPLA3 AD-67526.5 26.9 ± 8.8 41.8 ± 13.4 22 ± 2.2 AD-
 1636740.1 19.7 ± 3.7 19.4 ± 1.9 32.9 ± 4.9 SCN9A AD-1531725.2 26.8 ± 4.4 32.5 ± 3.4 37.6 ± 3.4
 AD-1636823.1 23.8 ± 2.4 32.5 ± 5.5 36.7 ± 4.8 SCN9A AD-1531737.2 40.1 ± 2.1 43.3 ± 3.6 57.4 ± 10
 AD-1636827.1 33.5 ± 2.8 38 ± 4.2 54.5 ± 6.9 PNPLA3 AD-1531673.2 42 ± 4.6 $78.6 \pm$

23.6 65.1 ± 4.5 AD-1636749.1 40.7 ± 3.3 31.6 ± 5.5 35.8 ± 3.3 Marc1 AD-1531717.2 16.1 ± 2.2
38.5 ± 2.8 46.9 ± 5.2 AD-1636799.1 11 ± 1.2 13.8 ± 2.7 25.9 ± 7.2 C3 AD-570644.2 15.1 ± 3.2
43.5 ± 6.8 61.4 ± 10.6 AD-1636700.1 12.8 ± 1.7 30.7 ± 6 35.8 ± 7.3 C3 AD-568962.3 24 ± 4.2
57.1 ± 12.1 69.8 ± 7.9 AD-1636706.1 21.1 ± 4.3 35.1 ± 2.9 50.5 ± 4.3 Marc1 AD-1531690.2 6.7
± 0.9 14.2 ± 2.8 8.7 ± 1.3 AD-1636771.1 6 ± 1.1 10.5 ± 1 10.8 ± 1.3 SCN9A AD-1531750.2
46.9 ± 2.7 58.3 ± 6.8 55.3 ± 4.3 AD-1636826.1 41 ± 5.9 52.9 ± 9.3 66.5 ± 9.4 Marc1 AD-
1531710.3 14.9 ± 2.2 34.3 ± 6.1 40.1 ± 7.6 AD-1636848.1 8.2 ± 1.1 25 ± 6 44.4 ± 3.3 PNPLA3
AD-67578.3 61 ± 15.4 54.9 ± 4.2 66.5 ± 4.9 AD-1636746.1 58.6 ± 4.5 60.5 ± 16.3 68.6 ±
14.1 PNPLA3 AD-67582.6 47.8 ± 7 71.9 ± 12.8 80.3 ± 12.1 AD-1636747.1 39.9 ± 1.7 51.8 ±
5.2 66.7 ± 12.8 C3 AD-1531666.2 39.7 ± 10.3 71.9 ± 4.8 85.9 ± 11.1 AD-1636723.1 15 ± 2.2
31.9 ± 6.2 67.3 ± 12.3 Marc1 AD-1531695.3 6.7 ± 0.6 11.8 ± 3.2 11 ± 1.4 AD-1636847.1 7.5
± 0.5 13.4 ± 2.7 19.9 ± 3.2 SCN9A AD-1531723.2 29.7 ± 5.8 37.8 ± 5.7 43.1 ± 3.3 AD-1636824.1
19.3 ± 1.5 28.9 ± 6.7 32.1 ± 3 PNPLA3 AD-75270.4 46.6 ± 12 62.6 ± 15.2 65.6 ± 12.2 AD-
1636748.1 42.2 ± 1.7 39.2 ± 9.5 54.7 ± 2.5 C3 AD-1531671.2 11.8 ± 2.7 33.2 ± 4.2 49.2 ± 7.8 AD-
1636704.1 24.1 ± 3.8 37 ± 8.3 56.6 ± 10.9 PNPLA3 AD-519350.6 35.8 ± 6.4 50.4 ± 4.8 49.4 ±
5 AD-1636724.1 45.8 ± 8.2 55 ± 8.8 50.7 ± 9 SCN9A AD-1531764.2 63.2 ± 4.6 66 ± 3.1
57 ± 10 AD-1636832.1 57.3 ± 4.8 69 ± 7.9 62.6 ± 5.9 SCN9A AD-1531739.2 39.7 ± 5.3 58.5 ±
5.4 60.7 ± 7.9 AD-1636835.1 24.5 ± 5.3 38.9 ± 1.9 57 ± 9.2 Marc1 AD-1531681.2 17.3 ± 3.6
55.2 ± 7.2 60.8 ± 11.6 AD-1636788.1 11.3 ± 2.6 21.4 ± 5.8 32.2 ± 9.2 Marc1 AD-1531704.2 7.4
± 0.7 36.8 ± 5 26.5 ± 2.2 AD-1636787.1 8.6 ± 2.2 14.5 ± 3.1 23.5 ± 3.8 PNPLA3 AD-67564.3
71.1 ± 5.4 74.7 ± 9 76.7 ± 7.6 AD-1636761.1 56.3 ± 2.7 55.7 ± 4.5 63.4 ± 6.7 Marc1 AD-
1531676.2 18.8 ± 4.6 57.6 ± 10 80 ± 12.9 AD-1636794.1 9.6 ± 1.9 22.2 ± 3.4 34.6 ± 7.4
Marc1 AD-1531700.2 30.1 ± 5.2 71.1 ± 6.8 76.9 ± 7.3 AD-1636804.1 12.8 ± 2.6 28.6 ± 2.7 51.1 ±
9.4 Marc1 AD-1531702.2 37.9 ± 6.4 72.7 ± 4.3 75.3 ± 5.5 AD-1636802.1 30.3 ± 4.9 55 ± 9.5
71.9 ± 20.4 SCN9A AD-1531730.2 39.4 ± 2.8 47.3 ± 1.2 42.4 ± 6.1 AD-1636814.1 30.6 ± 1.6 46.2
± 2.2 57.3 ± 0.4 SCN9A AD-1531738.2 32.3 ± 5.3 39.4 ± 1.9 45 ± 5.5 AD-1636812.1 30.4 ± 2.2
38.5 ± 7.1 43.3 ± 8 SCN9A AD-1531733.3 45.3 ± 2 53.3 ± 10.4 35.1 ± 6.9 AD-1636850.1
29.7 ± 3.6 44.5 ± 13.3 59.6 ± 9.9 PNPLA3 AD-67554.11 39.8 ± 4.4 45.5 ± 10 55.4 ± 7 AD-
1636738.1 48.8 ± 4.6 56.6 ± 13.9 56 ± 9.1 Marc1 AD-1531685.2 25.3 ± 1.4 41.9 ± 6.6 51.9 ±
2.1 AD-1636797.1 6 ± 0.9 14.9 ± 2.9 23.6 ± 1.9 PNPLA3 AD-75269.3 49.5 ± 10.5 66.5 ±
13.9 44.8 ± 8.1 AD-1636728.1 44.3 ± 2.7 40.4 ± 4 57.7 ± 11.3 SCN9A AD-1531752.2 87.9 ±
13.1 64.7 ± 10.1 70 ± 9.8 AD-1636842.1 62.9 ± 1.5 78.6 ± 8.9 87.1 ± 5.6 SCN9A AD-
1531754.3 42.1 ± 1.8 46.6 ± 3.8 39.8 ± 6.3 AD-1636852.1 25.4 ± 4 46 ± 2.2 52.6 ± 2.1
PNPLA3 AD-518942.2 59 ± 6.6 90.6 ± 20.3 106.6 ± 15.5 AD-1636764.1 37.6 ± 5.1 33.4 ± 4.9
40.6 ± 6.9 C3 AD-571932.2 22.1 ± 2.2 45.8 ± 6.9 62 ± 6.7 AD-1636698.1 17.8 ± 1.9 31 ± 5.9
49.4 ± 11.2 C3 AD-572022.2 14.4 ± 2.8 34 ± 3.8 44.4 ± 6.9 AD-1636697.1 15.9 ± 1.1 32.2 ± 5.4
52.1 ± 7.3 SCN9A AD-1531760.2 46.7 ± 3.2 68.7 ± 6.3 58.3 ± 1.6 AD-1636816.1 29.1 ± 5.4 51.5 ±
7.6 60.9 ± 8.7 SCN9A AD-1531766.2 50.2 ± 8.7 55.5 ± 5.8 48.9 ± 4.1 AD-1636846.1 55.2 ± 4.6
67.2 ± 14.9 87.1 ± 18.8 C3 AD-572577.3 35.8 ± 7.2 79.9 ± 14.7 95.7 ± 6.3 AD-1636711.1 51
± 12.3 74.5 ± 14.2 89.2 ± 5.4 Marc1 AD-1531697.2 21.5 ± 4.6 41 ± 1.2 26.5 ± 0.5 AD-
1636793.1 13.1 ± 1.9 17.1 ± 3.5 31.4 ± 5.3 PNPLA3 AD-67561.3 71.2 ± 14.1 81.5 ± 16.5 101.6
± 7.1 AD-1636742.1 65.7 ± 4.6 93.1 ± 24.5 74.2 ± 12.7 PNPLA3 AD-67586.3 73.6 ± 11.5
87.6 ± 12.8 86.7 ± 6.3 AD-1636755.1 57.5 ± 5.8 61.7 ± 12.1 64.6 ± 9.3 C3 AD-569763.5 18.2 ±
4.8 45.4 ± 3.6 52.7 ± 2.7 AD-1636696.1 19.2 ± 6.9 34.1 ± 5.9 43.5 ± 5.2 SCN9A AD-1531726.2
32.2 ± 2.5 42 ± 3.8 46.7 ± 4.1 AD-1636821.1 24.9 ± 1.7 39.4 ± 4.5 54.4 ± 2.4 SCN9A AD-
1531742.3 32.4 ± 5.7 52.4 ± 4.3 43.4 ± 4.2 AD-1636851.1 23.8 ± 4.1 56.9 ± 3.2 75.8 ± 13.1
PNPLA3 AD-67573.3 62.3 ± 10.5 74.5 ± 8.6 77.5 ± 10.2 AD-1636751.1 42.6 ± 2.7 41.8 ± 5.2
69.3 ± 10.7 SCN9A AD-1531761.2 58.2 ± 5.9 72.5 ± 5.2 60.6 ± 8.9 AD-1636818.1 26.3 ± 4.7 47.6
± 10 56.8 ± 6.5 Marc1 AD-1531712.2 8 ± 0.2 19.6 ± 2 14.5 ± 0.5 AD-1636781.1 11 ± 0.1
13.5 ± 0.7 19.5 ± 3.7 C3 AD-1531669.2 48.1 ± 3.2 87 ± 9.8 80.1 ± 5.5 AD-1636722.1 29.7 ± 3.1

50.1 ± 3.9 59.9 ± 5.4 SCN9A AD-1531745.2 41.3 ± 3.1 50.1 ± 2.4 42 ± 1.9 AD-1636820.1 29.3 ± 2.2 38.5 ± 2.8 47.9 ± 3.8 SCN9A AD-1531724.2 48.3 ± 2.7 55.4 ± 4.6 47.9 ± 0.9 AD-1636817.1 38 ± 6.1 50.9 ± 5 67.9 ± 5.4 SCN9A AD-1531731.2 43.6 ± 5.2 52.8 ± 1.2 66.9 ± 7 AD-1636825.1 33.6 ± 4.3 49.6 ± 5.4 56.3 ± 2 C3 AD-570132.3 26.7 ± 3.7 60.8 ± 9.6 87.8 ± 11.1 AD-1636708.1 19.5 ± 3.6 31 ± 3.8 45.3 ± 5.5 SCN9A AD-1531769.2 38.3 ± 4.6 53.1 ± 6.1 48.9 ± 3.6 AD-1636819.1 32.6 ± 2.1 43.7 ± 5.3 54.3 ± 1.1 Marc1 AD-1531694.2 6.4 ± 0.6 17.3 ± 2 11.6 ± 1.7 AD-1636775.1 8 ± 1.9 15.7 ± 2.8 19 ± 3.1 Marc1 AD-1531677.2 31.7 ± 3.8 59.5 ± 7.4 60.1 ± 15.2 AD-1636784.1 16.3 ± 1.8 25.4 ± 6.1 33 ± 4.8 SCN9A AD-1531727.2 50.1 ± 2.8 61.8 ± 5.6 74.4 ± 6.6 AD-1636829.1 42.8 ± 3.9 59.8 ± 7.2 68.7 ± 9 PNPLA3 AD-67575.11 62.6 ± 13.6 56.2 ± 5.8 58.4 ± 4.3 AD-1636768.1 49.4 ± 9.5 50.9 ± 6.4 48.9 ± 7.4 PNPLA3 AD-75274.6 72.6 ± 6 74.9 ± 13.3 96.4 ± 13.1 AD-1636745.1 66 ± 8.7 68.9 ± 7.1 86.5 ± 7.2 SCN9A AD-1531768.2 32.2 ± 1.8 46.7 ± 4.3 54.9 ± 5.9 AD-1636809.1 20 ± 0.9 30.9 ± 4.4 43.4 ± 4.9 C3 AD-570134.4 7.7 ± 1.1 12.8 ± 0.6 24.1 ± 7 AD-1636688.1 9 ± 1 20.9 ± 5.7 31.8 ± 8.4 Marc1 AD-1531707.2 6.6 ± 1.2 15.8 ± 0.7 9.2 ± 0.8 AD-1636773.1 7.9 ± 1 10 ± 0.4 12.1 ± 0.6 PNPLA3 AD-75272.3 56.9 ± 19.1 73.2 ± 12 68.6 ± 13.7 AD-1636752.1 41 ± 6 42 ± 7.5 69.6 ± 9.1 PNPLA3 AD-67567.3 82.7 ± 22.8 95.3 ± 15.3 85.4 ± 9.9 AD-1636759.1 56.3 ± 6.1 62.9 ± 15.9 72.3 ± 8.3 Marc1 AD-1531714.3 6.4 ± 0.9 18.1 ± 2.3 23.7 ± 5.3 AD-1636849.1 9 ± 1 16.6 ± 1.8 24.5 ± 1.2 Marc1 AD-1531720.2 5.4 ± 0.7 12.6 ± 1.7 10.8 ± 1.4 AD-1636776.1 5.3 ± 0.6 10.9 ± 1.2 12.6 ± 1.2 SCN9A AD-1531746.2 73.3 ± 4.7 86.7 ± 2.2 71.3 ± 10.6 AD-1636845.1 43.2 ± 5.8 91.2 ± 8.6 86.4 ± 19.1 SCN9A AD-1531755.2 37.8 ± 6.1 49.1 ± 5.4 46.9 ± 7.4 AD-1636810.1 36.2 ± 2.1 61.4 ± 3.2 73.5 ± 5.5 Marc1 AD-1531675.2 16.9 ± 3.3 32.5 ± 4 33 ± 5.7 AD-1636786.1 10.1 ± 1.6 16.9 ± 4.3 27 ± 4.3 SCN9A AD-1531767.2 28.3 ± 1.2 44 ± 2.7 56.2 ± 8.9 AD-1636808.1 20.5 ± 1.4 32 ± 3.3 39.4 ± 5.1 PNPLA3 AD-520053.7 30.9 ± 5.9 34.7 ± 8.4 52.9 ± 8.6 AD-1636736.1 31.1 ± 6.6 44 ± 4.2 52.5 ± 4 PNPLA3 AD-1010734.3 42.8 ± 16.5 58.8 ± 17.2 49.6 ± 9.8 AD-1636730.1 50.8 ± 10.1 38.6 ± 7.4 53.7 ± 6.5

TABLE-US-00014 TABLE 14 In vitro activity for various designs targeting Agt in Hep3B Parent D1 Duplex ID 10 nM 1 nM 0.1 nM Duplex ID 10 nM 1 nM 0.1 nM AD-1632799.1 88 ± 15 105 ± 23 90 ± 21 AD-1657992.1 55 ± 11 91 ± 9 73 ± 8 AD-1632801.1 6 ± 1 23 ± 4 33 ± 3 AD-1657994.1 2 ± 1 12 ± 3 15 ± 4 AD-1632805.1 82 ± 7 95 ± 7 83 ± 12 AD-1657998.1 43 ± 4 65 ± 16 66 ± 6 AD-1632838.1 58 ± 5 94 ± 7 92 ± 10 AD-1658030.1 64 ± 5 84 ± 7 82 ± 8 AD-1684490.1 50 ± 11 83 ± 26 80 ± 18 AD-1684491.1 38 ± 12 67 ± 14 73 ± 17 AD-1684492.1 28 ± 4 68 ± 7 71 ± 20 AD-1684493.1 19 ± 2 47 ± 12 70 ± 11 AD-1684494.1 92 ± 19 101 ± 10 116 ± 6 AD-1684495.1 18 ± 4 54 ± 9 60 ± 6 AD-1684496.1 24 ± 5 72 ± 8 99 ± 10 AD-1684497.1 4 ± 1 15 ± 3 42 ± 7 AD-1684498.1 8 ± 2 31 ± 9 66 ± 9 AD-1684499.1 34 ± 8 53 ± 10 79 ± 8 AD-1684500.1 4 ± 1 16 ± 5 33 ± 6 AD-1684501.1 2 ± 0 7 ± 2 24 ± 2 AD-1684502.1 98 ± 12 107 ± 8 98 ± 3 AD-1684503.1 78 ± 15 102 ± 10 112 ± 14 AD-1684504.1 12 ± 2 56 ± 6 78 ± 10 AD-1684505.1 1 ± 0 7 ± 3 18 ± 5 AD-1684506.1 16 ± 5 53 ± 12 58 ± 11 AD-1684507.1 10 ± 3 46 ± 9 54 ± 6 AD-1684508.1 31 ± 8 87 ± 10 101 ± 15 AD-1684509.1 5 ± 1 17 ± 2 52 ± 3 AD-1684510.1 95 ± 8 119 ± 9 131 ± 11 AD-1684511.1 30 ± 6 67 ± 5 96 ± 13 AD-1684512.1 2 ± 1 9 ± 3 24 ± 3 AD-1684513.1 4 ± 2 28 ± 3 49 ± 9 AD-1684514.1 39 ± 12 67 ± 8 82 ± 10 AD-1684515.1 44 ± 9 88 ± 10 100 ± 4 AD-1632840.1 17 ± 4 62 ± 5 77 ± 7 AD-1658032.1 7 ± 2 34 ± 3 52 ± 3 AD-1632841.1 9 ± 2 45 ± 8 70 ± 7 AD-1658033.1 2 ± 0 21 ± 4 32 ± 6 AD-1632842.1 62 ± 11 60 ± 4 67 ± 13 AD-1658034.1 29 ± 6 77 ± 17 72 ± 12 AD-1632843.1 6 ± 3 38 ± 6 45 ± 5 AD-1658035.1 3 ± 1 9 ± 1 18 ± 3 AD-1632844.1 7 ± 1 28 ± 5 45 ± 6 AD-1658036.1 3 ± 0 12 ± 2 20 ± 4 AD-1632846.1 7 ± 3 30 ± 7 54 ± 10 AD-1658038.1 5 ± 2 18 ± 2 38 ± 1 AD-1632847.1 9 ± 4 35 ± 7 47 ± 4 AD-1658039.1 2 ± 0 5 ± 1 15 ± 7 AD-1632848.1 67 ± 11 92 ± 8 108 ± 8 AD-1658040.1 5 ± 1 18 ± 2 34 ± 5 AD-1632849.1 4 ± 1 13 ± 2 22 ± 1 AD-1658041.1 7 ± 2 16 ± 0 31 ± 4 AD-1632836.1 1 ± 0 3 ± 2 7 ± 1 AD-1658042.1 1 ± 0 1 ± 0 3 ± 1 AD-1632850.1 26 ± 2 62 ± 7 70 ± 9 AD-1658043.1 2 ± 0 7 ± 1 14 ± 2 AD-1632851.1 17 ± 5 53 ± 8 74 ± 5 AD-1658044.1 1 ± 0 6 ± 2 12 ± 2 AD-1632852.1 4 ± 2 23 ± 4 46 ± 10 AD-1658045.1 2 ± 1 6 ± 2 16 ± 2 AD-

1632853.1 16 ± 7 55 ± 14 62 ± 7 AD-1658046.1 2 ± 1 6 ± 1 19 ± 5 AD-1632854.1 40 ± 12 68 ± 11 74 ± 9 AD-1658047.1 6 ± 3 17 ± 1 30 ± 7 AD-1632855.1 2 ± 1 9 ± 1 13 ± 1 AD-1658048.1 3 ± 1 9 ± 3 18 ± 4 AD-1632856.1 21 ± 6 47 ± 8 75 ± 9 AD-1658049.1 1 ± 0 7 ± 1 16 ± 3 AD-1632857.1 8 ± 3 40 ± 5 43 ± 11 AD-1658050.1 5 ± 1 20 ± 5 29 ± 4 AD-1632858.1 6 ± 1 23 ± 6 48 ± 4 AD-1658051.1 6 ± 2 14 ± 1 31 ± 2 AD-1632859.1 3 ± 2 12 ± 3 43 ± 8 AD-1658052.1 2 ± 0 6 ± 0 17 ± 3 AD-1632860.1 9 ± 3 27 ± 4 57 ± 5 AD-1658053.1 6 ± 1 22 ± 2 51 ± 4 AD-1632861.1 10 ± 2 38 ± 5 63 ± 8 AD-1658054.1 2 ± 1 8 ± 4 22 ± 3 AD-1632862.1 53 ± 10 69 ± 10 100 ± 11 AD-1658055.1 12 ± 4 40 ± 5 69 ± 6 AD-1632863.1 13 ± 2 29 ± 5 75 ± 14 AD-1658056.1 18 ± 5 51 ± 1 80 ± 5 AD-1632864.1 61 ± 11 74 ± 9 113 ± 24 AD-1658057.1 67 ± 6 66 ± 16 96 ± 18 AD-1632865.1 37 ± 4 63 ± 10 86 ± 17 AD-1658058.1 19 ± 5 40 ± 6 56 ± 12 AD-1632866.1 17 ± 1 42 ± 8 62 ± 8 AD-1658059.1 12 ± 4 27 ± 6 41 ± 7 AD-1632991.1 18 ± 5 50 ± 2 74 ± 10 AD-1658184.1 10 ± 2 38 ± 9 58 ± 9 AD-1632992.1 63 ± 9 100 ± 20 90 ± 11 AD-1658185.1 38 ± 6 67 ± 12 86 ± 13 AD-1632993.1 50 ± 4 87 ± 8 102 ± 20 AD-1658186.1 5 ± 1 29 ± 1 55 ± 9 AD-1632994.1 6 ± 1 28 ± 2 65 ± 14 AD-1658187.1 3 ± 0 11 ± 1 32 ± 4 AD-1632995.1 35 ± 9 47 ± 4 75 ± 17 AD-1658188.1 4 ± 2 11 ± 3 26 ± 4 AD-1632996.1 13 ± 2 42 ± 8 76 ± 6 AD-1658189.1 7 ± 2 27 ± 4 56 ± 12 AD-1632997.1 5 ± 1 23 ± 4 50 ± 6 AD-1658190.1 4 ± 1 19 ± 6 31 ± 6 AD-1632998.1 8 ± 2 23 ± 2 58 ± 4 AD-1658191.1 13 ± 4 23 ± 3 48 ± 11 AD-1632999.1 41 ± 5 75 ± 4 83 ± 16 AD-1658192.1 21 ± 6 46 ± 10 77 ± 9 AD-1633000.1 106 ± 9 126 ± 8 96 ± 10 AD-1658193.1 25 ± 3 60 ± 5 83 ± 11 AD-1633003.1 4 ± 1 33 ± 7 64 ± 9 AD-1658196.1 2 ± 0 6 ± 2 22 ± 6 AD-1633004.1 55 ± 10 83 ± 11 122 ± 18 AD-1658197.1 8 ± 1 24 ± 5 32 ± 3 AD-1633007.1 6 ± 1 19 ± 2 69 ± 8 AD-1658200.1 2 ± 0 7 ± 1 22 ± 4 AD-1633008.1 26 ± 7 53 ± 8 53 ± 3 AD-1658201.1 19 ± 3 63 ± 12 89 ± 9 AD-1633009.1 5 ± 1 12 ± 3 30 ± 5 AD-1658202.1 10 ± 1 25 ± 5 50 ± 9 AD-1633010.1 5 ± 0 9 ± 3 13 ± 2 AD-1658203.1 1 ± 0 1 ± 0 4 ± 1 AD-1633011.1 3 ± 1 10 ± 2 26 ± 1 AD-1658204.1 1 ± 0 5 ± 1 16 ± 3 AD-1633012.1 5 ± 1 33 ± 4 64 ± 9 AD-1658205.1 2 ± 0 4 ± 1 10 ± 2 AD-1633013.1 3 ± 1 10 ± 3 25 ± 5 AD-1658206.1 1 ± 0 5 ± 1 14 ± 3 AD-1633014.1 3 ± 1 23 ± 4 35 ± 5 AD-1658207.1 4 ± 1 12 ± 3 22 ± 5 AD-1633015.1 16 ± 4 58 ± 5 72 ± 7 AD-1658208.1 4 ± 0 26 ± 6 39 ± 6 AD-1633016.1 21 ± 5 90 ± 5 104 ± 13 AD-1658209.1 14 ± 3 51 ± 4 71 ± 4 AD-1633018.1 84 ± 5 118 ± 17 109 ± 7 AD-1658211.1 84 ± 18 105 ± 10 102 ± 18 AD-1633019.1 8 ± 1 28 ± 0 50 ± 5 AD-1658212.1 2 ± 0 4 ± 1 8 ± 3 AD-1633020.1 27 ± 5 70 ± 6 96 ± 13 AD-1658213.1 9 ± 2 28 ± 6 56 ± 6 AD-1633027.1 17 ± 2 46 ± 7 65 ± 12 AD-1658220.1 4 ± 1 6 ± 1 15 ± 4 AD-1633028.1 15 ± 2 49 ± 8 63 ± 7 AD-1658221.1 5 ± 1 28 ± 4 49 ± 7 AD-1633029.1 113 ± 4 120 ± 14 108 ± 15 AD-1658222.1 28 ± 2 47 ± 4 58 ± 9 AD-1633030.1 19 ± 3 70 ± 13 86 ± 7 AD-1658223.1 4 ± 1 25 ± 7 36 ± 3 AD-1633031.1 6 ± 3 14 ± 2 40 ± 7 AD-1658224.1 4 ± 1 4 ± 0 10 ± 2 AD-84731.2 8 ± 1 16 ± 1 28 ± 7 AD-1658225.1 3 ± 1 6 ± 1 12 ± 2 AD-1633032.1 6 ± 2 14 ± 4 29 ± 7 AD-1658226.1 4 ± 1 6 ± 1 19 ± 4 AD-1633033.1 6 ± 1 12 ± 2 29 ± 3 AD-1658227.1 3 ± 1 5 ± 1 10 ± 3 AD-1633034.1 82 ± 12 92 ± 11 95 ± 16 AD-1658228.1 42 ± 8 75 ± 17 95 ± 13 AD-1684516.1 42 ± 11 83 ± 18 77 ± 23 AD-1684517.1 10 ± 2 30 ± 9 70 ± 6 AD-1684518.1 80 ± 10 107 ± 22 108 ± 20 AD-1684519.1 100 ± 7 107 ± 21 111 ± 21 AD-1684520.1 24 ± 6 75 ± 13 87 ± 8 AD-1684521.1 6 ± 1 15 ± 5 21 ± 2 AD-1633048.1 46 ± 9 82 ± 8 98 ± 10 AD-1658242.1 20 ± 6 43 ± 6 65 ± 10 AD-1633049.1 10 ± 2 28 ± 6 39 ± 10 AD-1658243.1 9 ± 1 19 ± 2 45 ± 4 AD-1633094.1 19 ± 7 31 ± 3 56 ± 7 AD-1658288.1 24 ± 4 53 ± 10 63 ± 13 AD-1633095.1 48 ± 15 63 ± 13 66 ± 11 AD-1658289.1 16 ± 5 36 ± 7 55 ± 7 AD-1633119.1 43 ± 8 72 ± 17 96 ± 16 AD-1658313.1 25 ± 3 49 ± 8 54 ± 5 AD-1633121.1 18 ± 5 53 ± 10 74 ± 13 AD-1658315.1 3 ± 1 11 ± 2 34 ± 9 AD-1633122.1 66 ± 18 85 ± 17 96 ± 15 AD-1658316.1 13 ± 4 40 ± 8 63 ± 8 AD-1633254.2 11 ± 4 21 ± 6 32 ± 5 AD-1658448.2 7 ± 1 16 ± 4 21 ± 9 AD-1633257.2 10 ± 2 27 ± 4 39 ± 7 AD-1658451.2 9 ± 1 23 ± 3 28 ± 3 AD-1633269.2 10 ± 0 18 ± 4 22 ± 2 AD-1658463.2 7 ± 1 9 ± 2 11 ± 1 AD-1633270.2 47 ± 3 68 ± 2 75 ± 12 AD-1658464.2 48 ± 4 69 ± 8 81 ± 13 AD-1633271.2 12 ± 3 29 ± 8 44 ± 10 AD-1658465.2 8 ± 3 17 ± 3 22 ± 4 AD-1633272.2 27 ± 6 35 ± 6 57 ± 4 AD-1658466.2 20 ± 3 27 ± 4 43 ± 7 AD-1633273.2 76 ± 2 78 ± 2 98 ± 17 AD-1658467.2 83 ± 6 76 ± 11 101 ± 11

AD-1633290.2 42 ± 4 68 ± 11 75 ± 9 AD-1658484.2 5 ± 1 16 ± 6 28 ± 3 AD-1633291.2 30 ± 5
50 ± 13 64 ± 11 AD-1658485.2 73 ± 15 97 ± 8 93 ± 20 AD-1633324.2 6 ± 2 14 ± 3 22 ± 5 AD-
1658519.2 2 ± 0 4 ± 2 4 ± 1 AD-84739.5 2 ± 0 3 ± 1 3 ± 1 AD-1658520.2 2 ± 0 4 ± 1 8 ± 1
AD-1633325.2 2 ± 0 3 ± 1 5 ± 2 AD-1658521.2 1 ± 1 4 ± 1 5 ± 1 AD-1633326.2 3 ± 1 7 ± 1
17 ± 1 AD-1658522.2 1 ± 0 2 ± 0 3 ± 1 AD-1633327.2 20 ± 4 38 ± 5 49 ± 3 AD-1658523.2 12
 ± 1 29 ± 6 58 ± 3 AD-1633328.2 7 ± 1 16 ± 2 27 ± 3 AD-1658524.2 4 ± 1 11 ± 1 28 ± 2 AD-
1633329.2 7 ± 1 21 ± 4 44 ± 7 AD-1658525.2 2 ± 1 6 ± 1 12 ± 4 AD-1633330.2 14 ± 3 34 ± 2 58
 ± 1 AD-1658526.2 6 ± 1 24 ± 7 33 ± 6 AD-1633331.2 112 ± 5 115 ± 6 123 ± 19 AD-1658527.2
31 ± 2 59 ± 5 80 ± 9 AD-1633332.2 12 ± 3 40 ± 8 61 ± 6 AD-1658528.2 8 ± 2 25 ± 3 33 ± 3 AD-
1633333.2 2 ± 0 8 ± 2 15 ± 1 AD-1658529.2 3 ± 0 6 ± 1 12 ± 2 AD-1633334.2 9 ± 1 24 ± 4 50 $\pm$
3 AD-1658530.2 6 ± 1 11 ± 2 18 ± 4 AD-1633335.2 105 ± 18 127 ± 10 140 ± 12 AD-1658531.2
27 ± 2 50 ± 5 73 ± 8 AD-1633343.2 63 ± 5 87 ± 7 100 ± 14 AD-1658539.2 32 ± 4 44 ± 5 69 ± 4
AD-1633345.2 29 ± 3 56 ± 5 87 ± 11 AD-1658541.2 13 ± 3 27 ± 3 49 ± 5 AD-1633346.2 5 ± 1
13 ± 3 27 ± 6 AD-1658542.2 5 ± 1 16 ± 5 20 ± 4 AD-1633409.2 4 ± 0 12 ± 4 21 ± 4 AD-1658605.2
4 ± 1 13 ± 5 23 ± 1 AD-1633453.2 67 ± 7 81 ± 13 102 ± 6 AD-1658650.2 25 ± 1 53 ± 3 82 ± 6
AD-1633464.2 14 ± 1 38 ± 4 67 ± 3 AD-1658661.2 5 ± 1 14 ± 2 27 ± 3 AD-1633465.2 5 ± 1 15 $\pm$
3 39 ± 3 AD-1658662.2 4 ± 3 10 ± 6 12 ± 2 AD-1633466.2 11 ± 2 34 ± 10 61 ± 5 AD-1658663.2
3 ± 1 10 ± 1 19 ± 0 AD-1633467.2 6 ± 2 22 ± 1 39 ± 6 AD-1658664.2 3 ± 1 8 ± 2 13 ± 2 AD-
1633468.2 85 ± 8 104 ± 9 114 ± 3 AD-1658665.2 52 ± 6 67 ± 5 89 ± 7 AD-1633604.2 82 ± 13
103 ± 14 118 ± 17 AD-1658801.2 51 ± 6 75 ± 6 106 ± 12 AD-1633621.2 4 ± 1 8 ± 1 21 ± 2 AD-
1658818.2 3 ± 1 7 ± 2 16 ± 2 AD-1633622.2 24 ± 5 53 ± 5 84 ± 10 AD-1658819.2 25 ± 4 48
 ± 13 75 ± 7 AD-1633623.2 46 ± 4 75 ± 9 118 ± 6 AD-1658820.2 10 ± 2 35 ± 4 58 ± 6 AD-
1633627.2 6 ± 1 20 ± 4 34 ± 1 AD-1658824.2 4 ± 1 8 ± 2 19 ± 3 AD-1633628.2 4 ± 1 11 ± 2 19 $\pm$
2 AD-1658825.2 2 ± 1 7 ± 1 14 ± 3 AD-1633630.2 41 ± 7 56 ± 5 67 ± 15 AD-1658827.2 $15 \pm$
2 20 ± 5 53 ± 9 AD-1633631.2 5 ± 1 11 ± 2 22 ± 4 AD-1658828.2 3 ± 1 6 ± 1 10 ± 2 AD-
1633632.2 3 ± 1 8 ± 1 17 ± 2 AD-1658829.2 3 ± 1 9 ± 3 17 ± 5 AD-1633633.1 4 ± 1 9 ± 1 33 $\pm$
9 AD-1658830.1 3 ± 1 9 ± 2 20 ± 6 AD-1633634.1 3 ± 1 5 ± 1 19 ± 4 AD-1658831.1 1 ± 0 2 $\pm$
0 6 ± 2 AD-1633635.1 20 ± 3 36 ± 6 71 ± 8 AD-1658832.1 4 ± 1 10 ± 2 29 ± 4 AD-1633636.1 1
 ± 0 2 ± 0 6 ± 1 AD-1658833.1 1 ± 0 1 ± 0 4 ± 1 AD-1633637.1 9 ± 1 25 ± 7 54 ± 17 AD-
1658834.1 2 ± 1 2 ± 0 5 ± 2 AD-1633638.1 24 ± 5 36 ± 9 71 ± 10 AD-1658835.1 7 ± 1 13 ± 1
30 ± 6 AD-1633639.1 1 ± 0 2 ± 0 5 ± 1 AD-1658836.1 3 ± 1 5 ± 2 12 ± 4 AD-1633640.1 4 ± 1
6 ± 1 17 ± 1 AD-1658837.1 3 ± 0 5 ± 1 18 ± 5 AD-1633641.1 7 ± 3 17 ± 3 31 ± 8 AD-1658838.1
2 ± 0 3 ± 1 8 ± 1 AD-1633642.1 39 ± 7 65 ± 11 71 ± 12 AD-1658839.1 2 ± 1 4 ± 1 10 ± 2
AD-1633643.1 23 ± 1 39 ± 9 70 ± 14 AD-1658840.1 8 ± 2 10 ± 2 34 ± 8 AD-1633644.1 3 ± 0 8
 ± 2 25 ± 4 AD-1658841.1 2 ± 1 8 ± 1 16 ± 4 AD-1633645.1 9 ± 2 20 ± 3 37 ± 5 AD-1658842.1 2
 ± 1 3 ± 1 8 ± 2 AD-1633646.1 42 ± 6 48 ± 8 55 ± 8 AD-1658843.1 19 ± 5 28 ± 3 48 ± 4 AD-
1633647.1 87 ± 9 78 ± 10 89 ± 9 AD-1658844.1 66 ± 12 81 ± 11 93 ± 8 AD-1633648.1 8 ± 3
13 ± 3 32 ± 4 AD-1658845.1 2 ± 0 5 ± 0 14 ± 1 AD-1633649.1 27 ± 3 48 ± 8 87 ± 3 AD-
1658846.1 2 ± 1 7 ± 3 19 ± 5 AD-1633650.1 31 ± 8 54 ± 7 82 ± 8 AD-1658847.1 17 ± 8 30 ± 5
53 ± 4 AD-1633651.1 44 ± 6 63 ± 4 75 ± 8 AD-1658848.1 19 ± 3 32 ± 5 53 ± 4 AD-1633652.1
10 ± 1 18 ± 5 28 ± 3 AD-1658849.1 17 ± 3 25 ± 4 46 ± 6 AD-84707.2 2 ± 1 5 ± 2 12 ± 2 AD-
1321390.2 2 ± 0 3 ± 1 8 ± 2 AD-1633653.1 6 ± 1 15 ± 3 28 ± 7 AD-1658850.1 5 ± 1 10 ± 3 16 $\pm$
4 AD-1633678.1 113 ± 6 99 ± 13 96 ± 9 AD-1658875.1 67 ± 7 103 ± 12 95 ± 2 AD-1633683.1
114 ± 14 118 ± 4 110 ± 8 AD-1658880.1 88 ± 8 96 ± 11 98 ± 10 AD-1633732.1 1 ± 0 3 $\pm$
1 5 ± 2 AD-1658929.1 2 ± 1 5 ± 1 6 ± 2 AD-1633733.1 6 ± 2 10 ± 2 20 ± 2 AD-1658930.1 $4 \pm$
1 8 ± 3 15 ± 2 AD-1633734.1 20 ± 3 32 ± 4 51 ± 6 AD-1658931.1 2 ± 0 3 ± 0 5 ± 2 AD-
1633735.1 12 ± 4 21 ± 1 38 ± 6 AD-1658932.1 3 ± 1 9 ± 1 21 ± 5 AD-1633736.1 5 ± 1 8 ± 1 20
 ± 2 AD-1658933.1 1 ± 0 2 ± 1 4 ± 1 AD-1633737.1 27 ± 3 41 ± 2 63 ± 5 AD-1658934.1 3 ± 1
8 ± 2 19 ± 3 AD-1633738.1 36 ± 3 47 ± 2 66 ± 3 AD-1658935.1 9 ± 2 15 ± 2 31 ± 5 AD-
1633739.1 2 ± 1 4 ± 1 8 ± 2 AD-1658936.1 1 ± 0 2 ± 1 4 ± 1 AD-1633740.1 1 ± 0 2 ± 0 4 $\pm$

1 AD-1658937.1 2 ± 0 2 ± 0 3 ± 1 AD-1633741.1 5 ± 1 10 ± 2 22 ± 2 AD-1658938.1 2 ± 0 4 ± 0 6 ± 1 AD-1633742.1 12 ± 1 22 ± 2 37 ± 8 AD-1658939.1 5 ± 1 8 ± 1 16 ± 1 AD-1633743.1 5 ± 2 6 ± 2 14 ± 3 AD-1658940.1 4 ± 1 6 ± 1 8 ± 2 AD-1633759.1 1 ± 0 2 ± 1 4 ± 1 AD-1658954.1 1 ± 0 2 ± 1 3 ± 2 AD-1684522.1 4 ± 2 7 ± 2 13 ± 3 AD-1658955.1 3 ± 1 6 ± 1 9 ± 2 AD-1684523.1 3 ± 1 6 ± 1 7 ± 2 AD-1658956.1 3 ± 1 5 ± 1 7 ± 1 AD-1684524.1 2 ± 1 3 ± 1 8 ± 2 AD-1658957.1 2 ± 0 3 ± 1 5 ± 1 AD-1684525.1 15 ± 5 18 ± 4 34 ± 5 AD-1658958.1 2 ± 1 4 ± 2 5 ± 3 AD-1684526.1 2 ± 1 3 ± 1 7 ± 1 AD-1658959.1 1 ± 0 1 ± 0 2 ± 1 AD-1684527.1 1 ± 1 3 ± 1 4 ± 1 AD-1658960.1 1 ± 1 2 ± 1 3 ± 1 AD-1633777.1 14 ± 3 18 ± 3 38 ± 5 AD-1658992.1 2 ± 1 5 ± 1 16 ± 2 AD-1633779.1 11 ± 4 27 ± 7 41 ± 7 AD-1658994.1 3 ± 1 6 ± 1 19 ± 4 AD-1633780.1 30 ± 5 52 ± 8 61 ± 11 AD-1658995.1 51 ± 6 52 ± 7 55 ± 9 AD-1633840.1 66 ± 13 91 ± 7 99 ± 13 AD-1659055.1 7 ± 1 19 ± 3 25 ± 3 AD-1633841.1 96 ± 13 99 ± 25 88 ± 9 AD-1659056.1 93 ± 10 103 ± 9 92 ± 11 AD-1633842.1 4 ± 2 22 ± 6 49 ± 10 AD-1659057.1 2 ± 1 13 ± 3 31 ± 5 AD-1633843.1 6 ± 1 36 ± 2 55 ± 13 AD-1659058.1 3 ± 1 12 ± 2 37 ± 7 AD-1633844.1 22 ± 3 47 ± 10 81 ± 9 AD-1659059.1 5 ± 1 27 ± 6 54 ± 8 AD-1633845.1 28 ± 4 83 ± 9 71 ± 7 AD-1659060.1 14 ± 2 41 ± 5 65 ± 13 AD-1633846.1 10 ± 2 32 ± 6 58 ± 14 AD-1659061.1 1 ± 0 12 ± 3 22 ± 8 AD-84712.2 2 ± 0 8 ± 1 17 ± 4 AD-1659062.1 1 ± 0 6 ± 2 12 ± 3 AD-1633847.1 14 ± 3 69 ± 10 86 ± 5 AD-1659063.1 5 ± 1 28 ± 8 54 ± 4 AD-1633848.1 12 ± 4 40 ± 2 80 ± 20 AD-1659064.1 3 ± 1 22 ± 6 31 ± 1 AD-1633849.1 8 ± 1 34 ± 3 64 ± 6 AD-1659065.1 6 ± 2 21 ± 5 50 ± 12 AD-1633850.1 2 ± 0 12 ± 3 23 ± 1 AD-1659066.1 2 ± 1 9 ± 1 27 ± 2 AD-1633851.1 6 ± 2 37 ± 5 52 ± 7 AD-1659067.1 3 ± 0 24 ± 5 46 ± 8 AD-1633852.1 8 ± 2 39 ± 5 62 ± 8 AD-1659068.1 7 ± 1 20 ± 6 39 ± 4 AD-1633853.1 1 ± 0 7 ± 2 17 ± 4 AD-1659069.1 1 ± 0 3 ± 1 14 ± 3 AD-1633854.1 22 ± 3 59 ± 9 85 ± 9 AD-1659070.1 7 ± 2 27 ± 4 52 ± 12 AD-1633855.1 21 ± 3 82 ± 14 91 ± 12 AD-1659071.1 2 ± 0 13 ± 3 35 ± 5 AD-1684528.1 7 ± 2 30 ± 5 61 ± 8 AD-1684529.1 3 ± 0 12 ± 3 34 ± 7 AD-1684530.1 3 ± 1 18 ± 1 45 ± 4 AD-1684531.1 2 ± 0 9 ± 1 33 ± 5 AD-1684532.1 7 ± 1 28 ± 3 44 ± 5 AD-1684533.1 3 ± 1 19 ± 8 47 ± 11 AD-1684534.1 3 ± 1 15 ± 1 30 ± 3 AD-1684535.1 4 ± 3 4 ± 0 12 ± 2 AD-1633946.1 3 ± 1 10 ± 3 22 ± 3 AD-1659162.1 1 ± 0 6 ± 2 19 ± 4 AD-67328.2 1 ± 0 7 ± 1 20 ± 3 AD-1321384.2 3 ± 0 9 ± 1 22 ± 4 AD-1633947.1 3 ± 1 10 ± 2 23 ± 1 AD-1659163.1 2 ± 1 10 ± 1 22 ± 3 AD-1633948.1 2 ± 1 17 ± 9 32 ± 5 AD-1659164.1 1 ± 0 4 ± 1 11 ± 3 AD-1633949.1 3 ± 1 21 ± 1 51 ± 8 AD-1659165.1 3 ± 0 12 ± 2 23 ± 2 AD-84700.2 11 ± 2 37 ± 3 66 ± 4 AD-1659166.1 2 ± 0 12 ± 1 25 ± 4 AD-1633950.1 1 ± 0 9 ± 2 29 ± 5 AD-1659167.1 1 ± 0 6 ± 2 18 ± 2 AD-1633951.1 5 ± 1 29 ± 6 60 ± 10 AD-1659168.1 4 ± 1 15 ± 1 33 ± 7 AD-1633991.1 3 ± 1 19 ± 3 46 ± 4 AD-1659208.1 2 ± 0 11 ± 2 26 ± 4 AD-1633992.1 14 ± 1 45 ± 5 77 ± 9 AD-1659209.1 3 ± 1 8 ± 2 23 ± 5 AD-1633993.1 27 ± 5 57 ± 10 73 ± 3 AD-1659210.1 11 ± 1 43 ± 5 70 ± 8 AD-1634065.1 13 ± 3 48 ± 6 79 ± 1 AD-1659282.1 4 ± 1 18 ± 4 44 ± 6 AD-1634066.1 45 ± 5 77 ± 9 98 ± 12 AD-1659283.1 30 ± 2 69 ± 14 80 ± 11 AD-1634067.1 25 ± 5 68 ± 6 87 ± 11 AD-1659284.1 24 ± 4 60 ± 2 71 ± 3 AD-1634068.1 10 ± 2 38 ± 4 58 ± 3 AD-1659285.1 11 ± 1 38 ± 2 64 ± 5 AD-84730.2 7 ± 1 19 ± 4 30 ± 8 AD-1659286.1 5 ± 1 21 ± 7 41 ± 6 AD-1634069.1 5 ± 1 19 ± 4 34 ± 6 AD-1659287.1 2 ± 1 12 ± 2 22 ± 5 AD-1634070.1 2 ± 1 12 ± 4 24 ± 4 AD-1659288.1 3 ± 1 13 ± 3 22 ± 2 AD-1634071.1 29 ± 3 65 ± 6 98 ± 16 AD-1659289.1 24 ± 6 55 ± 8 79 ± 12 AD-1684536.1 54 ± 6 83 ± 5 84 ± 5 AD-1684537.1 75 ± 4 105 ± 5 109 ± 19 AD-1684538.1 64 ± 6 103 ± 6 102 ± 20 AD-1684539.1 61 ± 5 105 ± 8 83 ± 8 AD-1684540.1 73 ± 5 65 ± 4 87 ± 9 AD-1684541.1 57 ± 9 67 ± 10 96 ± 15 AD-1684542.1 89 ± 22 65 ± 5 97 ± 9 AD-1684543.1 85 ± 10 78 ± 1 104 ± 19 AD-1684544.1 98 ± 15 87 ± 8 117 ± 19 AD-1684545.1 56 ± 9 56 ± 6 94 ± 6 AD-1684546.1 79 ± 9 84 ± 9 94 ± 12 AD-1684547.1 23 ± 3 45 ± 3 64 ± 8 AD-1684548.1 58 ± 9 62 ± 12 86 ± 18 AD-1684549.1 16 ± 1 44 ± 8 57 ± 8 AD-1684550.1 47 ± 7 63 ± 15 100 ± 17 AD-1684551.1 37 ± 3 50 ± 7 80 ± 3 AD-1684552.1 55 ± 5 74 ± 14 112 ± 20 AD-1684553.1 31 ± 5 54 ± 7 76 ± 9 AD-1684554.1 75 ± 8 98 ± 17 107 ± 18 AD-1684555.1 25 ± 2 50 ± 9 53 ± 5 AD-1684556.1 76 ± 10 82 ± 12 115 ± 26 AD-1684557.1 26 ± 7 45 ± 8 75 ± 20 AD-1684558.1 57 ± 9 65 ± 7 101 ± 14 AD-1684559.1 21 ± 6 41 ± 5 69 ± 6 AD-1634072.1 8 ± 1 19 ± 3 32 ± 8 AD-1684560.1 6 ± 0

15 ± 3 25 ± 4 AD-1634073.1 5 ± 1 13 ± 3 18 ± 3 AD-1684561.1 3 ± 1 10 ± 1 11 ± 3 AD-1634074.1
3 ± 0 8 ± 1 15 ± 3 AD-1659290.1 3 ± 1 9 ± 1 15 ± 1 AD-1634075.1 5 ± 1 15 ± 2 32 ± 9 AD-
1659291.1 5 ± 1 15 ± 2 19 ± 3 AD-1634076.1 11 ± 2 36 ± 5 45 ± 4 AD-1659292.1 9 ± 1 25 ± 5 37
± 9 AD-1634077.1 12 ± 2 30 ± 5 59 ± 8 AD-1659293.1 3 ± 2 10 ± 2 16 ± 1 AD-1634078.1 11 ±
3 29 ± 7 55 ± 8 AD-1659294.1 5 ± 1 13 ± 3 24 ± 4 AD-1634079.1 3 ± 1 10 ± 3 21 ± 4 AD-
1659295.1 2 ± 0 4 ± 0 11 ± 1 AD-1634080.1 10 ± 3 34 ± 2 48 ± 7 AD-1659296.1 6 ± 2 15 ± 2 35
± 7 AD-1634081.1 5 ± 1 20 ± 2 38 ± 8 AD-1659297.1 14 ± 2 26 ± 2 52 ± 7 AD-1634082.1 27 ±
3 60 ± 4 79 ± 11 AD-1659298.1 17 ± 3 34 ± 6 60 ± 10 AD-1634105.1 2 ± 0 4 ± 1 12 ± 2 AD-
1659321.1 2 ± 1 5 ± 0 15 ± 2 AD-1634107.1 6 ± 2 21 ± 2 37 ± 9 AD-1659323.1 6 ± 1 13 ± 3 25 ±
2 AD-1634109.1 17 ± 2 40 ± 5 66 ± 8 AD-1659325.1 7 ± 2 17 ± 3 39 ± 6 AD-1634110.1 9 ± 2 26
± 5 42 ± 2 AD-1659326.1 6 ± 1 19 ± 1 41 ± 6 AD-1634111.1 5 ± 2 11 ± 3 25 ± 3 AD-1659327.1 3
± 1 14 ± 1 18 ± 2 AD-1634112.1 10 ± 1 32 ± 2 50 ± 9 AD-1659328.1 3 ± 1 10 ± 1 24 ± 6 AD-
1634113.1 7 ± 2 17 ± 3 37 ± 4 AD-1659329.1 4 ± 1 15 ± 1 29 ± 6 AD-1634114.1 17 ± 5 33 ± 1 68
± 8 AD-1659330.1 2 ± 1 5 ± 1 11 ± 2 AD-1634115.1 35 ± 5 63 ± 11 110 ± 7 AD-1659331.1 7
± 1 24 ± 4 43 ± 7 AD-84724.2 24 ± 2 45 ± 6 61 ± 8 AD-1659332.1 5 ± 1 11 ± 1 24 ± 2 AD-
1634116.1 5 ± 2 15 ± 3 33 ± 5 AD-1659333.1 3 ± 2 9 ± 1 17 ± 3 AD-1634117.1 9 ± 1 32 ± 3 45 ±
6 AD-1659334.1 10 ± 3 29 ± 1 42 ± 6 AD-1634118.1 25 ± 4 46 ± 5 79 ± 4 AD-1659335.1 6 ± 1
20 ± 3 31 ± 4 AD-68579.2 17 ± 5 41 ± 5 69 ± 6 AD-1659336.1 3 ± 1 10 ± 2 17 ± 3 AD-1634119.1
4 ± 1 14 ± 1 33 ± 4 AD-1659337.1 3 ± 0 10 ± 2 21 ± 2 AD-1634120.1 11 ± 3 33 ± 6 80 ± 5 AD-
1659338.1 3 ± 1 8 ± 3 20 ± 1 AD-1634121.1 8 ± 2 22 ± 6 44 ± 4 AD-1659339.1 2 ± 1 6 ± 1 18 ±
3 AD-1634122.1 18 ± 13 29 ± 2 60 ± 7 AD-1659340.1 2 ± 0 15 ± 3 24 ± 4 AD-1634123.1 4 ± 1 14
± 3 41 ± 1 AD-1659341.1 6 ± 1 23 ± 3 46 ± 8 AD-1634124.1 5 ± 1 17 ± 6 32 ± 5 AD-1659342.1 3
± 0 12 ± 2 19 ± 3 AD-1634125.1 4 ± 1 11 ± 2 22 ± 2 AD-1659343.1 4 ± 1 10 ± 3 26 ± 5 AD-
1634126.1 6 ± 1 16 ± 4 30 ± 4 AD-1659344.1 2 ± 0 8 ± 1 16 ± 3 AD-1634127.1 3 ± 0 7 ± 2 16 ±
1 AD-1659345.1 3 ± 0 6 ± 1 12 ± 2 AD-1634128.1 1 ± 0 1 ± 1 3 ± 2 AD-1659346.1 1 ± 0 2 ±
1 5 ± 1 AD-1634129.1 2 ± 1 6 ± 3 15 ± 2 AD-1659347.1 1 ± 0 1 ± 0 5 ± 2 AD-1634130.1 4 ±
1 13 ± 1 24 ± 4 AD-1659348.1 1 ± 1 1 ± 0 3 ± 1 AD-1684562.1 16 ± 4 33 ± 3 54 ± 7 AD-
1659349.1 9 ± 2 23 ± 2 40 ± 1 AD-1684563.1 1 ± 0 1 ± 1 3 ± 0 AD-1659350.1 1 ± 0 1 ± 1 2 ±
1 AD-1684564.1 1 ± 0 2 ± 1 2 ± 1 AD-1659351.1 1 ± 0 1 ± 0 1 ± 0 AD-1634135.1 2 ± 1 7 ±
1 13 ± 4 AD-1659371.1 1 ± 0 2 ± 1 4 ± 1 AD-1634136.1 2 ± 1 7 ± 3 12 ± 1 AD-1659372.1 3 ±
1 9 ± 1 21 ± 4 AD-1634137.1 1 ± 0 2 ± 0 2 ± 1 AD-1659373.1 1 ± 1 2 ± 1 3 ± 1 AD-
1634146.1 1 ± 0 2 ± 0 3 ± 1 AD-1659382.1 1 ± 0 1 ± 0 1 ± 0 AD-1634147.1 1 ± 0 4 ± 1 10 ±
3 AD-1659383.1 1 ± 1 2 ± 0 2 ± 0 AD-68585.2 5 ± 2 20 ± 2 29 ± 4 AD-1659384.1 2 ± 1 7 ± 1
11 ± 5 AD-1634148.1 2 ± 0 5 ± 1 9 ± 2 AD-1659385.1 1 ± 0 1 ± 0 1 ± 0 AD-1634149.1 1 ± 0
1 ± 0 3 ± 1 AD-1659386.1 1 ± 0 1 ± 0 3 ± 0 AD-1634150.1 1 ± 0 1 ± 0 4 ± 1 AD-1659387.1
1 ± 0 1 ± 0 2 ± 1 AD-1634151.1 0 ± 0 1 ± 0 2 ± 0 AD-1659388.1 0 ± 0 1 ± 0 1 ± 0 AD-
1634152.1 1 ± 0 1 ± 0 2 ± 1 AD-1659389.1 1 ± 0 1 ± 0 2 ± 0 AD-1634153.1 2 ± 1 2 ± 1 5 ±
1 AD-1659390.1 1 ± 0 2 ± 0 5 ± 2 AD-1634162.1 3 ± 0 5 ± 2 12 ± 2 AD-1659399.1 3 ± 1 5 ±
1 8 ± 1 AD-1634163.1 4 ± 0 11 ± 2 18 ± 3 AD-1659400.1 9 ± 2 18 ± 5 30 ± 3 AD-1634164.1 29 ±
6 35 ± 3 48 ± 7 AD-1659401.1 26 ± 1 23 ± 3 33 ± 4 AD-1634165.1 19 ± 3 19 ± 2 30 ± 5 AD-
1659402.1 20 ± 3 20 ± 5 24 ± 3 AD-1634169.1 65 ± 4 70 ± 7 79 ± 5 AD-1659406.1 63 ± 18 59
± 11 53 ± 6 AD-1634170.1 84 ± 13 74 ± 9 72 ± 1 AD-1659407.1 58 ± 5 69 ± 8 68 ± 6 AD-
1634171.1 81 ± 15 75 ± 8 85 ± 7 AD-1659408.1 86 ± 25 91 ± 9 81 ± 5 AD-1634172.1 50 ± 13 53 ±
7 67 ± 8 AD-1659409.1 58 ± 4 68 ± 6 65 ± 7 AD-1634173.1 60 ± 5 61 ± 5 56 ± 7 AD-1659410.1
49 ± 6 49 ± 2 52 ± 7 AD-1634174.1 69 ± 2 73 ± 5 77 ± 9 AD-1659411.1 36 ± 3 37 ± 4 39 ± 2
AD-1634175.1 61 ± 9 70 ± 9 81 ± 8 AD-1659412.1 36 ± 4 41 ± 7 42 ± 5 AD-1634176.1 70 ± 5
82 ± 5 88 ± 7 AD-1659413.1 64 ± 9 68 ± 4 74 ± 6 AD-1634177.1 57 ± 12 65 ± 9 67 ± 4 AD-
1659414.1 52 ± 10 67 ± 3 67 ± 3 AD-1634178.1 63 ± 8 76 ± 6 74 ± 4 AD-1659415.1 58 ± 5 76 ±
4 67 ± 9 AD-1634179.1 79 ± 11 83 ± 18 103 ± 9 AD-1659416.1 65 ± 6 86 ± 7 92 ± 7 AD-
1634180.1 47 ± 7 67 ± 10 72 ± 14 AD-1659417.1 58 ± 11 74 ± 7 73 ± 15 AD-1634181.1 71 ±

11 79 ± 6 72 ± 16 AD-1659418.1 54 ± 9 69 ± 7 56 ± 13 AD-1634182.1 62 ± 5 77 ± 3 86 ± 4
 AD-1659419.1 57 ± 17 68 ± 7 76 ± 7 AD-1634183.1 61 ± 11 70 ± 11 81 ± 11 AD-1659420.1 63
 ± 8 79 ± 11 86 ± 5 AD-1634184.1 64 ± 12 74 ± 9 80 ± 16 AD-1659421.1 33 ± 3 56 ± 8 62 ±
 10 AD-1634185.1 58 ± 9 71 ± 9 76 ± 12 AD-1659422.1 58 ± 7 72 ± 18 61 ± 6 AD-1634186.1
 60 ± 9 86 ± 12 86 ± 10 AD-1659423.1 62 ± 11 81 ± 15 82 ± 9 AD-1634187.1 43 ± 10 51 ± 9
 42 ± 8 AD-1659424.1 81 ± 9 92 ± 11 82 ± 27 AD-1634188.1 37 ± 5 53 ± 8 48 ± 10 AD-
 1659425.1 42 ± 8 52 ± 14 73 ± 4 AD-1634189.1 49 ± 5 70 ± 17 87 ± 8 AD-1659426.1 47 ± 9
 51 ± 9 67 ± 5 AD-1634190.1 50 ± 9 56 ± 10 72 ± 5 AD-1659427.1 46 ± 6 59 ± 16 76 ± 9 AD-
 1634191.1 44 ± 3 56 ± 7 65 ± 7 AD-1659428.1 56 ± 6 50 ± 8 80 ± 15 AD-1634192.1 45 ± 10
 49 ± 5 61 ± 5 AD-1659429.1 34 ± 5 42 ± 7 50 ± 5 AD-1634193.1 47 ± 10 42 ± 8 62 ± 10 AD-
 1659430.1 63 ± 6 47 ± 7 69 ± 11 AD-1634194.1 49 ± 2 53 ± 2 62 ± 7 AD-1659431.1 49 ± 8
 50 ± 7 63 ± 11 AD-1634195.1 52 ± 6 59 ± 11 80 ± 7 AD-1659432.1 59 ± 5 73 ± 7 82 ± 13
 AD-1634196.1 57 ± 7 49 ± 3 71 ± 11 AD-1659433.1 50 ± 12 43 ± 7 68 ± 11 AD-1634197.1 52
 ± 3 62 ± 4 70 ± 11 AD-1659434.1 52 ± 5 63 ± 5 72 ± 8 AD-1634199.1 61 ± 8 51 ± 5 60 ± 13
 AD-1659436.1 65 ± 5 53 ± 3 78 ± 8 AD-1634200.1 62 ± 8 50 ± 5 73 ± 15 AD-1659437.1 57 ±
 4 61 ± 3 55 ± 5 AD-1634203.1 77 ± 3 63 ± 7 86 ± 6 AD-1659440.1 65 ± 4 54 ± 6 64 ± 13 AD-
 1634209.1 91 ± 15 74 ± 8 112 ± 17 AD-1659446.1 78 ± 8 69 ± 3 88 ± 12 AD-1634210.1 73 ± 6
 65 ± 6 85 ± 9 AD-1659447.1 72 ± 5 68 ± 6 93 ± 10 AD-1634211.1 74 ± 13 69 ± 8 85 ± 8 AD-
 1659448.1 66 ± 12 50 ± 9 78 ± 9 AD-1634212.1 43 ± 3 45 ± 3 68 ± 8 AD-1659449.1 68 ± 3 61 ±
 5 91 ± 9 AD-1634213.1 71 ± 2 77 ± 17 98 ± 21 AD-1659450.1 73 ± 2 61 ± 6 103 ± 10 AD-
 1634214.1 85 ± 7 59 ± 5 85 ± 15 AD-1659451.1 52 ± 9 49 ± 9 56 ± 7 AD-1634215.1 82 ± 5
 79 ± 9 108 ± 9 AD-1659452.1 67 ± 8 59 ± 6 83 ± 6 AD-1634216.1 76 ± 9 70 ± 6 113 ± 7 AD-
 1659453.1 77 ± 10 64 ± 3 90 ± 12 AD-1634217.1 92 ± 6 83 ± 1 116 ± 15 AD-1659454.1 75 ± 4
 70 ± 6 101 ± 6 AD-1684565.1 85 ± 7 104 ± 15 84 ± 14 AD-1659481.1 49 ± 5 49 ± 5 61 ± 5
 AD-1684566.1 122 ± 27 74 ± 9 98 ± 5 AD-1659482.1 67 ± 6 51 ± 4 108 ± 15 AD-1684567.1 85
 ± 10 74 ± 7 98 ± 7 AD-1659483.1 89 ± 2 77 ± 9 102 ± 5 AD-1684568.1 71 ± 6 82 ± 12 107 ±
 14 AD-1659485.1 67 ± 4 60 ± 6 91 ± 11 AD-1684569.1 68 ± 2 80 ± 4 100 ± 5 AD-1659487.1
 68 ± 7 88 ± 15 91 ± 13 AD-1684570.1 78 ± 7 83 ± 6 123 ± 12 AD-1659488.1 79 ± 7 87 ±
 13 111 ± 9 AD-1634234.1 45 ± 8 73 ± 12 91 ± 5 AD-1659489.1 81 ± 5 75 ± 10 97 ± 7 AD-
 1634235.1 65 ± 9 72 ± 6 91 ± 4 AD-1659490.1 47 ± 2 68 ± 4 89 ± 15 AD-1634236.1 70 ± 8
 75 ± 9 96 ± 15 AD-1659491.1 32 ± 6 39 ± 3 48 ± 5 AD-1634237.1 75 ± 7 86 ± 14 113 ± 11
 AD-1659492.1 36 ± 7 40 ± 4 48 ± 4 AD-1634238.1 79 ± 13 91 ± 11 96 ± 12 AD-1659493.1 74
 ± 3 78 ± 4 104 ± 5 AD-1634282.1 64 ± 7 58 ± 8 72 ± 6 AD-1659537.1 63 ± 9 51 ± 8 61 ± 6
 AD-1634283.1 76 ± 8 66 ± 3 97 ± 18 AD-1659538.1 59 ± 2 55 ± 5 71 ± 4 AD-1634304.1 69 ±
 7 68 ± 8 102 ± 16 AD-1659559.1 70 ± 7 61 ± 11 85 ± 14 AD-1634305.1 51 ± 3 44 ± 4 50 ±
 12 AD-1659560.1 49 ± 3 59 ± 3 55 ± 4 AD-1634306.1 61 ± 5 65 ± 5 74 ± 4 AD-1659561.1 51 ±
 5 39 ± 5 73 ± 5 AD-1634307.1 83 ± 7 80 ± 7 103 ± 2 AD-1659562.1 74 ± 5 66 ± 8 93 ± 7 AD-
 1634308.1 61 ± 4 73 ± 7 82 ± 7 AD-1659563.1 69 ± 3 72 ± 4 98 ± 8 AD-1634327.1 61 ± 8 64 ±
 7 80 ± 15 AD-1659582.1 49 ± 9 52 ± 2 74 ± 10 AD-1634328.1 64 ± 6 64 ± 9 91 ± 4 AD-
 1659583.1 44 ± 3 56 ± 4 70 ± 5 AD-1634329.1 65 ± 3 64 ± 7 85 ± 5 AD-1659584.1 60 ± 5 61 ±
 9 88 ± 11 AD-1634330.1 56 ± 2 50 ± 6 54 ± 11 AD-1659585.1 72 ± 2 65 ± 7 81 ± 8 AD-
 1634331.1 44 ± 10 31 ± 6 49 ± 9 AD-1659586.1 53 ± 7 51 ± 9 72 ± 7 AD-1634332.1 46 ± 10 49
 ± 13 81 ± 3 AD-1659587.1 34 ± 6 57 ± 1 52 ± 17 AD-1634333.1 67 ± 10 59 ± 15 84 ± 13 AD-
 1659588.1 64 ± 3 62 ± 11 97 ± 13

In Vivo Mouse and Cyno Studies

[0441] All studies were conducted using protocols consistent with local, state and federal regulations as applicable and approved by the Institutional Animal Care and Use Committees (IACUCs) at Alnylam Pharmaceuticals.

[0442] In mouse pharmacodynamic studies, female C57BL/6 mice (Charles River Laboratories) were administered a single dose of a vehicle control (1× PBS or 0.9% sodium chloride) or siRNA

subcutaneously in the upper back. Bleeds were collected by retro-orbital bleeding. Serum were separated by centrifuging at 13000 rpm at room temperature for 10 mins. Mouse livers were collected and immediately snap frozen in liquid nitrogen, and stored at -80°C . for mRNA and siRNA analysis.

[0443] As shown in FIGS. 2A-2D and 10A-10D, the results demonstrate an improved or similar target knockdown in mice and an improved or similar target knockdown and duration of silencing in cyno.

Serum Protein Quantification

[0444] TTR protein was quantified by ELISA from serum isolated from whole blood. ELISA was performed according to manufacturer protocol (ALPCO, 41-PALMS-EO1) after a 3025-fold dilution of the serum samples. Data were normalized to pre-bleed TTR levels. All samples were assayed in duplicate and each data point is the average of all the mice within each cohort (n=3).

[0445] In some embodiments, the dsRNA molecule is not a dsRNA molecule listed in any one of Tables 15-25.

TABLE-US-00015 TABLE 15 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1353484.1
asasaag(Ahd)gaAfAfGfuguuuuuasasa 2141 VPusdTsaudAadAacacdTuUfcucuuuuscsu 2166 AD-
1353468.1 asasaga(Chd)ugAfUfAfcagaacgasusa 2142 VPusdAsucdGudTcugudAuCfagucuuuscsc
2167 AD-1353485.1 asasagagaaAfGfUfguuu(Uhd)auasusa 2143
VPusdAsuadTadAaacadCuUfucucuuususc 2168 AD-1353483.1
asascag(Uhd)gcUfAfAfuguuuauugsgsa 2144 VPusdCscadAudAacaudTaGfcacuguusgsg 2169 AD-
1353486.1 asagag(Ahd)aaGfUfGfuuuuauausasa 2145 VPusdTsaudAudAaaacdAcUfuucucuuusu
2170 AD-1353479.1 asasuuggauUfCfGfccau(Uhd)uuasusa 2146
VPusdAsuadAadAuggcdGaAfuccaauuscsc 2171 AD-1353473.1
ascsga(Ahd)caGfUfCfcuuauucscsa 2147 VPusdTsggdAudTaaggdAcUfguucuguscsg 2172 AD-
1353467.1 ascscaggaaAfGfAfcuga(Uhd)acasgsa 2148 VPusdCsugdTadTcagudCuUfuccuggusgsc
2173 AD-1353482.1 asgsauu(Ahd)gaGfAfGfuuuuauuuscscsa 2149
VPusdGsaadAudAaaacdTcUfcuaucucusc 2174 AD-1353480.1
asusugg(Ahd)uuCfGfCfcauuuuauusasa 2150 VPusdAsaudAadAauggdCgAfauccaauusc 2175 AD-
1353463.1 csasuca(Chd)caUfGfCfagauuauugscsa 2151 VPusdGscadTadAucugdCaUfggugaugsusu
2176 AD-1353478.1 cscsucu(Uhd)ggAfAfUfuggauucgscsa 2152
VPusdGscgdAadTccaadTuCfcaagaggsgsc 2177 AD-1353466.1
csusacagcaCfAfAfcaaaa(Uhd)gugsasa 2153 VPusdTscadCadTuugudTgUfgcuguagsgsg 2178 AD-
1353488.1 gsasaag(Uhd)guUfUfUfauauacggsusa 2154 VPusdAsccdGudAuauadAaAfcacuuucusc
2179 AD-1353487.1 gsasgaa(Ahd)guGfUfUfuuauauacgsa 2155
VPusdCsgudAudAuaaadAcAfcuuucucusc 2180 AD-1353481.1
gsasuucgccAfUfUfuuau(Uhd)uuuscscsa 2156 VPusdGsaadAadAuaaadAuGfgcgaaucscsg 2181 AD-
1353477.1 gsusccu(Uhd)aaUfCfCfagaaaccusgsa 2157 VPusdCsagdGudTucugdGaUfuaaggacsug
2182 AD-1353489.1 gsusguu(Uhd)uaUfAfUfacgguacuscusa 2158
VPusdAsagdTadCcguaadTaUfaaaacacsusu 2183 AD-1353491.1
usasgac(Ahd)uuGfCfUfauucuguususa 2159 VPusdAsaadCadGaauadGcAfaugucuasusu 2184 AD-
1397058 csusacagcaCfAfAfcaaaa(Uhd)gugaaL96 2160 VPusdTscadCadTuugudTgUfgcuguagsgsg
2178 AD-1397059 asasaga(Chd)ugAfUfAfcagaacgauaL96 2161
VPusdAsucdGudTcugudAuCfagucuuuscsc 2167 AD-1397064
asusugg(Ahd)uuCfGfCfcauuuuauuaL96 2162 VPusdAsaudAadAauggdCgAfauccaauusc 2175
AD-1397065 gsasuucgccAfUfUfuuau(Uhd)uuucaL96 2163
VPusdGsaadAadAuaaadAuGfgcgaaucscsg 2181 AD-1397066
gsasgaa(Ahd)guGfUfUfuuauauacgaL96 2164 VPusdCsgudAudAuaaadAcAfcuuucucusc 2180
AD-1397067 gsusguu(Uhd)uaUfAfUfacgguacuuaL96 2165
VPusdAsagdTadCcguaadTaUfaaaacacsusu 2183

TABLE-US-00016	16 Exemplary dsRNA	Molecules	SEQ	SEQ ID	ID Duplex	ID
Sense sequence (5'.fwdarw.3')	NO	Antisense sequence (5'.fwdarw.3')	NO	AD-1397070.1		
ascsgug(Ahd)ccCfAfAfgcucgcaugaL96	2185	VPusdCsaudGcdGagcudTgGfgucacgusgsa	2290	AD-1397073.1		
usgsacc(Chd)aaGfCfUfcgcauggucaL96	2186	VPusdGsacdCadTgcgadGcUfugggucascsg	2291	AD-1397075.1		
ascscga(Ahd)gcUfCfGfcauggucagaL96	2187	VPusdCsugdAcdCaugcdGaGfcuuggguscsa	2292	AD-1397090.1		
usgsgca(Ghd)caAfCfAfaaggauuugaL96	2188	VPusdCsaadAudCcuuudGuUfgcugccascsu	2293	AD-1397091.1		
gsgscag(Chd)aaCfAfAfaaggauuugaaL96	2189	VPusdTscadAadTccuudTgUfugcugccsasc	2294	AD-1397094.1		
asascaa(Ahd)ggAfUfUfugaaacuugaL96	2190	VPusdCsaadGudTucaadAuCfcuuuguusgsc	2295	AD-1397095.1		
ascsaag(Ghd)gaUfUfUfgaaacuuggaL96	2191	VPusdCscadAgdTuucadAaUfccuuugusug	2296	AD-1397096.1		
csasaag(Ghd)auUfUfGfaaacuugguaL96	2192	VPusdAsccdAadGuuucdAaAfuccuuugsus	2297	u AD-1397097.1		
asasagg(Ahd)uuUfGfAfaacuuggugaL96	2193	VPusdCsacdCadAguuudCaAfauccuuusgsu	2298	AD-1397098.1		
asasgga(Uhd)uuGfAfAfacuugguguaL96	2194	VPusdAscadCcdAaguudTcAfaauccuusug	2299	AD-1397102.1		
asgsacg(Ahd)ugUfCfAfaccuuguguaL96	2195	VPusdAscadCadAgguudGaCfaucgucugsc	2300	AD-1397107.1		
usasggg(Chd)uaAfCfCfaguucucuuaL96	2196	VPusdAsagdAgdAacugdGuUfagcccuasasa	2301	AD-1397108.1		
gsgsgcu(Ahd)acCfAfGfuucucuuaL96	2197	VPusdCsaadAgdAgaacdTgGfuuagcccsusa	2302	AD-1397109.1		
gsgscua(Ahd)ccAfGfUfucucuuaL96	2198	VPusdAscadAadGagaadCuGfguuagccscsu	2303	AD-1397110.1		
asascca(Ghd)uuCfUfCfuuuguaaggaL96	2199	VPusdCscudTadCaaagdAgAfacugguusag	2304	AD-1397116.1		
csuscuu(Uhd)guAfAfGfgacuugugcaL96	2200	VPusdGscadCadAguccdTuAfcaaagagsasa	2305	AD-1397118.1		
asusacu(Ghd)agGfGfUfgaaauuaagaL96	2201	VPusdCsuudAadTuucadCcCfucaguausgsg	2306	AD-1397119.1		
ascsuga(Ghd)ggUfGfAfaauuaagggaL96	2202	VPusdCscddTudAauuudCaCfccucagusasu	2307	AD-1397120.1		
csusgag(Ghd)guGfAfAfauuaagggaL96	2203	VPusdTscddCudTaauudTcAfccucagsusa	2308	AD-1397126.1		
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asgscug(Ahd)acAfUfAfuaacauagauaL96	2206	VPusdAsucdTadTguaudAuGfuucagcugsc	2311	AD-1397133.1		
gscsuga(Ahd)caUfAfUfacauagaugaL96	2207	VPusdCsaudCudAuguadTaUfguucagcsusg	2312	AD-1397135.1		
gsasaca(Uhd)auAfCfAfuaagauguugaL96	2208	VPusdCsaadCadTcaudGuAfuauguucsasg	2313	AD-1397138.1		
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VPusdCsaadGudTucaadAuCfcuuuguusgsc 2295 AD-1397185.1
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VPusdAscadCadAgguudGaCfaucgucugsc 2300 AD-1397196.1
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1423299.1 gsgsaua(Ahd)uaUfCfAfaacacguccaL96 2289 VPusdGsgadCgdTguuudGaUfauuaucsus
2382 u AD-1397266.2 asusaau(Ahd)ucAfAfAfcacgucccgL96 2261
VPusdCsggdGadCgugudTuGfauuuauuscs 2354 c AD-1397268.2
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AD-1397270.2 usasuca(Ahd)acAfCfGfucccggaL96 2263
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1397277.2 csascgu(Chd)ccGfGfGfaggcggaL96 2265
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AD-1397306.2 asuscca(Uhd)caUfAfAfaccaggaggaL96 2271
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1397073.3 usgsacc(Chd)aaGfCfUfcgcauggucaL96 2186 VPusdGsacdCadTgcgadGcUfugggucascsg
2291 AD-1397252.2 gscsucg(Chd)auGfGfUfcaguuuuagaL96 2256
VPusdCsuudTudAcugadCcAfugcgagcsusu 2349 AD-1397263.2
csasgua(Ahd)aaGfCfAfaagacgggaaL96 2260 VPusdTscddCgdTcuuudGcUfuuuacugsasc 2353 AD-
1397309.2 csasuca(Uhd)aaAfCfCfaggagguggaL96 2273 VPusdCscadCcdTccugdGuUfuaugaugsgsa
2366
TABLE-US-00017 TABLE 17 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1251268.1
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2471 augscsc AD-1251275.1 csasuga(Uhd)cuUfCfUfuugucguagaL96 2383
VPudCuadCgdAcaaadGadAgdAuc 2472 gsusg AD-1251275.2
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AD-1251282.1 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384 VPusdCsagdTadAaagudGudAcdTcga
2473 casusu AD-1251357.1 gsasgaa(Uhd)UfcaCfUfuuuucuuuL96 2385
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1331348 asusgau(Chd)UfuCfUfUfugucguagsusa 2389 VPudAcudAcdGacaadAgdAadGauca 2475
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1251358.1 cscsugaagcAfUfAfaaug(Uhd)uuuL96 2393 VPusdGsaadAadCauuudAudGcUfuc
2482 aggsusu AD-1251378.1 usasaaugUfuUfuCfgaau(Uhd)ucacaL96 2394
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uscsuuugUfcgUfAfguga(Uhd)uuuL96 2395 VPudGaadAadTcacudAcdGaCfaaaga 2484 sgsg AD-
1251476.1 ususua(Chd)aacaUfCfcguuauuacaL96 2396 VPudGuadAudAacggdAudGuUfguaa 2485

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AD-1479603 usasaugUfuUfuCfgaag(Uhd)ucascsa 2398 VPudGugdAadTuucgdAadAaCfauuu
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1479612 uscsuuugUfcgUfAfguga(Uhd)uuuscsa 2399 VPudGaadAadTcacudAcdGaCfaaaga 2484
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2488 uuasusg AD-1481961 uscsuuugUfcgUfAfguga(Uhd)uuuscsa 2399
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1251279.2 csasaca(Chd)aaUfUfUfcuucuuagcaL96 2401 VPudGcadAadGaagadAaUfudGugu 2491
ugsusu AD-1251278.1 ascsacaaUfUfUfcuuc(Uhd)uagcaL96 2402
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1251348.1 usgsuaggagdAaUfucuu(Uhd)uuuacL96 2404 VPudGsaadAadAugaadTuCfuCfcua 2494
cascs AD-1251354.1 asgsgagaauUfcdAcuuu(Uhd)cuucaL96 2405
VPudGaadGadAaagudGaAfuUfcucc 2495 usgsc AD-1251365.1
asgsca(Uhd)aaaUfgUfuuuucgaaauaL96 2406 VPudAuudTcdGaaaadCaUfuUfaugc 2496 ususc AD-
1251419.1 gsusaga(Uhd)CfuUfgCfaauuaccuaL96 2407 VPudAugdGudAauugdCaAfgAfucua
2497 csgsg AD-1251474.1 asusuuuacadAcdAuccg(Uhd)uauuaL96 2408
VPudAaudAadCggadGuUfgUfaaaa 2498 usgsu AD-1251419.2
gsusaga(Uhd)CfuUfgCfaauuaccuaL96 2407 VPudAugdGudAauugdCaAfgAfucua 2497 csgsg
AD-1479622 gsusaga(Uhd)CfuUfgCfaauuaccasusa 2409 VPudAugdGudAauugdCaAfgAfucua
2497 csgsg AD-1479593 asgsca(Uhd)aaaUfgUfuuuucgaaasusa 2410
VPudAuudTcdGaaaadCaUfuUfaugc 2496 ususc AD-1479622
gsusaga(Uhd)CfuUfgCfaauuaccasusa 2409 VPudAugdGudAauugdCaAfgAfucua 2497 csgsg AD-
1481938 gsusaga(Uhd)CfuUfgCfaauuaccasusa 2409 VPudAsugdGudAauugdCaAfgAfucu 2499
acsgsg AD-1481952 asgsca(Uhd)aaaUfgUfuuuucgaaasusa 2410
VPudAsuudTcdGaaaadCaUfuUfaug 2500 cususc AD-1481938
gsusaga(Uhd)CfuUfgCfaauuaccasusa 2409 VPudAsugdGudAauugdCaAfgAfucu 2499 acsgsg
AD-1479586 usgsuaggagdAaUfucuu(Uhd)uuuscsa 2411 VPudGsaadAadAugaadTuCfuCfcua
2494 cascs AD-1251304.1 ascsaaagggAfAfAfacia(Uhd)cuucaL96 2412
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csasaagggAfAfAfcaau(Chd)uuccaL96 2413 VPudGgadAgdAuuguuuUfcCfcuuugs 2502 usg AD-
1251321.1 gsgsaaa(Chd)aaUfCfuuccguuucL96 2414 VPudGaadAcdGgaagauUfgUfuuuuccs 2503
csu AD-1251345.1 usgsuaggAfgAfAfUfucuu(Uhd)uuuacL96 2415
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1251353.1 usasggagaaUfUfCfauu(Uhd)ucuaL96 2417 VPudAsagdAadAagugaaUfuCfuccu
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ususgauaguUfAfCfcuag(Uhd)uugcaL96 2419 VPudGcadAadCuagguaAfcUfauc 2508 gsg AD-
1251257.1 usugau(Ahd)guUfAfCfcuaguugcaL96 2420 VPudGscadAadCuagguaAfcUfauc
2509 asgsg AD-1251264.1 usugauaguUfAfCfcuaa(Uhd)uugcaL96 2421
VPudGscadAadTuagguaAfcUfauc 2510 asgsg AD-1251258.1
ususgauaguUfAfCfcuag(Uhd)uugcaL96 2419 VPudGscadAadCuagguaAfcUfauc 2509 asgsg
AD-1251261.1 gsasuaguUfAfCfcuag(Uhd)uugcaL96 2422 VPudGscadAadCuagguaAfcUfauc
2511 gsg AD-1479593 asgsca(Uhd)aaaUfgUfuuuucgaaasusa 2410

VPudAuudTcdGaadCaUfaugc 2496 ususc AD-1251283.1
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AD-1251384.1 csasuga(Uhd)CfuUfCfUfuugucguagaL96 2423
VPuCfuadCgdAcaaadGadAgdAucau 2513 gsusg AD-1251274.2
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AD-1251269.1 csasuga(Uhd)cuUfCfUfuugucguagaL96 2383 VPusCfsuadCgdAcaaadGadAgdAuca
2514 ugsusg AD-1251274.1 csasuga(Uhd)cuUfCfUfuugucguagaL96 2383
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csasuga(Uhd)cuUfCfUfuugucguagaL96 2383 VPusCfsuadCgdAcaaadGadAgdAuca 2515 ugscsc
AD-1251274.3 csasuga(Uhd)cuUfCfUfuugucguagaL96 2383 VPuCfuadCgdAcaaadGadAgdAucau
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1331354 csasuga(Uhd)cuUfCfUfuugucguasgsa 2424 VPuCfuadCgdAcaaadGadAgdAucau 2513
gsusg AD-1479606 csasuga(Uhd)CfuUfCfUfuugucguasgsa 2425
VPuCfuadCgdAcaaadGadAgdAucau 2513 gsusg AD-1481942
csasuga(Uhd)cuUfCfUfuugucguasgsa 2424 VPusCfsuadCgdAcaaadGadAgdAuca 2514 ugsusg
AD-1481955 csasuga(Uhd)CfuUfCfUfuugucguasgsa 2425 VPusCfsuadCgdAcaaadGadAgdAuca
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AD-1251325.1 asasaacaauCfUfUfccgu(Uhd)ucaaaL96 2427 VPuUfugdAadAcggadAgdAuUfguuu
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1251457.1 gsasua(Uhd)aUfuUfudAcaacauccgaL96 2430 VPuCfggdAudGuugudAadAaUfauau
2521 csgsc AD-1251485.1 csasagugUfuCfCfUfacug(Uhd)caugaL96 2431
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1481945 asasaacaauCfUfUfccgu(Uhd)ucasasa 2432 VPusUfsugdAadAcggadAgdAuUfguu 2523
uuscsc AD-1481948 gsgsagaaUfuCfaCfuuuu(Chd)uucsgsa 2433
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AD-1251286.1 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384
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1251290.1 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384 VPuCfagdTadAaagudGuAfcdTcgaca
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1251284.2 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384 VPusCfsagdTadAaagudGuAfcdTcgac
2525 asusu AD-1331352 usgsucgaguAfCfAfcuuu(Uhd)acusgsa 2437
VPusCfsagdTadAaagudGuAfcdTcgac 2525 asusu AD-1251322.1
gsasaaa(Chd)aaUfCfUfuccauuucaL96 2438 VPuUfgadAadTggaagaUfudGuuuuucs 2531 csc AD-
1479581 gsasaaa(Chd)aaUfCfUfuccauuuucsasa 2439 VPuUfgadAadTggaagaUfudGuuuuucs 2531 csc
AD-1479581 gsasaaa(Chd)aaUfCfUfuccauuuucsasa 2439 VPuUfgadAadTggaagaUfudGuuuuucs

2531 csc AD-1481943 gsasaaa(Chd)aaUfCfUfuccauuucsa 2439
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gsgsagaaUfuCfaCfuuuu(Chd)uucsgsa 2433 VPuCfsgadAgdAaaagdTgdAaUfucucc 2519 susg AD-
1251303.1 csascaaagggAfAfaacaa(Uhd)cuuaL96 2440 VPusAfsagdAudTguuuucCfcUfuugu 2533
gsusu AD-1251306.1 asasagggAfaAfAfCfaauc(Uhd)uccgaL96 2441
VPusCfsggdAadGauuguuUfuCfccuu 2534 usgsu AD-1251307.1
asasaggggaadAaCfaauc(Uhd)uccgaL96 2442 VPuCfsggdAadGauugdTuUfuCfccuuu 2535 sgsu AD-
1251318.1 asgsggaaAfaCfAfAfucuu(Chd)cguuaL96 2443 VPusAfsacdGgdAagauugUfuUfuccc
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1251254.1 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384 VPuCfsgadTadAaaguguAfcUfcgacasc
2539 sc AD-1251250.1 usgsucgaguAfCfAfcuuu(Uhd)acugaL96 2384
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AD-1251359.1 csusgaagCfaUfAfAfaugu(Uhd)uucgaL96 2448
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1251362.1 gsasagcaUfaAfAfUfguuu(Uhd)cgaalL96 2450 VPusUfsucdGadAaacauuUfaUfgcuu
2547 csasg AD-1251369.1 asgscauaaaUfgUfuuu(Uhd)gaaauaL96 2451
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1251446.1 gsusgaaaCfaAfAfCfcuaa(Chd)gugaaL96 2456 VPusUfscadCgdTaagguuUfgUfuucac
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AD-1251470.1 asusauu(Uhd)UfaCfAfAfcauccguuaaL96 2459 VPusUfsaadCgdGauguugUfaAfaa
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VPusAfsacdGgdAagauugUfuUfuccc 2536 ususu AD-1251363.1
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1251377.2 asusaaa(Uhd)guUfUfUfcgaaauucaalL96 2452 VPusUfsgadAudTucgaaaAfcAfuuuau
2550 sgsu AD-1331355 usgsucgaguAfCfAfcuuu(Uhd)acusgsa 2437
VPusCfsagdTadAaaguguAfcUfcgaca 2538 susu AD-1479582
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1479587 gsgsagaaUfuCfAfCfuuuu(Chd)uucsgsa 2462 VPusCfsgadAgdAaaagugAfaUfucucc 2544

susg AD-1479589 csusgaagCfaUfAfAfaugu(Uhd)uucsgsa 2463
 VPusCfsgadAadAcauuuaUfgCfuucag 2545 sgsu AD-1479591
 gsaagcaUfaAfAfUfguuu(Uhd)cgasasa 2464 VPusUfsucdGadAaacaUfaUfgcuu 2547 csasg AD-
 1479596 asgscauaaaUfgUfuuu(Uhd)gaaasusa 2465 VPusAfsuudTcdAaaaadCaUfuUfaug 2548
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 asusaaa(Uhd)guUfUfUfcgaaauucsasa 2466 VPusUfsgadAudTucgaaaAfcAfuuuau 2550 sgsu AD-
 1479604 usasca(Uhd)gAfuCfUfUfcuuugucgsusa 2467 VPusAfscgdAcAaagaagAfuCfaugua 2551
 sgsg AD-1479605 csasuga(Uhd)CfuUfCfUfuugucguasgsa 2425
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 2468 VPuUfucdGadAaacadTuUfaUfgcuuc 2546 sasg AD-1481950
 gsaagcauadAaUfguuu(Uhd)cgasasa 2468 VPusUfsucdGadAaacadTuUfaUfgcu 2561 ucsasg AD-
 1251251.1 uscsgaguAfCfAfcuuu(Uhd)acugaL96 2434 VPusCfsagdTadAaguguAfcUfcgasc 2541
 g AD-1479580 uscsgaguAfCfAfcuuu(Uhd)acusgsa 2469 VPusCfsagdTadAaguguAfcUfcgasc
 2541 g AD-1331349 gsaagcauadAaUfguuu(Uhd)cgasasa 2468
 VPuUfucdGadAaacadTuUfaUfgcuuc 2546 sasg
 TABLE-US-00018 TABLE 18 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1395794.1
 csuscuccaagUfAfUfgaucgucuuL96 2562 asdAsgadCgdAucaudAcUfuggagagsg 2567 AD-
 1395803.1 csasgugaugCfUfCfuccaaguauuL96 2563 asdAsuadCudTggagdAgCfaucacugsusg 2568
 AD-1395805.1 cscsaaguauGfAfUfcgucugcaguL96 2564 asdCsugdCadGacgadTcAfuacuuggsasg
 2569 AD-1395816.1 gsgsguuuguUfudGUfuucauuuL96 2565
 asdGsaadAudGaaacdAaAfcaaacccsusg 2570 AD-1395823.1 gsasguauuuCfUfCfagcauuaauL96
 2566 asdTsgdAadTgcugdAgAfaauacuscsc 2571
 TABLE-US-00019 TABLE 19 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1465901.1
 csasgcuaagGfCfAfguucuacguuL96 2572 asdAscgdTadGaacudGcCfuuagcugsusg 2690 AD-
 1465906.1 ususcuugggCfCfUfacuuaauauuL96 2573 asdAsuadTadAaguadGgCfccaagaasgsu 2691
 AD-1465907.1 usascuuuauAfUfGfcugaagucguL96 2574 asdCsgadCudTcagcdAuAfuaaaguasgs
 2692 AD-1465908.1 ascscuaugaAfUfGfagagaagucguL96 2575
 asdCscgdAcdTucagdCaUfauaaagusasg 2693 AD-1465909.1 asgsuaauuuAfUfCfagaaggugcuL96
 2576 asdGscadCcdTucugdAuAfauuuacusgsu 2694 AD-1465911.1
 asuscagaagGfUfGfcuucuaccuL96 2577 asdGsgudAadGaagcdAcCfuucugausasa 2695 AD-
 1465912.1 uscsagaaggUfGfCfuucuaccuL96 2578 asdAsggdTadAgaagdCaCfcuucugasusa 2696
 AD-1465913.1 csasgaagguGfCfUfucuuaccuuL96 2579 asdAsagdGudAagaadGcAfccuucugsasu
 2697 AD-1465915.1 ascscuaugaAfUfGfagagaagucguL96 2580
 asdCsugdAudAcuccdAuUfcuauaggusgsu 2698 AD-1465916.1 cscsuaugaaUfGfGfagauaucaguL96
 2581 asdAscudGadTacudCaUfucuaaggusgsu 2699 AD-1465917.1
 asusgaauggAfGfUfaucagugaguL96 2582 asdCsucdAcdTgauadCuCfcauucausasg 2700 AD-
 1465918.1 asusgccucaCfAfCfacaucuaauuuL96 2583 asdAsaudAgdAugugdTgUfgaggcausgsu 2701
 AD-1465919.1 usgscucacAfCfAfcaucuaauuL96 2584 asdTsaadTadGaugudGuGfugaggcasusg
 2702 AD-1465920.1 gscscucacaCfAfCfcaucuaauuL96 2585
 asdGsuadAudAgaugdTgUfgugaggcsasu 2703 AD-1465921.1 cscsucacacAfCfAfucuaauuL96
 2586 asdAsgudAadTagaudGuGfugagggsca 2704 AD-1465922.1
 csuscacacaCfAfUfcuauuacucuL96 2587 asdGsagdTadAuagadTgUfgugaggsgsc 2705 AD-
 1465927.1 gsascguuugAfCfAfagcaaaucguL96 2588 asdCsgadTudTgcuudGuCfaaacgucsusu 2706
 AD-1465928.1 gscscagucaUfCfAfuccuaaanguL96 2589 asdCsaudTadGggaudGaUfgacuggcsusc
 2707 AD-1465932.1 asasuguacaCfAfGfucaauuggauuL96 2590
 asdAsucdCadTugacdTgUfguacauusasg 2708 AD-1465935.1 gscscagauaUfAfAfcaguuuguguL96
 2591 asdCsacdAadAcugudTaUfaucuggcsasu 2709 AD-1465936.1

cscsagaauAfAfCfaguugugcuL96 2592 asdGscadCadAucugdTuaAfuaucuggscsa 2710 AD-1465937.1 gsgascagaaCfCfAfucauaagguuL96 2593 asdAsccdTudAugaudGgUfucugcucscsa 2711 AD-1465942.1 csasuccacuAfCfCfCgcaaaauauguL96 2594 asdCsaudAudTugcgdGuAfguggaugsusa 2712 AD-1465945.1 gscsugggauGfCfAfggcuuacauuL96 2595 asdAsugdTadAgccudGcAfucccagcsusu 2713 AD-1465946.1 csusgggaugCfAfGfgcuuacauuuL96 2596 asdAsaudGudAagccdTgCfaucccagcsu 2714 AD-1465947.1 usgsggaugcAfGfGfcuuacauuguL96 2597 asdCsaadTgdTaagcdCuGfcaucccagsgsc 2715 AD-1465949.1 gsgsaugcagGfCfUfuacauugacuL96 2598 asdGsucdAadTguaadGcCfugcauccscsa 2716 AD-1465951.1 asusgcaggcUfUfAfcauugacauuL96 2599 asdAsugdTcdAugudAaGfccugcauscsc 2717 AD-1465955.1 asgsgcuuacAfUfUfgacauuaauL96 2600 asdTsuudAadTgucadAuGfuaagccusgsc 2718 AD-1465956.1 gsgscuuacaUfUfGfacauuaaaauL96 2601 asdTsuudTadAugudAaUfguaagccsusg 2719 AD-1465957.1 gscsuuacauUfGfAfcauuaaaaauL96 2602 asdTsuudTudAugudCaAfuguaagcsu 2720 AD-1465958.1 csusuacauuGfAfCfauuaaaaaacuL96 2603 asdGsudTudTaaugdTcAfauguaagcsu 2721 AD-1465959.1 ususacauugAfCfAfuaaaaaacuL96 2604 asdAsgudTudTuaudGuCfaauguaasgsc 2722 AD-1465960.1 usascauugaCfAfUfuaaaaaacuL96 2605 asdCsagdTudTuuaadTgUfcaauguasag 2723 AD-1465961.1 ascsauugacAfUfUfaaaaaacugcuL96 2606 asdGscadGudTuuaadAuGfucaaugusasa 2724 AD-1465968.1 csasccuguaAfUfAfccagcgaaauL96 2607 asdAsuudCgdCuggudAuUfacaggugscsa 2725 AD-1465969.1 usgsuaauacCfAfGfcaauaugguL96 2608 asdCscadTadTucgdcTgGfuaauacagsg 2726 AD-1465970.1 gsusaauaccAfGfCfcauuauggauL96 2609 asdTscadAudAuucgdCuGfguaauacsag 2727 AD-1465972.1 gsusauguaCfAfCfcauuaagcuL96 2610 asdTscgdTadCugugdTgUfacaauacsusu 2728 AD-1465973.1 ususauguacAfCfAfcaguacgaauL96 2611 asdTscudGudAcugudGuGfuacauaascu 2729 AD-1465976.1 gsusacacacAfGfUfacgaagauguL96 2612 asdCsaudCudTcguadCuGfuguguacsasu 2730 AD-1465986.1 asasaccuauAfCfUfuauaagugguL96 2613 asdCscadCudTaaadGuAfuagguuuscsu 2731 AD-1465992.1 asusgaaucaAfAfCfaucaugagcuL96 2614 asdGscudCadTgaugdTuUfgauucausasa 2732 AD-1465997.1 cscsaguggcAfCfUfucuguaguguL96 2615 asdCsacdTadCagaadGuGfccacuggsasc 2733 AD-1465998.1 asgsuggcacUfUfCfuguaguguguL96 2616 asdCsacdAcdTacagdAaGfugccacusgsg 2734 AD-1465999.1 csusgggcacUfCfAfuaucuaauuL96 2617 asdAsuadGadTgaudGaGfugcccagsusg 2735 AD-1466002.1 ususaacuucCfAfUfgaaucuaguL96 2618 asdCsudGadAuucadTgGfaaguuascsa 2736 AD-1466004.1 usgsaugaagAfCfUfcauugagauL96 2619 asdTscudCadTaugadGuCfuucaucasu 2737 AD-1466005.1 asasacucauAfAfUfugaauacagguL96 2620 asdCscudGadTucaadTgAfugaguuuscsu 2738 AD-1466006.1 asasacacagAfUfAfuaauuguuguL96 2621 asdCsaadCadAuuaudAuCfuguguuusgsa 2739 AD-1466007.1 csascagauaUfAfAfuauguuguuuL96 2622 asdAsacdCadAcaudTaUfaucugugsusu 2740 AD-1466009.1 asusucugaaGfAfCfccuauagaguL96 2623 asdCsudTadTaggdTcUfucagaausasu 2741 AD-1466010.1 csgsucuaUfUfCfacuuggugcuL96 2624 asdGscadCcdAagudAaAfguagacgsusa 2742 AD-1466012.1 asasuuacuaGfCfAfcauaaaguuuL96 2625 asdAsacdTudTaugudGcUfaguuuususc 2743 AD-1466013.1 usascuagcaCfAfUfaaaguuggguL96 2626 asdCsccdAadCuuaadTgUfgcuaguasasu 2744 AD-1466014.1 gsgsauggcAfUfUfuggcuucuguL96 2627 asdCsagdAadGccadAuGfccaucucscu 2745 AD-1466019.1 asasguuuccUfAfGfaguagacauL96 2628 asdTsgudCudAacudTaGfgaaacuusug 2746 AD-1466020.1 cscsuagaguUfAfGfacauaaaacuL96 2629 asdGsaudTudAugudTaAfcucuaggsasa 2747 AD-1466021.1 usascaaguaAfGfAfcaggauggauL96 2630 asdTscadAudCcugudCuUfacuuguasgsa 2748 AD-1466024.1 asgsgaccuuUfCfAfcccucuaaguL96 2631 asdCsudAgdAgggudGaAfggucuscsg 2749 AD-1466026.1 usgscuucuaAfAfAfuccaauaauL96 2632 asdTscudAudTggaudTuAfugaagcascu 2750 AD-1466027.1 cscsaugaaAfCfAfucucuucccuL96 2633 asdGsggdAadGagudGuUfcauuggsas 2751

AD-1466034.1 gsaaccgaagUfCfAfcaaguccuuL96 2634 asdAsagdGadCuugudGaCfuucggucsasu
2752 AD-1466039.1 ususcuacccUfUfCfugaaucauguL96 2635
asdCsudGadTucagdAaGfgguagaasusa 2753 AD-1466040.1 csasucuccuAfCfUfcucaauguL96
2636 asdAsudAudTgagadGuAfggagaugsasa 2754 AD-1466043.1
ususuaauccAfCfUfgguauaguL96 2637 asdAscudAudAaccadGuGfgauuaasusu 2755 AD-
1466044.1 usuaauccaCfUfGfguuauaguguL96 2638 asdCsacdTadTaaccdAgUfggauuaasasu 2756
AD-1466046.1 asusgguacaGfAfUfuacauugaguL96 2639 asdCsudAadTguaadTcUfguaccauscsu
2757 AD-1466047.1 ascsugauguUfAfGfgacaaacauL96 2640
asdAsugdTudTgucddTaAfaucagususu 2758 AD-1466050.1 usgsaagacuCfUfGfaugauauucuL96
2641 asdGsaadTadTcaudAgAfgucuucasasu 2759 AD-1466051.1
gsusaugaagAfGfCfaucucggaauL96 2642 asdTsudCgdAgaugdCuCfuucacuacsusc 2760 AD-
1466052.1 asgsagcaucUfCfGfgaauucuuguL96 2643 asdCsaadGadAuuccdGaGfaugcucususc 2761
AD-1466055.1 asasuucuugGfUfCfcuauuaucauL96 2644 asdTsgadTadAuaggdAcCfaagaauuscsc
2762 AD-1466058.1 usgsaaguggAfUfGfauguauuccuL96 2645
asdGsgadTadAcaudAuCfcacuucasgsc 2763 AD-1466059.1 gsasaguggaUfGfAfuguauuccauL96
2646 asdTsggdAudAcaudCaUfccacuucsasg 2764 AD-1466060.1
asuscagaggGfAfAfagacuuauguL96 2647 asdCsudAadGucuudTcCfcucugausgsa 2765 AD-
1466062.1 asgsccaaauAfGfCfaguauuaccuL96 2648 asdGsgudAudAacugdCuAfuuuggcusgsa 2766
AD-1466063.1 asgscaguuaUfAfCfcuacguaguL96 2649 asdCsudAcdGuaggdTaUfaacugcusasu
2767 AD-1466070.1 gsasgagaauUfUfGfucuuacuauuL96 2650
asdAsudGudAagacdAaAfuucucucsasu 2768 AD-1466081.1 gsasaaagaaGfAfGfcugguacuauL96
2651 asdTsgdTadCcagcdTcUfucuuuucsasu 2769 AD-1466083.1
asasagaagaGfCfUfgguacuauL96 2652 asdCsudAgdTaccadGcUfcuucuuususc 2770 AD-
1466085.1 asgsaagagcUfGfGfuacuauaauL96 2653 asdTsudAudAguacdCaGfcucuucususu 2771
AD-1466089.1 gsasgcugguAfCfUfaugaaaagauL96 2654 asdTscudTudTcaudGuAfccagcucusu
2772 AD-1466091.1 gscsugguacUfAfUfgaaaagaaguL96 2655
asdCsudCudTuucadTaGfuaccagcsusc 2773 AD-1466095.1 ususucacgcCfAfUfuaaugggauL96
2656 asdAsudCcdAuuaadTgGfcgugaaascu 2774 AD-1466096.1
asusuaauggGfAfUfgaucuacaguL96 2657 asdCsugdTadGaucadTcCfcauuaausgsg 2775 AD-
1466097.1 gscsucclaaGfAfCfauucacguguL96 2658 asdCsacdGudGaaugdTcUfugggagcscsg 2776
AD-1466100.1 asusgcaaacGfCfCfauuucuuauuL96 2659 asdAsudAgdAaagdGcGfuugcauscsc
2777 AD-1466101.1 gscsaaacgcCfAfUfuucuuaucauL96 2660
asdTsgdTadAgaadTgGfcguuugcsasu 2778 AD-1466104.1 usasagcacuGfGfUfaucuaucuuL96
2661 asdAsgdTadTgauadCcAfgugcuuasgsu 2779 AD-1466108.1
asasacaugGfUfGfgaucuauauL96 2662 asdTsaadAadGaucddAcCfauuguuusasa 2780 AD-
1466109.1 csasauggugGfAfUfcuuaaanguL96 2663 asdCsudTadTaagdTcCfaccuugcsusu 2781
AD-1466110.1 gsgsuggaucUfUfAfuaaugcuuguL96 2664 asdCsaadGcdAuuaudAaGfauccaccsas
2782 AD-1466111.1 asuscuuaaAfUfGfcuuggaguguL96 2665
asdCsacdTcdCaagcdAuUfuaaagauscsc 2783 AD-1466115.1 usasuaccacAfGfAfguucuauguL96
2666 asdAscadTadGaacudCuGfugguauasgsc 2784 AD-1466116.1
usasccacagAfGfUfucuauguaguL96 2667 asdCsudCadTagaadCuCfugugguasusa 2785 AD-
1466118.1 csascagaguUfCfUfauguagcuuuL96 2668 asdAsagdCudAcaudGaAfcucugugsgsu 2786
AD-1466120.1 asgsuucuaUfUfAfgcuuacaguL96 2669 asdAscudGudAagcdAcAfuagaacuscu
2787 AD-1466126.1 uscsaguugAfCfCfcaccuauuguL96 2670
asdCsaadTadGguggdGuCfaaacugasusu 2788 AD-1466127.1 csasguugaCfCfCfaccuauuguL96
2671 asdAscadAudAggugdGgUfcaaacugsasu 2789 AD-1466128.1
csusauugugGfCfUfagauauauuuL96 2672 asdAsaudAudAucudGcCfacaauagsgsu 2790 AD-
1466131.1 asgscaaaucAfCfAfgcuucucguL96 2673 asdCsgadAgdAagcdGuGfauuugcsusg 2791
AD-1466133.1 asasaugauCfUfAfcuagaucuuL96 2674 asdGsaudCudTgagudAgAfucauuuscu
2792 AD-1466139.1 uscsaugguGfGfAfcaaguuuuuL96 2675

asdAsaadAudCuugudCcAfcauggasgsg 2793 AD-1466140.1 cscsauggugGfAfCfaagauuuuuuL96
 2676 asdAsaadAadTcuugdTcCfaccauggsasg 2794 AD-1466141.1
 csasuggugGfAfAfagauuuuuuguL96 2677 asdCsaadAadAucuudGuCfcaccaugsgsa 2795 AD-
 1466142.1 asusggugGfAfAfagauuuuuuguL96 2678 asdTscadAadAaucudTgUfccaccausgsg 2796
 AD-1466143.1 usgsguggacAfAfGfauuuuugaauL96 2679 asdTsucdAadAaaucdTuGfuccaccasusg
 2797 AD-1466144.1 gsgsuggacaAfGfAfuuuuugaaguL96 2680
 asdCsuudCadAaaaadCuUfguccaccsas 2798 AD-1466145.1 gsusggacaaGfAfUfuuuugaagguL96
 2681 asdCscudTcdAaaaadTcUfuguccacscsa 2799 AD-1466152.1
 asgsauuuuuGfAfAffggaaauacuL96 2682 asdAsgudAudTuccudTcAfaaaaucususg 2800 AD-
 1466153.1 gsasuuuuugAfAfGfgaaauacuauL96 2683 asdTsgdTadTuuccdTuCfaaaaaucsusu 2801
 AD-1466157.1 ususugaaggAfAfAfuaacuaauacuL96 2684 asdGsuadTudAguadTuCfcuucaaasasa
 2802 AD-1466158.1 ususgaaggaAfAfUfacuaauaccuL96 2685
 asdGsgudAudTaguadTuUfccuucaaasasa 2803 AD-1466160.1 csusaauaccAfAfAfggacauguguL96
 2686 asdCsacdAudGuccudTuGfguauuagsusa 2804 AD-1466161.1
 usasauaccaAfAfGfgacaugugauL96 2687 asdTscadCadTguccdTuUfgguauuasgsu 2805 AD-
 1466162.1 csasaucauuUfCfCfagguuuuacuL96 2688 asdGsaudAadAccugdGaAfaugauugsgsg 2806
 AD-1466163.1 asuscauuucCfAfGfguuuauccguL96 2689 asdCsggdAudAaaccdTgGfaaauaususg
 2807

TABLE-US-00020 TABLE 20 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1395757.1
 csasggu(Chd)cuCfAfCfuuuuauccsusa 2808 VPusdAsggdAudTaaagdTgdAgdGaccugscsg 2811 AD-
 1395758.1 csasggu(Chd)cuCfaCfUfuuuuauccsusa 2809 VPusdAsggdAudTaaagdTgdAgdGaccugscsg
 2811 AD-1395763.1 csascuu(Uhd)aaUfCfCfucuauccasgsa 2810
 VPusdCsugdGadTagagdGadTudAaagugsasg 2812 AD-1395756.1
 csasggu(Chd)cuCfAfCfuuuuauccsusa 2808 VPusdAsggdAudTaaagdTgAfggaccugscsg 2813 AD-
 1395762.1 csascuu(Uhd)aaUfCfCfucuauccasgsa 2810 VPusdCsugdGadTagagdGaUfuuaagugsasg
 2814

TABLE-US-00021 TABLE 21 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1331279.1
 asasuguucaCfAfAfuaaagcuccuL96 2815 asdGsgadGcdTuauadTgUfgaacausus 2839 AD-
 1331280.1 asusuugcuaUfGfUfuagacgauguL96 2816 asdCsaudCgdTcuaadCaUfagcaaauscsu 2840
 AD-1331281.1 ususgcuaugUfUfAfgacgauguaaL96 2817 usdTscadAudCgucudAaCfauagcaasasu
 2841 AD-1331282.1 usgscuauguUfAfGfacgauguaaaL96 2818
 usdTsuadCadTcgucdTcAfcauagcasasa 2842 AD-1331283.1 asascugagaAfGfAfacuacauauaL96
 2819 usdAsuadTgdTaguudCuUfcucaguuscsc 2843 AD-1331284.1
 asascaacaGfCfAfuaagucuaaL96 2820 usdAsuudTgdAcuaudGcUfguugguususa 2844 AD-
 1331285.1 cscscacagaAfAfUfuucucuaucuL96 2821 asdGsaudAgdAgaaadTuUfcugugggsusu 2845
 AD-1331286.1 csasgguaguCfCfAfuggacauuaaL96 2822 usdTsaadTgdTccaudGgAfcuaccugsasu
 2846 AD-1331287.1 gsgsuaguccAfUfGfgacauuaauL96 2823
 asdAsuudAadTguccdAuGfgacuaccsusg 2847 AD-1331288.1 asgsuuggaaGfAfCfuggaaagacaL96
 2824 usdGsudTudTccagdTcUfuccaacuscsa 2848 AD-1331289.1
 usgsgaaagaCfAfAfcaaacaauauL96 2825 asdTsaadTgdTuugudTgUfcuuuccasgsu 2849 AD-
 1331290.1 ususuacuugGfGfAfaaucacgaaaL96 2826 usdTsucdGudGauuudCcCfaaguaaasasa 2850
 AD-1331291.1 gsgsgaaucAfCfGfaaaccaacuL96 2827 usdAsgudTgdGuuudGuGfauuuccsasa
 2851 AD-1331292.1 gsasaaucacGfAfAfaccaacuauaL96 2828
 usdAsuadGudTgguudTcGfugauuucscsc 2852 AD-1331293.1 csgsaaaccaAfCfUfauacgcuacaL96
 2829 usdGsuadGcdGuauadGuUfgguuucgsusg 2853 AD-1331294.1
 asuscaaccaAfAfAfuguugaaccaL96 2830 usdGsgadTcdAcaudTuUfgguugausus 2854 AD-
 1331295.1 ususaaaacuCfUfAfaacuugacuaL96 2831 usdAsgudCadAguuudTgAfguuuaascsa 2855
 AD-1331296.1 csasaaacuGfAfAfagccuccuauL96 2832 asdTsgdTadGgcuudTcAfaguuuugsasg

2856 AD-1331297.1 uscsaaacGfAfAfugauggauL96 2833
 asdAsucdCadTcuaudTcGfauguugasasu 2857 AD-1331298.1 csasaaacuuCfAfAfugaaacguguL96
 2834 asdCsacdGudTucadTgAfaguuuugsusg 2858 AD-1331299.1
 asasucacgaAfAfCfcaacuauacuL96 2835 asdGsuadTadGuuggdTUfUfcgugauususc 2859 AD-
 1331300.1 gsgsgaaucAfUfUfuuagaugguuL96 2836 asdAsccdAudCuaaadAuUfgauucccsasc 2860
 AD-1331301.1 csasaaauguUfGfAfuccaaucauL96 2837 asdTsugdGadTggaudCaAfcauuuugsusu
 2861 AD-1331302.1 usgsgacauuAfAfUfucaacaucgaL96 2838
 usdCsgadTgdTugaadTuAfauguccasusg 2862 AD-1331328.1 asasuguucaCfAfAfuaaagcuccuL96
 2815 asdGsgadGcdTuauadTgdTgdAacauususu 2863 AD-1331329.1
 asusuugcuaUfGfUfuagacgauguL96 2816 asdCsaudCgdTcuaadCadTadGcaaauscsu 2864 AD-
 1331330.1 ususgcuaugUfUfAfgacgauguaaL96 2817 usdTsaadAudCgucudAadCadTagcaasasu 2865
 AD-1331306.1 usgscuauguUfAfGfacgauguaaL96 2818 usdTsuadCadTcgucdTadAcdAuagcasasa
 2866 AD-1331331.1 asascugagaAfGfAfacuacauauaL96 2819
 usdAsuadTgdTaguudCudTcdTcaguuscsc 2867 AD-1331332.1 asascaacaGfCfAfuauguaaL96
 2820 usdAsuudTgdAcuaudGcdTgdTugguususa 2868 AD-1331333.1
 cscscacagaAfAfUfuucucuaucuL96 2821 asdGsaudAgdAgaadTudTcdTgugggsusu 2869 AD-
 1331334.1 csasgguaguCfCfAfuggacauuaaL96 2822 usdTsaadTgdTccaudGgdAcdTaccugsasu 2870
 AD-1331311.1 gsgsuaguccAfUfGfgacauuaauuL96 2823 asdAsuudAadTguccdAudGgdAcuaccsusg
 2871 AD-1331335.1 asgsuuggaaGfAfCfuggaaagacaL96 2824
 usdGsucdTudTccagdTcdTudCcaacuscsa 2872 AD-1331336.1 usgsgaaagaCfAfAfcaaacauuauL96
 2825 asdTsaadTgdTuugudTgdTcdTuuccasgsu 2873 AD-1331314.1
 ususuacuugGfGfAfaucacgaaaL96 2826 usdTsucdTudGauuudCcdCadAguaaasasa 2874 AD-
 1331337.1 gsgsgaaucAfCfGfaaaccaacuL96 2827 usdAsgudTgdGuuudCudGadTuucccsasa 2875
 AD-1331316.1 gsasaaucacGfAfAfaccaacuauaL96 2828 usdAsuadGudTgguudTcdGudGauuucscsc
 2876 AD-1331338.1 csgsaaaccaAfCfUfauacgcuaL96 2829
 usdGsuadGcdGuauadGudTgdGuuucgsusg 2877 AD-1331339.1 asuscaaccaAfAfAfuguugaucuaL96
 2830 usdGsgadTcdAcaudTudTgdGuugaususu 2878 AD-1331340.1
 ususaaaacuCfUfAfaacuugacuL96 2831 usdAsgudCadAguuudTgdAgdTuuuaascsa 2879 AD-
 1331320.1 csasaaacuuGfAfAfagccuccuauL96 2832 asdTsgdTadGgcuudTcdAadGuuuugsasg 2880
 AD-1331341.1 uscsaacacGfAfAfuaugaugauuL96 2833 asdAsucdTadTcuaudTcdGadTguugasasu
 2881 AD-1331322.1 csasaaacuuCfAfAfugaaacguguL96 2834
 asdCsacdGudTucadTgdAadGuuuugsusg 2882 AD-1331342.1 asasucacgaAfAfCfcaacuauacuL96
 2835 asdGsuadTadGuuggdTudTcdGugauususc 2883 AD-1331343.1
 gsgsgaaucAfUfUfuuagaugguuL96 2836 asdAsccdAudCuaaadAudTgdAuucccsasc 2884 AD-
 1331325.1 csasaaauguUfGfAfuccaaucauL96 2837 asdTsgdTadTggaudCadAcdAuuuugsusu 2885
 AD-1331344.1 usgsgacauuAfAfUfucaacaucgaL96 2838 usdCsgadTgdTugaadTudAadTguccasusg
 2886

TABLE-US-00022 TABLE 22 Exemplary dsRNA molecules SEQ ISEQ ID ID Duplex ID
 Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-1554875
 gscscugugaGfGfAfcuccaagaguL96 2887 asdCsucdTudGgagudCcUfcacaggcsusu 3013 AD-
 1554909 gsgsugcuacUfCfUfgguauuuccuL96 2888 asdGsgadAadTaccadGaGfuagcaccscsc 3014
 AD-1554910 gsusgcuaCuCfUfGfguauuuccuL96 2889 asdAsggdAadAuaccdAgAfguagcaccsc
 3015 AD-1554911 usgscuacucUfGfGfuauuuccuauL96 2890 asdTsgdTadAauacdCaGfaguagcaccsc
 3016 AD-1554912 gscsuacucuGfGfUfauuuccuaguL96 2891
 asdCsuaadGgdAaaudCcAfgaguagcsasc 3017 AD-1554913 csusacucugGfUfAfuuuccuagguL96
 2892 asdCscudAgdGaaudAcCfagaguagcsa 3018 AD-1554914
 usascucuggUfAfUfuuccuagguuL96 2893 asdCsccdTadGgaaadTaCfcagaguasgsc 3019 AD-1554915
 ascsucugguAfUfUfuccuagguuL96 2894 asdAsccdCudAggaadAuAfccagagusasg 3020 AD-
 1554916 csuscugguaUfUfUfccuagguuauL96 2895 asdTsaadCcdTaggadAaUfaccagagsusa 3021
 AD-1554917 uscsugguauUfUfCfcuagguacuL96 2896 asdGsuadCcdCuaggdAaAfuaccagagsu

3022 AD-1554951 gsgsucagccAfGfGfuguacucaguL96 2897 asdCsgcdCudTguacdCcUfaggaausasc
3023 AD-1554951 gsgsucagccAfGfGfuguacucaguL96 2898
asdCsugdAgdTacacdCuGfgcugaccsas 3024 AD-1554955 asgscagguGfUfAfcucaggcaguL96 2899
asdCsugdCcdTgagudAcAfccuggcusgsa 3025 AD-1554992 gscscacuucUfCfCfcaggaucuuuL96
2900 asdAsagdAudCcuggdGaGfaaguggcsgsa 3026 AD-1554997
ususucccaGfGfAfucuuacccguL96 2901 asdCsggdGudAagaudCcUfgggagaasgsu 3027 AD-
1555000 uscsccaggaUfCfUfuacccgccguL96 2902 asdCsggdCgdGguaadGaUfccugggagsgsa 3028
AD-1555030 gscscuuccgCfAfGfugaaaccgcuL96 2903 asdGscgdGudTucacdTgCfggaggcsasc 3029
AD-1555106 csasacuccaGfCfUfccgucuuuuL96 2904 asdAsaudAgdAcggadGcUfggaguugsusa
3030 AD-1555112 csasgcuccgUfCfUfauuccuuuguL96 2905
asdCsaadAgdGaauadGaCfggagcugsgsa 3031 AD-1555114 csuscaccugCfUfUfcuucugguuuL96
2906 asdAsacdCadGaagadAgCfaggugagsgsg 3032 AD-1555115
uscsaccugcUfUfCfuucugguucuL96 2907 asdGsaadCcdAgaagdAaGfcaggugagsgsg 3033 AD-
1555117 ascscugcuUfUfUfcugguucauuL96 2908 asdAsugdAadCagadAgAfagcaggusgsa 3034
AD-1555118 cscsugcuucUfUfCfugguucauuuL96 2909 asdAsaudGadAccagdAaGfaagcaggsusg
3035 AD-1555120 usgscuucuuCfUfGfuguucauuuL96 2910
asdAsgadAudGaaccdAgAfagaagcasgsg 3036 AD-1555121 gscsuucuuUfGfGfuucauuucuuL96
2911 asdGsagdAadTgaacdCaGfaagaagcsasg 3037 AD-1555122 csusucuucUfGfUfucauucuccuL96
2912 asdGsgadGadAugaadCcAfgaagaagscsa 3038 AD-1555123
ususcuucugGfUfUfcauucuccauL96 2913 asdTsggdAgdAaugadAcCfagaagaasgsc 3039 AD-
1555128 csusguucaUfUfCfuccaaauccuL96 2914 asdGsgadTudTggagdAaUfgaaccagsasa 3040
AD-1555184 ascsgggccGfAfGfuaagaguguL96 2915 asdCsacdTudCguacdTcGfgcccugusagsg 3041
AD-1555185 csasgggcccGfGfUfacgaagugguL96 2916 asdCscadCudTcguadCuCfggcccugsusa
3042 AD-1555212 cscsagugugAfAfAfgacauagcuuL96 2917 asdAsgcdTadTgucudTuCfacacuggscsu
3043 AD-1555213 csasgugugaAfAfGfacauagcuguL96 2918
asdCsagdCudAugudTuUfcacacugsgsc 3044 AD-1555234 asusugaauUfCfAfcgucggguuuL96
2919 asdAsacdCcdAgcgudGgAfauucaausgsc 3045 AD-1555235
ususgaauucCfAfCfugcugguuguL96 2920 asdCsaadCcdCagcgdTgGfaauucaasusg 3046 AD-
1555236 usgsaauccAfCfGfcugguuguuL96 2921 asdAscadAcCdCagcdGuGfgaauucasasu 3047
AD-1555238 asasuuccacGfCfUfgguuguuauL96 2922 asdTsaadCadAcccadGcGfuggaauuscsa 3048
AD-1555241 uscsacgcUfGfGfuguuaccguL96 2923 asdCsggdTadAcaacdCcAfgcguggasasu
3049 AD-1555242 cscsacgcugGfGfUfuguuaccguL96 2924
asdGscgdGudAacaadCcCfagcguggsasa 3050 AD-1555243 csascgcuggGfUfUfguuaccgcuL96
2925 asdAsgcdGgdTaacadAcCfcagcgugsgsa 3051 AD-1555247
csusggguugUfUfAfccgcuacaguL96 2926 asdCsugdTadGcgudAaCfaaccagsgscsg 3052 AD-
1555342 gsgsgaccgaCfUfGfgccauguauuL96 2927 asdAsuadCadTggccdAgUfcgguccsgsg 3053
AD-1555343 gsgsaccgacUfGfGfccauguauguL96 2928 asdCsaudAcCdAuggcdCaGfucgguccscsg
3054 AD-1555345 ascscgacugGfCfCfauguaugacuL96 2929
asdGsucdAudAcaugdGcCfagucgguscsc 3055 AD-1555346 cscsgacuggCfCfAfuguaugacguL96
2930 asdCsugdCadTacaadGgCfcagucggsusc 3056 AD-1555348
gsascuggccAfUfGfuaugacguguL96 2931 asdCsacdGudCauacdAuGfgccagucsgsg 3057 AD-
1555349 ascsggccaUfGfUfaugacgugguL96 2932 asdCscadCgdTcauadCaUfggcccaguscsg 3058
AD-1555350 csusggccauGfUfAfugacguggcuL96 2933 asdGscddAcCdGucaudAcAfuggccagsusc
3059 AD-1555366 asgsgcucaUfAfCfcucgguguaL96 2934 asdTsaadAcCdCgaggdTgAfugagccuscsc
3060 AD-1555428 gscscugcacAfGfCfuacuacgacuL96 2935
asdGsucdGudAguagdCuGfugcaggcscsc 3061 AD-1555429 cscsugcacaGfCfUfacuacgaccuL96
2936 asdGsgudCgdTaguadGcUfgugcaggscsc 3062 AD-1555535
cscsucucugGfAfCfuacggcuuguL96 2937 asdCsaadGcdCguagdTcCfagagaggsgsc 3063 AD-1555537
uscsucuggaCfUfAfcggcuuggcuL96 2938 asdGscddAadGccgudAgUfccagagagsgsg 3064 AD-
1555546 usascggcuUfGfGfcccucugguuuL96 2939 asdAsacdCadGagggdCcAfagccguasgsu 3065

AD-1555547 ascsggcuugGfCfCfucugguuuuL96 2940 asdAsaadCcdAgaggdGcCfaagccgusag
3066 AD-1555548 csgsgcuuggCfCfCfucugguuuuL96 2941
asdCsaadAcdCagagdGgCfcaagccgsusa 3067 AD-1555549 gsgscuuggcCfCfUfcugguuuugauL96
2942 asdTscadAadCagadGgGfccaagccsgsu 3068 AD-1555581
gsasggaggcAfGfAfaguaugauuuL96 2943 asdAsaudCadTacuudCuGfccuccucsasg 3069 AD-
1555583 gsgsaggcagAfAfGfuaugauuuuL96 2944 asdCsaadAudCauacdTuCfugccuccsusc 3070
AD-1555584 gsasggcagaAfGfUfaugauuuugcuL96 2945 asdGscadAadTcauadCuUfcugccucscsu
3071 AD-1555585 asgsgcagaaGfUfAfugauuuugccuL96 2946
asdGsgcdAadAucaudAcUfucugccuscsc 3072 AD-1555586 gsgscagaagUfAfUfgauuuugccguL96
2947 asdCsggdCadAaucadTaCfuucugccsusc 3073 AD-1555587
gscsagaaguAfUfGfauuuugccguuL96 2948 asdAscgdGcdAaaucdAuAfcuucugcscsu 3074 AD-
1555588 csasgaaguaUfGfAfuuugccguguL96 2949 asdCsacdGgdCaaudCaUfacuucugscsc 3075
AD-1555589 asgsaaguauGfAfUfuugccgugcuL96 2950 asdGscadCgdGcaaadTcAfuacuucusgsc
3076 AD-1555590 gsasaguaugAfUfUfugccgugcauL96 2951
asdTsgcdAcdGgcaadAuCfauacuucsusg 3077 AD-1555615 csasguggacGfAfUfccagaacaguL96 2952
asdCsugdTudCuggadTcGfuccacugsgsc 3078 AD-1555616 asgsuggacGfAfUfcagaacagguL96 2953
asdCscudGudTcuggdAuCfguccacusgsg 3079 AD-1555626 cscsagaacaGfGfAfggcuguguguL96
2954 asdCsacdAcdAgccudCcUfguucuggsasu 3080 AD-1555628
asgsaacaggAfGfGfcuguguggcuL96 2955 asdGscddAcdAcagcdCuCfcuguucusgsg 3081 AD-
1555706 usgsugcgggUfGfCfacuauggcuuL96 2956 asdAsgcdCadTagugdCaCfccgcacascsc 3082
AD-1555707 gsusgcggguGfCfAfcuauggcuuuL96 2957 asdAsagdCcdAuagudGcAfcccgcacsasc
3083 AD-1555709 gscsgggugcAfCfUfauggcuuuL96 2958
asdAscadAgdCcauadGuGfcacccgcsasc 3084 AD-1555711 gsgsgugcacUfAfUfggcuuguacuL96
2959 asdGsuadCadAgccadTaGfugcaccscgsc 3085 AD-1555717 ascsuaggcUfUfGfuacaaccaguL96
2960 asdCsugdGudTguacdAaGfccauagusgsc 3086 AD-1555723
gscsuuguacAfAfCfcagucggacuL96 2961 asdGsucdCgdAcuggdTuGfuacaagcscsa 3087 AD-
1555725 csusgcccugGfAfGfaguuccucuuL96 2962 asdAsgadGgdAacudTcCfagggcagsgsg 3088
AD-1555768 gscscuggauGfAfGfagaaacugcuL96 2963 asdGscadGudTucudTcAfuccaggcscsg 3089
AD-1555771 usgsaugagAfGfAfaacugcguuuL96 2964 asdAsacdGcdAguuudCuCfucauccasgsg
3090 AD-1555772 gsgsaugagaGfAfAfacugcguuuuL96 2965
asdAsaadCgdCaguudTcUfcucauccsasg 3091 AD-1555776 gsasgagaaaCfUfGfcguuugcaguL96 2966
asdCsugdCadAacgcdAgUfuucucucsasu 3092 AD-1555789 ususgcagaGfCfCfacauuccaguL96
2967 asdCsugdGadAugugdGcUfcugcaaascsg 3093 AD-1555894
gsusgggacaUfUfCfaccuuccaguL96 2968 asdCsugdGadAggugdAaUfguccacsasu 3094 AD-
1555895 usgsggacauUfCfAfccuuccaguL96 2969 asdAscudGgdAaggudGaAfuguccascsa 3095
AD-1555897 gsgsacauucAfCfCfuuccaguguuL96 2970 asdAscadCudGgaagdGuGfaauguccscsa
3096 AD-1555898 gsascauucAcCfCfUfuccaguguguL96 2971 asdCsacdAcdTggaadGgUfgaaugucscsc
3097 AD-1555899 ascsauucacCfUfUfccagugugauL96 2972 asdTscadCadCuggadAgGfugaauguscsc
3098 AD-1555900 csasuucaccUfUfCfcagugugaguL96 2973
asdCsucdAcdAcuggdAaGfgugaugsusc 3099 AD-1556052 asuscgcugaCfCfGfcugggugauuL96
2974 asdAsucdAcdCcagcdGgUfcagcgausgsa 3100 AD-1556057
usgsaccgcUfGfGfugauaacaguL96 2975 asdCsugdTudAucacdCcAfgcggucasgsc 3101 AD-
1556126 csgsuguuccUfGfGfgcaagguguuL96 2976 asdAscadCcdTugccdCaGfgaacacgsgsu 3102
AD-1556127 gsusguuccuGfGfGfcaagguguguL96 2977 asdCsacdAcdCuugcdCcAfggaacacsgsg
3103 AD-1556137 gscsaaggugUfGfGfcagaacucguL96 2978 asdCsgadGudTcugcdCaCfaccuugcscsc
3104 AD-1556139 asasggugugGfCfAfgaacucgcguL96 2979
asdCsgcdGadGuucudGcCfacaccuusgsc 3105 AD-1556163 csusggagagGfUfGfuccuuaaguL96
2980 asdCsuudGadAggacdAcCfucuccagsgsc 3106 AD-1556164
usgsgagaggUfGfUfccuuaagguL96 2981 asdCscudTgdAaggadCaCfcucuccasgsg 3107 AD-
1556166 gsasgaggugUfCfCfuucaagguguL96 2982 asdCsacdCudTgaagdGaCfaccucucscsa 3108

AD-1556167 asgsagguguCfCfUfcaagguguL96 2983 asdTugadGgAfcaccuscsc 3109
AD-1556319 asuscccacaGfGfAfccugugcaguL96 2984 asdCsugdCadCaggudCcUfgugggauscsa
3110 AD-1556359 usgsacgcccaCfGfCfaugcuguguL96 2985
asdAscadCadGcaugdCgUfggcgucascsc 3111 AD-1556360 gsascgccacGfCfAfugcuguguguL96
2986 asdCsacdAcdAgcaudGcGfuggcgucsasc 3112 AD-1556382
gscsuaccgcAfAfGfggcaagaaguL96 2987 asdCsuudCudTgcccTuGfcgguagcscsg 3113 AD-1556383
csusaccgcaAfGfGfgcaagaaguL96 2988 asdCscudTcdTugccdCuUfgcgguagcscsc 3114 AD-1556465
gsgscuaacUfAfCfuucggcgucuL96 2989 asdGsacdGcdCgaagdTaGfuuaggccsgsg 3115 AD-
1556466 gscscuaacuAfCfUfucggcgucuuL96 2990 asdAsgadCgdCgaadGuAfguuaggcscsg 3116
AD-1556484 csusacacccGfCfAfucacagguguL96 2991 asdCsacdCudGugaudGcGfgguguagsasc
3117 AD-1556510 gscsuggaucCfAfGfcaagugguguL96 2992
asdCsacdCadCuugcdTgGfauccagcsusg 3118 AD-1556584 usgsgcaggaGfGfUfggcaucuuguL96
2993 asdCsaadGadTgccadCcUfccugccascsc 3119 AD-1556585 gsgscaggagGfUfGfgcaucuuguL96
2994 asdAscadAgdAugccdAcCfuccugccsasc 3120 AD-1556586
gscsaggaggUfGfGfcaucuugucuL96 2995 asdGsacdAadGaugcdCaCfcuccugcscsa 3121 AD-
1556587 csasggagguGfGfCfaucuugucuuL96 2996 asdAsgadCadAgaugdCcAfccuccugscsc 3122
AD-1556613 usgsaugucuGfCfUfccagugauguL96 2997 asdCsaudCadCuggadGcAfgacaucasgsg
3123 AD-1556677 csasauucucUfCfUfccuccgucuuL96 2998
asdGsgadCgdGaggadGaGfagaauugsgsg 3124 AD-1556709 gsgscucagcAfGfCfaagaauugcuL96
2999 asdAsgcdAudTcuugdCuGfcugagccsasc 3125 AD-1556710
gscsucagcaGfCfAfagaauugcuguL96 3000 asdCsagdCadTucuudGcUfgcugagcscsa 3126 AD-1556789
csusggucuaAfCfUfugggaucuguL96 3001 asdCsagdAudCccaadGuUfagaccagsgsg 3127 AD-
1556790 usgsgucuaaCfUfUfgggaucugguL96 3002 asdCscadGadTcccadAgUfuagaccasgsg 3128
AD-1556791 gsgsucuaacUfUfGfggaucugguL96 3003 asdCscddAgdAucccdAaGfuuagaccsasg
3129 AD-1556795 usasacuuggGfAfUfcugggaauuguL96 3004 asdCsaudTcdCcagadTcCfcaaguuasgsa
3130 AD-1556799 ususgggaucUfGfGfgaauggauguL96 3005
asdCsuudCcdAuuccdCaGfaucceaasgsu 3131 AD-1556802 gsgsaucuggGfAfAfuggaagguguL96
3006 asdCsacdCudTccaudTcCfcagaucscsa 3132 AD-1556908 usgsagcucaGfCfUfgcccuuugguL96
3007 asdCscadAadGggcadGcUfgagcucascsc 3133 AD-1556909
gsasgcucagCfUfGfcccuuuggauL96 3008 asdTscddAadAgggcdAgCfugagcucsasc 3134 AD-
1556911 gscsucagcuGfCfCfcuuuggaauuL96 3009 asdAsuudCcdAaaggdGcAfgcugagcsusc 3135
AD-1556915 asgscugcccUfUfUfggaauaauguL96 3010 asdCsuudTadTuccadAaGfggcagcsgsa 3136
AD-1556917 csusgcccUfGfGfaauaaagcuuL96 3011 asdAsgcdTudTauudCaAfagggcagcsu 3137
AD-1556918 usgscuuuuGfGfAfauaaagcuguL96 3012 asdCsagdCudTuauudCcAfaagggcagsgsc
3138

TABLE-US-00023 TABLE 23 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
Sense sequence (5'.fwdarw.3') NO: Antisense sequence(5'.fwdarw.3') NO: AD-
ascsgagacAfaGfAfCfcaucuacacuL96 3139 asdGsugdCadGauggucUfudGucugusgsc 3153 1181513.1
AD- cscsgagcCfGfUfCfucuaaauuuL96 128 asdAsaudTgdTagagaaCfGdGcucggsasu 3154
1181527.1 AD- gsasugccAfaGfAfAfcacuaugauuL96 3140 asdAsucdAudAguguucUfudGgcaucscsu
3155 1181564.1 AD- ascsgagadAaaUfUfcuacuacauuL96 3141
asdAsugdCadGuagaauUfuCfucugusgsg 3156 1181612.1 AD- gsasaauuUfuCfUfacaucuauuL96
3142 asUfsaudAgdAuguadGudAgdAauuucsusc 3157 1181618.1 AD-
csgsagccGfuUfCfUfcuacaauuuauL96 3143 asUfsaadTudGuagagadAcdGgcucgsgsc 3158 1181537.1
AD- csasagccUfuGfGfCfucaauaccuL96 3144 asUfsggdTadTugagccdAadGgcuugsgsc 3159
1181552.1 AD- asgsaaauUfcUfAfCfuacaucuauuL96 84
asAfsuadGadTguagdCadGaAfuuucscsu 3160 1181601.1 AD- gsasgaaaUfuCfUfdAcuacaucuauL96
3145 asUfsagdAudGuagudAgdAaUfuucscsusg 3161 1181614.1 AD-
asgsauccdGagCfCfuacuauaauL96 3146 asUfsucdAudAguaggcUfcdGgaucususc 3162 1181524.1
AD- cscsgagcCfGfUfCfucuaaauuuL96 128 asAfsaudTgdTagagaaCfGdGcucggsasu 3163

1181528.1 AD- cscsgagcCfGfUfCfucuaaauL96 128 asAfsaudTgdTagagaaCfGdGcucggsgsc
3164 1181529.1 AD- asgsagaaAfuUfCfUfacuauauL96 3147
asAfsagdTadTaguagaAfuUfucucusgsu 3165 1181598.1 AD- csasagguCfuUfCfUfcucuggcguguL96
3148 asCfsagdCcdAgagagaAfgAfccuugsgsc 3166 1181604.1 AD-
cscsuacadGagAfAfauucuacuauL96 3149 asUfsagdTadGaauuucUfcUfguaggscsu 3167 1181609.1
AD- ascsagagAfaAfUfUfcuacuacauL96 3150 asAfsugdTadGuagaauUfuCfucugusgsg 3168
1181611.1 AD- gsasaauuCfuAfCfUfacaucuaauL96 3151 asUfsaudAgdAuguagaAfgAfauuucsusc
3169 1181617.1 AD- ascscuaCfuCfUfdGuuguucgaauL96 3152
asUfsucdGadAcaacagAfgUfagggusgsg 3170 1181619.1 AD- csasagguCfuUfCfUfcucuggcguguL96
3148 asCfsagdCcdAgagagaAfgAfccuugsgsc 3166 1181604.2
TABLE-US-00024 TABLE 24 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-
csusuaaaAfgGfGfAfcagauuucuaL96 3171 usdAsgadAudAcuguccCfuUfuuaagscsc 3179 1143243.1
AD- gscsuuaaAfaGfGfGfacagauuucuL96 3172 asdGsaadTadCuguccCfuUfuuaagcsasa 3180
1143252.1 AD- gscsuuaaAfaGfGfGfacagauuucuL96 3172 asdGsaadTadCuguccCfuUfuuaagcsasa 3180
1143252.1 AD- gscsuuaaAfaGfGfGfacagauuucuL96 3172 asdGsaadTadCuguccCfuUfuuaagcsasa 3180
3181 1143254.1 AD- csusuaaaAfgGfGfAfcagauuucuaL96 3173
asdAsgadAudAcuguccCfuUfuuaagscsc 3182 1143278.1 AD- usasaaAfgGfGfAfcagauuucuaL96
3174 usdAsgadAudAcuguccCfuUfuuaasag 3183 1143245.1 AD- usasaaAfaGfGfGfacagauuucuL96
3175 asdGsaadTadCuguccCfuUfuuaasgsc 3184 1143256.1 AD- usasaaAfgGfGfAfcagauuucuaL96
3176 asdAsgadAudAcuguccCfuUfuuaasag 3185 1143280.1 AD-
asasaaggGfaCfAfGfuauucucaguL96 3177 asCfsugdAgdAauacugUfcCfcuuuusasa 3186 1143286.1
AD- asasaaggGfaCfAfGfuauucucaguL96 3177 asCfsugdAgdAauacugUfcCfcuuuusgsc 3187
1143287.1 AD- asasaaggGfaCfAfGfuauucucaguL96 3177 asCfsugdAgdAauacugUfcCfcuuuusgsc 3187
3188 1143288.1 AD- asasggGfaCfAfGfuauucucaguL96 3178 asCfsugdAgdAauacugUfcCfcuususu
3189 1143289.1
TABLE-US-00025 TABLE 25 Exemplary dsRNA molecules SEQ SEQ ID ID Duplex ID
Sense sequence (5'.fwdarw.3') NO: Antisense sequence (5'.fwdarw.3') NO: AD-
csusuc(Uhd)gGfuCfUfAfaacaaacusasa 3190 VPusUfsagdTudTguuuagAfcCfagaagsasu 3212
1231468 AD- usasaau(Chd)UfcAfCfAfuagaaucusa 3191
VPusAfsagdTudAcuauguGfaGfauuaasag 3213 1231470 AD-
usasacu(Uhd)UfaAfUfCfucuaauagscsa 3192 VPusGfscudAudAugagauUfaAfaguuasasc 3214
1231471 AD- cscsuga(Uhd)CfuUfCfUfgguuuuaascsa 3193
VPusGfsuudTadAaccagaAfgAfucaggsasa 3215 1231472 AD-
usgsua(Chd)aGfuAfAfGfugauaacasgsa 3194 VPusCfsugdTudAucacuuAfcUfguacasag 3216
1231473 AD- ususcgugGfuAfUfUfcuua(Chd)uagsusa 3195
VPusAfscudAgdTaagaauAfcCfacgaasag 3217 1231478 AD-
uscsuuc(Uhd)GfgUfCfUfaaaugaacsusa 3196 VPusAfsugdTcdAuuuagaCfcAfgaagasusc 3218
1231479 AD- uscgug(Chd)UfaCfAfAfcuuucucasasa 3197
VPusUfsugdAgdAaaguugUfaGfcacgasusu 3219 1231485 AD
gsusugugUfaCfAfCfacau(Uhd)ggusasa 3198 VPusUfsacdCadAugugugUfaCfacaacsasu 3220
1231492 AD- csgsugc(Uhd)AfcAfAfCfuucuuaasgsa 3199
VPusCfsuudGadAgaaguGfuAfgcacgsasu 3221 1231496 AD-
usgsaaa(Uhd)UfgUfUfGfacauugugsasa 3200 VPusUfscadCadAugucaCfaAfuuucasgsc 3222
1231497 AD- ascsaau(Chd)uGfCfUfuuuuaaacsusa 3201
VPusAfsugdTudAaaaagcAfaCfauugusasa 3223 1231498 AD-
asasaugUfuGfAfCfaca(Uhd)gagsusa 3202 VPusAfscudCadTagugucAfaCfaauuuscsa 3224
1231499 AD- gscsugaaAfuUfGfUfugaua(Chd)ugsusa 3203
VPusAfscadGudAucaacaAfuUfucagcsag 3225 1231502 AD-
usgsuugaCfaCfUfGfugaa(Uhd)gcusasa 3204 VPusAfsagdCadTucacagUfgUfcaacasasu 3226
1231505 AD- ascsauugCfcAfAfAfggu(Uhd)ucusasa 3205

VPusUfsagdAadAccuuuuGfgCfaaugusug 3227 1231507 AD-
 ususcug(Chd)UfaAfUfCfuuguugcusasa 3206 VPusUfsagdCadAcaagauUfaGfcagaasgsc 3228
 1231513 AD- asuscuugUfuUfUfCfucua(Uhd)ucasasa 3207
 VPusUfsugdAadTagagaaAfaCfaagausgsa 3229 1231519 AD-
 csasgucaUfaAfUfCfuaua(Uhd)uaasasa 3208 VPusUfsuudAadTauagauUfaUfgacugsug 3230
 1231520 AD- usgsguaaCfaCfUfAfauaa(Uhd)aaasasa 3209
 VPusUfsuudTadTuauuagUfgUfuaccascsa 3231 1231521 AD- cscsucauGfgAfAfGfagaagcagauL96
 3210 asUfscudGcdTucucuuCfcAfugaggscsu 3232 1523180.1 AD-
 cscsggu(Chd)AfgGfUfUfcugcuuuusasa 3211 VPusUfsaadAadGcagaacCfudGaccggscsc 3233
 1498528

[0446] All of the U.S. patents, U.S. patent application publications, foreign patents, foreign patent applications and non-patent publications referred to in this specification are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0447] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. A dsRNA agent comprising a sense strand and antisense, each strand independently having a length of 15-35 nucleotides wherein each nucleotide is independently modified or unmodified, wherein the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand, and wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand.
2. The dsRNA agent of claim 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand.
3. (canceled)
4. The dsRNA of claim 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5-end of the sense strand.
5. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9 or positions 11 and 12, counting from 5-end of the sense strand.
6. (canceled)
7. The dsRNA agent of claim 1, wherein the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.
8. The dsRNA of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.
9. The dsRNA agent of claim 1, wherein the sense strand comprises at least one 2'-OMe nucleotide.
10. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.
11. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense

strand.

12. (canceled)

13. (canceled)

14. The dsRNA agent of claim 1, wherein the antisense strand comprises a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

15. (canceled)

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

20. (canceled)

21. (canceled)

22. (canceled)

23. (canceled)

24. (canceled)

25. (canceled)

26. The dsRNA agent of claim 1, wherein the dsRNA agent comprises a ligand.

27. The dsRNA agent of claim 1, wherein the sense strand comprises a ligand.

28. The dsRNA agent of claim 27, wherein the ligand is at 3'-end of the sense strand.

29. The dsRNA agent of claim 27, wherein the ligand is at 5'-end of the sense strand.

30. The dsRNA agent of claim 27, wherein the ligand comprises an ASGPR ligand.

31. The dsRNA agent of claim 27, wherein the ligand is lipophilic group.

32. The dsRNA agent of claim 31, wherein the ligand is a C.sub.10-30aliphatic group.

33. The dsRNA agent of claim 32, wherein the C.sub.10-30aliphatic group is a C.sub.10-30alkyl group.

34. The dsRNA agent of claim 33, wherein the C.sub.10-30alkyl group is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group.

35. The dsRNA agent of claim 23, wherein the ligand is conjugated to a non-terminal nucleotide of the sense strand.

36. The dsRNA agent of claim 35, wherein the ligand is conjugated to the 2'-position of a non-terminal nucleotide of the sense strand, optionally conjugated to one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5' end).

37. The dsRNA agent of claim 26, wherein the ligand comprises an abasic nucleotide, optionally the abasic nucleotide is an inverted nucleotide and linked via a 5'->5' or a 3'->3' linkage to a strand of the dsRNA agent.

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)

42. (canceled)

43. (canceled)

44. The dsRNA agent of claim 1, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand.

45. The dsRNA agent of claim 1, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand.

46. The dsRNA agent of claim 1, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs.

47. (canceled)

48. (canceled)

49. A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand, wherein: the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand.

50. The dsRNA agent of claim 49, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand.

51. The dsRNA agent of claim 1, comprising a phosphate mimic at the 5'-end of the antisense strand.

52. The dsRNA agent of claim 51, wherein the phosphate mimic is a 5'-E-vinyl phosphonate.

53. (canceled)

54. The dsRNA agent of claim 49, wherein remaining nucleotides in the sense strand are unmodified nucleotides or modified nucleotides, optionally selected from the groups consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C.sub.10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C.sub.10-30aliphatic group; and the remaining nucleotides in the antisense strand are unmodified nucleotides or modified nucleotides, optionally selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

55. The dsRNA agent of claim 54, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C.sub.10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C.sub.10-30aliphatic group; and the remaining nucleotides in the antisense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA, and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

56. (canceled)

57. (canceled)
